DIPLOMATIC WORKING DOCUMENT

The Provisional Steps Framework

Phase I Stabilization Framework — Version 1.4

A Calculated, Flexible Foundation for US-Iran Peace, Nuclear Stabilization, and Regional Maritime Governance

Calibrated for the Late May 2026 Negotiation Environment

AUTHORIZED NEGOTIATORS ONLY — CONFIDENTIAL

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QUICK REFERENCE — CRISIS CONTACTS

Instructions: Contact numbers to be designated per Article 28 before Framework activation. No placeholders permitted. See Crisis Contact Registry on Dashboard.

IRGC-Navy Hotline: ______________________ (tested daily 06:00 UTC | incident response: 15 min)

Swiss Facilitator (24h): ______________________ | Dashboard: ______________________

Omani Backchannel (secure): ______________________ | Political Consultation Channel

JIG Duty Officer (24h): ______________________ | ______________________

Emergency Bulletin: Published within 24h of Tier 3 event | Dashboard: ______________________

Designated per Article 28 | Updated per Crisis Contact Registry | Last updated: __________

Appendices

Page-Accurate Index

All page numbers reflect the actual pagination of this document

Preamble: The Last Window

"We do not inherit the earth from our ancestors; we borrow it from our children." In the same spirit, the negotiators who sign this Framework do not inherit the right to let this moment pass.

As of late May 2026, the United States and the Islamic Republic of Iran stand at the closest point to either durable peace or catastrophic escalation since hostilities began. After nearly twelve weeks of military confrontation, both sides have exhausted the military escalation ladder and returned to the negotiating table. The temporary ceasefire, brokered by Pakistan and observed since early April, remains technically in force but is fraying. President Trump has stated he will give diplomacy "a few days" or "one shot." Supreme Leader Khamenei has declared that Iran will protect its nuclear and missile capabilities at all costs. The world holds its breath.

From Version 4.0 to the Provisional Steps Framework v1.0

The Provisional Steps Framework represents a fundamental philosophical shift from its predecessor, The Final Steps Accord v4.0. Where v4.0 prioritized automated, airtight mechanisms designed to eliminate human discretion, v1.0 introduces **calculated political flexibility** — the very mechanisms designed to make v4.0 "airtight" were the ones paralyzing leadership in both capitals.

The following targeted structural modifications transform the document from an uncompromising, automated machine into a flexible diplomatic tool — one that protects both leaders from political ambush at home while maintaining the core stabilization objectives:

Table 1: v4.0 to v1.0 Structural Modifications Summary

Area	v4.0 Mechanism	v1.0 Reform	Location
Lebanon Firewall	10-casualty hair-trigger with JIG majority vote	Three-tier escalation ladder (10/25/50) + 72-hour diplomatic consult window	Appendix E
Legislative Ratification	Hard 180-day deadline with punitive damages	Rolling Executive Option: 90-day extensions via decree	Appendix D

Day 45 Nuclear Default	Automatic financial freeze	In-Iran downblending default (material stays); Optional Oman Escrow by mutual agreement; clarified Day 75 timeline	Article 1.6
Sanctions Relief Pacing	15-day micro-cycles with asymmetric front-loading	30-Day Stabilization Blocks with Mutual Escrow Deposit	Article 7
Telemetry Protocol	Real-time continuous streaming to Western-accessible hubs	Batch-Verification and Air-Gapped protocol	Parts I & II
MMMF Deadlock Default	Automatic 40% payout to Iran	Revenue frozen in escrow until consensus budget signed	Article 5.2
Day-7 Blockade Lift	Conditional Joint Review Board certification	Unconditional automated 24-hour countdown trigger	Article 6
SWIFT Compliance Rolls	Cascading tier rollbacks for entire banking sector	Targeted Bank Isolation through pre-approved clearing hubs	Article 9
Reconstruction Credits	Iran funds its own reconstruction with frozen assets	US Dollar-for-Dollar Matching Guarantee via DFC	Article 10
Automated Penalties	Technical teams auto-trigger Tier 2 benefit reductions	Mandatory 48-hour Executive Consultation Window	Article 13

Phase I Stabilization Framework — Operative Preamble Language

This document is the **Provisional Steps Framework Phase I Stabilization Accord** — an initial, time-limited framework designed to end active hostilities and establish verified baselines. Phase II (Days 60–180) and Phase III (Year 2+) shall address deeper nuclear, missile, and regional issues through successor frameworks. This Framework creates no presumption that Phase I outcomes will dictate Phase II terms. Both parties retain full negotiating positions for subsequent phases.

This document draws upon the Final Steps Accord v4.0, the Crescent Bridge Accord, the Stability First Enhanced Accord v5.7, the Reciprocal Opening Window, the Reciprocal Stabilization and Custody Framework v3, the Graduated Reciprocal Assurance Framework, and the Stability First Phase II Maritime Annex — integrating their most viable mechanisms into a single, coherent, executable whole, now recalibrated with the political flexibility necessary for signature.

This Framework fully incorporates the reality of Iran's newly established Persian Gulf Strait Authority (PGSA), founded on 5 May 2026, not by treating it as an obstacle to be dismantled but as an institutional fact to be integrated into a cooperative governance framework. The PGSA represents Iran's assertion of maritime sovereignty; this Framework provides the mechanism through which that assertion becomes compatible with freedom of navigation, international law, and global economic stability.

The philosophy governing every provision is **Synchronized Reciprocal Motion**: neither side moves alone. Every Iranian commitment triggers a corresponding US commitment on the same clock cycle. Verification is shared. Both governments can defend the sequencing as responding only when the other has moved first or simultaneously. When the negotiation becomes about data instead of history — about sensor readings, escrow confirmations, and physical lock verifications — both sides can verify compliance without trusting each other.

No Precedent Clause (Operative and Binding)

> **PREAMBLE NO-PRECEDENT PROVISION (V1.1 OPERATIVE ADDITION)**
>
> This Framework creates **no legal or political precedent** for future negotiations. Each provision is *sui generis*, tailored to the May 2026 negotiating environment and the specific circumstances giving rise to this Framework. No inference may be drawn from any provision regarding positions on any matter not expressly addressed herein. Either Party may assert in any domestic, international, or diplomatic forum that this Framework has no precedential value. This clause is self-executing and shall be cited by both Parties whenever defending the Framework against claims that it establishes binding precedent for unrelated matters.

No Regime Change Objective (Operative and Binding)

> **PREAMBLE NO-REGIME-CHANGE PROVISION (V1.2 OPERATIVE ADDITION)**
>
> The United States affirms that this Framework is exclusively a **stabilization and verification mechanism** designed to prevent nuclear escalation and establish maritime security. It is **not**, and shall not be interpreted as, a precursor to, component of, or facilitation mechanism for any regime change strategy, internal political transformation, or governmental transition in the Islamic Republic of Iran. Neither the

Framework nor any implementation mechanism under it — including verification access, telemetry protocols, escrow arrangements, or diplomatic channels — may be used to gather intelligence for non-Framework purposes, support domestic opposition movements, or facilitate internal political disruption.

(a) Non-Interference Binding Commitment: The United States commits that no Party, guarantor state, or Framework institution shall use Framework access, data, or mechanisms for purposes unrelated to the specific verification and stabilization objectives defined herein. This includes: intelligence collection beyond stated verification requirements; contact with domestic political groups under cover of verification activities; and economic pressure designed to precipitate governmental instability rather than address specific Framework compliance concerns.

(b) Verification-Only Access: All IAEA and JIG inspections, telemetry access, and facility visits are strictly limited to the nuclear and maritime verification objectives of this Framework. Inspectors shall not photograph, document, or report on non-nuclear facilities, military installations unrelated to the Framework, or civilian infrastructure absent specific Framework authorization. Any inspection activity exceeding its authorized scope may be terminated by the host Party immediately, with no penalty to the host.

(c) No Precondition for Broader Transformation: This Framework does not require, presuppose, or anticipate any specific form of Iranian domestic political arrangement, economic system, or social structure as a condition of continued operation. The Framework's Rolling Horizon Architecture depends solely on verified compliance with the specific technical and operational commitments defined herein — not on any assessment of Iran's internal political character.

(d) Good-Faith Interpretation: Both Parties shall interpret all Framework provisions in good faith as addressing only the specific nuclear, maritime, and economic stabilization objectives defined in the operative Articles. Neither Party may invoke ambiguous language to justify actions outside the Framework's stated scope. Disputes over scope interpretation are resolved through the standard JIG dispute process under Article 13.

Why This Matters: Without this provision, Iranian hardliners will inevitably interpret the Framework as a stabilization-first, strangulation-second strategy designed to weaken Iran internally before applying decisive pressure. This provision makes explicit what the architecture already implies: the Framework is bounded, technical, and scope-limited. It survives only if both sides believe it addresses only what it says it addresses.

Domestic Framing Autonomy (Operative and Binding)

PREAMBLE DOMESTIC FRAMING AUTONOMY PROVISION (V1.2 OPERATIVE ADDITION)

Each Party retains **complete and sole authority** over how it describes, explains, frames, and presents this Framework to its domestic audience, legislature, media, and political institutions. Neither Party may require the other to use specific terminology, acknowledge specific characterizations, or adopt specific narrative frames.

(a) Prohibited Required Terminology: Neither Party shall require the other to use terms including but not limited to: "concession," "recognition," "normalization," "capitulation," "surrender," "defeat," "victory," "diplomatic breakthrough," "historic reconciliation," or "strategic partnership." Each Party describes the Framework using terminology consistent with its domestic political requirements.

(b) Domestic Description Freedom: The United States may describe the Framework as it sees fit to Congress, the American public, and allied governments — including framing it as a non-proliferation achievement, a security measure, or an economic stabilization tool. The Islamic Republic of Iran may describe the Framework as it sees fit to the Majlis, the Iranian public, and regional partners — including framing it as a sovereignty-preserving technical arrangement, a hostile-power management mechanism, or an economic necessity. Neither description invalidates the other.

(c) No Mutual Recognition Implication: Participation in this Framework creates no implication of diplomatic recognition, political endorsement, or strategic alignment beyond the specific technical obligations defined herein. The Framework is a transactional stabilization mechanism, not a relationship-defining treaty. Both Parties may continue all existing bilateral and multilateral relationships without restriction.

(d) Joint Statements Limited: Joint public statements by both Parties shall use only mutually agreed **technical language** — describing specific operational achievements ("the IAEA verified X," "escrow release Y was completed") — and shall avoid political characterization. All joint statements require advance textual approval by both Parties. Either Party may unilaterally decline to participate in any joint statement without penalty.

(e) Internal Documentation Protection: Neither Party may leak, publish, or reference the other Party's internal framing documents, talking points, or legislative presentations. Internal political messaging is protected as confidential diplomatic communication under Article 16.

> **Why This Matters:** Political survival in both capitals depends on the ability to sell the same agreement using different language. If the US President must call it a "non-proliferation victory" while the Iranian President calls it a "sovereignty-preserving technical arrangement," both are correct within their domestic contexts. Forcing either leader to adopt the other's framing creates a political vulnerability that could kill the deal at home.

ONE-PAGE EXECUTIVE SUMMARY

Orientation Only — Not Legally Binding — For Heads of Government and Senior Advisors

This Framework is a **180-day stabilization agreement** between the United States and the Islamic Republic of Iran. It does four things:

1. Secures Iran's nuclear material. All high-risk enriched uranium is placed under IAEA custody. The material stays in Iran by default. Twenty-three disposition pathways are available for Phase II negotiation. No title transfer. No foreign ownership.

2. Opens the Strait of Hormuz. The US naval blockade lifts on Day 7 after IAEA verification of HEU consolidation — but only after both capitals confirm through their defense ministers. Human confirmation, not automatic execution.

3. Releases frozen assets in 30-day blocks. Sanctions relief is tied to verified compliance, not political promises. A $500M mutual escrow and up to $5B annually in US dollar-for-dollar matching provide tangible economic incentives. A $50M Political Discretion Fund rewards partial compliance and goodwill.

4. Prevents accidental war. Direct military-to-military hotlines, mandatory consultation windows, cooling-off periods before escalatory actions, and a binding No Automatic Military Escalation Clause ensure that compliance disputes become diplomatic conversations, not casus belli.

Core Principles:

- **No one moves alone.** Every Iranian commitment triggers a simultaneous US commitment.
- **No automated penalties.** Every penalty requires human review at the executive level.

- **No permanent surrender.** Either party can withdraw with 30 days' notice and a capped exit payment.
- **No regime change objective.** This is stabilization, not strangulation.
- **No sunset clauses.** Rolling horizon architecture replaces fixed-term traps.
- **Partial success survives.** No single module failure collapses the entire Framework.
- **Hotline tested daily at 06:00 UTC.** Failure to answer is a Tier 1 dispute. Crisis contacts must be real, designated, and certified before activation (Article 28).

Leadership Protections: Domestic framing autonomy (describe the deal however you need to). Executive consultation veto over penalties. Political override of technical findings. Direct leader-to-leader channel through Oman. Narrative dispute resolution for framing conflicts. Institutional continuity through leadership transitions.

The remaining 200+ pages are the operational details. This summary is for orientation only. It is not legally binding and creates no interpretive obligations. For executable provisions, refer to the operative Articles, Appendices, and Nuclear Operative Annex.

"Trust is good. Verification is better. But a Framework that survives without either — that is architecture."

Executive Summary: The Sixteen Bridges

The Three Immutable Principles

1. **Telemetry-First Diplomacy:** Every commitment must be expressible as a sensor reading, a data feed, a physical lock confirmation, or a financial transaction record. If it cannot be verified by a neutral machine, it does not belong in the operational text.
2. **Rollback, Not Snapback:** If either party breaches, the Framework does not collapse. It pauses in graduated tiers: discretionary measures halt first, core constraints survive longest, and humanitarian protections remain in force regardless.
3. **Modular Architecture:** Failure in one module does not cascade. The compiler continues executing all other loaded modules. Maritime disputes cannot terminate

humanitarian channels. Nuclear verification gaps do not affect Hormuz transit. Lebanon incidents cannot collapse the core US-Iran agreement.

Executive Summary: The Sixteen Bridges

Table 2: All 16 Issues & Resolutions

Arena	Issue	Resolution Mechanism	Article
Nuclear	HEU Stockpile Disposition	IAEA Dual-Lock Custody Exchange; 14-day consolidation; legal title decoupled from physical custody; in-Iran custody default; in-Iran downblending default at Day 75; Optional Oman Escrow by mutual agreement; **twelve-pathway menu** including Regional LEU Bank, Joint Conversion Facility, Uranium-for-Isotopes Swap, Scientific Research Custody, Reconstruction Bond, and Sovereign Fuel Leasing Trust; limited US veto on Pathways D, F, K, and L only	1
	Enrichment Cap & Timeline	Sliding-Scale 12–15 Year Moratorium	2
	Underground Site Hardening	Optional civilian conversion; mothball protocol with air-gapped telemetry deployment	3
Maritime	PGSA vs. Freedom of Navigation	JHTA advisory governance with sovereignty recalibration; 12-month pilot phase; PGSA recognized as sole coastal enforcement within Iranian territorial waters	4
	Transit Security Fees	Multilateral Maritime Maintenance Fund with frozen-revenue deadlock protection	5
	US Naval Blockade	Unconditional automated Day-7 synchronized lift with 24-hour countdown trigger	6

Economic	Sanctions Relief Sequencing	30-Day Stabilization Block Model with Mutual Escrow Deposit; partial sanctions freeze; stand-off tied to HEU consolidation start	7
	Frozen Assets	Neutral Escrow under Sovereign Immunity Shielding	8
	SWIFT Reconnection	Phased Multi-Tier Virtual Consolidation with Targeted Bank Isolation hubs	9
	War Reparations	Regional Stability Reconstruction Credits with US Dollar-for-Dollar Matching Guarantee	10
Durability	Negotiation Window	Mutual Letter of Intent with Binding Timelines	11
	Institutional Guarantees	Distributed Guarantor Matrix (6 States)	12
	Internal Regime Fractures	Executive Consultation Governance with mandatory 48-hour consultation window	13
	Missile Reconstitution	Deployment Pause & Transfer Freeze	14
	Global Agrifood Crisis	Energy & Agricultural Release Valve	15
	Israeli Security Coordination	US-Israel Bilateral Coordination through Tel Aviv Channel; non-impediment clause	16

Architecture	Framework Duration	Rolling Horizon Architecture with conditional auto-renewal	17
	Military Escalation Firewall	Explicit decoupling of compliance penalties from use of force authorization; independent legal effect; severability	18
Cyber Protection	Critical Infrastructure protection; cryptographic hash chains; anti-spoofing; encryption standards (TLS 1.3, AES-256-GCM); sensor integrity; no backdoors; post-quantum readiness	19-D, 3.4, 19-E, 4.8, 3.5	
Spoiler & Cyber Resilience	Spoiler Actor Definition, Attribution Levels, and Remediation Safe Harbor	19	
	False Flag Attack Protocol — presumption, investigation, consequences	19-A	
	Spoiler Financing and Supply Chain Tracking; Proscribed Spoiler Organization designation	19-B	
	Information Warfare and Disinformation Protocol; WBI point system	19-C	
	Critical Infrastructure Cyber Protection; prohibited cyber operations; 90%+ attribution standard	19-D	
	Cyber Incident Response and Escalation; 4-level classification; no offensive retaliation without JIG authorization	19-E	

Part I

Nuclear Stabilization Architecture

The nuclear file represents both the origin of the current crisis and its most technically complex dimension. Three interconnected issues have created a deadlock: the disposition of Iran's highly enriched uranium stockpile, the duration and scope of any enrichment moratorium, and the fate of Iran's hardened underground facilities. This Part resolves all three through an integrated architecture that strips Iran of rapid breakout capability while preserving sovereign dignity and NPT rights.

Article 1: HEU Custody Exchange Protocol

Issue 1: The Nuclear Deadlock. President Trump demands the complete physical recovery and removal of Iran's roughly 900-pound (400 kg) highly enriched uranium (HEU) stockpile. Secretary Rubio maintains that Iran must physically surrender and hand over its HEU up front. Iran has communicated, as a binding sovereign position through its authorized negotiating delegation, that its enriched uranium must remain under Iranian legal title and within Iranian territory during the Phase 1 verification period. Iranian Foreign Ministry Spokesperson Esmail Baghaei has publicly framed the current text as a "framework agreement" meant to defer core nuclear details into a secondary 30-to-60 day window after baseline sanctions relief. The chasm between physical removal and domestic retention has collapsed every previous negotiation.

v1.0 Resolution: The IAEA Custody Exchange with Phase 1/Phase 2 Structural Decoupling

> **V1.0 SURGICAL CHANGE — DECOUPLED OWNERSHIP AND CUSTODY WITH EXTENDED TIMELINE, IN-IRAN DEFAULT, AND DAY 45 IAEA ESCROW**

TRANSITION

This solution structurally decouples legal ownership from physical custody. The Central Bank of Iran retains full sovereign legal title to the HEU stockpile at all times. Physical custody is transferred to a secure, IAEA-monitored facility as a baseline verification metric for Phase 1. The consolidation timeline is extended from 72 hours to 14 days, with in-Iran dual-lock custody as the default arrangement. Final disposition is deferred to the 30–60 day broader nuclear negotiating window. A new **Day 45 IAEA Escrow Transition** mechanism replaces the automatic financial freeze default, ensuring material security without triggering an immediate economic chokehold.

1.1 The Custody Architecture

ARTICLE 1.1 — HEU CONSOLIDATION AND CUSTODY (OPERATIVE)

The entire stockpile of uranium enriched to 60% U-235 and above — estimated at approximately 400 kg HEU equivalent — shall be consolidated, inventoried, and placed under IAEA custody within the first 14 days of Framework activation. From Day 1, Iran declares a freeze on all movement and enrichment of the stockpile. Two custody pathways are available:

Pathway A (In-Iran — Default): Dual-lock custody at the Esfahan Nuclear Technology Center or another IAEA-declared facility within Iranian territory, selected by mutual agreement. The IAEA holds exclusive digital encryption keys to all monitoring arrays, tamper-evident seal controls, and access verification systems. Iran holds physical perimeter security keys. Neither party can access the sealed containers without the other's physical presence. Pathway A shall be the default first step upon Framework activation.

Pathway B (Oman — Conditional): Physical custody transferred to IAEA custody facility in Muscat, Oman (not the United States) only if the Parties mutually agree or if the JIG certifies by unanimous vote that in-Iran custody cannot be adequately verified. The stockpile is transported in IAEA-certified secure containers under armed IAEA guard. Continuous CCTV monitoring covers the entire transport chain. Satellite surveillance tracks the transport vessels. The stockpile remains under IAEA dual-lock custody at the Muscat facility. **Oman is a neutral facilitator, not a US proxy. No US personnel may be present during transfer unless mutually agreed by both Parties in writing.**

Political Will Preservation Clause: If either Party determines that it cannot complete its Phase 1 consolidation obligations within the 14-day window due to verified administrative, technical, or logistical constraints (including but not limited to: IAEA inspector availability, transportation delays, or domestic holiday periods), it may request a one-time extension of up to 7 additional days through the JIG. Such request shall not be unreasonably denied. Delays attributable to this provision shall not constitute a material breach or trigger step-back.

Legal Title Structure (Binding for Both Pathways): The Central Bank of Iran retains full sovereign legal title to the HEU throughout the custody period. The IAEA holds physical custody only. The Custody Services Agreement specifies that: (a) Iran retains full legal title; (b) the IAEA may not transfer, sell, or dispose of the material without written consent of the Central Bank of Iran; (c) the material is held for the purpose of baseline verification and civilian conversion negotiation, not for destruction or permanent removal from Iranian ownership; (d) the final disposition is subject to the outcome of the 30–60 day nuclear negotiating window; (e) if the Framework is terminated by mutual agreement, the material is returned to Iran under IAEA supervision.

IAEA Custody Personnel Clarification (Pathway A — In-Iran): For purposes of Pathway A, "IAEA custody" means the IAEA's existing inspection and verification personnel already present in Iran under the Safeguards Agreement may perform custody verification functions. No additional IAEA personnel shall be required to enter Iran beyond the existing inspector cadre unless mutually agreed. The dual-lock system shall be installed and maintained by Iranian technicians under IAEA technical supervision. No foreign military or security personnel shall enter Iranian territory for custody purposes. This provision directly addresses the Iranian concern that "IAEA custody" might require foreign intrusion on Iranian soil.

> **PREAMBLE — NARRATIVE DISPUTE RESOLUTION PROCESS (V1.2 HUMAN-IN-THE-LOOP ADDITION)**
>
> **(a) When Framing Becomes Misrepresentation:** The Domestic Framing Autonomy provision above protects each Party's right to describe the Framework using its own terminology. However, if a Party believes that the other Party's domestic framing has crossed from protected interpretation into **material misrepresentation** that: undermines public support for the Framework in a way that threatens compliance; creates confusion about what the Framework actually requires; or deliberately contradicts mutually agreed technical facts in a way that sabotages implementation — the complaining Party may request a **Narrative Dispute Resolution session**.
>
> **(b) Procedure:** The session shall be: facilitated by the Swiss mediator; include senior communications advisors from both Parties (not lawyers or negotiators — communications professionals); convened within 14 days of the request; held in private; and limited to 4 hours. The Swiss mediator may issue a **non-binding finding**

as to whether the challenged framing exceeds protected Domestic Framing Autonomy. The finding shall identify: which specific statements are protected interpretation; which, if any, cross into material misrepresentation; and what clarifying language would resolve the dispute.

(c) Clarifying Statement: If the mediator finds that framing exceeds protected autonomy, the violating Party shall issue a **clarifying statement** within 7 days. The statement need not retract the original framing — it need only clarify the mutually agreed technical facts. The clarifying statement shall be published on the Swiss Dashboard. No penalties attach to framing disputes. This is a **political process, not a legal one**.

(d) Good-Faith Standard: Both Parties shall approach Narrative Dispute Resolution in good faith. A pattern of frivolous complaints (three or more complaints in 180 days that the mediator finds entirely without merit) may be referred to the JIG as a potential gaming action under Article 13.6. Conversely, a pattern of material misrepresentation (three or more findings of excessive framing in 180 days) may be referred to the JIG as a potential bad-faith action.

Why This Matters: The Domestic Framing Autonomy provision creates necessary political space for both sides to sell the deal at home. But what happens when one side's messaging actively sabotages the deal? Without a resolution process, framing disputes fester into compliance disputes. Narrative Dispute Resolution creates a political (not legal) process for addressing genuine conflicts while preserving the autonomy principle. Political problems get political solutions.

Mutual Disappointment Acknowledgment (Non-Binding but Recommended)

PREAMBLE — MUTUAL DISAPPOINTMENT ACKNOWLEDGMENT (V1.2 POLITICAL SMOOTHING ADDITION)

The Parties acknowledge that **neither side has achieved everything it sought** in this Framework. Each side has made concessions that will be criticized at home. Each side has accepted provisions it would have preferred to reject. This is the nature of a negotiated compromise, not a dictated peace.

The Parties further acknowledge that **implementation will be imperfect**, that disputes will arise, and that the Framework's resilience mechanisms will be tested. The measure of this Framework is not whether it prevents every dispute — it is whether disputes, when they arise, are resolved without escalation to war. Both Parties accept

> this imperfect reality and commit to making the Framework work despite its imperfections.
>
> This acknowledgment is **non-binding and interpretive**. It creates no legal obligations, no enforcement mechanisms, and no dispute resolution procedures. Its sole purpose is to set realistic expectations, reduce the political cost of signing, and remind both Parties that compromise, however uncomfortable, is preferable to the alternative. Both Parties may cite this acknowledgment in domestic fora when defending the Framework against claims that it is one-sided or incomplete.

1.2 Phase 1 vs. Phase 2 Clarity

Phase 1 secures the material; Phase 2 decides its fate. This is the fundamental architecture: physical consolidation and verification occur in Phase 1, while the political decision on final disposition is deferred to the dedicated Phase 2 nuclear negotiating window.

Table 3: Phase 1 and Phase 2 Commitments

Phase	Timeline	Iranian Commitment	US Reciprocal	Final Disposition
Phase 1: Baseline Verification	Days 1–30	Physical consolidation, inventory, and IAEA custody activation (Pathway A default, or B if certified). Physical custody + mass spectrometry confirms isotopic composition. Dual-lock seals applied.	$250M humanitarian tranche; OFAC general license; 5th Fleet stand-off posture (triggered upon IAEA confirmation that HEU consolidation has begun); partial sanctions designation freeze	Baseline satisfies Phase 1 trigger. No final disposition required.
Phase 2: Nuclear Negotiating Window	Days 30–60	Participation in dedicated nuclear negotiating track to determine final HEU disposition from the **twenty-three-pathway menu**.	Tranche 2 release (25% of remaining frozen assets); petrochemical export licenses	Phase 2 negotiated outcome. Iran selects pathway. US verifies execution with limited veto on Pathways D, F, K, and L only.

US Narrative Frame: "Weapons-grade uranium secured — the entire Iranian stockpile of near-weapons-grade material has been placed under IAEA custody with US monitoring access within two weeks. Breakout time extended from weeks to over a year. Physical verification is verified and real."

Iranian Narrative Frame: "Iran's enriched uranium remains Iranian property on Iranian soil. The Central Bank of Iran retains sovereign title. IAEA custody is a temporary verification measure, not a transfer of sovereign control. Physical custody to IAEA is a verification step in a broader negotiation — not surrender. Final disposition will be decided in the nuclear negotiating window on terms that respect our NPT rights."

Article 1.3 — No Hidden US Custody Clause (Operative and Binding)

At no stage may the United States receive, store, or exercise direct physical custody over the Iranian enriched-uranium stockpile. All custody remains with the IAEA and, where transferred, the designated third-party facilitating state. However, US technical personnel may be present at IAEA custody facilities as observers with joint data access **only upon prior written consent of the Islamic Republic of Iran**, which shall not be unreasonably withheld. Consent once granted may be revoked with 14 days' notice. Unreasonable denial or revocation of consent shall be subject to JIG dispute resolution under Article 13.4. If the JIG determines that consent was unreasonably denied or revoked, the Swiss facilitator may mediate. If mediation fails, the matter is referred to the Guarantor Matrix. This provision ensures Iranian sovereignty over foreign presence on its soil while preserving US access for legitimate verification purposes. If the Parties mutually agree in Phase 2 (Days 30–60) to a disposition pathway requiring temporary US technical involvement, such involvement shall be governed by a separate implementing agreement. The blanket prohibition on US custody applies only to Phase 1 baseline verification.

US personnel shall not exercise operational control over consolidation, packaging, transit, or receipt operations during Phase 1. This clause is self-executing and non-waivable for Phase 1.

1.3 Verification Timeline

Table 4: Phase 1 Verification Timeline

Phase	Clock Time	Action	Verifying Party
T+0 (Day 1)	0000 GMT	Framework activates; Parties activate Day-One commitments; Iran declares freeze on HEU movement and enrichment	Joint Secretariat

T+12h (Day 1)	1200 GMT	IAEA inspectors arrive at declared facility; begin physical inventory	IAEA
T+24h (Day 2)	0000 GMT	Inventory phase 1 complete; mass spectrometry confirms isotopic composition	IAEA
T+7d	Day 7	Interim custody status report; Pathway A default activated or JIG certification for Pathway B	JIG + IAEA
T+14d	Day 14	Day-Fourteen deadline: dual-lock custody initiated and verified; consolidation complete	Joint Secretariat

1.4 Synchronized US Commitment

Simultaneously with Iranian custody activation, the United States shall: (1) Release a USD 250 million humanitarian tranche via Swiss escrow; (2) Issue an OFAC general license for humanitarian trade (medicine, food, medical equipment, aviation safety parts) for 90 days; (3) Transition US naval forces to a stand-off posture: no patrols within 5 nautical miles of the Hormuz transit corridor — triggered only upon IAEA confirmation that HEU consolidation has begun, not upon Framework activation alone; (4) Freeze all new nuclear-related sanctions designations for the duration of the Framework; existing non-nuclear sanctions (terrorism, human rights) remain enforceable; (5) Publish a formal statement acknowledging Iran's sovereign right to peaceful nuclear energy under NPT Article IV.

1.5 The Twenty-Three Pathways for Final Disposition (Phase 2, Days 30–60)

ARTICLE 1.5 — FINAL DISPOSITION PATHWAY SELECTION (V1.1 EXPANSION)

Within the 30–60 day nuclear negotiating window, Iran shall select one of **twenty-three** IAEA-verified disposition pathways. Iran shall select the final disposition pathway from the twenty-three-option menu. The United States may not veto the selection except for:

- (a) Pathway D (Russian Federation), which requires joint US-Iran concurrence;
- (b) Pathway F (hybrid/multiple destinations), which requires joint US-Iran concurrence on the specific destination countries;
- (c) Pathway K (Uranium-Backed Reconstruction Bond), which requires joint US-Iran concurrence on valuation and fund structure;
- (d) Pathway L (Sovereign Fuel Leasing Trust), which requires joint US-Iran concurrence on credit purchasers and drawdown schedule;

> - (e) Pathway M (Sovereign Trustee Export — Oman Trust), which requires joint US-Iran concurrence on trust charter terms and return conditions.
>
> For all other pathways, the United States has no veto — only the right to verify execution.

Table 5: The Twenty-Three Disposition Pathways

Path	Description	Destination/Timeline	US Veto
A	In-place downblending to medical-grade LEU at Esfahan	Medical isotope production in Iran; 142 days	No veto
B	HEU export to IAEA custody in Oman	IAEA facility, Oman (NOT US); 14 days	No veto
C	Downblended in Iran, transferred to IAEA LEU Bank	Oskemen, Kazakhstan; 75 days	No veto
D	Transfer to Rosatom under IAEA safeguards	Russian Federation; 60–90 days	Joint concurrence required
E	Conversion to reactor-grade LEU fuel assemblies	Bushehr reactor supply; 180 days	No veto
F	Hybrid: stockpile divided into batches through different pathways	Multiple destinations; concurrent	Joint concurrence on destinations required
G	Regional LEU Bank hosted in Iran — Iran becomes custodian of a neutral fuel bank for the region under IAEA supervision; 60% HEU downblended to LEU as initial capital	Regional LEU Bank, Esfahan; 210 days to operational	No veto
H	Joint Uranium Conversion Facility (Iran + Gulf states) — HEU downblended and fabricated into fuel for Gulf states' future civilian reactors; creates economic interdependence	Esfahan or dedicated new facility; 365 days to first fuel rods	No veto

I	Uranium-for-Isotopes Swap — HEU downblended to medical-grade LEU and fabricated into targets; Iran receives guaranteed long-term supply of medical isotopes from international partners	Isotope targets at Esfahan; isotope supply from EU/Japan/Canada; 120 days	No veto
J	Scientific "Custody for Research" — HEU placed under IAEA custody but made available for legitimate scientific research (neutron activation, nuclear forensics); Iran receives priority access and co-authorship on research	Research facility, Iran; indefinite with 5-year review	No veto
K	Uranium-Backed Reconstruction Bond — HEU valued as strategic asset ($5–10 billion equivalent); verified downblending releases funds from international reconstruction fund for Iran	Asset-backed escrow, Oman; valuation + phased release over 5 years	Joint concurrence required
L	Sovereign Fuel Leasing Trust — HEU capitalized into sovereign trust owned by Central Bank of Iran; trust issues "nuclear fuel credits" purchased by international partners; Iran receives immediate liquidity; graduated 10–15 year drawdown	Trust domiciled in Oman; credits issued to Gulf states, EU, China, Japan; 10–15 year drawdown	Joint concurrence required
M	Sovereign Trustee Export (Oman Trust) — HEU transferred to IAEA custody in Oman under sovereign trust; Iran retains legal title and beneficial ownership as trustee; immediate liquidity via trust certificates; return permitted only for peaceful downblending	IAEA-Omani triple-lock facility, Muscat; 14-day transport; indefinite trust with annual drawdown	Joint concurrence required
N	Reactor Fuel Fabrication and Export — HEU downblended to LEU, fabricated into certified fuel assemblies, exported to IAEA-safeguarded reactors worldwide. Iran becomes commercial nuclear fuel supplier.	In-Iran fabrication; export to global reactors; 24 months	No veto

O	Stable Isotope Production — HEU processed into stable isotopes for medical diagnostics, cancer therapy, semiconductor manufacturing. Iran becomes regional isotope supply hub.	In-Iran production; delivery to hospitals and manufacturers; 18 months	No veto
P	Betavoltaic Battery Manufacturing — HEU converted into long-life nuclear batteries for pacemakers, satellites, sensors. Cutting-edge technology export.	In-Iran manufacturing; global export; 30 months	No veto
Q	Fluorochemical Feedstock Production — HEU depleted and converted to UF_6 for HF, WF_6, $LiPF_6$ production. Iran becomes fluorochemical industry hub.	In-Iran production; export to semiconductor/battery industries; 20 months	No veto
R	Rare Earth Element Co-Production — HEU feedstock for rare earth extraction (Nd, Dy, Tb, Eu). Iran becomes strategic rare earth supplier.	In-Iran solvent extraction; export to EV/wind turbine manufacturers; 30 months	No veto
S	Uranium Silicide Fuel for Research Reactors — HEU downblended to U_3Si_2 fuel for global research reactor fleet. RERTR program participation.	In-Iran fabrication; export to ~240 research reactors; 22 months	No veto
T	Depleted Uranium for Industrial Armor/Counterweights — HEU depleted to 0.3% U-235, fabricated into DU metal for aircraft, shielding, ballast.	In-Iran production; industrial export; 16 months	No veto
U	Uranium-Carbon Nanotube Composites — Depleted UO_2 nanoparticles in CNT matrices for aerospace shielding, thermal management, EMI protection.	In-Iran nanomaterial synthesis; aerospace/medical export; 36 months	No veto
V	Uranium Catalysts for Petrochemicals — Depleted uranium catalysts for petroleum refining, plastics production, emissions reduction.	In-Iran catalyst manufacturing; regional refinery supply; 18 months	No veto

W	Uranium-Phosphate Fertilizer — HEU depleted to natural levels, combined with phosphate for agricultural fertilizer. Material becomes food.	In-Iran fertilizer production; domestic and regional agriculture; 14 months	No veto — irreversible

Article 1.5-A — Joint Disposition Fund (Operative)

To encourage selection of mutually acceptable disposition pathways, a Joint Disposition Fund is established with initial capitalization of $500 million. **The Fund is a stabilization incentive designed to accelerate safe civilian conversion and reduce breakout risk — not a payment or bribe. Fund Capitalization:** The Fund shall be capitalized by drawing 10% from Tranche 2 and 10% from Tranche 3 asset releases, with each Party's contribution capped at 50% of total Fund capitalization ($250 million each). Funds shall be held in a separate sub-account of the Swiss escrow, administered by the Swiss facilitator, and disbursed only upon IAEA verification of completed downblending or transfer under the selected pathway. If the selected pathway does not qualify for a disbursement (per the schedule below), the capitalized funds shall be returned proportionally to Tranche 2 and Tranche 3 upon completion of Phase I.

Fund Drawdown Schedule — Progress-Based, Not Completion-Based (v1.1 Fix):

The Joint Disposition Fund is a **$500 million drawdown facility**, not a one-time completion bonus. Iran may draw funds according to the schedule below **upon IAEA verification of pathway commencement**, not merely upon completion. This funds progress, not just endpoints.

- **$100 million:** Pathways B (Oman), C (Kazakhstan), E (Bushehr fuel), G (Regional LEU Bank), H (Joint Conversion Facility), I (Uranium-for-Isotopes Swap), **M (Sovereign Trustee Export)**, N (Reactor Fuel Export), O (Stable Isotopes), Q (Fluorochemicals), S (Silicide Fuel), and V (Petrochemical Catalysts) — 50% on commencement, 50% on completion;
- **$80 million:** Pathways P (Betavoltaic Batteries), R (Rare Earth Co-Production), and U (CNT Composites) — 50% on commencement, 50% on completion;
- **$75 million:** Pathways J (Scientific Custody for Research) and L (Sovereign Fuel Leasing Trust) — 50% on commencement, 50% on completion;
- **$60 million:** Pathway T (Depleted Uranium Industrial) — 50% on commencement, 50% on completion;
- **$50 million:** Pathway A (in-Iran downblending) — 50% on commencement, 50% on completion;
- **$40 million:** Pathway W (Uranium-Phosphate Fertilizer) — 50% on commencement, 50% on completion;

- **$150 million:** Pathway K (Uranium-Backed Reconstruction Bond) — released in three $50 million tranches tied to verified downblending milestones;
- **No drawdown:** Pathways D (Russia) and F (hybrid with Russian involvement).

Pathway Switching Protection: If Iran switches pathways mid-stream, drawn funds are credited against the new pathway's schedule, with no clawback. This rewards genuine effort and allows pivoting without penalty. The Fund creates a positive incentive for pathways that advance non-proliferation objectives while supporting Iran's civilian nuclear and economic interests — without imposing a veto on any pathway.

US Narrative: "We reward cooperation. Iran keeps full formal choice across twenty-three options, but has a $500 million reason to pick pathways that give everyone confidence. The new pathways create regional interdependence, scientific collaboration, and economic integration — making a return to enrichment politically costly for Iran's own partners." **Iranian Narrative:** "The Fund is a financial bonus for pathways that serve Iran's own economic interests. No pathway is blocked. Some are simply more lucrative. The twenty-three options include pathways that keep material in Iran, create jobs, establish Iran as a regional nuclear fuel hub, and monetize our asset without surrendering it."

Article 1.5-B — The Six New Pathways: Detailed Rationales (v1.1 Expansion: Pathways G–L)

> **PATHWAY G — REGIONAL LEU BANK HOSTED IN IRAN (OPERATIVE DETAIL)**
>
> **Concept:** Iran becomes the custodian of a neutral fuel bank for the region, under IAEA supervision. The 60% HEU is downblended to LEU and becomes the initial capital of the bank.
>
> **Why it works:** Establishes Iran as a trusted regional nuclear fuel supplier. Gulf states with future civilian reactor programs gain a local fuel source. The IAEA LEU Bank in Kazakhstan provides precedent — a non-nuclear weapon state hosting a neutral fuel reserve. Iran retains custodial prestige and economic benefit without retaining weapons-capable material.
>
> **Operational details:** The Regional LEU Bank shall be established at Esfahan under IAEA safeguards. LEU stock shall be available for purchase by any NPT-compliant state in the region. Iran receives a custodial fee (1.5% of fuel value annually) and first-right-of-refusal on any bank management contracts.
>
> **Narrative: US:** "Iran transitions from proliferation risk to regional fuel supplier. The LEU Bank is under IAEA lock and key." **Iran:** "We are the Switzerland of the Middle East — a neutral custodian of fuel for peace."

PATHWAY H — JOINT URANIUM CONVERSION FACILITY (OPERATIVE DETAIL)

Concept: Iran's HEU is downblended and fabricated into fuel for Gulf states' future civilian reactors. Creates economic interdependence between Iran and its neighbors.

Why it works: Similar to Russia's fuel take-back model, but inverted — Iran becomes the fuel fabricator, not the recipient. Gulf states (UAE, Saudi Arabia, Egypt) with announced nuclear programs need fuel. Iran has the infrastructure and expertise. A joint facility creates mutual economic dependence that makes conflict costly for all parties.

Operational details: Joint venture structure: Iran 45%, Gulf consortium 40%, IAEA Technical Support 15%. Facility located at a dedicated site selected by mutual agreement. All fuel fabricated under IAEA safeguards and sold only to IAEA-safeguarded reactors.

Narrative: US: "Iran's nuclear expertise is channeled into peaceful commerce with its neighbors. That's the best non-proliferation outcome possible." **Iran:** "Our nuclear capability becomes a source of regional leadership and economic partnership, not isolation."

PATHWAY I — URANIUM-FOR-ISOTOPES SWAP (OPERATIVE DETAIL)

Concept: Iran's HEU is downblended to medical-grade LEU and fabricated into targets. In exchange, Iran receives a guaranteed long-term supply of medical isotopes from international partners (EU, Japan, Canada).

Why it works: Similar to existing "cash-for-fuel" arrangements but structured as a barter. Iran's medical community currently relies on imported isotopes for cancer treatment and diagnostics. A guaranteed 20-year supply addresses a genuine humanitarian need. The isotope-producing partners (Belgium, Netherlands, Japan, Canada) gain a reliable LEU source. This is the most humanitarian of all twenty-three pathways.

Operational details: LEU targets fabricated at Esfahan under IAEA supervision. Isotope supply contracts guaranteed by the Joint Disposition Fund (up to $200 million in supply guarantees). IAEA monitors both sides of the swap.

Narrative: US: "Weapons-grade material becomes cancer treatments. That's a transformation worth supporting." **Iran:** "Our enriched uranium saves Iranian lives. Every gram downblended is a dose of medicine for our people."

PATHWAY J — SCIENTIFIC "CUSTODY FOR RESEARCH" (OPERATIVE DETAIL)

Concept: The HEU is placed under IAEA custody but made available for legitimate scientific research (neutron activation analysis, nuclear forensics training, materials science). Iran receives priority access and co-authorship on all research.

Why it works: The CERN model of shared international infrastructure demonstrates that sensitive facilities can serve global science while remaining secure. Iranian scientists gain access to cutting-edge research. The international scientific community gains access to unique research capabilities. The material remains under IAEA lock but produces knowledge, not waste.

Operational details: Research program managed by a Scientific Consortium (Iran 40%, IAEA 30%, international partners 30%). All research is unclassified and published in peer-reviewed journals. Iran receives: priority beam time; guaranteed co-authorship on all publications; training for 50 Iranian scientists annually; and shared patent rights on any commercializable discoveries.

Narrative: US: "Iranian scientists join the global research community. That's how you build lasting scientific norms." **Iran:** "Our nuclear program transforms into a center of global scientific excellence. Iranian names on world-class research."

PATHWAY K — URANIUM-BACKED RECONSTRUCTION BOND (OPERATIVE DETAIL)

Concept: The HEU is valued as a strategic asset ($5–10 billion equivalent). Its verified downblending and disposal releases funds from an international reconstruction fund for Iran — an asset-backed lending model for peace.

Why it works: Precedent exists in asset-backed lending and escrow-for-peace models. The HEU has intrinsic value as processed nuclear material. By "securitizing" it, Iran unlocks immediate reconstruction capital without surrendering the material upfront. Downblending occurs in phases, each phase releasing a corresponding tranche of reconstruction funds. This creates a direct, verifiable link between disarmament and development.

Operational details: Independent valuation by a neutral financial panel (World Bank, IMF, Omani Sovereign Wealth Fund). Reconstruction fund capitalized by: 40% unfrozen Iranian assets; 30% Gulf state contributions; 20% EU development finance; 10% US Development Finance Corporation guarantees. Funds released in 5 tranches, each tied to IAEA-verified downblinding milestones.

Narrative: US: "Every kilogram downblended becomes a hospital, a school, a power plant. That's a trade worth making." **Iran:** "Our nuclear asset becomes the foundation of our reconstruction. We capitalize our own future."

PATHWAY L — SOVEREIGN FUEL LEASING TRUST (OPERATIVE DETAIL)

Concept: Iran's 60% enriched uranium stockpile is not destroyed, not downblended, and not transferred out of Iran. Instead, it is deposited into a sovereign trust that leases its value over time — a financial innovation that preserves sovereignty while creating economic value.

How it works:

Table 5-A: Sovereign Fuel Leasing Trust — Six-Step Mechanism

Step	Action	Verification
1. Legal Re-characterization	The HEU is not "surrendered." It is capitalized. A sovereign trust is established, owned by the Central Bank of Iran, with the HEU as its initial capital asset.	Trust charter registered with IAEA and Swiss facilitator; legal opinion from all six guarantor states
2. IAEA Custody (Physical)	The HEU remains in Iran but under IAEA physical seal and continuous monitoring. Iran retains legal title and beneficial ownership.	Dual-lock custody activated; Dashboard monitoring live
3. Leasing the "Value"	The trust issues "nuclear fuel credits" or "isotope bonds" backed by the HEU asset. These credits are purchased by international partners (Gulf states, EU, China, Japan) for cash.	Credit issuance verified by Swiss financial cell; purchaser identities disclosed to both Parties
4. Cash Released to Iran	Iran receives immediate liquidity from the sale of credits, without the HEU moving or being destroyed.	Escrow release confirmed on Dashboard; funds deposited to Iranian Civilian Reconstruction Fund

5. Graduated Drawdown	Over 10–15 years, the trust releases the HEU for downblending or transfer in small, verified increments, each release triggered by a corresponding release of frozen assets or sanctions relief.	Annual IAEA verification of drawdown schedule; JIG certifies compliance with graduated plan
6. Final Residual	At the end of the trust period, the remaining HEU is downblended to medical-grade LEU or transferred to a third-party fuel bank. Iran has already received the economic value upfront.	Final IAEA inventory; residual material disposition by mutual agreement

Why it works: This is the most financially sophisticated of the twenty-three pathways. It transforms a liability (weapons-capable material) into an income-producing asset without requiring Iran to surrender the material immediately. The trust structure is legally robust, drawing on asset-backed securities law, sovereign wealth fund governance, and IAEA custody precedent. The 10–15 year drawdown gives Iran long-term economic visibility while the graduated structure provides the US with continuous verification leverage.

Narrative: US: "Iran's nuclear material is locked, monitored, and gradually converted — while Iran receives the economic benefit upfront. That's verification plus incentive in one structure." **Iran:** "We did not surrender our nuclear asset. We monetized it. The material remains Iranian property under IAEA seal, and we receive the development funds our people deserve."

Article 1.5-C — The Ten In-Iran Transformation Pathways (v1.4 Expansion: Pathways N–W)

GOVERNING PRINCIPLE: DON'T DESTROY THE HEU. TRANSFORM IT INTO SOMETHING VALUABLE THAT CANNOT BE QUICKLY TURNED BACK.

The In-Iran Transformation Philosophy: Pathways N through W represent a fundamental innovation in nuclear disposition strategy. Rather than exporting Iran's 60% HEU stockpile to foreign destinations (Pathways B–D), or downblending it in place (Pathway A), these pathways convert the HEU into commercially valuable products manufactured inside Iran. The material stays on Iranian soil. Iranian workers process it. Iranian companies sell the products. And yet — the transformation is designed so that reversal to weapons-usable material requires months or years of detectable industrial work, not a quick flip of a switch.

Shared Legal Framework (all ten pathways): (a) Central Bank of Iran retains legal title to all uranium and derived products until point of sale; (b) IAEA dual-lock custody throughout all processing stages within Iran; (c) reversal requires months of detectable work across multiple facilities; (d) economic benefit creates Iranian commercial stakeholders who have financial incentive to protect the pathway, not circumvent it; (e) all products verified as depleted to ≤19.75% U-235 (or ≤0.3% for depleted uranium products) before commercial export; (f) end-user tracking via IAEA database for all exported products containing uranium.

PATHWAY N — REACTOR FUEL FABRICATION AND EXPORT

What happens: HEU is downblended to LEU (≤19.75% U-235), fabricated into certified fuel assemblies at an IAEA-monitored facility in Iran, and exported to approved reactor operators worldwide. Iran becomes a commercially competitive nuclear fuel supplier.

Reversibility: Low. Fuel assemblies must be chemically reprocessed to recover U-235 — a months-long, detectable process. IAEA detection probability near 100%.

Economic benefit: Fuel fabrication fees ($2M–$5M per reactor reload); long-term supply contracts ($50M–$200M annually).

PATHWAY O — STABLE ISOTOPE PRODUCTION

What happens: HEU is downblended and processed into targets for isotope enrichment and separation. Output: stable isotopes for PET scan calibration, cancer therapy, semiconductor manufacturing. Iran becomes a regional hub for stable isotope supply.

Reversibility: Very Low. Stable isotopes are chemically distinct from uranium feedstock. Reversal requires extensive reprocessing and re-enrichment (12+ months).

Economic benefit: Isotope sales ($500K–$2M per batch); hospital supply contracts ($20M–$80M annually). Global market: $3–5B annually.

PATHWAY P — BETAVOLTAIC BATTERY MANUFACTURING

What happens: HEU is converted into betavoltaic batteries that power devices for 20–100 years without charging: cardiac pacemakers, deep-space satellites, undersea sensors. Cutting-edge nuclear battery technology.

Reversibility: Negligible. Radioisotopes are chemically bound in semiconductor substrates and sealed in tamper-proof housings. Recovery would require destroying batteries and dissolving semiconductors.

Economic benefit: Premium battery sales ($5K–$50K per unit); aerospace contracts ($30M–$150M annually).

PATHWAY Q — FLUOROCHEMICAL FEEDSTOCK PRODUCTION

What happens: HEU is depleted and converted to UF_6 for industrial fluorine extraction. Output: HF for semiconductor etching, WF_6 for chip fabrication, $LiPF_6$ for lithium-ion battery electrolytes. Iran becomes a regional fluorochemical hub.

Reversibility: Negligible. Uranium is depleted to ≤0.3% U-235 and chemically transformed into gaseous/liquid industrial products. Collection from dispersed users is logistically impossible.

Economic benefit: Fluorochemical sales ($500K–$2M per batch); semiconductor supply contracts ($40M–$120M annually).

PATHWAY R — RARE EARTH ELEMENT CO-PRODUCTION

What happens: HEU is used as processing feedstock for rare earth element extraction (neodymium, dysprosium, terbium, europium) — critical inputs for EV motors, wind turbines, MRI machines. Iran becomes a strategic rare earth supplier.

Reversibility: Negligible. Uranium is depleted to ≤0.3% U-235 and dispersed into leach solutions and tailings. Rare earth oxides contain no weapons-usable material.

Economic benefit: REE oxide sales ($50–$500/kg); supply contracts ($50M–$200M annually). Addresses Chinese monopoly.

PATHWAY S — URANIUM SILICIDE FUEL FOR RESEARCH REACTORS

What happens: HEU is downblended to ≤19.75% U-235 and fabricated into U_3Si_2 fuel plates for the global research reactor fleet (~240 reactors). Advances global HEU-to-LEU conversion (non-proliferation).

Reversibility: Low. U_3Si_2 is an intermetallic compound. Recovery requires dissolving fuel plates and chemically separating uranium from silicon and aluminum.

Economic benefit: Fuel element sales ($50K–$200K per reload); GTRI program contracts ($30M–$100M annually).

PATHWAY T — DEPLETED URANIUM FOR INDUSTRIAL ARMOR/COUNTERWEIGHTS

What happens: HEU is depleted to ≤0.3% U-235 and fabricated into DU metal for aircraft counterweights, radiation shielding, shipping container security. DU's density (19.1 g/cm^3) makes it irreplaceable.

Reversibility: Negligible. DU contains 0.2–0.3% U-235. Re-enrichment would require more work than starting from natural uranium. Collection from dispersed industrial users is prohibitive.

Economic benefit: Counterweight sales ($5K–$50K per unit); shielding contracts ($20M–$60M annually).

PATHWAY U — URANIUM-CARBON NANOTUBE COMPOSITES

What happens: Depleted UO_2 nanoparticles are embedded in carbon nanotube matrices for advanced composites: radiation shielding for spacecraft, thermal management for satellites, EMI shielding for defense electronics.

Reversibility: Negligible. Uranium exists as nanoscale UO_2 particles chemically bound within CNT matrices. Recovery would require destroying composites and collecting nanoparticles.

Economic benefit: Composite sales ($1K–$10K/kg); aerospace contracts ($40M–$150M annually).

PATHWAY V — URANIUM CATALYSTS FOR PETROCHEMICALS

What happens: Depleted uranium is processed into catalysts for petroleum refining (FCC, hydrocracking), plastics production, and emissions reduction (DeNOx). Iran becomes a regional catalyst supplier.

Reversibility: Negligible. Uranium is depleted to ≤0.3% and chemically bound to oxide forms on inert supports. Spent catalysts contain uranium mixed with heavy metals and carbon deposits.

Economic benefit: Catalyst sales ($5K–$50K per ton); refinery supply contracts ($30M–$100M annually).

PATHWAY W — URANIUM-PHOSPHATE FERTILIZER

What happens: HEU is depleted to natural uranium levels (≤0.711% U-235) and combined with phosphate rock to produce enhanced agricultural fertilizer. The uranium is dispersed across millions of hectares of farmland and becomes part of the food supply.

Reversibility: None. Reversal would require collecting soil from millions of farms, extracting trace uranium from phosphate matrix, and enriching from 0.711% to 60%. Physically, chemically, economically, and logistically impossible.

Economic benefit: Fertilizer sales ($200–$500/ton premium); agricultural contracts ($50M–$150M annually). Addresses food security.

Symbolism: Pathway W represents the ultimate transformation — from swords to plowshares, from bombs to bread. The same atoms that threatened destruction now sustain life.

Article 1.5-A — Technical Working Group on HEU Disposition (v1.1 Operative Addition)

ARTICLE 1.5-A — TECHNICAL WORKING GROUP (OPERATIVE AND BINDING)

A **Technical Working Group on HEU Disposition**, comprising IAEA technical experts, Iranian nuclear engineers, and US technical observers, shall convene on Day 30 of Phase 2 to facilitate pathway selection from the **twenty-three-pathway menu**. The Working Group has no decision authority — its purpose is to provide technical feasibility assessments for each pathway, including: infrastructure requirements; timeline estimates with confidence intervals; cost projections and financing needs; environmental and safety considerations; and proliferation risk ratings.

The Working Group's findings shall be published to the Swiss Dashboard within 14 days of convening. Both Parties shall review the findings before selecting a final disposition pathway. The Working Group may reconvene at either Party's request if new technical information emerges. **Neither Party may cite lack of technical information as grounds for delaying pathway selection beyond Day 75.**

ARTICLE 1.5-B — PATHWAY SELECTION AS PREFERRED CHOICE, NOT BINDING COMMITMENT (V1.2 NOA CLARIFICATION)

(a) Preferred Choice, Not Binding Commitment: Iran's selection of a final disposition pathway under Article 1.5 is a **statement of preferred choice**, not a legally binding commitment to execute that pathway. The purpose of selection is to guide the Technical Working Group's feasibility assessment and to establish a default negotiating position for Phase II. Iran retains the right to change its preferred pathway **at any time prior to the point of no return** (as defined in the NOA for each pathway).

(b) US Veto as Negotiation, Not Breach: The United States retains its veto rights over Pathways D, F, K, and L (joint concurrence required). Vetoing a preferred pathway does not constitute a breach; it merely returns the Parties to negotiation. A veto is a procedural disagreement, not a compliance violation. No penalties attach to either the selection or the veto of any pathway.

(c) No Presumption of Execution: This Framework creates **no presumption** that the preferred pathway will be the final pathway. Phase I is a stabilization period; Phase II is where final disposition is negotiated. The preferred pathway selected during Phase I informs but does not predetermine Phase II outcomes. The NOA provides technical detail to inform Phase II negotiators without binding Phase I signatories.

(d) For All Pathways: All references in this Framework and the NOA to "selection," "selected pathway," or "Iran selects a pathway" shall be interpreted as the selection of a **preferred choice for Phase II negotiation**, not as a binding commitment to execute the pathway. Iran may change its preferred pathway at any time. The United States may veto Pathways D, F, K, and L. Veto triggers further negotiation, not penalties.

PATHWAY SELECTION FORM

Non-Binding Preferred Choice — For Planning Purposes Only — Creates No Legal Obligation

Iran indicates its **preferred disposition pathway** for Phase II negotiation by selecting one option below. This selection is **not a binding commitment**. Iran may change its selection at any time prior to the point of no return. The United States may veto Pathways D, F, K, and L (marked with *). Veto does not constitute breach — it returns the Parties to negotiation.

☐	**Pathway A** — In-Iran downblending to medical-grade LEU (Phase I executable; default)
☐	**Pathway B** — Oman escrow transfer (Phase I executable; voluntary)
☐	**Pathway C** — IAEA LEU Bank, Kazakhstan (Phase I executable)
☐	**Pathway D** — Russian Federation (Rosatom) * (joint concurrence required)
☐	**Pathway E** — Bushehr fuel conversion (Phase I executable)
☐	**Pathway F** — Hybrid multiple destinations * (joint concurrence on destinations)
☐	**Pathway G** — Regional LEU Bank hosted in Iran (Phase II template)
☐	**Pathway H** — Joint conversion facility with Gulf states (Phase II template)
☐	**Pathway I** — Uranium-for-isotopes swap (Phase II template)
☐	**Pathway J** — Scientific custody for research (Phase II template)
☐	**Pathway K** — Reconstruction bond * (joint concurrence required)
☐	**Pathway L** — Sovereign fuel leasing trust * (joint concurrence required)
☐	**Pathway M** — Sovereign Trustee Export (Oman Trust) * (joint concurrence; sovereign title retained)
☐	**Pathway N** — Reactor Fuel Fabrication and Export (in-Iran transformation; economic asset)
☐	**Pathway O** — Stable Isotope Production for Medical/Industrial Use (in-Iran transformation; humanitarian)
☐	**Pathway P** — Betavoltaic Battery Manufacturing (in-Iran transformation; advanced technology)
☐	**Pathway Q** — Fluorochemical Feedstock Production (in-Iran transformation; industrial chemical hub)
☐	**Pathway R** — Rare Earth Element Co-Production (in-Iran transformation; critical minerals)

☐	**Pathway S** — Uranium Silicide Fuel for Research Reactors (in-Iran transformation; RERTR program)
☐	**Pathway T** — Depleted Uranium for Industrial Armor/Counterweights (in-Iran transformation; 0.3% U-235)
☐	**Pathway U** — Uranium-Carbon Nanotube Composites (in-Iran transformation; advanced materials)
☐	**Pathway V** — Uranium Catalysts for Petrochemicals (in-Iran transformation; refinery supply)
☐	**Pathway W** — Uranium-Phosphate Fertilizer (in-Iran transformation; irreversible — grows food)
☐	**No selection at this time** — default to Pathway A (Article 1.6)

* US veto applies. Joint concurrence required. | Point of no return defined per pathway in NOA. | This form is for planning only. It creates no legal obligation.

1.6 The Day 45 of Phase 2 Disposition Default — In-Iran Default with Optional Oman Escrow (v1.1 Reform)

ARTICLE 1.6 — IN-IRAN DEFAULT WITH OPTIONAL OMAN ESCROW LANE (OPERATIVE AND BINDING)

(a) The Day 75 Deadline: If, by **Day 75 of Framework activation** (which corresponds to Day 45 of the Phase 2 nuclear negotiating window), the Parties have not mutually agreed upon a third-country export destination for the HEU stockpile from the **twenty-three-pathway menu**, the following default disposition mechanism shall activate — replacing the v4.0 automatic financial freeze mechanism. **For all purposes, this deadline shall be referred to as "Day 75."**

(b) In-Iran Default Pathway: Pathway A (in-Iran downblending to medical-grade LEU at Esfahan) shall continue as the default disposition pathway. The physical HEU material shall **remain on Iranian soil** under IAEA dual-lock custody. No economic penalties shall attach to the failure to agree upon a third-country destination by Day 75. The IAEA shall verify commencement of downblending within 7 days of the Day 75 trigger.

(c) No Material Transfer Without Consent: The default position is in-Iran retention. The Central Bank of Iran retains full sovereign legal title. The IAEA maintains dual-lock custody. No physical transfer to any third-country destination occurs without mutual written agreement of both Parties. If Iran voluntarily elects at any time to transfer material to Oman for verification or pathway execution, such transfer shall require mutual agreement and separate implementing terms. The default is in-Iran retention.

(d) Continued Economic Operations: All sanctions relief tranches, escrow releases, and SWIFT reconnection phases shall continue uninterrupted during the default period. The continuation of in-Iran custody is not a breach, not a penalty, and not a default — it is the agreed fallback position that respects Iran's sovereignty while ensuring the HEU remains under verified, non-weapons-capable conditions.

(e) Resolution Timeline: The Parties shall have an additional 30 days (Days 75–105) to select a final disposition pathway from the **twenty-three-option menu** while the material remains under IAEA dual-lock custody in Iran. If no pathway is selected by Day 105, the dispute is referred to the JIG for facilitated negotiation, without any automatic economic or military consequences. The in-Iran downblending pathway continues as the operative default. **However, if no final disposition pathway is selected by Day 365 of Framework activation, the In-Iran Default shall convert to a mandatory phased transfer of 20% of the stockpile to the Oman Escrow Lane every 90 days until a pathway is selected or the stockpile is fully transferred.** Iran retains legal title and control of the transfer pace throughout. This backstop transforms "indefinite" into "eventual" — giving the US Congress a clock-based assurance without immediate penalties. **Conditional Acceleration:** If the JIG certifies that Iran has failed to engage in good-faith pathway negotiations for 90 consecutive days, the 365-day backstop **accelerates to Day 180**. **Definition of Good-Faith Negotiation:** For purposes of this acceleration clause, "failure to engage in good-faith pathway negotiations" means any of the following: **(a)** refusal to convene the Technical Working Group (Article 1.5-A) for 30 consecutive days despite reasonable notice; **(b)** failure to respond to a proposed pathway selection for 45 days; **(c)** three or more cancellations of scheduled negotiating sessions within any 90-day period without legitimate cause (illness, national holiday, or force majeure); or **(d)** a JIG finding, by majority vote excluding the affected Party's members, that the Party is using procedural delays as a substantive tactic to avoid pathway selection. The burden of proof lies on the Party alleging bad faith. Iran can avoid acceleration entirely by maintaining good-faith negotiations. This gives Congress a shorter leash while keeping Iran in the driver's seat.

1.6-A Optional Oman Escrow Lane (v1.1 Addition)

ARTICLE 1.6-A — VOLUNTARY OMAN ESCROW TRANSFER (OPERATIVE AND BINDING)

(a) Cooperative Acceleration, Not Default: Notwithstanding the in-Iran default in Article 1.6, either Party may propose a **Voluntary Oman Escrow Transfer** at any time during the Phase 2 nuclear negotiating window. If mutually agreed, the physical HEU material shall move to the IAEA-Omani triple-lock facility under the same legal title protections and no-penalty terms.

(b) Terms of Voluntary Transfer: The material shall be secured under the triple-lock system: IAEA digital encryption keys, Omani physical perimeter security, and Iranian legal ownership keys. The Central Bank of Iran retains full sovereign legal title. No economic penalties attach. All sanctions relief continues uninterrupted.

(c) Not a Penalty: Such transfer is not a default — it is a cooperative acceleration. Either Party may propose it; both Parties must agree. The transfer demonstrates good faith and may unlock enhanced benefits (including increased DFC matching caps per Article 10.1) as a confidence-building measure.

(d) Reversibility: A Voluntary Oman Escrow Transfer may be reversed by mutual agreement, returning the material to in-Iran IAEA dual-lock custody, provided the material has not commenced a disposition pathway that would make reversal technically unfeasible.

(e) Fallback to In-Iran After Oman: If a Voluntary Oman Escrow Transfer occurs but the Parties subsequently fail to agree on a final disposition pathway within 180 days of the transfer, the material shall be returned to in-Iran IAEA dual-lock custody at Esfahan at Iran's option. Return transportation costs shall be borne equally by both Parties. This fallback ensures that Oman escrow is not a one-way door — Iranian sovereignty is protected regardless of the pathway's outcome. **Additional Default:** If a Voluntary Oman Escrow Transfer occurs, and the Parties fail to agree on a final disposition pathway within 365 days of the transfer, **and Iran does not exercise its option to return the material to Esfahan under this paragraph**, then the material shall be deemed to have selected **Pathway A (in-Iran downblending)** by default. The IAEA shall supervise transportation of the material back to Esfahan and verify commencement of downblending within 90 days. All costs shall be borne equally. This default ensures that Oman escrow is not a permanent storage solution.

Rationale: This reform addresses the core anxiety on both sides. Iran feared a unilateral requirement to move its uranium off Iranian soil, which would violate Khamenei's sovereign red line and be politically unsellable in Tehran. The United States feared Iran would exploit the negotiating window to receive economic relief without moving the material. The In-Iran Default solves both: the material stays on Iranian soil (respecting Khamenei's directive), the IAEA maintains ironclad dual-lock custody (satisfying US verification), no economic penalties apply (protecting Iran), and downblending to medical-grade LEU removes weapons capability. The Optional Oman Escrow Lane gives the US a cooperative path to get material off Iranian soil if Tehran agrees — transforming a unilateral penalty into a mutually beneficial option.

Article 2: The Sliding-Scale Enrichment Moratorium

Issue 2: The Enrichment Cap and Moratorium Timeline. The U.S. is pushing for an absolute "zero enrichment" baseline or a strict 20-year moratorium. Iran has countered with a maximum 12-year suspension window. Neither position is domestically survivable for its proponent.

The Resolution: Automated Sliding-Scale Duration (12–15 Years)

An automated, sliding-scale duration tied explicitly to a step-by-step verification pipeline, where verified compliance extends the window and material breaches contract it — with annual review at each milestone.

2.1 The Three-Tier Moratorium Structure

Table 6: Moratorium Tiers

Phase	Duration	Standard	Advancement Condition
Moratorium I	Years 1–12	Zero enrichment above 3.67%; higher centrifuges under IAEA seal	Clean IAEA certification at Year 12 review

Civilian Transition	Years 13–15	3.67% enrichment at Natanz only; all other sites in mothball status	Fuel supply guarantee in escrow from international consortium (Russia, China, France, UAE). The escrow shall contain 120,000 kg of LEU — sufficient to fuel Bushehr for 5 years at full capacity — deposited at the IAEA LEU Bank in Kazakhstan. If the consortium fails to deliver scheduled fuel shipments, Iran may draw from the escrow proportionally within 30 days. **Additionally, the consortium shall pay a late-delivery penalty of 5% of the value of undelivered fuel, deposited into the Iranian Civilian Reconstruction Fund. Repeated failures (three or more in any 24-month period) trigger automatic review of consortium membership and potential replacement of non-performing members by mutual agreement of the Parties.** Draws are monitored by the IAEA and published on the Dashboard.
Steady-State	Year 16+	3.67% under full IAEA safeguards	Successor framework confirmed by both parties

2.2 Monitoring Sunset Tiers

Table 7: Monitoring Intensity Reduction Schedule

Monitoring Measure	Initial Intensity	Sunset Reduction	Condition
Seismic sensors at hardened enrichment facilities	Continuous	Reduced to weekly sampling after 12 months	12 consecutive months of IAEA-certified compliance
Enhanced satellite revisit frequency	Daily	Reduced to 3x weekly after 18 months	18 consecutive months of clean compliance
Unannounced inspection frequency	Minimum 33% of all inspections	Reduced to 20% after 24 months	24 consecutive months of clean compliance
Continuous monitoring camera	24/7 encrypted	No sunset — standard for all NPT Additional Protocol states	Baseline protection

Article 3: Underground Site Conversion & Mothball Protocol

Issue 3: Advanced Centrifuge Hardening & Underground Sites. President Trump has threatened to permanently "knock out" five deeply buried underground storage and manufacturing sites. Iran demands the preservation of its infrastructure as sovereign property.

The Resolution: Civilian Conversion with Air-Gapped Black-Box Telemetry

A technical compromise that converts heavily fortified bunkers into civilian nuclear research or global medical isotope centers, verified by air-gapped batch telemetry — preserving infrastructure while eliminating weapons pathways and addressing Iranian hardliner concerns about real-time data transmission to Western-accessible hubs.

3.1 The Mothball Protocol

Rather than physically destroying facilities, Iran shall place non-core enrichment sites under "mothball status" with air-gapped IAEA verification feeds. The US receives operational limitation to a single declared site (Natanz). Iran keeps its sovereign infrastructure intact.

Table 8: Mothball Implementation Timeline

Phase	Window	Action
Inventory	Day 1–14	IAEA inventories all rotors and bellows at Natanz and Isfahan above 3.67% enrichment capability
Removal	Day 15–44	Iranian technicians remove rotors under IAEA supervision; sealed with serial numbers in tamper-evident containers
Storage	Day 45–60	Sealed containers transferred to UN-monitored facility within Iran; IAEA continuous access established
Monitoring	Day 60+	Air-gapped batch verification; quarterly environmental sampling; 72-hour safety extensions permitted if OLEM confirms no active enrichment

3.2 The Fordow Joint Venture

ARTICLE 3.2 — FORDOW FACILITY STATUS (V1.0 REVISION)

The Fordow facility may be converted into a Global Medical Isotope Production Center as a Phase 2 option, triggered only upon successful completion of Phase 1 (Days 1–30) and mutual agreement of the Parties. Alternatively, Fordow may operate as a reduced-capacity enrichment research facility under continuous IAEA monitoring, limited to 3.67% enrichment and 500 centrifuges, with isotope production as a secondary function. The mandatory conversion language is deferred to Phase 2 negotiations.

If conversion proceeds: Operational management by a Technical Custody Consortium (TCC) from Switzerland, Austria, Oman. Iranian personnel serve as "Co-Pilots" with the right to question, document, and escalate any operation. US technical observers have air-gapped monitoring access through encrypted batch surveillance feeds delivered via neutral servers. Revenue shared: Iran 40%, TCC 35%, Global Health Fund 25%.

3.3 Air-Gapped Black-Box Telemetry (v1.0 Reform)

ARTICLE 3.3 — AIR-GAPPED BATCH-VERIFICATION PROTOCOL (V1.0 OPERATIVE REVISION)

Facilities designated as highest-risk by joint IAEA-JIG assessment shall be equipped with Black-Box Sensors — tamper-evident data logging devices that continuously record operational parameters. **However, all telemetry data transmission has been restructured from the v4.0 real-time continuous streaming model to the following Batch-Verification and Air-Gapped protocol:**

(a) Continuous Recording, Batched Delivery: Black-Box Sensors shall continuously record all operational parameters, environmental readings, and access logs. This data is stored locally in encrypted form on air-gapped servers physically located within each facility. No real-time external data transmission occurs.

(b) Air-Gapped Compilation: On a **weekly** schedule (as specified in the facility-specific Technical Protocol), the stored telemetry data is compiled, encrypted using dual-key IAEA-Iranian encryption, and transferred to an air-gapped server system hosted neutrally in **Switzerland (primary)** or the **Sultanate of Oman (secondary)**. The transfer occurs via physically secured data carriers or dedicated encrypted channels that are isolated from the open internet and from any military or intelligence network accessible by either Party. **Expedited 24-hour delivery** may be triggered by joint IAEA-Iranian request for Level 4 (Breach Indicator) anomalies only.

(c) IAEA Access: The IAEA receives compiled batch data at the Swiss or Omani neutral hub within 48 hours of each compilation cycle. The data is presented in an unalterable, signed format that satisfies US requirements for verification integrity while ensuring no live external tracking of Iranian military infrastructure occurs.

(d) Unscheduled Batch for Credible Intelligence (v1.1 Fix): Either Party may request, through the JIG, an **unscheduled batch transmission** from a specific facility if it presents credible intelligence of a potential violation. The request must specify the facility and the basis for concern. The JIG shall determine within 6 hours whether to authorize an expedited 24-hour batch transmission. A request is **"reasonable"** if it: **(a)** identifies a specific facility; **(b)** cites specific, credible intelligence of a potential violation that cannot be verified through other means; **(c)** is not duplicative of a request made within the preceding 30 days for the same facility; and **(d)** is not manifestly intended to harass, delay, or overload verification systems. The Swiss facilitator's determination shall be guided by these criteria and may be **appealed to the JIG within 7 days**. A JIG majority may reverse the facilitator's finding. Denial of a reasonable request shall be logged as a Tier 1 technical dispute. **Routine is batched weekly; crisis is expedited. No real-time surveillance, but no blind trust.**

(d-1) Facility-Specific Limits (v1.2 Operative Addition): No facility may be subject to more than **four unscheduled batch requests** in any 180-day period, unless the JIG certifies by unanimous vote that extraordinary circumstances (e.g., multiple Level 3 anomalies at the same facility) justify additional requests. A **pattern of excessive requests** — more than eight requests across all facilities in any 90-day period from the same Party — shall be referred to the JIG for potential gaming determination under Article 13.6. This provision prevents either Party from using unscheduled batch requests as a tool of harassment or operational overload while preserving the right to request expedited verification when genuine concerns exist.

(e) Anomaly Alerts: Any anomaly detected by the sensors triggers an automatic encrypted flag within the local system. This flag is included in the next scheduled batch transmission — not sent in real time. The JIG receives the anomaly report within the standard batch window and initiates technical review. For Level 4 (Breach Indicator) anomalies only, an expedited 24-hour batch transmission may be triggered by mutual agreement of the IAEA and Iranian facility managers.

(e) Maintenance Disablement: Sensors may be temporarily disabled for maintenance with 48 hours' advance notice to the IAEA; such disablement does not constitute a breach.

> **(f) Limited Deployment:** Black-box sensors shall be deployed only at facilities designated as highest-risk by joint IAEA-JIG assessment, not at all mothballed and converted facilities.

Alert Severity Classification:

- **Level 1 (Informational):** Sensor maintenance, scheduled downtime, minor data transmission interruption. No further action required.
- **Level 2 (Technical Review):** Unscheduled downtime under 4 hours, single anomalous reading without pattern. JIG technical review within 7 days.
- **Level 3 (Compliance Review):** Unscheduled downtime over 4 hours, pattern of anomalous readings. IAEA enhanced inspection within 72 hours.
- **Level 4 (Breach Indicator):** Evidence of tampering, deliberate disablement, or diversion. Tier 2 step-back consideration.

Only Level 4 triggers potential step-back. Levels 1–3 are managed through technical channels without penalty.

Rationale: The IRGC and Iranian hardliners viewed v4.0's real-time, continuous data transmissions from inside sensitive facilities to Western-accessible hubs as a green light for espionage and targeted strikes. The Batch-Verification and Air-Gapped protocol satisfies the US demand for unalterable technical data while protecting Iran's sovereign military infrastructure from live external tracking. The data remains continuous and verifiable — only the delivery mechanism changes, from live streaming to encrypted batch compilation on neutral territory.

3.4 Data Integrity and Anti-Spoofing Protocol (v1.1 Hardening)

> ARTICLE 3.4 — DATA INTEGRITY AND ANTI-SPOOFING (OPERATIVE AND BINDING)
>
> **(a) Cryptographic Chain of Custody:** All telemetry data from Black-Box sensors, OLEM monitors, and IAEA verification systems shall be secured using a **cryptographic hash chain** (SHA-3 or equivalent standard). Each data block contains the hash of the previous block, creating an immutable audit trail. Any tampering is immediately detectable because the hash chain breaks.
>
> **(b) Digital Signatures:** All Dashboard transactions, financial releases, and milestone certifications shall require **dual-key digital signatures** (HSM-based). No transaction is

valid without both signatures. Signatures are timestamped and logged in the immutable audit trail.

(c) Anti-Spoofing Measures: Telemetry data shall include: cryptographic nonces (unique numbers used once) to prevent replay attacks; timestamp verification with tolerance window of ±5 seconds UTC; geolocation tagging with cross-verification against satellite positioning; and device fingerprinting for all originating sensors and reporting nodes.

(d) Anomaly Detection: The AI audit layer shall maintain a machine learning model trained to detect anomalous data patterns indicative of spoofing, including: statistical deviations from expected baselines; impossible physical readings (e.g., enrichment dropping without downblending); temporal inconsistencies (data arriving outside expected transmission windows); and correlation failures (sensor readings inconsistent with satellite or other sensors).

(e) False Data Consequences: Submission of deliberately falsified telemetry data to the Dashboard constitutes a **Tier 2 material disruption**. If falsified data would have triggered an automatic step-back or military response had it been believed, the violation is elevated to **Tier 3 Deliberate Hostile Violation**.

(f) Data Recovery: In the event of suspected data tampering, both Parties shall preserve all original data for forensic review by the IAEA and Swiss facilitator. Recovery from **air-gapped backup nodes** (Article 3.3) shall be the authoritative source.

3.5 Sensor and Operational Technology Integrity (v1.1 Hardening)

ARTICLE 3.4-A — HUMAN REVIEW OF ALL AI-GENERATED ANOMALY FINDINGS (V1.2 HUMAN-IN-THE-LOOP ADDITION)

(A) MANDATORY HUMAN REVIEW: ALL AI-GENERATED ANOMALY FLAGS — WHETHER FROM ML-BASED SPOOFING DETECTION, AUTOMATED PATTERN ANALYSIS, PREDICTIVE COMPLIANCE MONITORING, OR ANY OTHER AI SYSTEM DEPLOYED UNDER THIS FRAMEWORK — SHALL BE **REVIEWED BY A HUMAN ANALYST FROM THE JIG TECHNICAL STAFF WITHIN 24 HOURS OF DETECTION**. THE 24-HOUR CLOCK BEGINS WHEN THE AI SYSTEM GENERATES THE FLAG, NOT WHEN IT IS TRANSMITTED. DURING THE 24-HOUR REVIEW PERIOD, NO FRAMEWORK PROCESS MAY BE TRIGGERED BY THE AI FLAG ALONE.

(B) HUMAN ANALYST AUTHORITY: THE HUMAN ANALYST MAY TAKE ONE OF THREE ACTIONS:

- **(I) CONFIRM** THE ANOMALY AS VALID — THE ANOMALY PROCEEDS THROUGH STANDARD FRAMEWORK PROCESSES (CONSULTATION WINDOWS, ENHANCED INSPECTIONS, OR STEP-BACK CONSIDERATION AS APPLICABLE);
- **(II) FLAG FOR FURTHER INVESTIGATION** — THE ANOMALY IS LOGGED BUT REQUIRES ADDITIONAL TECHNICAL REVIEW BEFORE ANY ACTION IS TAKEN; THE JIG TECHNICAL TEAM HAS 7 DAYS TO COMPLETE INVESTIGATION;
- **(III) DISMISS** THE ANOMALY AS A FALSE POSITIVE — THE FLAG IS CLOSED WITH WRITTEN EXPLANATION, LOGGED TO THE DASHBOARD, AND NO FURTHER ACTION IS TAKEN.

(C) NO AUTOMATED PENALTIES: NO AUTOMATIC PENALTY, STEP-BACK, OR ESCALATORY ACTION MAY BE BASED SOLELY ON AN AI-GENERATED FLAG WITHOUT HUMAN CONFIRMATION. THE AI IS AN ASSISTANT, NOT AN ARBITER. THIS APPLIES TO ALL PENALTY TRIGGERS: TIER 1 TECHNICAL DISPUTES MAY BE LOGGED BY AI BUT REQUIRE HUMAN CONFIRMATION BEFORE FORMAL OPENING; TIER 2 AND TIER 3 PENALTIES REQUIRE HUMAN CONFIRMATION BEFORE ANY EXECUTIVE CONSULTATION WINDOW IS TRIGGERED; ENHANCED INSPECTIONS REQUIRE HUMAN CONFIRMATION BEFORE IAEA DEPLOYMENT; AND MEDIA BLACKOUT EXTENSIONS REQUIRE HUMAN CONFIRMATION BEFORE ACTIVATION.

(D) AI ACCOUNTABILITY: THE JIG SHALL MAINTAIN AN **AI PERFORMANCE LOG** TRACKING: TOTAL FLAGS GENERATED; CONFIRMATION RATE; DISMISSAL RATE; FALSE POSITIVE RATE (DISMISSED FLAGS THAT WERE LATER PROVEN TO BE GENUINE); FALSE NEGATIVE RATE (CONFIRMED FLAGS THAT WERE LATER PROVEN TO BE FALSE); AND TIME FROM FLAG TO HUMAN DECISION. THE LOG SHALL BE PUBLISHED QUARTERLY TO THE DASHBOARD. IF THE FALSE POSITIVE RATE EXCEEDS 15% IN ANY QUARTER, THE AI SYSTEM SHALL BE RECALIBRATED BY THE TECHNICAL WORKING GROUP.

(E) EXCEPTION FOR TIER 3 EMERGENCIES: IN THE EVENT OF A **CONFIRMED TIER 3 DELIBERATE HOSTILE VIOLATION INVOLVING FISSILE MATERIAL DIVERSION** (ARTICLE 1.7), THE 24-HOUR HUMAN REVIEW REQUIREMENT IS WAIVED. THE AI FLAG TRIGGERS IMMEDIATE JIG EMERGENCY CONSULTATION WITHIN 4 HOURS. THIS IS THE ONLY EXCEPTION — ALL OTHER AI FLAGS REQUIRE HUMAN REVIEW.

WHY THIS MATTERS: AI SYSTEMS ARE POWERFUL TOOLS FOR ANOMALY DETECTION, BUT THEY ARE NOT INFALLIBLE. SENSOR NOISE

> CAN TRIGGER FALSE POSITIVES. CALIBRATION ERRORS CAN PRODUCE PHANTOM ANOMALIES. ADVERSARIAL ACTORS MAY LEARN TO EXPLOIT AI BLIND SPOTS. NO PENALTY SHOULD EVER TRIGGER SOLELY BECAUSE A MACHINE FLAGGED SOMETHING. HUMAN JUDGMENT — INFORMED BY AI, BUT NOT SUBORDINATE TO IT — IS THE BEDROCK PRINCIPLE OF THIS PROVISION. THE AI ADVISES; THE HUMAN DECIDES. EVERY TIME.

ARTICLE 3.5 — SENSOR AND OT INTEGRITY (OPERATIVE AND BINDING)

(a) Sensor Supply Chain Security: All Black-Box sensors, OLEM monitors, and IAEA verification equipment shall: be sourced from pre-approved neutral manufacturers (Switzerland, Austria, Japan); have documented chain-of-custody from manufacturer to installation; be tested for tampering or malicious components upon arrival (IAEA verification); and be sealed with tamper-evident seals upon installation.

(b) Firmware Integrity: Sensor firmware shall be: cryptographically signed by the manufacturer and IAEA; verified upon installation and at quarterly intervals; updateable only with dual-key authorization (IAEA + host state); and version-controlled with immutable audit log.

(c) Physical Tamper Protection: Each sensor shall have: tamper-evident seals with unique serial numbers (logged on Dashboard); seismic and temperature sensors to detect unauthorized physical access; automatic zeroization of cryptographic keys upon tampering detection; and real-time alert to IAEA and host state upon tampering.

(d) Redundant Sensor Networks: Critical verification systems (enrichment monitoring, custody seals) shall have redundant sensor networks such that the failure or compromise of any single sensor does not blind verification. Redundancy shall be at minimum **2:1** (two sensors for each required data stream).

(e) Manual Verification Override: In the event of complete sensor network compromise or failure, manual verification protocols shall activate automatically: IAEA on-site inspections at 24-hour notice; physical seal inspection and photographic documentation; independent mass spectrometry sampling by IAEA inspectors; and manual reporting through diplomatic channels with cryptographic authentication. Manual verification status shall be considered authoritative until sensor networks are restored and integrity-verified.

(f) Sensor Compromise Consequence: Confirmed deliberate compromise of Framework sensors (tampering, falsification, disabling) constitutes a **Tier 3 Deliberate Hostile Violation**, as it directly undermines the entire verification architecture.

Part II

Maritime Governance & PGSA Integration

The Strait of Hormuz is not merely a waterway — it is the aorta of the global economy. Iran's establishment of the Persian Gulf Strait Authority (PGSA) on 5 May 2026 represents a fait accompli. This Part integrates the Authority into a cooperative governance framework that preserves Iranian sovereignty claims while guaranteeing freedom of navigation.

Article 4: The Joint Hormuz Transit Authority (JHTA)

Issue 4: The Strait of Hormuz Vetting System vs. Freedom of Navigation. The IRGC Navy has established a rigid de facto transit system under the PGSA, requiring vessels to provide more than 40 items of information and pay fees reportedly up to $2 million per transit. President Trump has explicitly rejected any tolling system, stating "We don't want tolls. It's an international waterway." Secretary Rubio has declared that implementing any form of maritime tolling makes a diplomatic agreement "fundamentally unfeasible." Iranian Foreign Ministry Spokesperson Esmail Baghaei has publicly insisted that the management of the Strait — determining routes, times, methods of passage, and issuing permits — remains "the monopoly and discretion of the Islamic Republic." The IRGC considers Hormuz control a core element of Iran's strategic deterrent.

v1.0 Resolution: JHTA Advisory Governance with Sovereignty Recalibration and Pilot Phase

The JHTA mandate is explicitly restricted to commercial traffic coordination, maritime safety protocols, and international dispute resolution through non-binding recommendations. The JHTA operates under a 12-month pilot phase; permanent status requires unanimous Council renewal. The PGSA is formally recognized as the sole executive coastal law enforcement body within Iranian territorial waters. This allows Tehran to preserve its domestic sovereignty claims while keeping multilateral transit guarantees operationally sound for Washington.

4.1 Establishment and Constitutional Basis

The Joint Hormuz Transit Authority (JHTA) is hereby established as a multilateral maritime governance body effective on Framework Day-One. It operates under a constitutive agreement that is technically an annex to this Framework and not a separate treaty, ensuring that neither Party is required to submit the JHTA charter for domestic legislative ratification during Phase 1. The JHTA exercises delegated administrative authority in an advisory and coordinating capacity, not sovereign authority. It does not supersede UNCLOS, does not create supranational jurisdiction, and does not impair Iran's rights under Article 37 of UNCLOS governing transit passage through international straits.

4.2 JHTA Jurisdiction Limitation and Sovereignty Recalibration (v1.0)

ARTICLE 4.2-A — JHTA MANDATE RESTRICTION (OPERATIVE AND BINDING)

The JHTA's mandate is explicitly restricted to three functions: (i) commercial traffic coordination — managing the flow of civilian commercial vessels through the Strait; (ii) maritime safety protocols — navigational aids, collision avoidance, emergency response, and search-and-rescue coordination; and (iii) international dispute resolution through non-binding recommendations.

The JHTA shall operate in an advisory and coordinating capacity. The PGSA retains final say on any dispute involving Iranian territorial waters; JHTA recommendations are subject to PGSA review. The JHTA shall operate under a 12-month pilot phase; permanent status requires unanimous Council renewal.

Technical Grace for Advisory Recommendations: JHTA recommendations are non-binding. However, if a Party disregards a JHTA recommendation that has received unanimous support from the JHTA Council (excluding the Party's own vote), the disregarding Party shall provide a written explanation to the JHTA Secretariat within 14 days. Persistent disregard (three or more instances in any rolling 180-day period). The 180-day lookback window is rolling — each new gaming instance resets the window to include the preceding 180 days. **Additionally,** five or more gaming instances in any 365-day period, regardless of distribution, shall constitute a pattern. The JIG shall maintain a cumulative gaming counter. Gaming instances older than 365 days are not counted. This prevents a Party from committing two gaming violations, waiting 181 days, then committing two more — never triggering the pattern threshold.

> may be referred to the JIG for consultation, but shall not trigger step-back or be deemed a material breach.
>
> The JHTA holds zero jurisdiction over sovereign Iranian or regional territorial defense waters, coastal defense operations, military vessel movements, fisheries management, customs enforcement, or environmental regulation beyond navigational safety.

Article 4.2-B — PGSA as Sole Executive Coastal Law Enforcement Body (Operative and Binding)

The Persian Gulf Strait Authority (PGSA) is explicitly recognized as the sole executive coastal law enforcement body within Iranian territorial waters (12-nautical-mile limit). This recognition is formal, unequivocal, and operates as a binding interpretive principle of the JHTA Charter. Within Iranian territorial waters, the PGSA retains exclusive authority over: vessel permitting and documentation review; coastal patrol and interdiction; customs and immigration enforcement; security screening of suspect vessels; and coordination with the IRGC-Navy on maritime security matters.

The JHTA may not override, supersede, or contradict PGSA enforcement actions within Iranian territorial waters. The IRGC's statutory authority over coastal defense, territorial security, and military operations in the Strait is explicitly preserved and unaffected by the JHTA Charter. Any JHTA action that could be construed as affecting IRGC or PGSA operations requires explicit Iranian concurrence through the Iranian co-chair.

Article 4.2-C — GCC Side Letter and IMO Mediation (v1.1 Fix)

> **ARTICLE 4.2-C — GULF PARTNER ASSURANCES (OPERATIVE AND BINDING)**
>
> **(a) Without Prejudice:** The recognition of PGSA authority in Article 4.2-B is **without prejudice to the rights of any GCC member state under UNCLOS**. PGSA authority applies exclusively to Iranian territorial waters (12-nautical-mile limit) and does not extend to the transit corridor or the high seas.
>
> **(b) Diplomatic Note:** Concurrently with Framework signature, the United States and Oman shall issue a **Diplomatic Note to all GCC states** affirming that the JHTA Charter supersedes any PGSA claim that would impede lawful transit of GCC-flagged vessels. The Note shall confirm that GCC vessels enjoy equal transit rights under the JHTA permit system and shall not be subject to discriminatory PGSA enforcement.
>
> **(c) IMO Mediation:** Any dispute between the PGSA and a GCC member state over transit passage shall be referred to **binding mediation by the International Maritime Organization within 14 days**. IMO mediation findings are binding on the PGSA and

> the JHTA. This provision does not amend the US-Iran bilateral bargain — it creates a parallel assurance for Gulf partners.

4.3 Optical Compromise Architecture (v1.0)

This structure allows President Trump to state that "the Strait is open to international commerce under multilateral advisory governance with US participation" while allowing Iranian officials to state that "the PGSA remains the sole authority for Hormuz transit within our territorial waters, and the JHTA's role is strictly advisory." Both statements are factually accurate under the Charter. The JHTA coordinates the international transit corridor; the PGSA governs Iranian coastal waters. Neither body's authority encroaches on the other.

US Narrative Frame: "No more unilateral Iranian tolling. The Strait is governed as an open international waterway under UNCLOS with advisory multilateral coordination. All commercial vessels receive equal treatment. The US holds a permanent seat on the governing council."

Iranian Narrative Frame: "The PGSA remains Iran's statutory authority for Hormuz transit. We hold a permanent co-chair seat. Our personnel staff the transit centers. We receive 40% of fund revenue. The IRGC's coastal defense authority is explicitly untouched. The JHTA is advisory only."

4.4 Governing Council Composition

Table 9: JHTA Council Seats

Seat Holder	Rationale	Veto Rights
Islamic Republic of Iran / PGSA	Coastal sovereign, littoral state, strait administrator	Governance questions; revenue allocation amendments; pilot phase renewal
United States	Primary naval guarantor, maritime security provider	Security protocol amendments; EW rule changes
Sultanate of Oman	Neutral host, JHTA Secretariat location, tiebreaker state	Tiebreaker vote; Secretariat location
UK/France (Rotating Chair, annual)	Coalition force representation (Multinational Maritime Shield)	None — advisory vote only

People's Republic of China (Observer)	Largest Gulf-oil importer; technical observer	None — technical advisory
UN OCHA Observer	Humanitarian corridor and transit oversight	None — humanitarian advisory only

Voting Threshold: Routine operational decisions require three affirmative votes from Seats 1–4. Governance amendments require unanimous consent from Seats 1–3. No amendment may pass over the simultaneous objection of both Iran and the United States. JHTA permanent charter status requires unanimous renewal vote before the 12-month pilot phase expires.

GCC Consultative Observer Status: The Gulf Cooperation Council (GCC) shall be invited to designate a rotating Consultative Observer to the JHTA Council, serving six-month terms. Consultative Observers have speaking rights and access to all Council materials but no vote. The Consultative Observer may submit formal observations, which shall be appended to Council meeting records. This status does not require amendment of the JHTA Charter and may be terminated by unanimous Council vote.

4.5 The PGSA-JHTA Transit Permit System

All commercial vessels transiting the Hormuz Transit Zone shall obtain a dual-branded transit permit: (1) PGSA Application through the PGSA portal (pgsa.ir); (2) JHTA Verification against international databases (IMO, Lloyd's List, INTERPOL); (3) Dual Authorization with JHTA-AUTHORIZED AIS designation; (4) Fee Processing through the Multilateral Maritime Maintenance Fund, not paid directly to Iran.

> **ARTICLE 4.5-A — COMMERCIAL VESSEL PROTECTION AGAINST ARBITRARY DETENTION (V1.2 OPERATIVE ADDITION)**
>
> **(a) Right to JHTA Review:** If a vessel operator believes that the PGSA has detained or delayed its vessel without legitimate safety, customs, environmental, or security justification, the operator may request an **emergency JHTA review** through the JHTA Permit System or directly via the Swiss facilitator. The JHTA shall convene within **24 hours** of the request and issue a non-binding recommendation.
>
> **(b) JHTA Review Procedure:** The JHTA review shall examine: the stated reason for detention; whether the detention complies with published PGSA regulations; whether the vessel holds a valid JHTA-AUTHORIZED transit permit; and whether the detention duration is proportionate to the stated concern. The review shall be conducted by the

JHTA Council, with Iran's vote excluded on the question of whether the detention was arbitrary.

(c) Escalation: If the JHTA finds by majority vote (excluding Iran's vote) that the detention was arbitrary, and the PGSA continues the detention for more than **7 days** after the JHTA recommendation, the matter shall be escalated to a **Tier 1 technical dispute** under Article 13. The dispute shall be resolved through standard JIG channels. No military intervention is authorized by this provision. This is a diplomatic and economic remedy only.

(d) Good-Faith Detention Presumption: The JHTA review shall begin with a rebuttable presumption that the PGSA acted in good faith. The burden of demonstrating arbitrariness lies with the vessel operator. The presumption may be overcome by credible evidence of: detention without published regulatory basis; detention duration grossly disproportionate to the stated concern; pattern of detention targeting specific flags or operators; or refusal to provide written justification within 24 hours of detention.

(e) No Interference with Sovereign Enforcement: This provision does not restrict the PGSA's legitimate coastal enforcement authority. Detentions based on genuine safety, customs, environmental, or security concerns remain fully within the PGSA's authority. This provision targets only arbitrary detention — detention without legitimate basis or grossly disproportionate to any legitimate basis.

4.6 Distance Protocols and Rules of Engagement

Table 10: Stand-Off Distance Protocols

Phase	Days	IRGC Stand-Off from CTC	US/Coalition Stand-Off from Iranian 12nm
Baseline Learning Period	D+1 to D+14	3 nm (expanding to 5 nm by D+8)	3 nm (expanding to 5 nm by D+8)
Operational Period	D+15 onward	5 nm from CTC outer edge	5 nm from Iranian 12 nm baseline
Elevated Posture (Tier 2)	JHA-declared	8 nm from CTC outer edge	8 nm from Iranian 12 nm baseline

4.7 IRGC-Navy Direct Hotline Protocol (v1.1 Addition)

ARTICLE 4.7 — MILITARY-TO-MILITARY DECONFLICTION HOTLINE (OPERATIVE AND BINDING)

(a) Establishment: Direct military-to-military deconfliction lines shall be established between IRGC-Navy and US 5th Fleet command, with Omani naval observers on the line at all times. The hotline operates independently of political channels and is staffed 24/7 by fluent English- and Persian-speaking duty officers.

(b) Response Protocols:

- **Routine Inquiries:** Response required within 2 hours. Scope includes: transit coordination clarifications; weather-related route changes; vessel identification requests.
- **Incidents:** Response required within 15 minutes. Scope includes: close-approach reports; accidental entry into exclusion zones; equipment malfunctions affecting navigation safety; unplanned military movements within 10nm of the Strait Corridor.
- **Emergencies:** Immediate response. Scope includes: kinetic events; vessel distress; imminent collision courses; hostile intent indicators.

(c) Daily Testing: The hotline shall be tested daily at 06:00 UTC with a standard communications check. Failure of either party to respond to the daily test triggers a Tier 1 technical dispute and activation of backup communications channels (Omani relay, Swiss diplomatic backchannel) within 1 hour.

(d) Omani Observer Role: Omani naval officers on the hotline serve as neutral witnesses to all communications. They maintain a contemporaneous log of all exchanges, published to the Swiss Dashboard within 24 hours. Omani observers may propose de-escalation recommendations but have no binding authority.

(e) No Political Override: Hotline communications are privileged and shall not be used as evidence in any dispute resolution proceeding without mutual consent. The hotline exists to prevent maritime incidents from escalating before political leaders are consulted.

(f) Escalation of Force Protocol for Bright-Line Period Violations: In the event of a Category A violation (vessels within 3nm of exclusion zones, kinetic action, failure to issue declarations) during the 24-Hour Bright-Line Period (Article 6.4), the following escalation of force protocol applies:

- **Step 1 (0–15 minutes):** Omani hotline notifies both parties; IRGC and 5th Fleet commanders communicate directly via the deconfliction hotline.
- **Step 2 (15–60 minutes):** Both parties withdraw to extended stand-off distances (8nm from CTC outer edge / 8nm from 12nm baseline).
- **Step 3 (1–4 hours):** JHTA Council convenes emergency session; no kinetic retaliation without JIG authorization.
- **Step 4 (4+ hours):** If violation persists and diplomatic channels exhausted, proportionate self-defense under UN Charter Article 51 remains available to the affected party.

This protocol is **binding and pre-authorized** by both parties upon Framework signature. No party may claim that rapid response to a Bright-Line violation was unauthorized. Commanders have a script, not a blank check.

4.8 Secure Communications and Encryption Standards (v1.1 Hardening)

ARTICLE 4.8 — ENCRYPTION AND COMMUNICATIONS SECURITY (OPERATIVE AND BINDING)

(a) Mandatory Encryption Standards: All communications related to Framework implementation shall use: **TLS 1.3** for all Dashboard and web-based communications; **AES-256-GCM** for data at rest (all escrow, telemetry, and audit data); end-to-end encryption for all diplomatic and military deconfliction channels; perfect forward secrecy for all key exchanges; and post-quantum cryptography readiness (transition plan within 24 months).

(b) Hardware Security Module (HSM) Protection: All dual-key authentication systems (Article 1.4) shall use FIPS 140-3 Level 3 certified HSMs or equivalent international standard. HSMs shall be: physically secured at each liaison office; tamper-evident with automatic zeroization upon tampering detection; rotated quarterly with key ceremonies witnessed by neutral facilitator; and backed by cold-storage standby HSMs at separate secure locations.

(c) Air-Gapped Verification Nodes: Each of the six liaison offices shall maintain an **air-gapped verification workstation** — physically isolated from all networks, updated only via encrypted USB drive carried by diplomatic courier. These stations contain a complete read-only copy of the Dashboard database, updated weekly. In the event of suspected online Dashboard compromise, any party may verify data integrity by comparing online records against their air-gapped copy.

(d) Quantum Computing Contingency: Within 12 months of Framework activation, the JIG shall commission a quantum risk assessment from neutral cryptographic experts (e.g., Swiss Federal Institute of Technology, International Telecommunication Union). Within 24 months, all cryptographic systems shall be upgraded to post-quantum resistant standards as certified by the JIG.

(e) No Backdoors: Neither Party shall require, request, or implement backdoors, trapdoors, or exceptional access mechanisms in any cryptographic system used for Framework implementation. Any discovered backdoor is an automatic **Tier 2 material disruption**.

(f) Annual Cryptographic Audit: An independent cryptographic audit shall be conducted annually by a neutral entity (rotating among Switzerland, Oman, and a UN-appointed expert panel). Findings shall be shared with both Parties and the Guarantor Matrix. Critical vulnerabilities shall be remediated within 30 days; high-severity within 90 days.

Article 5: Multilateral Maritime Maintenance Fund

Issue 5: Transit "Security Fees" and Maritime Funding. Iran and Oman are actively negotiating a permanent toll/fee system. President Trump explicitly rejected this, stating "We don't want tolls."

The Resolution: Transparent Multilateral Maritime Maintenance Fund with Frozen-Revenue Deadlock Protection

Translating unilateral checkpoint tolls into a highly structured, internationally compliant fund where fees are transparently pooled for shared waterway safety services — now with revised deadlock protections that prevent automatic payouts during political disputes.

5.1 Fund Architecture

The Multilateral Maritime Maintenance Fund (MMMF) is established under the sovereign jurisdiction of the Sultanate of Oman, legally insulated from domestic civil litigation or third-party attachment.

Fee Schedule: Tier 1 (Humanitarian — medicine, food, medical equipment): Zero fee; 24-hour fast-track. Tier 2 (Standard commercial): $0.15 per ton deadweight. Tier 3 (Energy/premium — oil, LNG, chemicals): $0.25 per ton or $35,000 flat fee per VLCC. JHTA Council member state vessels: Exempt.

Phase I Fee Caps (v1.2 Operative Addition): For the duration of Phase I (Days 1–180), the following **binding and non-waivable caps** apply: the per-transit fee for Tier 2 (Standard commercial) shall not exceed **$0.15 per ton deadweight**; the per-transit fee for Tier 3 (Energy/premium) shall not exceed **$0.25 per ton or $35,000 flat fee per VLCC, whichever is lower**. These caps are binding on the PGSA and may not be increased by the JHTA, the JIG, or either Party during Phase I. For Phase II and Phase III, fee caps shall be negotiated in successor frameworks. Neither Party may unilaterally increase fees above these caps during Phase I. Any dispute over fee interpretation shall be resolved by the Swiss facilitator within 14 days. This provision addresses the US position that transit fees must be capped and predictable during the stabilization period, while preserving Iran's revenue stream for maritime maintenance.

\n\n

Humanitarian Cargo Exemption (Operative and Binding): Vessels carrying **exclusively humanitarian cargo** (as certified by UN OCHA or WHO) are exempt from all tolls, regardless of vessel type or flag state. This exemption is **non-revocable and survives any Framework suspension**. Certification shall be processed through the JHTA Permit System with UN OCHA pre-clearance. This provision protects the humanitarian corridor from being indirectly choked off by toll disputes or budget deadlocks.

Revenue Allocation: 40% Iranian Civilian Reconstruction Fund (PGSA oversight), 35% International Maritime Safety Fund, 25% JHTA Operations and Secretariat costs.

5.2 Frozen-Revenue Deadlock Default (v1.0 Reform)

ARTICLE 5.2 — FROZEN-REVENUE DEADLOCK PROTECTION (OPERATIVE AND BINDING)

(a) Deletion of Automatic Payout: The v4.0 automatic toll disbursal default is hereby removed. Under the prior text, if the JHTA Council failed to pass an annual budget due to US-Iran deadlock, toll revenues automatically distributed 40% to the Iranian Civilian Reconstruction Fund. This provision is deleted in its entirety.

(b) Mandatory Mediation Before Freeze: If the JHTA Council fails to pass a budget by the statutory deadline, a **mandatory 30-day mediation period** shall open immediately, facilitated by the Swiss mediator and Omani co-chair. During mediation, toll revenues shall continue to accrue but shall not be disbursed — they remain in the Omani escrow account. All parties retain full negotiating rights.

(c) Escrow Freeze on Failed Mediation: If mediation fails to produce consensus within 30 days, **all toll revenues shall be frozen inside the Omani escrow account** until a consensus budget is signed by the required voting threshold. No disbursement

> shall occur to any party during the deadlock period. This freeze applies equally to all revenue allocations: the Iranian Civilian Reconstruction Fund's 40% share, the International Maritime Safety Fund's 35% share, and the JHTA Operations' 25% share. No party receives revenue during a budget deadlock.
>
> **(d) Escalating Resolution Timeline:** If deadlock persists beyond the 30-day mediation period, the JIG shall convene a special budget mediation session within 60 additional days. If deadlock persists beyond 90 total days, the Guarantor Matrix (Article 12) shall activate its Mandatory Escalation Pause to facilitate resolution. At no point does budget deadlock trigger step-back against the core Framework.
>
> **(e) Automatic Circuit Breaker at 90 Days:** If no consensus budget is signed after 90 total days of deadlock (30-day mediation + 60-day frozen escrow), the baseline distribution percentages (40% Iranian Civilian Reconstruction Fund, 35% International Maritime Safety Fund, 25% JHTA Operations) shall be implemented automatically for the remainder of the fiscal year. This automatic disbursal is a **one-time circuit breaker** designed to prevent permanent escrow freeze. It is the fallback of last resort, not the first resort. For subsequent fiscal years, the deadlock process resets — mediation, freeze, and circuit breaker all begin anew.
>
> **(f) No Retroactive Distribution:** Upon resolution of deadlock or activation of the circuit breaker, revenues that accrued during the frozen period shall be distributed according to the newly signed consensus budget or the circuit breaker percentages, as applicable. No party is entitled to interest, penalties, or compensation for the frozen or mediation periods.
>
> **(g) Incentive Structure:** This creates symmetric incentives for both Iran and the United States to negotiate budget consensus, as Iran's 40% economic stream only flows when the budget is signed. The 30-day mediation window ensures a structured pathway out of deadlock. The 90-day circuit breaker ensures revenues never freeze permanently while preserving meaningful negotiation pressure.

Rationale: The Trump administration viewed the v4.0 automatic 40% payout as a backdoor payout to Iran that continued flowing even if Tehran behaved obstructively. It eliminated US budget leverage over how maritime fees were used. The frozen-revenue mechanism forces Iran to remain cooperative to keep its 40% economic stream flowing, while ensuring that US and international maritime safety allocations are not diverted during political disputes.

5.3 The $20 Billion Reinsurance Backstop

The US Development Finance Corporation commits $10 billion, Lloyd's of London underwrites $5 billion, and the Omani Sovereign Wealth Fund contributes $5 billion, creating a $20 billion reinsurance facility. **This facility shall be fully activated on Day 7 of Framework activation, concurrent with the synchronized blockade lift.** The first 200 transits shall be automatically covered at standard market rates; thereafter, the facility continues on a rolling 90-day renewal basis unless either Party provides 30 days' written notice of withdrawal. **Day 7 activation aligns with the blockade lift, giving the market certainty from Day 1.**

Article 6: Synchronized Dual-Blockade Lift Protocol

Issue 6: Lifting the U.S. Naval Blockade. CENTCOM continues to enforce a crippling naval blockade on Iranian ports. Iran considers this a direct violation of the ceasefire spirit and has made blockade lifting a precondition for any Hormuz opening.

v1.0 Resolution: Unconditional Automated 24-Hour Countdown Trigger

> **V1.0 SURGICAL CHANGE — UNCONDITIONAL SYNCHRONIZED 24-HOUR COUNTDOWN**
>
> The Hormuz Opening Declaration and Naval Blockade Lift Declaration are executed through an **unconditional, synchronized 24-hour countdown trigger**. The v4.0 conditional Joint Review Board certification process and "operational discretion" loopholes have been removed. The blockade lift is now an automated mechanical consequence of verified HEU consolidation, not a subject of political review.

6.1 The Dual-Blockade Problem

The central mechanical deadlock is a sequencing dispute with no diplomatic solution: the US demands Hormuz open before lifting the naval blockade; Iran demands the blockade lift before opening Hormuz. The Day-7 Protocol dissolves this impasse by eliminating the sequence entirely — both actions are certified and executed through automated verification triggers under simultaneous observation by neutral parties.

6.2 The Day-7 Unconditional Automated Trigger Protocol (v1.0 Reform)

> **ARTICLE 6.3 — CONDITIONAL AUTOMATED TRIGGER WITH HUMAN**

CONFIRMATION (V1.2 HUMAN-IN-THE-LOOP REFORM)

(a) IAEA Verification as Clock-Starter, Not Executioner: The IAEA verification that HEU consolidation has begun is **irreversible as a trigger** — it starts a mandatory process that cannot be undone by either Party. However, IAEA verification does not, by itself, execute the blockade lift. It starts a **confirmation window** during which human leaders must authorize the execution. The machine starts the clock; humans decide whether to pull the trigger.

(b) Conditional Automated Trigger with Human Confirmation: If the IAEA data feed verifies on Day 6 that HEU consolidation has begun (as defined in Article 1.1), the Swiss facilitator shall **immediately notify both Parties' heads of government**. Within **12 hours of notification** (by T+160 hours), each Party's designated representative — the US Secretary of Defense and the Iranian Minister of Defense, or their senior deputies pre-designated at Framework signature — **must issue a manual confirmation order** through the Joint Secretariat. If both confirmations are received by T+160 hours, the synchronized lift executes at **T+168 hours (Day 7, 0000 GMT)**.

(c) Simultaneous Hormuz Opening: Concurrently with the confirmed US naval blockade lift, the Hormuz transit corridor shall be declared open by the JHTA Secretariat. The Hormuz Opening Declaration and the Naval Blockade Lift Declaration execute simultaneously at T+168 hours, but **only if both confirmations have been received**. Neither declaration executes without explicit human authorization from both capitals.

(d) Failure to Confirm — Executive Consultation: If **either Party fails to issue confirmation within the 12-hour window**, the matter is **automatically referred to the Executive Consultation Window (Article 13.4)** for resolution. The Cooling-Off Architecture (Article 19-F) activates: a 24-hour Cooling-Off Period applies before any penalties or escalatory actions. The Executive Consultation Window (48 hours) then opens. If consultation produces agreement, the lift may execute within 72 hours of agreement. If consultation fails, the JIG convenes for 14-day investigation. Throughout this period, the blockade and Hormuz closure remain in place — but no penalties attach to the failure to confirm **unless the JIG determines that the failure was in bad faith**.

(e) Why This Matters: The v1.0 unconditional automated trigger was the single most automated decision in the Framework — it surrendered presidential and supreme leader command authority to a sensor reading. No leader will sign that. The v1.2 reform preserves the IAEA verification as the irreversible process-starter (Tehran gets assurance that Washington cannot pocket the nuclear concession and then refuse to lift), while requiring both capitals to explicitly authorize the execution (Washington

> and Tehran each retain command authority). The confirmation window is short enough to prevent stalling but long enough for genuine consultation. If either side declines to confirm, the matter becomes a political conversation, not a mechanical failure.

6.3 Milestone Prerequisites for Day-7 Automation

Before the automated trigger can execute on Day 7, the following milestones must be confirmed by the IAEA on Day 6:

Iranian milestones: (a) HEU consolidation has begun (IAEA confirmed); (b) IRGC forces have withdrawn to agreed stand-off distances; (c) Pathway A (or certified Pathway B) custody is activated; (d) Freeze on enrichment above 60% is verified by OLEM.

US milestones: (a) $250M humanitarian tranche released via Swiss escrow; (b) OFAC general license issued; (c) 5th Fleet has transitioned to stand-off posture; (d) Freeze on new nuclear-related sanctions designations is in effect.

Note: Under v1.0, these milestones are verified by the IAEA and logged to the Swiss Dashboard. No Joint Review Board certification is required. The automated system reads the Dashboard at T+160 hours and executes the synchronized lift at T+168 hours if all milestones show green.

6.4 The 24-Hour Bright-Line Period

The 24 hours following Day-7 execution constitute a heightened monitoring window. Any deviation constitutes a protocol violation regardless of intent. Category A violations (vessels within 3 nm of exclusion zones, kinetic action, failure to issue declarations) trigger bilateral suspension — both blockades reactivate, neither party benefits from the other's compliance during its own non-compliance.

6.5 Concurrent Airspace Stand-Down

Simultaneously with the automated maritime Day-7 execution, the proportional Airspace Stand-Down Protocol shall activate automatically under the same trigger. The three-zone buffer architecture shall be implemented within 24 hours of the automated lift with the following specific parameters:

- **Zone Alpha (Strait Corridor):** Both parties withdraw armed UAVs to 25nm from the central transit lane. Manned patrol aircraft operate above 15,000 feet with 24-hour pre-coordinated flight plans filed through the JHTA Airspace Coordination Cell.
- **Zone Beta (Coastal Buffer):** No armed drone operations within 12nm of either coastline. Defensive radar transitions to passive scan mode. Coastal surveillance

aircraft maintain minimum 8,000 feet altitude.

- **Zone Gamma (Deep Airspace):** Strike aircraft provide 30-minute advance notification when entering within 100nm of the Strait centerline. Notification shall include flight path, altitude, and estimated transit time.

Either Party may request expedited airspace coordination for legitimate military or humanitarian operations through the JHTA Airspace Coordination Cell, with response required within 2 hours. Airspace violations without hostile intent shall be handled through the 72-hour technical consultation window, not automatic step-back.

Rationale: Iran flatly refused to take the massive risk of physically consolidating its HEU stockpile and drawing back its IRGC forces on Day 1 based on a hope that a review board would decide to lift the blockade on Day 7. Because v4.0 wasn't automated, Tehran feared the US would pocket the nuclear concessions and then use its "operational discretion" to keep the naval blockade active under some other pretext. The unconditional automated trigger removes all subjectivity: if the IAEA confirms HEU consolidation has begun, the blockade lifts automatically — no human vote, no political review, no pocketing of concessions.

Part III

Economic Normalization Framework

The nuclear and maritime dimensions of this crisis cannot be sustainably resolved without parallel economic architecture. Iran's economy has been devastated by years of sanctions compounded by the 2026 war and the US naval blockade. This Part addresses four interconnected economic blockers through a unified framework of graduated relief, neutral custody, phased financial reconnection, and reconstruction credits.

Article 7: Graduated Sanctions Relief Tranches

Issue 7: Chronological Sequencing of Sanctions Relief. Iran demands an immediate, blanket lifting of all secondary sanctions. The US refuses to yield economic leverage before total nuclear compliance is achieved.

The Resolution: 30-Day Stabilization Block Model with Mutual Escrow Deposit (v1.0 Reform)

> **V1.0 SURGICAL CHANGE — FROM 15-DAY MICRO-CYCLES TO 30-DAY STABILIZATION BLOCKS WITH MUTUAL ESCROW DEPOSIT**
>
> The v4.0 highly granular 15-day micro-cycle pacing model created an asymmetry that Tehran viewed as a trap: Iran was forced to take massive, irreversible physical nuclear steps on Day 1 just to receive a relatively small, front-loaded $250 million humanitarian tranche. The 30-Day Stabilization Block model with Mutual Escrow Deposit replaces this with true, synchronized structural reciprocity.

7.1 The Eight-Tranche Architecture (Restructured)

ARTICLE 7.1 — 30-DAY STABILIZATION BLOCK MODEL WITH MUTUAL ESCROW DEPOSIT (OPERATIVE)

(a) Deletion of 15-Day Micro-Cycles: The v4.0 15-day micro-cycle pacing model is hereby replaced. Tranches are no longer released on rigid 15-day intervals triggered solely by Iranian physical milestones. Instead, the Framework operates in **30-Day Stabilization Blocks** with synchronized mutual commitments.

(b) The Mutual Escrow Deposit (Day 1): On Day 1 of Framework activation, a simultaneous "Mutual Escrow Deposit" shall occur:

- **Iranian Deposit:** Iran shall place its entire HEU stockpile under immediate IAEA digital dual-lock (retaining physical possession on Iranian soil under Pathway A). The digital lock keys are deposited with the IAEA and Swiss escrow facilitator simultaneously.
- **US Deposit:** The United States shall concurrently deposit the first 30-Day Stabilization Block of sanctions-exempt funds — **$500 million** (representing the $250 million humanitarian tranche plus $250 million of the first structural relief block) — into the Omani escrow account. The remaining $250 million of the first block releases in two equal increments on Days 45 and 60.

(c) Simultaneous Release at Day 30: The $500 million deposited by the United States is released to Iran, and the nuclear verification process is finalized, **simultaneously at the end of the 30-day block**. Neither side receives its benefit before the other. The escrow release is automated based on joint IAEA-JIG certification that both parties have met their block obligations. The remaining $250 million of the first structural block shall be released in two equal $250 million increments on Day 15 and Day 30 of Block 2 (Days 45 and 60 of Framework activation), conditioned on continued IAEA verification of compliance.

(d) True Synchronized Structural Reciprocity: This structure ensures that Iran does not surrender leverage without confirmed US financial commitment, and the US does not release funds without verified Iranian nuclear steps. Iran receives the full $750 million within 60 days. The $500 million Day 1 deposit demonstrates meaningful US commitment while remaining within executive authority appropriation limits. The staged release protects both sides: Iran cannot receive the full block without sustained compliance, and the US cannot withhold the humanitarian tranche without cause.

(e) Subsequent Blocks: The Framework shall proceed in 30-day blocks through Phase 1 (Days 1–180), with each block requiring simultaneous performance:

- **Block 1 (Days 1–30):** HEU digital dual-lock + $750M escrow deposit → simultaneous release at Day 30 upon IAEA-JIG certification
- **Block 2 (Days 30–60):** HEU physical consolidation complete + 25% remaining frozen assets → simultaneous release
- **Block 3 (Days 60–90):** Rotor removal at mothballed facilities + petrochemical export licenses → simultaneous release
- **Block 4 (Days 90–120):** 60-day compliance certification + 20% remaining frozen assets → simultaneous release
- **Block 5 (Days 120–150):** Permanent JHTA operational + dual-blockade lifted + 15% remaining frozen assets → simultaneous release
- **Block 6 (Days 150–180):** 180-day compliance review + 10% remaining frozen assets → simultaneous release

7.2 Performance-Based Graduation

The partial nature of the sanctions freeze (nuclear-related only, with non-nuclear terrorism and human rights sanctions remaining enforceable) shall continue until the JIG certifies in writing that Iran has achieved all of the following Phase I performance benchmarks:

- 90 consecutive days of IAEA-verified compliance (no Tier 3 violations, and no more than one Tier 2 violation lasting less than 7 days during the 90-day period);
- Verified HEU consolidation and custody activation under Article 1.1;
- Successful execution of the automated Day-7 synchronized lift protocol;
- No material breach of the missile/drone deployment pause under Article 14.

De Minimis Error Buffer (v1.1 Addition): In calculating the 90 consecutive days of IAEA-verified compliance, the following de minimis errors shall not count as violations and shall not reset the compliance clock:

- Single-day OLEM data transmission interruption under 2 hours, resolved within 24 hours;
- Administrative reporting delay under 12 hours;
- Seal tamper alert subsequently determined to be equipment malfunction (not deliberate tampering);
- Scheduled maintenance with 48 hours' advance notice as permitted under Article 3.3.

This buffer protects against mechanical or administrative errors that do not reflect bad faith. **Neither Party may invoke a de minimis error as grounds for step-back or penalty.**

Rolling Compliance Window (v1.1 Fix): Compliance is measured on a **rolling 90-day window**, not a fixed period with resets. A Tier 2 violation reduces the compliance percentage by 10 days for each day the violation persists, capped at a 30-day reduction. The clock does not reset to zero. Full compliance is restored when the rolling 90-day window contains no uncured Tier 2 violations lasting more than 7 days, and no Tier 3 violations. This structure makes gaming mathematically difficult: a single violation is a setback, not a catastrophe, and repeated violations compound rather than restart the clock.

Upon JIG certification, the sanctions freeze shall automatically expand to cover all nuclear-related sanctions designated under this Framework, with non-nuclear sanctions (terrorism, human rights, conventional weapons) reviewed separately through the JIG for potential Phase II resolution. This is not a sunset — it is a performance-based graduation. If compliance falters after graduation, step-back under Article 13 applies proportionately.

US Narrative: "This isn't a sunset — it's a merit badge. Iran earns its way into broader relief by proving compliance. If they cheat afterward, we take it back."

7.3 Eight-Tranche Sanctions Relief Architecture (Restructured)

Table 11: Eight-Tranche Sanctions Relief Architecture (30-Day Blocks)

Tranche	Timing	Relief Granted	Physical Milestone Trigger	Reversibility
Pilot + Block 1	Day 1 (deposit) → Day 30 (release)	$500M release (humanitarian + partial structural); medicine, food, medical equipment licenses; partial sanctions freeze (nuclear-related only)	Accord signature; ceasefire confirmed; HEU digital dual-lock deposited; $500M US escrow deposited	Non-reversible after Day 30 release
Block 1 Supplement	Day 45 and Day 60	$250M in two equal increments ($125M each)	Continued IAEA verification of compliance; no Tier 2+ violations	Conditional on continued compliance

1	Day 30	25% of remaining frozen assets; civilian aircraft parts, agricultural goods	Stage 1 closed; IAEA deployed; stockpile custody verified	Non-reversible
2	Day 60	25% of remaining frozen assets; petrochemical exports authorized	60-day compliance; rotor removal verified	Non-reversible
3	Day 120	20% of remaining frozen assets; metals and minerals trade	Permanent JHTA operational; dual-blockade lifted	30-day remediation
4	Day 180	15% of remaining frozen assets; automotive parts, textiles	180-day compliance review	JIG breach review
5	Day 365	10% of remaining frozen assets	12 months clean certification	Material-breach hold
6	Year 2–3	Remainder of frozen assets	Full successor framework compliance	Successor risk appetite

Article 7.3-A — Over-Compliance Credit (v1.1 Operative Addition)

> **ARTICLE 7.3-A — ACCELERATION FOR OVER-COMPLIANCE (OPERATIVE AND BINDING)**
>
> If Iran completes a milestone ahead of schedule and maintains IAEA-verified compliance during the accelerated period, the corresponding economic tranche shall be **released proportionally earlier**. Over-compliance credits may be **banked and applied to future tranche conditions**, effectively allowing Iran to "earn ahead" rather than waiting for rigid deadlines.
>
> The JIG shall certify over-compliance credits quarterly. Credits are calculated as: (days ahead of schedule / days in milestone window) × tranche value. Banked credits may be applied to any future tranche at Iran's election, or redeemed for accelerated release of the current tranche. **Over-compliance credit does not waive any verification requirement — all IAEA and JIG certification procedures remain mandatory. Scope-Based Over-Compliance Credit:** Over-compliance credits also apply to **scope expansion. If Iran exceeds a milestone requirement by 20% or more (e.g., mothballing 120 rotors instead of 100, or downblending 120 kg instead of 100 kg), the JIG may certify a scope-based credit of up to 10% of the associated tranche value, calculated as: (actual scope − required scope) / required scope × 10%,**

> **capped at 10% per milestone. These credits are bankable and may be applied to any future tranche.**

7.4 Currency Protection

All downstream asset releases from Tranche 3 onward are denominated in IMF Special Drawing Rights (SDR) or a fixed basket of Swiss Francs and Omani Rials (60/40). This protects real economic value from dollar inflation or monetary shocks. **Exchange Rate Floor:** Notwithstanding the SDR or basket denomination, the real value of each asset release shall be no less than the equivalent value in 2026 US dollars as of the date of Framework activation, adjusted for US CPI-U inflation. If the SDR depreciates by more than 10% against the 2026 US dollar baseline, either Party may request JIG review and adjustment of the conversion mechanism within 30 days. This protects Iran from currency volatility without creating a one-way bet against the dollar.

Rationale for 30-Day Block Reform: Tehran viewed the v4.0 15-day micro-cycles as a trap that stripped them of leverage. They took massive, irreversible physical nuclear steps on Day 1 (consolidating HEU, freezing enrichment, declaring mothball status) but received only a relatively small $250 million humanitarian tranche — with the bulk of sanctions relief held behind multiple future milestones they might never reach. The 30-Day Stabilization Block with Mutual Escrow Deposit creates true, synchronized structural reciprocity: both parties deposit their commitments on Day 1, and both receive their benefits simultaneously at Day 30. Neither side is asked to move first or alone.

Article 7.5 — Political Discretion Fund (v1.2 Human-in-the-Loop Addition)

> ARTICLE 7.5 — POLITICAL DISCRETION FUND (OPERATIVE AND BINDING)
>
> **(a) Establishment:** A **Political Discretion Fund** of **$50 million** shall be established, capitalized equally by both Parties (**$25 million each**) from the Mutual Escrow Deposit. The Fund is held in a dedicated sub-account of the Swiss escrow and is **exempt from all step-back penalties** under Article 13. No Tier 2 or Tier 3 penalty may reduce, freeze, or delay disbursement from the Political Discretion Fund.
>
> **(b) Disbursement Authority:** Disbursement from the Political Discretion Fund requires **unanimous agreement** of: both Parties' heads of government (or their pre-designated representatives at the Deputy Secretary/Deputy Minister level or above); and the Swiss facilitator. All three must agree. No single Party may unilaterally

disburse Fund resources. The Swiss facilitator's role ensures that disbursement is genuinely mutual and not coerced.

(c) Permitted Uses: The Political Discretion Fund may be used for:

- **(i) Goodwill projects:** Joint infrastructure (bridges, power lines, water treatment) that benefits both Parties' populations; cultural exchange programs; scientific cooperation initiatives; environmental restoration projects;
- **(ii) Partial compliance rewards:** If a Party is in substantial but not full compliance, a discretionary payment may be authorized as a positive incentive for continued progress. This acknowledges that compliance is rarely binary — there is a spectrum, and political judgment is required to navigate it;
- **(iii) Confidence-building measures:** Payments that facilitate cooperation outside the strict compliance schedule, such as humanitarian surge funding, emergency medical supply purchases, or disaster relief coordination.

(d) Prohibited Uses: The Political Discretion Fund shall not be used for: military purposes; intelligence activities; payment of penalties or liquidated damages; or any purpose that would violate the No Regime Change Objective (Preamble). The Swiss facilitator shall review all proposed disbursements for compliance with these prohibitions.

(e) Reporting: All disbursements from the Political Discretion Fund shall be: published on the Swiss Dashboard within 7 days; included in the JIG Monthly Bulletins (Article 12.3-B); and audited annually by an independent auditor approved by both Parties.

(f) Replenishment: The Fund shall be replenished annually to its $50 million cap from escrow interest accruals (Article 8.1-A). If interest is insufficient, both Parties shall contribute equally. If either Party declines to replenish, the Fund operates at its current balance until exhausted.

Why This Matters: The current Framework treats compliance as binary: you are either compliant (full benefits) or non-compliant (step-back penalties). Real compliance is a spectrum. The Political Discretion Fund gives politicians a tool for the gray zone — a way to reward partial progress, build goodwill, and create positive incentives that the automatic compliance architecture cannot provide. At $50 million, it is small enough to be non-strategic but large enough to matter politically. The unanimous requirement ensures neither Party can abuse it. It is political discretion, not mechanical rigidity.

Article 8: Neutral Escrow & Asset Shielding Architecture

Issue 8: Liquidating and Shielding Frozen Assets. Iran demands the immediate unfreezing of its sovereign assets worldwide. The US requires ironclad performance reserves to prevent diversion.

The Resolution: Joint Sovereign Custody Protocol with Neutral Escrow

A slice of frozen funds (30%) is transferred to an independent third-party account at the Central Bank of Oman (Muscat), legally designated for civilian infrastructure reconstruction and verified humanitarian relief. The remaining 70% remains in existing frozen accounts, released tranche-by-tranche upon verified milestones.

8.1 Escrow Allocation

Table 12: Escrow Fund Allocation

Category	Allocation	Administrator	Rolling Back
Humanitarian (medicine, food)	25%	UN OCHA / WHO	Protected — Humanitarian Firewall
Critical Infrastructure	45%	World Bank / Swiss Financial Cell	Protected — Tranches 1–2
Iranian Civilian Reconstruction	20%	JHTA with Iranian programmatic control	Subject to Tier 3 review
Maritime Safety	10%	International Maritime Safety Fund	Protected

Article 8.1-A — Escrow Interest Accrual (v1.1 Operative Addition)

> ARTICLE 8.1-A — INTEREST ON ESCROWED FUNDS (OPERATIVE AND BINDING)
>
> Funds held in escrow shall accrue interest at the **greater of (a) the 6-month LIBOR rate, or (b) the average annual inflation rate of the United States (CPI-U) and Iran (official CPI)** as jointly certified by the IMF. This dual-floor protects the real value of escrowed assets for both parties. Interest is credited to the Iranian Civilian Reconstruction Fund upon release and is not subject to clawback or step-back under any circumstances. This provision ensures that Iran benefits from the time value of

> assets held in escrow, creating a positive incentive for funds to remain in the escrow structure rather than demanding immediate release.

8.2 Sovereign Immunity Wrapper

The escrow fund is legally structured under sovereign immunity protections within Oman. The US issues a binding national security waiver protecting all fund assets from domestic judicial seizure, private civil litigation, or secondary sanctions enforcement. No US court may attach, freeze, or garnish Fund assets. This shield applies retroactively from first deposit and survives any framework suspension.

8.3 OFAC Comfort Letters and Pre-Clearance

The US Treasury shall issue comprehensive, irrevocable OFAC general licenses and formal Comfort Letters to all participating European and Gulf clearinghouses. These letters provide that all transactions routed through the designated Central Bank of Oman escrow account for humanitarian or civilian purposes are legally pre-cleared and entirely exempt from secondary sanctions exposure, asset forfeiture mechanisms, and civil or criminal penalties. Non-US financial institutions, shipping companies, insurers, and commodity traders are explicitly shielded from secondary sanctions designation. Immunization is legally binding and cannot be revoked by executive discretion without 180 days' prior written notice. No notice of revocation may be issued during the first 365 days.

> **ARTICLE 8.3-A — HUMANITARIAN CARVE-OUT FOR FUTURE UNSC SANCTIONS (V1.2 OPERATIVE ADDITION)**
>
> **(a) Commitment to Seek Exemptions:** Notwithstanding any future UN Security Council resolution imposing sanctions on the Islamic Republic of Iran, the United States commits to using its diplomatic powers at the Security Council to **secure humanitarian exemptions** for: escrow funds held under this Framework; all humanitarian transactions (medicine, food, medical equipment, aviation safety); and Survival Core operations (Article 11.1-A). This commitment is a best-efforts obligation, not a guarantee against future sanctions. It acknowledges that the US cannot control UNSC action, but will actively work to protect Framework humanitarian mechanisms.
>
> **(b) Survival Core Protected Status:** The $50M Survival Core Escrow Fund (Article 11.1-A) and all Survival Core operations (hotline, humanitarian corridor, diplomatic channel) shall be designated as **humanitarian-protected status** in any US diplomatic representation to the UNSC. The US shall argue that Survival Core operations are humanitarian in nature and should be exempt from any sanctions regime under UN

Charter Article 103 (humanitarian exception) and customary international humanitarian law.

(c) Escrow Protection: The US shall argue that escrow funds held by a neutral third party (Switzerland/Oman) for the purpose of verified compliance incentives are not Iranian state assets subject to sanctions. This legal argument shall be based on the trust structure of the escrow and the IAEA's verification role.

(d) Honest Acknowledgment: This provision does not guarantee that future UNSC sanctions will exempt Framework mechanisms. It is a binding commitment to try. If the US fails to secure exemptions despite good-faith efforts, neither Party may treat that failure as a Framework violation. The Guarantor Matrix may review the circumstances and recommend adjustments, but no penalties attach to an unsuccessful diplomatic effort.

8.4 Spoiler Litigation Grandfathering (v1.1 Hardening)

ARTICLE 8.4 — SPOILER LITIGATION GRANDFATHERING (OPERATIVE AND BINDING)

(a) Pre-Framework Contract Protection: All civilian commercial contracts signed prior to Framework activation that are otherwise compliant with international law shall be **protected against spoiler litigation** designed to attach escrowed assets.

(b) Spoiler Litigation Definition: "Spoiler litigation" is legal action in any jurisdiction that: seeks to attach, freeze, or seize assets released under this Framework; and is funded or encouraged by a Party in violation of Appendix D.1-A, or is brought by entities with no good-faith claim to the assets.

(c) Stay of Proceedings: Upon notification by the Joint Secretariat that litigation constitutes spoiler litigation, the affected Party may seek a **stay of proceedings** in the relevant court based on the Framework's sovereign immunity and act of state provisions.

(d) Indemnification: A Party that suffers losses due to spoiler litigation shall be **indemnified** by the Party found to have supported or encouraged such litigation, up to the amount of losses plus reasonable legal fees. Indemnification is enforceable through the Swiss escrow mechanism.

Article 9: Phased SWIFT Reconnection Protocol

Issue 9: SWIFT Reconnection and Banking Isolation. Iran's total disconnection from the global financial grid paralyzes its commercial sector.

The Resolution: Phased Multi-Tier Virtual Consolidation with Targeted Bank Isolation (v1.0 Reform)

The v4.0 cascading tier rollback mechanism is replaced with Targeted Bank Isolation through pre-approved clearing hubs, preventing billions in stranded mid-transit transactions while preserving compliance leverage.

9.1 SWIFT Reconnection Phases

Table 13: SWIFT Reconnection Phases

Phase	Timing	Access	IAEA Milestone Trigger
Phase 0 (Pre-SWIFT)	Days 1–30	No SWIFT. Qatari riyal bridge facility for humanitarian payments only	OLEM confirms 60% enrichment freeze within 72h
Phase 1 (Restricted)	Days 30–60	5 designated Iranian banks; MT 103/202 only; transaction cap $1M per day per bank	Moratorium declared; standard monitoring confirmed
Phase 2 (Expanded)	Days 60–120	All clean-record Iranian banks; full message types; transaction caps lifted for humanitarian trade	Continuous monitoring; no diversion detected for 60 days
Phase 3 (Full)	Day 120+	Full SWIFT membership; all Iranian financial institutions in good standing	Year 1 clean certification

9.2 Targeted Bank Isolation Protocol (v1.0 Reform)

ARTICLE 9.2 — TARGETED BANK ISOLATION SYSTEM (OPERATIVE AND BINDING)

(a) Deletion of Tier Rollbacks: The v4.0 SWIFT Remediation Fuse and cascading tier rollback mechanism is hereby removed in its entirety. Under v4.0, if a compliance issue was detected, a 14-day "Remediation Fuse" triggered, and if unresolved, the entire financial tier rolled back exactly one level, cutting off banking messages across multiple institutions. This provision is deleted.

(b) Targeted Bank Isolation — Institution Only: If the IAEA or financial monitors flag a compliance issue at a specific Iranian bank, the response shall be **targeted and isolated**, not sweeping. **Isolation applies to the specific legal entity found to be non-compliant only — not to transaction streams, sectors, affiliates, or subsidiaries.**

(b-1) "Clean-Record Iranian Bank" Definition (v1.1 Fix): A "clean-record Iranian bank" is defined as a financial institution that has: **(i)** no outstanding OFAC sanctions designation; **(ii)** no Tier 2 or Tier 3 compliance violation in the preceding 180 days; and **(iii)** certified to the JIG that it has implemented AML/CFT controls consistent with FATF standards as verified by an independent audit. The JIG may suspend a bank's clean-record status by majority vote pending investigation of a credible compliance concern. This creates an objective standard, not a political one.

- **Step 1 — Flagged Institution Hold:** The specific bank flagged for compliance concern shall be immediately routed to an isolated, pre-approved clearing hub: the **Swiss Humanitarian Escrow** or the **Omani Financial Bridge Facility**.
- **Step 2 — Continued Broad Access:** All clean-record Iranian banks that are not the subject of the compliance flag shall continue uninterrupted SWIFT access at their current tier level. The broader SWIFT network remains fully active for compliant institutions.
- **Step 3 — Isolated Clearing:** Disputed transactions from the flagged institution(s) shall clear through the isolated hub under enhanced monitoring (48-hour settlement delay, dual-key authorization, IAEA financial monitor oversight). Clean transactions from the flagged institution may continue through the hub with standard monitoring.
- **Step 4 — Resolution Pathways:** The flagged institution has 14 days to remediate the compliance concern. Upon IAEA certification of remediation, full standard SWIFT access is restored. If remediation fails, the institution remains on the isolated hub, with potential escalation to individual sanctions designation under existing OFAC authorities — but this does not affect other Iranian banks. **(c) Rehabilitation Review (v1.2 Operative Addition):** A bank that remains on the isolated hub for more than **90 days** may petition the JIG for a rehabilitation review. The JIG shall convene within 30 days of petition. If the JIG determines by majority vote (excluding the affected Party's members) that the bank has implemented corrective measures and no longer poses a compliance risk, the bank shall be returned to standard SWIFT access. A bank may petition for rehabilitation review **once every 180 days**. Unreasonable denial of a meritorious rehabilitation

> petition may be appealed to the Guarantor Matrix. This provision ensures that no bank faces indefinite isolation without a path to redemption.
>
> **(c) No Cascading Disconnect:** Under no circumstances may a compliance issue at one Iranian bank trigger a cascading disconnect for the entire Iranian banking sector. Each bank is assessed and managed individually. A single technical bookkeeping error at one institution does not affect any other institution's SWIFT access.
>
> **(d) Western Bank Protection:** The isolated clearing hub structure protects Western banks from the risk of billions of dollars in global commercial transactions becoming stranded mid-transit due to a sudden country-wide tier rollback. Transactions already in flight when a flag is raised complete through the hub with guaranteed settlement.

Rationale: Western banks refused to participate in v4.0 because a sudden compliance "roll back" threatened to leave billions of dollars in global commercial transactions instantly stranded mid-transit. Conversely, Iran feared that a single technical bookkeeping error at one rogue bank would trigger a cascading disconnect for their entire banking sector. Targeted Bank Isolation solves both: clean banks stay connected, flagged banks clear through a protected hub, and no transaction is left stranded.

Article 10: Regional Stability Reconstruction Credits

Issue 10: War Reparations vs. Economic Reconstruction Credits. Tehran formally requested US-funded war reparations. The Trump administration flatly rejected the concept of reparations.

The Resolution: Regional Stability Reconstruction Credits with US Dollar-for-Dollar Matching Guarantee (v1.0 Reform)

Framing the financial packages as development credits rather than liability-based reparations — keeping both sides politically safe at home while delivering equivalent economic benefit — now enhanced with a US matching guarantee to address Iranian hardliner concerns about asymmetric burden-sharing.

10.1 The No-Fault Architecture with US Matching Guarantee (v1.0 Reform)

> **ARTICLE 10.1 — REGIONAL STABILITY RECONSTRUCTION CREDITS WITH US DOLLAR-FOR-DOLLAR MATCHING GUARANTEE (OPERATIVE)**

(a) Fund Structure: The International Gulf Reconstruction Fund — established under sovereign immunity in Oman, managed by the World Bank with Iranian programmatic input. Funding base: 30% of Iranian frozen assets (already Iranian money) plus contributions from Gulf states, EU, and international development banks.

(b) US Dollar-for-Dollar Matching Guarantee: The prior v4.0 limitation of US contribution to "technical assistance, engineering expertise, and development finance guarantees — not direct cash" is hereby supplemented. **Amendment:** For every dollar Iran draws from its own unfrozen assets to spend on certified civilian infrastructure (including but not limited to water treatment, power generation, telecommunications, healthcare facilities, and educational institutions), the US Development Finance Corporation (DFC) **must** match it with an equivalent, low-interest long-term credit line or development guarantee.

(c) Matching Caps and Conditions:

- The US matching commitment is **capped at $5 billion per calendar year** during Phase 1 and Phase 2, with a **minimum annual commitment of $1 billion**. If the DFC fails to provide the minimum annual matching commitment, the cap lifts and Iran may draw an equivalent amount from its own frozen assets without matching requirements.
- Matching funds are released only upon IAEA-JIG verification that the Iranian-drawn funds have been disbursed to certified civilian infrastructure projects (not military, dual-use, or IRGC-linked entities).
- **Disbursement Timeline:** DFC matching funds shall be disbursed within 90 days of IAEA-JIG verification that Iranian-drawn funds have been expended on certified projects. Delays beyond 90 days trigger automatic interest accrual at LIBOR + 2%.
- DFC credit lines carry a maximum interest rate of 2% above the 10-year US Treasury yield, with 15-year repayment terms and 3-year grace periods.
- Matching funds may be structured as direct DFC loans, loan guarantees enabling private bank participation, or blended finance structures — at Iran's election.
- **Private Sector Bridge:** If DFC direct lending capacity is exhausted, the matching commitment may be fulfilled through: (a) loan guarantees to private banks (enabling private capital at reduced risk); (b) political risk insurance for private investors; (c) syndicated loans arranged by DFC with commercial bank participation. The minimum $1 billion annual floor applies to total matching value (direct loans + guarantees + insurance), not solely direct DFC loans. This expands the pool of capital available for matching and protects against DFC budget constraints.

> **(d) Certified Expenditure Categories:** Matching applies to: civilian power generation; water treatment and sanitation; telecommunications infrastructure; hospitals and medical equipment; schools, universities, and vocational training centers; environmental remediation (oil spill cleanup, desertification programs); and agricultural modernization.
>
> **(e) Excluded Categories:** Military facilities, dual-use research, IRGC-linked entities. Automatic exclusion; no appeal. No matching funds may be directed to any entity designated on the SDN list.

10.2 Eligible Expenditure Categories

Table 14: Eligible Expenditure Categories

Category	Examples	Conditions	US Matching
Civilian Infrastructure	Power generation, water treatment, telecommunications	Open tender; IAEA diversion clearance	Yes — DFC credit line
Healthcare	Hospitals, medical equipment, pharmaceutical production	WHO verification of civilian end-use	Yes — DFC guarantee
Education	Schools, universities, vocational training centers	UNESCO oversight	Yes — DFC loan
Environmental Remediation	Oil spill cleanup, desertification programs	UNEP technical assessment	Yes — blended finance
Military facilities, dual-use research, IRGC-linked entities	—	Automatic exclusion; no appeal	No matching; prohibited

Rationale: Tehran viewed v4.0 as deeply asymmetric — they were asked to fund their own "reparations" using their own frozen money while the US provided nothing but "technical advice." Hardliners in Tehran labeled this an insult to Iranian sovereignty. The Dollar-for-Dollar Matching Guarantee gives Tehran the massive financial influx needed to justify the deal domestically ($5 billion annual DFC commitment represents real, tangible US skin in the game) while keeping the US safe from paying direct "reparations" — the matching is structured as development finance, not liability payment. Both sides can defend the arrangement at home.

Part IV

Treaty Durability & Regional Architecture

An agreement that collapses upon the first breach is worse than no agreement at all — it destroys trust and makes future negotiation impossible. This Part addresses the structural challenges that determine whether the Framework endures beyond the first crisis, introduces a rolling horizon alternative to fixed sunset clauses, and explicitly decouples compliance penalties from military authorization.

Article 11: The 30-Day Letter of Intent & Binding Timeline

Issue 11: The 30-Day Negotiation Window. The current ceasefire is highly fragile. President Trump has stated he will only give diplomacy "a few days" or "one shot." The absence of a binding timeline means every day of negotiation is a day of existential risk.

The Resolution: Binding Letter of Intent with Auto-Renewing Ceasefire

Transitioning the unstable temporary halt into a binding 30-day negotiation timeline locked via a mutual Letter of Intent, underpinned by a self-renewing 72-hour ceasefire that eliminates the "negotiation about negotiation" trap.

11.1 The Letter of Intent

Upon signature, both parties shall simultaneously deposit a Letter of Intent with the Joint Secretariat affirming: commitment to the full 180-day implementation timeline; authorization for the Joint Secretariat to convene emergency consultations within 4 hours of any reported violation; agreement that no military action shall be taken during the 180-day period without first exhausting dispute resolution procedures; acceptance that the 72-hour ceasefire auto-renews at Hour 71 provided verification teams report compliance.

11.2 The Escrow Ceasefire Architecture

The ceasefire is a self-renewing 72-hour module. If verification teams report compliance at Hour 71, the module auto-renews for another 72 hours. Peace is maintained without either side signing away long-term leverage upfront. Separating the ceasefire clock (72-hour loops) from the implementation clock (30-day phases) neutralizes the "negotiation about negotiation" trap.

11.3 Attribution Deferral Window for Proxy Incidents

If a regional proxy strike occurs, neither side may unilaterally declare a material breach. The Joint Coordination Secretariat has 48 hours to conduct forensic telemetry and attribute the incident. During this window, the 72-hour military freeze remains active, provided Iran issues an immediate public condemnation and freezes matching local militia supply logistics. Only the Secretariat's attribution finding can trigger Tier escalation. Rogue actors and false-flag operations are excluded from bilateral penalties.

Article 12: Distributed Guarantor Matrix

Issue 12: Institutional Mediation and Guarantee Frameworks. A profound trust deficit exists between Washington and Tehran. Neither country believes the other will honor a signed document.

The Resolution: Six-State Distributed Guarantor Matrix

A formal guarantor consortium where regional mediators and global powers serve as co-depositaries and technical verifiers — distributing trust across multiple independent parties.

12.1 The Matrix Structure

Table 15: Distributed Guarantor Matrix

State/Entity	Function	Core Guarantee
Oman	Lead facilitator, host nation, JHTA maritime Secretariat location, diplomatic backchannel	Physical hosting of JHTA Secretariat; observer
Pakistan	Procedural mediation, crisis summits, leadership relay	Emergency convening authority within 24 hours

Qatar	Financial escrow and bridging liquidity	Emergency liquidity for humanitarian payments
China (PRC)	Economic guarantor, Hormuz oil importer, non-military technical observer	Guarantee of energy transit; non-enforcement
Russian Federation	Nuclear technical guarantor (HEU disposition Path D)	Technical custody and conversion verification
European Union (E3+)	Sanctions relief coordinator, civilian nuclear partner	Blocking statute protection against secondary sanctions

12.2 The Mandatory Escalation Pause

When any facilitating state determines escalation is imminent, it issues a Mandatory Escalation Pause Recommendation. This activates a 30-minute review window. Neither party may take escalatory action. Both must designate senior officials for consultation within 30 minutes. Either party may request extended Cooling-Off (24–72 hours). If either party supports the pause, the freeze continues.

12.3 Joint Implementation Group

The JIG is a standing technical body composed of one senior official from each guarantor state plus the US and Iran. It meets monthly by secure video and within 72 hours of any reported discrepancy. The JIG does not have binding arbitration authority but serves as the primary compliance assessment body.

12.3-B Political Judgment Override of JIG Technical Findings (v1.2 Human-in-the-Loop Addition)

> **ARTICLE 12.3-B — POLITICAL JUDGMENT OVERRIDE (OPERATIVE AND BINDING)**
>
> **(a) Override Authority:** Notwithstanding any technical finding by the JIG, **either Party may refer a dispute to the Guarantor Matrix for political review** if the Party believes that the JIG's technical finding is based on incomplete evidence, reflects technical error, or fails to account for political context. The referral must be in writing, filed with the Joint Secretariat, and must specify the grounds for political override.
>
> **(b) Guarantor Matrix Political Review:** The Guarantor Matrix shall convene within **14 days** of referral and issue a **political judgment**. The Guarantor Matrix's judgment **supersedes the JIG's technical finding for purposes of Tier 2 or Tier 3 step-back**

decisions. The JIG's technical finding remains the authoritative technical record, but political leaders may override its consequences.

(c) Scope of Political Review: The Guarantor Matrix may: **(i)** affirm the JIG's technical finding and its consequences; **(ii)** modify the consequences (reduce penalties, extend timelines, add conditions); **(iii)** suspend consequences pending further compliance; or **(iv)** overturn the consequences while accepting the technical finding. The Guarantor Matrix applies **political judgment, not just technical metrics** — it may consider: the bilateral political relationship; regional security context; humanitarian implications; and the Party's overall compliance record.

(d) Technical Record Preserved: The JIG's original technical finding remains on the Dashboard as the authoritative technical record. The Guarantor Matrix's political judgment is filed alongside it. Both are public. The distinction between "what happened" (technical) and "what to do about it" (political) is preserved. Technical experts inform; politicians decide.

(e) Limitations: Political override may not: reverse a confirmed Tier 3 Deliberate Hostile Violation involving fissile material diversion (Article 1.7 emergency procedures apply); override the No Automatic Military Escalation Clause (Article 18); or override a mandatory Cooling-Off Period (Article 19-F). These are hardwired protections that no political body may circumvent.

(f) No Precedent: Each political override is *sui generis*. The Guarantor Matrix's reasoning in one case does not bind future cases. This prevents the override mechanism from becoming a routine nullification tool.

Why This Matters: The JIG is a technical body of experts. Technical findings are often ambiguous — sensor readings can be misinterpreted, context can be missed, political implications can be overlooked. Political leaders, not technical experts, are accountable for the consequences of compliance decisions. This provision ensures that technical experts inform, but do not control, political decisions. The machine advises; the humans command.

12.3-A JIG Annual Public Report (v1.1 Operative Addition)

ARTICLE 12.3-A — ANNUAL TRANSPARENCY REPORT (OPERATIVE AND BINDING)

No later than 90 days after each anniversary of Framework activation, the JIG shall issue a **public annual report** summarizing:

- **(a)** Compliance status for each Party against Framework benchmarks;
- **(b)** Any Tier 2 or Tier 3 violations, their resolution, and any pending remediation;
- **(c)** Economic benefits delivered to each Party, including escrow releases, DFC matching disbursements, and SWIFT phase achievements;
- **(d)** Regional security incidents and their disposition under the Lebanon escalation ladder and Dossier Isolation Firewall;
- **(e)** Recommendations for Framework improvement, including technical upgrades and procedural refinements.

The report shall be unclassified and published on the Swiss Dashboard. **Nothing in this provision requires disclosure of intelligence sources, methods, or classified operational details.** Either Party may request redaction of specific sensitive information; disputes over redaction are resolved by the JIG Chair within 7 days.

(f) Israeli Caveat Right (v1.1 Fix): The State of Israel, as an **affected non-party**, may submit a caveat of up to **500 words** to be appended to the unclassified JIG Annual Public Report, provided the caveat: does not disclose intelligence sources or methods; does not name Iranian officials; and is limited to factual corrections or alternative attribution assessments. The caveat shall be labeled *"Submitted by the State of Israel pursuant to Article 16 of the Framework."* **Iran shall have no right to edit or veto the caveat**, but may submit a **response caveat of equal length**, labeled as an Iranian response. Neither caveat is binding on the JIG or either Party.

ARTICLE 12.3-B — MONTHLY COMPLIANCE BULLETINS (V1.2 OPERATIVE ADDITION)

ARTICLE 12.3-A(G) — EXECUTIVE SUMMARY OF POLITICAL CONSEQUENCES (V1.2 HUMAN-IN-THE-LOOP ADDITION)

(g) Political Consequences Summary: Every JIG report — annual, monthly, or emergency — shall include, on the **first page**, a one-paragraph **"Executive Summary of Political Consequences"** written in plain, non-technical language accessible to heads of government and senior political staff. The summary shall answer four questions:

- **(i)** Does this report identify any compliance issue that could trigger Tier 2 or Tier 3 penalties?

- **(ii)** Does any pending decision require political-level consultation (Executive Consultation Window, Guarantor Matrix review, or Political Consultation Channel)?
- **(iii)** Has any regional incident crossed a Tier threshold under Appendix E (Lebanon Escalation Ladder)?
- **(iv)** Has any Party invoked Temporary Operational Suspension (Article 20), Force Majeure (Article 22), or other emergency measures?

The summary is **not binding** — it is a political briefing tool. It does not replace the technical report; it translates the technical report into political language. The summary ensures that political leaders can grasp the operational status of the Framework without reading 200 pages of technical detail. The JIG Chair is responsible for ensuring the summary is accurate, concise, and non-alarmist. Either Party may request revision of the summary if it believes the summary mischaracterizes the technical findings; revision disputes are resolved by the Swiss facilitator within 48 hours.

Why This Matters: Political leaders have limited time and attention. A 200-page technical report with detailed sensor readings, forensic data, and forensic analysis is essential for technical staff but useless for a president or supreme leader who needs to know: "Is there a crisis? Do I need to act? Who do I call?" The Executive Summary of Political Consequences answers those questions in one paragraph. It respects political time and attention span while preserving technical rigor for those who need it.

(a) Monthly Publication: No later than the 15th day of each month, the JIG shall issue a **public compliance bulletin** of no more than four pages, summarizing:

- **(i)** Framework modules currently Active, Under Review, or in Temporary Suspension;
- **(ii)** Compliance milestones achieved in the preceding 30 days;
- **(iii)** Escrow releases, DFC disbursements, or SWIFT phase transitions completed;
- **(iv)** Any Tier 1 technical disputes opened or resolved;
- **(v)** Regional incidents logged under the Dossier Isolation Firewall (without attribution);
- **(vi)** Upcoming milestones for the next 30 days.

(b) Format and Distribution: Bulletins shall be: written in plain language accessible to non-technical readers; published simultaneously on the Swiss Dashboard and distributed to both Parties' diplomatic missions; available in English, Farsi, and Arabic;

and archived in a searchable public repository. No classified information, intelligence sources, or operational security details shall be included.

(c) Factual Only — No Attribution: Monthly bulletins shall report **facts and milestones only**. They shall not attribute responsibility for disputes, characterize Party conduct, or recommend policy actions. The tone is neutral, technical, and descriptive — not evaluative. If a dispute exists, the bulletin states only that "a dispute is under review by the JIG."

(d) Emergency Bulletins: The JIG may issue **extraordinary bulletins** within 24 hours of: a Tier 3 Deliberate Hostile Violation; activation of Temporary Operational Suspension (Article 20); a Guarantor Matrix emergency session; or any event that materially alters the Framework's operational status. Emergency bulletins are factual and time-limited — they report what happened, not why or who is at fault.

(e) Correction Protocol: Either Party may request correction of a monthly bulletin within 7 days of publication. Factual errors shall be corrected within 48 hours and republished with a dated erratum. Disputed characterizations (as distinct from facts) are resolved by the JIG Chair. If the Chair determines a characterization is disputed, the bulletin is amended to remove the characterization while retaining underlying facts.

Why This Matters: Information asymmetry is a weapon in domestic politics. If only one side tells its public what the Framework is doing, that side controls the narrative. Monthly bulletins create a neutral factual record that both sides can cite, reference, and defend. They prevent either Party from claiming the Framework is "failing" or "succeeding" in ways that contradict the actual data.

12.4 Scenario Testing and Spoiler Wargaming (v1.1 Hardening)

ARTICLE 12.4 — ANNUAL SPOILER WARGAME AND RED TEAMING (OPERATIVE AND BINDING)

(a) Annual Spoiler Wargame: The JIG shall convene an annual scenario testing exercise, known as the **"Spoiler Wargame,"** to identify potential spoiler tactics and test Framework responses. Participation is mandatory for both Parties at the technical level (Deputy JIG representatives). The wargame is classified but a sanitized summary is published to the Dashboard.

(b) Scenario Library: The JIG shall maintain a classified Scenario Library of potential spoiler actions, including: rogue commander scenarios; false flag attacks; information warfare campaigns; economic sanctions evasion techniques; supply chain diversion tactics; and cyber intrusion against verification systems.

(c) Red Team Function: A neutral **Red Team**, appointed by the Swiss facilitator and approved by both Parties, shall propose spoiler scenarios and assess Framework vulnerabilities. Red Team findings are shared with both Parties and the Guarantor Matrix. The Red Team has no decision authority — its role is adversarial testing.

(d) Remedial Updating: Based on wargame findings, the JIG may propose **technical amendments to spoiler countermeasures** by unanimous consent of the JIG (excluding the affected Party's members for specific vulnerabilities). Such amendments do not reopen the core political bargain — they are technical hardening measures only.

12.5 Consensus Minus One Decision Rule (v1.2 Human-in-the-Loop Addition)

ARTICLE 12.5 — CONSENSUS MINUS ONE DECISION RULE (OPERATIVE AND BINDING)

(a) General Rule: For all **significant decisions** under this Framework — including but not limited to: Tier 2 and Tier 3 determinations; cross-module spillover certifications; Temporary Operational Suspension approvals; pathway selection disputes; and gaming pattern certifications — the JIG shall decide by **"consensus minus one"**: agreement of all JIG members **except** the Party whose conduct is under review, or except the Party invoking the decision.

(b) How It Works: In a JIG with US, Iranian, and guarantor state representatives, "consensus minus one" means: if the decision concerns Iranian conduct, Iran's vote is excluded and all other members must agree; if the decision concerns US conduct, the US vote is excluded and all other members must agree; if the decision concerns a general Framework matter, the Party that invoked the decision has its vote excluded. This ensures that **no Party can be outvoted on decisions about its own conduct**, while preventing a single Party from blocking consensus.

(c) Deadlock Escalation: If consensus minus one cannot be achieved within the applicable timeline for the decision (14 days for Tier 2, 48 hours for Tier 3 emergency, 7 days for TOS), the matter is **automatically referred to the Guarantor Matrix** for final political judgment. The Guarantor Matrix operates by majority vote of its members. There is no appeal from a Guarantor Matrix decision on procedural matters.

(d) Routine Technical Matters: Routine technical matters — including: telemetry anomaly logging; inspection scheduling adjustments; batch transmission timing;

> equipment calibration; Dashboard maintenance; and report formatting — remain subject to **majority vote** of the JIG. Consensus minus one does not apply to routine technical matters. This preserves operational efficiency for day-to-day technical management.
>
> **(e) Jurisdictional Determination:** If there is a dispute about whether a matter is "significant" (requiring consensus minus one) or "routine technical" (majority vote), the Swiss facilitator shall make a preliminary determination within 24 hours. Either Party may appeal to the Guarantor Matrix within 7 days. Pending appeal, the Swiss facilitator's determination governs.
>
> **Why This Matters:** In a body with the US, Iran, and guarantor states aligned with one side or the other, simple majority vote means one Party plus its guarantors can outvote the other Party. That creates a perception of bias and undermines trust. "Consensus minus one" means Iran cannot be outvoted on decisions about Iran's conduct, and the US cannot be outvoted on decisions about US conduct. If consensus can't be reached, the matter escalates to the Guarantor Matrix — preventing both bias and paralysis. Fairness through structure.

Article 13: Executive Consultation Governance (v1.0 Reform)

Issue 13: Internal Regime Fractures and Automated Escalation. Top US officials have highlighted intense ideological fractures within Tehran's leadership. Automated deal triggers could easily be derailed by political hardliners on either side. The v4.0 framework allowed technical enforcement teams on the ground to automatically trigger Tier 2 "benefit reductions" if a deadline was missed by more than 72 hours — handing too much destructive power to low-level inspectors and automated telemetry feeds.

The Resolution: Executive Consultation Governance with Mandatory 48-Hour Consultation Window (v1.0 Reform)

> **V1.0 SURGICAL CHANGE — FROM AUTOMATED PENALTIES TO EXECUTIVE CONSULTATION**
>
> Automation informs; humans decide — including the highest-ranking diplomatic officials. A governance system that requires direct executive-level consultation before any economic penalty can activate, giving political leaders the ultimate veto over whether an anomaly is a genuine breach or just a technical glitch.

13.1 The Three-Tier Breach System (Retained)

Table 16: Three-Tier Breach Classification

Tier	Classification	Telemetry Signature	Base Response
1	Technical Dispute (De Minimis)	<0.5% above enrichment ceiling; ≤2 un-mothballed rotors; ≤48h reporting delay	Engineering resolution within 14 days; no rollback; confidential handling
2	Material Disruption	>0.5% above ceiling; >2 rotors; >72h IAEA obstruction	Up to 25% benefit reduction; 30-day remediation; earn-back via compliance (proportionality rules apply per 13.3)
3	Deliberate Hostile Violation	Nuclear weapons activity; fissile diversion; IAEA attack	Up to 50% reduction; 90-day remediation; full earn-back after sustained compliance (proportionality rules apply per 13.3)

13.2 De Minimis Thresholds

Enrichment ceiling: 3.67% U-235, with ±0.5% De Minimis tolerance. Un-mothballed centrifuges: Up to 2 rotors permitted for **legitimate research** — defined as activities exclusively related to: **(a)** stable isotope production for medical, industrial, or agricultural purposes; **(b)** nuclear safety or materials science unrelated to enrichment; **(c)** IAEA-supervised research under the Additional Protocol; or **(d)** training of IAEA inspectors. Research with direct application to uranium enrichment above 3.67% is not legitimate. The JIG shall maintain a public registry of approved research activities. Any unregistered centrifuge operation is presumptively a Tier 2 violation. IAEA reporting delays: Up to 48 hours for administrative or technical reasons. **(e) De Minimis Research Exemption (v1.2 Operative Addition):** Notwithstanding the above categories, the JIG may authorize by majority vote a **specific, time-limited, low-enrichment research activity** not covered by categories (a) through (d) if: **(i)** the activity has a legitimate scientific or medical purpose; **(ii)** the activity is conducted under IAEA supervision at an IAEA-monitored facility; **(iii)** the activity does not exceed 30 days in duration; **(iv)** the enrichment level does not exceed 3.67% U-235; and **(v)** both Parties are notified 14 days in advance of the activity commencing. No more than **two such authorizations** may be granted in any 365-day period. Unreasonable denial of a request meeting all five criteria may be appealed to the Guarantor Matrix.

13.3 The Step-Back Architecture

Tier 1 Step-Down: No benefit reduction. 14-day resolution. JIG technical assistance. Quiet compliance corrections confidential.

Tier 2 Step-Down: Base 25% reduction in non-humanitarian benefits. Humanitarian tranches protected. SWIFT phase holds (Targeted Bank Isolation active). 30-day remediation. Full restore on IAEA clean certification within 14 days.

Tier 3 Step-Down: Base 50% reduction in non-humanitarian benefits. Tier 3 modules enter hold. 90-day remediation. Full earn-back requires: 90-day sustained compliance + IAEA clean certification + JIG majority confirmation.

Article 13.3 — Proportional Penalty Clause (Operative)

Step-back reductions are proportionate to the severity and duration of the breach. A Tier 2 violation lasting less than 7 days shall trigger a 10% reduction, not 25%. A Tier 3 violation that is immediately remediated (within 72 hours) shall trigger a 25% reduction, not 50%. Full reductions (25% for Tier 2, 50% for Tier 3) apply only after the 30-day (Tier 2) or 90-day (Tier 3) remediation window has expired without compliance. This proportionality rule prevents minor or quickly-cured violations from triggering disproportionate penalties. **Cross-Module Spillover Protection:** Step-back reductions shall be applied **only within the module where the violation occurred**, unless the JIG certifies by a **two-thirds majority vote** (excluding the affected Party's members) that the violation directly and materially undermines a different module. Unanimity is not required. The Swiss facilitator's determination of "direct and material undermining" shall be guided by: **(a)** whether the violation affects verification systems used in another module; **(b)** whether the violation demonstrates a pattern of bad faith across modules; or **(c)** whether the violation has caused economic or military consequences in another module. For example, a Tier 2 nuclear violation triggers step-back only on nuclear-related tranches (Articles 1, 2, 3, 7), not on maritime benefits (Articles 4, 5, 6) or SWIFT access (Article 9), absent unanimous JIG certification. This "module isolation" principle preserves the Framework's modular architecture and prevents a single-file breach from cascading.

> **ARTICLE 13.3-A — PARTIAL SUCCESS DOCTRINE (V1.2 OPERATIVE ADDITION)**
>
> **(a) Binding Doctrine:** The Parties affirm that **no single unresolved module invalidates the remainder of the Framework**. This Partial Success Doctrine is a binding interpretive principle: each module (Nuclear: Articles 1-3; Maritime: Articles 4-6; Economic: Articles 7-10; Durability: Articles 11-28) operates as an independent

contractual track. Progress, compliance, or failure in one track does not automatically determine the status of any other track.

(b) Module Survival Schedule: The following table defines minimum module survival conditions — the baseline that continues even if all other modules fail:

Table X1: Partial Success Module Survival Conditions

Module	Minimum Survival Condition	Protected Elements
Nuclear (Articles 1-3)	Article 18 (No Automatic Military Escalation) survives all nuclear disputes	IAEA verification access; HEU custody; no military escalation
Maritime (Articles 4-6)	Article 4.7 (IRGC-Navy Hotline) survives all maritime disputes	Deconfliction protocols; freedom of navigation baseline
Economic (Articles 7-10)	Article 8.1-A (Escrow Interest) survives all economic disputes	Escrow principal protection; humanitarian tranches
Survival Core (Article 11)	Article 11.1-A (Survival Core Escrow) survives complete Framework suspension	Hotline operations; humanitarian corridor; diplomatic channel

(c) Progressive Reactivation: If one module enters suspension (under Article 20), the remaining modules continue in full force. When the suspended module is reactivated, it resumes from its pre-suspension state — no compliance clock reset, no penalty for the suspension period itself (unless suspension was caused by that module's breach). This prevents module failure from creating permanent compliance debt.

(d) No Aggregate Termination: Neither Party may terminate the entire Framework solely because one or more modules are underperforming, in dispute, or in Temporary Operational Suspension. Termination under Article 17 requires either: (i) comprehensive material breach affecting a majority of modules; (ii) explicit written notice under the Rolling Horizon Architecture; or (iii) invocation of the Expedited Withdrawal Option with liquidated damages payment. No single-module failure constitutes grounds for aggregate termination.

(e) JIG Module-Status Registry: The JIG shall maintain a public "Module Status Dashboard" showing the operational status of each module: Active, Under Review, Temporary Suspension (Article 20), or Escalated. The Dashboard updates within 24 hours of any status change. All Parties and guarantor states have access. This transparency prevents misunderstandings about which elements remain in force.

Why This Matters: Without the Partial Success Doctrine, a single catastrophic failure — a nuclear dispute, a maritime incident, a banking crisis — could unravel the entire Framework even when 80% of it is working perfectly. This doctrine creates legal resilience: the Framework is a portfolio of independent tracks, not a single chain whose weakest link destroys everything. It transforms the architecture from chain to web.

ARTICLE 13.3-B — MANDATORY "SECOND LOOK" REVIEW AT 30 DAYS (V1.2 HUMAN-IN-THE-LOOP ADDITION)

(a) Automatic Review: Any Tier 2 or Tier 3 step-back penalty shall be subject to a mandatory **"Second Look" review** conducted by the Guarantor Matrix **30 days after the penalty activates**. The Second Look is automatic — neither Party needs to request it. The Guarantor Matrix shall convene within 7 days of the 30-day mark.

(b) Second Look Options: The Guarantor Matrix may, by majority vote:

- **(i) Affirm** the penalty — it continues at full strength;
- **(ii) Reduce** the penalty — by up to 50% if the Party has shown substantial compliance progress;
- **(iii) Suspend** the penalty — pending further compliance verification within 30 days;
- **(iv) Terminate** the penalty — if the Party has restored compliance and the IAEA certifies clean status.

(c) Political Review, Not Technical: The Second Look is a **political review**, not a technical one. The Guarantor Matrix applies political judgment: Has the Party made genuine efforts to comply? Are there extenuating circumstances? What are the political costs of continuing the penalty? What signal does reduction or termination send? The IAEA provides technical input, but the Guarantor Matrix makes the political decision.

(d) Repeat Reviews: Either Party may request an additional Second Look review every **30 days thereafter**. The Guarantor Matrix shall convene within 14 days of any such request. There is no limit on the number of Second Look reviews. This ensures that penalties are not perpetual and that restored compliance leads to restored benefits.

> **(e) No Penalty During Review:** While a Second Look review is pending, the penalty remains in effect unless the Guarantor Matrix orders a suspension. The request for review does not automatically stay the penalty.
>
> **Why This Matters:** The current Framework has no automatic penalty review mechanism. Once a penalty is applied, it continues until the Party achieves full compliance and the JIG certifies restoration — a process that could take months. The Second Look creates a regular checkpoint where political judgment is applied. It prevents perpetual punishment and ensures that penalties serve their purpose (incentivizing compliance) rather than becoming permanent sanctions in disguise. Restored compliance should lead to restored benefits — that is the core principle of the Framework.

13.4 Executive Consultation Governance Provisions (v1.0 Reform)

ARTICLE 13.4 — MANDATORY 48-HOUR EXECUTIVE CONSULTATION WINDOW (OPERATIVE AND BINDING)

(a) No Automatic Economic Penalties: The v4.0 provision allowing the JIG and IAEA technical teams to automatically trigger Tier 2 "benefit reductions" (cutting economic tranches up to 25%) if a deadline is missed by more than 72 hours is **hereby deleted**. No economic benefit reduction, asset freeze, or SWIFT restriction may activate automatically based on telemetry readings, missed deadlines, or technical anomaly flags alone.

(b) Mandatory Executive Consultation Window: Any flagged telemetry anomaly, missed deadline, or compliance concern that would previously have triggered an automatic Tier 2 or Tier 3 penalty shall instead trigger a **mandatory 48-hour Executive Consultation Window**. During this window:

- The US Secretary of State and the Iranian Foreign Minister (or their designated deputies at the Deputy Secretary/Ambassador level) shall communicate directly through the Omani diplomatic backchannel.
- Both sides shall present their technical assessments of the anomaly: the IAEA presents its findings; Iran presents its explanation; the JIG presents its preliminary analysis.

- The Parties shall jointly determine whether the anomaly is: (i) a genuine material breach requiring step-back; (ii) a technical glitch or logistical delay not constituting bad faith; or (iii) a disputed matter requiring extended JIG investigation.

(c) Political Leaders' Discretionary Review (v1.2 Human-in-the-Loop Reform): The US Secretary of State and the Iranian Foreign Minister each hold **discretionary review authority** over any proposed Tier 2 or Tier 3 economic penalty during the 48-hour consultation window. Upon review, the official has four options — not an absolute veto:

- **(i) Approve** the penalty as proposed by the JIG;
- **(ii) Reduce** the penalty by up to 50% for documented extenuating circumstances (e.g., the violation was caused by genuine technical failure, natural disaster, or force majeure);
- **(iii) Defer** the penalty for up to 14 days pending further investigation by the JIG or IAEA;
- **(iv) Reject** the penalty only upon a **written finding, shared with the JIG**, that the proposed penalty is: manifestly disproportionate to the alleged breach; based on a demonstrably incorrect technical finding; or would cause demonstrable humanitarian harm to civilians.

Rejection without a written finding meeting one of these three criteria is itself a **Tier 1 technical dispute**. The written finding becomes part of the JIG record and may be reviewed by the Guarantor Matrix upon either Party's request. This discretionary authority ensures political judgment while preventing arbitrary nullification of the Framework's compliance mechanisms. Leaders have options, not a blank check.

(d) Relationship to Political Judgment Override and Second Look (v1.2 Sequential Clarification): The discretionary review in this section applies to **proposed penalties before they activate** — it is the first line of political judgment. The political judgment override in Article 12.3-B applies to **JIG technical findings that have already led to active penalties** — it is the second line. The Second Look in Article 13.3-B is the **mandatory 30-day review** of all active penalties. These three mechanisms are **sequential, not conflicting**:

- **First line (before activation):** Discretionary review under this Article — the Secretary of State or Foreign Minister may approve, reduce, defer, or reject a proposed penalty with written finding;
- **Second line (after activation):** Political judgment override under Article 12.3-B — either Party may refer an active penalty to the Guarantor Matrix for political review at any time;

- **Third line (mandatory checkpoint):** Second Look under Article 13.3-B — the Guarantor Matrix must review all active penalties at 30 days.

Neither mechanism is redundant. All three require written justification for rejection or override. None may circumvent the Tier 3 fissile diversion exception (Article 1.7) or the No Military Escalation Clause (Article 18). The sequential architecture ensures that political judgment is available at every stage — before penalties activate, after they activate, and at regular intervals — while preventing any single mechanism from nullifying the Framework's compliance architecture.

(e) Post-Veto Procedures: If either executive vetoes the proposed penalty, the matter is referred to the JIG for a 14-day technical investigation. During this period: no economic penalties apply; humanitarian tranches continue uninterrupted; the Framework remains in full force. The JIG shall issue a binding technical finding at the end of the 14-day period. If the JIG confirms a material breach by majority vote (excluding the affected Party's members), the penalty may be reactivated — but either executive may invoke a second veto, triggering a 30-day Guarantor Matrix mediation.

(e) Good-Faith Presumption: During the 48-hour Executive Consultation Window, there shall be a rebuttable presumption of good faith. The burden of proof lies on the party alleging breach to demonstrate deliberate non-compliance, not merely technical delay or administrative difficulty.

(f) Good-Faith Consultation Credit: If either Party initiates executive consultation in good faith within the first 24 hours of an anomaly being flagged, any subsequent step-back penalty shall be **reduced by 75% for the first 14 days**, and **no public attribution of breach shall be issued until the 48-hour window expires**. This "early engagement credit" applies once per 90-day period and is **bankable for future compliance milestones**. It makes early engagement dramatically cheaper than delay — a behavioral incentive for immediate diplomatic response.

(g) Automatic Stay for Non-Cooperation: If either Party invokes the Executive Consultation Window but the other Party fails to designate a representative or refuses to communicate within the first 12 hours, the 48-hour window shall be **automatically stayed for an additional 24 hours**, and the Swiss facilitator shall issue a notice of non-cooperation. If the non-cooperating Party continues to refuse engagement after the 24-hour stay, the complaining Party may proceed to the post-consultation procedures under Article 13.4(d) without further delay. A **pattern of refusal (three or more instances in any rolling 180-day period). The 180-day lookback window is rolling — each new gaming instance resets the window to include the preceding 180 days. Additionally, five or more gaming instances in any 365-day period, regardless of distribution, shall constitute a pattern. The JIG shall maintain a cumulative**

> **gaming counter. Gaming instances older than 365 days are not counted. This prevents a Party from committing two gaming violations, waiting 181 days, then committing two more — never triggering the pattern threshold.** constitutes a Tier 2 material disruption. This prevents a party from running out the clock by refusing to talk.
>
> **(h) Expedited Exception:** The Executive Consultation Window does not apply to: (i) Tier 1 technical disputes, which are resolved through normal JIG technical channels; (ii) confirmed fissile material diversion or nuclear weapons activity (Tier 3 Deliberate Hostile Violation), which triggers immediate JIG emergency consultation within 4 hours; (iii) violations of the No Automatic Military Escalation Clause (Article 18), which are handled through separate procedures.

13.5 Attribution Deliberation Media Blackout (Operative and Binding)

When the JIG or SEAC is conducting attribution deliberations for any incident that could trigger Tier 2 or Tier 3 step-back, a mandatory 72-hour media blackout applies to all Parties and guarantor states. During this period:

- No official or on-record statements may be issued attributing responsibility for the incident.
- Leaks of attribution deliberations shall be investigated by the Joint Secretariat; responsible officials shall be subject to Tier 1 technical dispute proceedings.
- The blackout lifts automatically upon JIG issuance of a preliminary attribution finding, or after 72 hours if no finding is issued. The JIG may vote by majority to extend the blackout for an additional 72 hours. Maximum one extension.
- The blackout does not apply to unclassified operational reporting (e.g., "an incident occurred at location X at time Y").

Violation of the media blackout by a Party's official spokesperson or designated representative shall result in a Tier 1 technical dispute demerit and a public corrective statement issued through the Joint Secretariat.

Senior Official Leak Escalation: A deliberate leak attributable to a senior official (minister-level or above, or their immediate designee) shall be referred to the JIG for potential Tier 2 step-back consideration if the leak materially prejudices attribution deliberations. The JIG shall determine within 14 days whether the leak was deliberate and materially harmful. If so, the violating Party shall be subject to a 5% benefit reduction in the affected module for 30 days, in addition to the Tier 1 demerit. Low-level administrative leaks remain at Tier 1. This provision is not a sunset; it is a performance-based consequence for bad-faith conduct.

Rationale: The v4.0 text handed too much destructive power to low-level inspectors and automated telemetry feeds. If a technical sensor glitched or a logistical delay occurred at an underground site, the agreement automatically started punishing Iran economically before the leaders in Washington or Tehran even had a chance to talk. The Executive Consultation Window ensures that no economic benefit reductions or asset freezes can activate without the direct, personal involvement of the US Secretary of State and the Iranian Foreign Minister. Political leaders — not machines — hold the ultimate veto over war and peace.

13.6 Anti-Gaming and Bad Faith Prohibition (v1.1 Hardening)

> **ARTICLE 13.6 — ANTI-GAMING AND BAD FAITH EXPLOITATION (OPERATIVE AND BINDING)**
>
> **(a) Prohibited Gaming Actions:** The following actions constitute "gaming" and are prohibited:
>
> - Deliberately timing compliance actions to exploit grace period boundaries;
> - Using de minimis allowances as a pattern of systematic avoidance;
> - Deliberately triggering technical disputes to delay obligations;
> - Exploiting the 48-hour Executive Consultation Window to stall without good faith;
> - Using "remediation window" resets as a repeated cycle of temporary compliance followed by violation.
>
> **(b) Pattern of Gaming:** If the JIG determines, by majority vote, that a Party has engaged in a **pattern of gaming** (three or more instances in any rolling 180-day period). The 180-day lookback window is rolling — each new gaming instance resets the window to include the preceding 180 days. **Additionally,** five or more gaming instances in any 365-day period, regardless of distribution, shall constitute a pattern. The JIG shall maintain a cumulative gaming counter. Gaming instances older than 365 days are not counted. This prevents a Party from committing two gaming violations, waiting 181 days, then committing two more — never triggering the pattern threshold., the following apply:
>
> - The de minimis buffer is suspended for that Party for 90 days;
> - The 30-day compliance clock reset for minor Tier 2 violations is reduced to 14 days;
> - The Party's JIG representative may be subject to enhanced disclosure requirements.
>
> **(c) Gaming Register:** The JIG shall publish a "Gaming Register" identifying specific prohibited patterns as they are identified, providing notice for all Parties. The Register

> is updated quarterly and published on the Swiss Dashboard.
>
> **(d) Burden Shift:** Once a pattern of gaming is certified, the burden shifts to the gaming Party to demonstrate that each contested action was **not gaming**. The presumption of good faith (Article 13.4(e)) is suspended for that Party until clean compliance is restored for 90 consecutive days.

Article 14: Ballistic & Drone Program Pause Protocol

Issue 14: Drone and Missile Program Reconstitution. Intelligence reports indicate Iran is rapidly using the ceasefire window to reconstitute its drone and missile industrial base. The US wants strict limitations, which Iran has previously declared non-negotiable as sovereign defense capabilities.

The Resolution: Deployment Pause and Transfer Freeze

Deferring heavy ballistic limitations to later, mature dialogue stages while implementing an immediate, verifiable pause on offensive deployment and cross-border transfers — freezing the threat without demanding disarmament.

14.1 The Three-Pillar Structure

Pillar 1 — Deployment Freeze: Iran commits to no new offensive deployment of ballistic missiles or combat drones to regional proxy forces for the duration of the Framework. Verified through satellite surveillance, signals intelligence, and regional observer reports.

Pillar 2 — Transfer Suspension: All cross-border transfers of missile components, drone assemblies, and related technology to non-state actors are suspended. Verification through intelligence sharing among guarantor states and JHTA maritime monitoring of cargo manifests.

Pillar 3 — Industrial Activity Monitoring: Iran's domestic missile and drone production facilities are subject to baseline satellite monitoring. New facility construction or significant production expansion triggers a Tier 1 consultation but is not itself a violation.

14.2 Iranian Safeguards

This Protocol explicitly does NOT: require dismantlement of existing missile or drone inventories; prohibit domestic testing of short-range defensive systems; impose limits on Iran's sovereign right to maintain conventional military forces; subject military facilities to IAEA inspection.

14.3 Rogue Commander Accountability Protocol (v1.1 Hardening)

> **ARTICLE 14.3 — ROGUE COMMANDER ACCOUNTABILITY (OPERATIVE AND BINDING)**
>
> **(a) Rogue Commander Definition:** A "rogue commander" is a military or proxy commander who takes action contrary to the explicit orders of their state leadership, with the effect of undermining the Framework.
>
> **(b) State Responsibility Safe Harbor:** A state is **not responsible** for rogue commander actions if it:
>
> - Issues public condemnation within 24 hours of the action;
> - Takes verifiable disciplinary action against the commander within 14 days;
> - Implements measures to prevent recurrence within 30 days;
> - Provides compensation for any damages (where applicable).
>
> **(c) Burden of Proof:** The burden is on the state claiming rogue commander status to demonstrate that: the commander lacked authorization; the state took timely remedial action; and the action was contrary to state policy.
>
> **(d) Repeat Rogue Pattern:** If a state experiences **three or more rogue commander incidents** in any 180-day period from forces under its effective control, a **rebuttable presumption** arises that the state is not exercising adequate command authority, triggering Tier 1 technical dispute and enhanced monitoring.
>
> **(e) Safe Harbor Incentive:** If a Party takes prompt, verifiable remedial action against a rogue commander within 14 days of the incident, **no step-back shall apply**. Remedial actions include: public condemnation; asset freezes on the commander; arrest or removal of the individual; or severance of supply lines.

Article 15: Energy & Agricultural Release Valve

Issue 15: Systemic Agrifood Shocks and Global Stabilization. The prolonged closure of the Strait of Hormuz has choked off a third of the world's fertilizer supply, causing the UN to warn of a systemic global agrifood crisis.

The Resolution: Immediate Energy and Agricultural Release Valve

Creating an immediate separation of global humanitarian supply lines from the underlying political dispute.

15.1 The Zero-Fee Humanitarian Corridor

From Day-One: Eligible cargo (wheat, rice, fertilizer, medical supplies, infant formula, agricultural seeds, humanitarian energy) fast-tracked through UN OCHA pre-clearance. All JHTA transit fees waived. Vessels may request neutral naval escort (Omani or UN-flagged) at no cost. Insurance covered under the $20 billion reinsurance backstop.

15.2 Agricultural Supply Guarantee

The JHTA shall publish a monthly Agricultural Transit Forecast. The United States and EU shall commit to expedited export licensing for agricultural inputs to Iran during the 180-day implementation period.

15.3 Energy Price Stabilization Mechanism

The JHTA shall publish daily transit statistics and the International Energy Agency shall release weekly strategic petroleum reserve drawdown plans if prices exceed agreed thresholds. Transparency is the antidote to panic.

Article 16: US-Israel Bilateral Coordination

Issue 16: Israeli Security Concerns and the Tel Aviv Channel. The State of Israel has expressed profound concern that a comprehensive US-Iran agreement could proceed without adequate consideration of Israeli security interests, including the potential for Iran to redirect resources toward proxy forces threatening Israel's northern border.

The Resolution: Bilateral Coordination with Non-Impediment Guarantee

Establishing a formal mechanism for US-Israel bilateral security consultation that operates entirely outside the US-Iran bilateral obligations, ensuring that Israeli security concerns are heard while preserving the integrity of the Framework's implementation.

Article 16.1 — Bilateral Coordination Through the Tel Aviv Channel (Operative and Binding)

The United States affirms that it will maintain bilateral security coordination with Israel through the existing Tel Aviv Channel. This coordination is separate from and does not affect the bilateral obligations set forth in this Framework between the United States and Iran.

Article 16.2 — Consultation on Framework-Related Security Matters (Operative and Binding)

The United States will consult with Israel regarding any matter arising under this Framework that implicates Israeli security interests, provided that such consultation does not delay or condition the implementation of any provision of this Framework.

Article 16.2-A — Pre-Decision Consultative Window for Lebanon Incidents (v1.1 Fix)

ARTICLE 16.2-A — 24-HOUR PRE-DECISION NOTIFICATION (OPERATIVE AND BINDING)

(a) Binding Notification: Notwithstanding the separation of channels in Article 16.2, **prior to any US decision to exercise its executive veto under Article 13.4 (Executive Consultation Window) regarding a Lebanon-origin incident**, the United States shall provide Israel with a **24-hour pre-decision notification** through the Tel Aviv Channel. This notification shall include: the US preliminary assessment of attribution; the factual basis for the assessment; and the proposed US vote or veto.

(b) Israeli Written Assessment: Upon receipt of the 24-hour notification, Israel may submit a **written assessment** of up to 2,000 words, which the US shall **append to the JIG record** without editing. The assessment may include: alternative attribution analysis; intelligence-based corrections; and recommended US course of action. The assessment is advisory only and does not bind the US decision.

(c) Procedural Invalidity for Failure to Notify: Failure to provide the 24-hour notification shall render the US veto procedurally invalid for that incident. The matter shall be referred to the Guarantor Matrix for an additional 72-hour consultation before any vote or veto may be exercised. This procedural safeguard is automatic and non-waivable.

(d) What Israel Gets: A binding 24-hour look before any US veto on a Lebanon incident. Transparency with teeth. **What Iran Sees:** Nothing. This is a US internal process. Israel has no veto. The US retains final decision authority. **What the US Gets:** Codification of what the US would do anyway — consult Israel before a major Lebanon decision — without giving Israel a paper veto.

Article 16.3 — Non-Veto Clause (Operative and Binding)

Nothing in this Article grants Israel any right to veto, suspend, or otherwise impede the implementation of this Framework. This Article is a unilateral affirmation by the United States and does not create any obligation on the part of Iran. **Nothing in Article 16.2-A shall be construed as granting Israel a veto over US decisions.** The 24-hour pre-decision notification is a procedural transparency measure, not a substantive right.

16.1 Operational Parameters of the Tel Aviv Channel

The Tel Aviv Channel shall operate as a bilateral US-Israel mechanism distinct from the Joint Implementation Group, the JHTA Council, and all other multilateral bodies established under this Framework. Its functions are limited to: (a) advance notification to Israeli counterparts of implementation milestones that may affect regional force posture; (b) secure exchange of intelligence assessments related to Iranian proxy activity; and (c) coordination of US and Israeli naval and air assets in the Eastern Mediterranean and Red Sea corridors outside the Hormuz Transit Zone.

16.2 Separation of Channels

The Tel Aviv Channel shall not serve as a conduit for Israeli objections to Framework implementation. No communication through the Tel Aviv Channel may be cited by either the United States or Iran as grounds for delay, suspension, or renegotiation of any provision of this Framework. The Tel Aviv Channel is a coordination mechanism, not a decision-making body.

16.3 Non-Creation of Obligations for Iran

This Article constitutes a unilateral statement of policy by the United States. It imposes no obligation, direct or indirect, on the Islamic Republic of Iran. Iran is not required to recognize the Tel Aviv Channel, to participate in its proceedings, or to accept any output or recommendation emanating from it. The absence of Iranian obligation in this regard shall not be construed as a waiver of any Iranian right under this Framework.

Article 17: Rolling Horizon Architecture — The Sunset Alternative

Recognizing that fixed-duration agreements create perverse incentives for both parties to posture for the expiration date, this Framework replaces traditional sunset clauses with a conditional rolling horizon architecture that auto-renews based on verified compliance while preserving either party's right to withdraw with adequate notice.

17.1 The Problem with Fixed Sunset Clauses

Traditional sunset clauses create three structural problems: (a) both parties begin positioning for the expiration date years in advance, degrading compliance quality; (b) hardliners in both capitals can simply wait out the clock rather than engaging constructively; (c) renegotiation leverage is artificially concentrated at the expiration moment rather than distributed across the agreement lifecycle. The Rolling Horizon architecture solves all three.

17.2 The Rolling Horizon Mechanism

Phase I Horizon (Days 1–180): The initial Phase I Stabilization Framework operates for 180 days. At Day 150, a mandatory Joint Review Conference assesses compliance, identifies remaining gaps, and determines whether conditions exist for Phase II activation. This assessment is not a renegotiation; it is a certification of readiness for deeper commitments.

Phase II Horizon (Days 180–540 / Months 6–18): Phase II addresses deeper nuclear constraints (enrichment moratorium extensions, permanent HEU disposition, full SWIFT normalization), missile and drone program limitations, and regional security architecture. Phase II activates automatically upon: (a) JIG certification of 90% Phase I compliance; (b) deposit of legislative Instruments of Ratification for Phase II Commitments (per Appendix D) by both parties; (c) IAEA clean compliance certification for 150 consecutive days.

Phase III Horizon (Months 18–36+): Phase III addresses successor framework negotiations for a comprehensive, permanent peace agreement including potential full diplomatic normalization. Phase III activates automatically upon: (a) JIG certification of 90% Phase II compliance; (b) successful completion of the first Regional Security Architecture Review; (c) mutual Letter of Intent for permanent framework negotiations.

Successor Framework Negotiation Mandate: Negotiations for Phase II shall commence no later than Day 90 of Phase I, regardless of whether Phase I certification is complete. These negotiations shall run in parallel to Phase I implementation. If Phase I certification is delayed beyond Day 180, the Phase II negotiations shall continue on an informational basis, with no binding commitments until certification occurs. Either Party may propose draft Phase II text at any time; the other Party shall respond within 30 days. Failure to engage in Phase II negotiations in good faith shall be logged as a Tier 1 technical dispute but shall not delay Phase I benefits.

17.3 Conditional Auto-Renewal

Each phase auto-renews for an identical period upon:

- IAEA certification of sustained compliance (no Tier 2 or 3 violations in the preceding 90 days);
- JIG majority confirmation that both parties are implementing in good faith;
- Absence of a Formal Non-Renewal Notice from either party.

If auto-renewal conditions are not met, the Framework does not terminate; it enters Stabilized Hold — humanitarian and Survival Core provisions remain active, but new tranche releases and phase advancements pause until compliance is restored or the parties agree on modified terms.

17.4 Orderly Withdrawal Protocol

Either party may withdraw from any phase by depositing a Formal Non-Renewal Notice with the Joint Secretariat. The notice must specify: (a) the grounds for non-renewal; (b) specific compliance concerns if applicable; (c) requested remedial measures.

Withdrawal timing:

- **Phase I:** 90 days' notice required, reducible to 30 days upon payment of liquidated damages equal to 10% of remaining escrowed funds (capped at $250M). Shorter notice does not trigger military escalation or other penalties. HEU remains under IAEA custody during the notice period. Stand-off distances remain in effect.
- **Phase II:** 180 days' notice required, reducible to 60 days upon payment of liquidated damages equal to 10% of remaining escrowed funds (capped at $500M). All escrow funds already released remain accessible. SWIFT access phases down over 90 days.
- **Phase III:** 365 days' notice required, reducible to 120 days upon payment of liquidated damages equal to 10% of remaining escrowed funds (capped at $1B). Full successor framework terms govern the transition.

No Clawback of Released Funds: Funds already released from escrow prior to the deposit of a Formal Non-Renewal Notice shall not be subject to return, clawback, or recission under any circumstances. The withdrawing Party retains full title and control over all previously released assets. Only future tranches not yet released as of the notice date are affected by withdrawal. This "no clawback" provision is absolute and non-waivable.

17.5 The "No Cliff" Principle

Under no circumstances does non-renewal or withdrawal trigger an immediate collapse of stability mechanisms. The Survival Core (ceasefire hotline, humanitarian corridor, emergency diplomatic channel) remains active for **365 days** following any withdrawal notice, and shall **automatically renew for successive 365-day periods unless both Parties agree in writing to terminate it**. The Guarantor Matrix may extend the Survival

Core by majority vote even if one Party objects. The Survival Core is a standalone humanitarian arrangement independent of the Framework's political status. HEU custody arrangements remain under IAEA supervision until a successor agreement or orderly return protocol is executed. The stand-off posture in Hormuz remains in effect for 90 days post-withdrawal unless military exigencies require adjustment with 48 hours' notice.

17.6 Political Narrative Frames

US Narrative Frame: "This agreement has no expiration date that Iran can wait out. It continues as long as Iran continues to comply. If Iran cheats, the benefits pause while the security architecture holds. If we both comply, the relationship deepens. The ball is always in Tehran's court."

Iranian Narrative Frame: "This is not a permanent surrender. Either party can withdraw with adequate notice. But as long as the United States honors its commitments, the relationship advances. The auto-renewal is conditioned on verified US compliance, not Iranian submission."

17.3 Phase I Completion Ceremony (v1.2 Political Smoothing Addition)

> **ARTICLE 17.3 — PHASE I COMPLETION CEREMONY (NON-BINDING, OPTIONAL)**
>
> **(a) Trigger:** If the JIG certifies **180 consecutive days of IAEA-verified compliance** with no Tier 3 violations and no more than one Tier 2 violation, the Parties shall jointly convene a **public ceremony** at a neutral location (Switzerland or Oman) to mark the successful completion of Phase I.
>
> **(b) Ceremony Elements:** The ceremony shall include: a **joint statement** acknowledging the Parties' commitment to the Framework and the region's stability; a **moment of silence** for those lost in the 2026 conflict; and a **symbolic act of cooperation** — such as joint planting of a tree, joint ribbon-cutting at a reconstruction project funded by the Political Discretion Fund, or joint dedication of a humanitarian facility.
>
> **(c) What This Is Not:** The ceremony is **not a treaty**, not a recognition of diplomatic relations, not a precedent for Phase II, and not a commitment to negotiate further. It is a **human moment acknowledging that peace, however fragile, is worth celebrating**. It recognizes that 180 days of verified compliance in one of the world's most volatile regions is an achievement deserving of acknowledgment.
>
> **(d) Voluntary Participation:** Either Party may participate or decline **without penalty**. If one Party declines, the other may hold a unilateral acknowledgment ceremony if it

> so chooses. The JIG shall issue a formal Phase I Completion Certificate regardless of ceremony participation.
>
> **(e) Media and Public:** The ceremony shall be open to international media. Both Parties' domestic media shall have equal access. The ceremony shall be broadcast live on the Swiss Dashboard. This is a public acknowledgment, not a secret deal.
>
> **Why This Matters:** The Framework is excellent at describing what happens when things go wrong — penalties, step-backs, suspensions, withdrawals. It says almost nothing about what happens when things go right. The Completion Ceremony gives both sides a reason to want Phase I to succeed: not just the absence of penalties, but the presence of a moment. It is low-stakes, optional, and symbolic — but symbolism matters in diplomacy. Leaders need something to celebrate, not just something to enforce.

Article 18: No Automatic Military Escalation Clause

Article 18 — Breach-Military Escalation Firewall (Operative and Binding)

Step-back, Review Pause, or even full framework suspension shall not be construed as authorization for military action. Any party contemplating military measures against the other must:

a. first exhaust all dispute resolution mechanisms under Article 13;
b. provide 14 days' advance notice through the Swiss facilitator;
c. secure UN Security Council authorization unless responding to an ongoing armed attack under UN Charter Article 51.

Breach of the Framework is a diplomatic and economic matter, not a military authorization.

18.1 Purpose and Rationale

Without this Article, hardliners on either side could argue that a "material breach" justifies airstrikes or kinetic retaliation. This Article explicitly decouples compliance penalties from the use of force. The step-back architecture (Article 13) provides graduated diplomatic and economic responses to breaches; it does not create a casus belli. Military action requires independent justification under international law, separate and apart from the Framework's compliance framework.

18.2 The Exhaustion Requirement

Before any party may take military measures against the other in connection with Framework-related matters, it must exhaust:

- Tier 1 technical dispute resolution (14 days);
- Tier 2 material disruption remediation (30 days);
- JIG emergency consultation (72 hours);
- Swiss facilitator mediation (7 days);
- Any applicable Mandatory Escalation Pause (30 minutes to 72 hours).

This exhaustion requirement does not apply to: (a) responses to ongoing armed attacks under UN Charter Article 51; (b) protection of nationals under internationally recognized doctrines; (c) UN Security Council-authorized military action.

18.3 The 14-Day Advance Notice Requirement

Any party contemplating military measures must provide 14 days' advance notice through the Swiss facilitator to: (a) the other party; (b) all guarantor states; (c) the UN Secretary-General. The notice must specify: the legal basis for the proposed action; the targets and scope; the expected duration; and the steps that would avert the action. During the 14-day period, the Mandatory Escalation Pause is automatically in effect.

18.4 UN Security Council Authorization

Military measures taken in connection with Framework matters (as opposed to independent Article 51 self-defense) require UN Security Council authorization. The absence of UNSC authorization does not automatically render the military action unlawful under general international law, but it does constitute a violation of this Framework's dispute resolution architecture, triggering proportional step-back.

18.5 Independent Legal Effect and Severability

(a) Severability: The No Automatic Military Escalation Clause is **severable** and survives any determination that other provisions of this Framework are invalid or unenforceable. This Clause creates independently enforceable obligations under international law, separate from the Framework's political commitments.

(b) Independent Enforcement: The Guarantor Matrix may enforce the military escalation prohibition under this Article even if the Framework has been formally suspended or terminated by one or both Parties. The prohibition on automatic military escalation from compliance breaches is a standalone obligation under international law.

(c) Survival: This Article survives total framework suspension. Even if all other provisions are in hold, the prohibition on automatic military escalation from compliance breaches remains operative for 365 days following suspension, enforceable through the Guarantor Matrix.

US Narrative Frame: "This clause doesn't prevent us from defending ourselves or our allies. It simply says that a compliance dispute over enrichment levels or SWIFT access doesn't justify bombing. Self-defense under Article 51 remains fully available."

Iranian Narrative Frame: "This prevents the United States from using a minor compliance pretext to justify military strikes. Breach is met with diplomatic step-back, not war. Our right to self-defense under international law remains untouched."

Article 19: Spoiler Actor Management (v1.1 Hardening)

Issue 19: Spoiler Actors and Framework Destabilization. The Framework's flexibility mechanisms, while essential for signature, create vulnerabilities that determined adversaries — state or non-state — can exploit to undermine, delay, or collapse the agreement. Rogue commanders, false flag operators, information warfare campaigns, and economic saboteurs all pose existential threats to the Framework's durability. Neither Party benefits from a system that punishes the state for actions it did not authorize, or that allows bad actors to game the very protections designed to preserve peace.

The Resolution: Comprehensive Spoiler Actor Architecture

A four-article hardening suite (Articles 19 through 19-C) that defines spoiler actors, establishes attribution protocols, creates remediation safe harbors, and mandates institutional adaptation — all without reopening the core political bargain.

> ARTICLE 19 — SPOILER ACTOR MANAGEMENT (OPERATIVE AND BINDING)
>
> **(a) Definition:** A "spoiler actor" is any state, non-state actor, military commander, political faction, commercial entity, or individual that takes action with the **primary purpose** of undermining, delaying, or collapsing this Framework, or forcing either Party into a position of non-compliance.
>
> **(b) Attribution Protocol:** Upon reasonable suspicion that a spoiler actor is operating, either Party may refer the matter to **SEAC** (Shadow Escalation Attribution Cell). SEAC shall complete attribution assessment within 14 days, with a preliminary finding within 7 days.
>
> **(c) Attribution Levels:**

- **Level 1 (Rogue Actor):** Action by an individual or small group not directed by state leadership. **Consequence:** Party whose territory or command structure hosted the actor shall take reasonable remedial measures within 14 days.
- **Level 2 (State-Tolerated):** Action by an actor operating with state tolerance (failure to prevent despite knowledge). **Consequence:** Tier 1 technical dispute; enhanced monitoring.
- **Level 3 (State-Directed):** Action directed or approved by state leadership. **Consequence:** Treated as Tier 2 material disruption by the directing Party.

(d) Safe Harbor for Remediation: If a Party takes prompt, verifiable remedial action against a spoiler actor within 14 days of attribution, **no step-back shall apply**. Remedial actions include: public condemnation; asset freezes; arrest of individuals; or severance of supply lines. This safe harbor incentivizes remediation rather than defensiveness.

(e) SEAC Composition: SEAC comprises one intelligence analyst from each guarantor state (Oman, Pakistan, Qatar, China, Russia, EU) plus one technical observer from the IAEA. SEAC operates under the Joint Secretariat with classified information sharing protocols. SEAC findings are advisory — attribution levels are determined by the JIG.

Article 19-A: False Flag Attack Protocol (v1.1 Hardening)

ARTICLE 19-A — FALSE FLAG ATTRIBUTION PROTOCOL (OPERATIVE AND BINDING)

(a) Definition: A "false flag attack" is an operation designed to appear as if it originated from one Party when it was actually conducted by a third party, rogue actor, or the other Party itself.

(b) Presumption of Good Faith: Upon any attack, there shall be a **rebuttable presumption** that the attack is not a false flag absent clear and convincing evidence. This presumption protects victims from having to prove their innocence before responding to an actual attack.

(c) False Flag Investigation: If either Party claims an attack is a false flag, SEAC shall conduct an **expedited investigation** with 48-hour preliminary finding and 14-day final attribution. During the investigation, all standard attribution procedures (media blackout, executive consultation) remain in effect. **Expedited Pathway for High-Confidence False Flags:** If, within the first 72 hours, SEAC certifies by **unanimous vote of its permanent members** (Oman, US, UK, France) that there is **"high**

confidence" (90%+ probability) that the attack was a false flag, the investigation may be concluded at 72 hours with a preliminary final finding. The accused Party retains the right to submit counter-forensics within 7 days, but the provisional finding shall be **presumptively binding** for Tier step-back purposes unless rebutted by clear and convincing evidence. This expedited pathway is available only for attacks causing 10+ casualties or significant damage to Critical Cyber Infrastructure.

(d) Consequence for Confirmed False Flag: If a Party is confirmed to have conducted or directed a false flag attack, the following apply:

- The attack is treated as a **Tier 2 material disruption** by that Party;
- The false flag operation itself constitutes an **additional Tier 2 violation**;
- The attacking Party **forfeits any right to claim self-defense** based on the attack;
- The matter is referred to the **Guarantor Matrix** for potential sanctions coordination.

(e) No Automatic Retaliation: During any false flag investigation, the responding Party **may not take military action** based on the attacked status unless the attack itself meets UN Charter Article 51 self-defense thresholds **independent of attribution**. The Framework's 14-day notice requirement (Article 18.3) still applies unless independent Article 51 criteria are satisfied.

Article 19-B: Spoiler Financing and Supply Tracking (v1.1 Hardening)

ARTICLE 19-B — SPOILER FINANCING AND SUPPLY TRACKING (OPERATIVE AND BINDING)

(a) Supply Chain Monitoring Cell: The JIG shall establish a **Supply Chain Monitoring Cell** to track: weapons and drone component transfers to non-state actors in the region; financial flows from either Party to entities designated as spoiler actors; and dual-use technology transfers that could enable spoiler operations.

(b) Reporting Obligations: Each Party shall provide **quarterly reports** to the JIG on: known supply routes to proxy forces under its influence; financial transfers to entities that could be used for spoiler operations; and steps taken to disrupt spoiler supply chains.

(c) Spoiler Designation: The JIG may, by majority vote, designate an entity as a **"Proscribed Spoiler Organization."** Upon designation, both Parties commit to:

freezing any assets of the organization within their jurisdiction; prohibiting financial transactions with the organization; and denying safe passage or operational support.

(d) Non-Designation as Admission: Failure to designate an organization does not constitute admission that the organization is not a spoiler. Designation is a remedial tool, not a finding of liability. The burden of proof for designation is "reasonable suspicion" — not "beyond reasonable doubt."

Article 19-C: Information Warfare and Disinformation Protocol (v1.1 Hardening)

ARTICLE 19-C — INFORMATION WARFARE AND DISINFORMATION (OPERATIVE AND BINDING)

(a) Prohibited Conduct: The following information warfare actions are prohibited:

- State-sponsored disinformation campaigns designed to collapse domestic support for the Framework in the other Party;
- Fabricated intelligence dossiers alleging Framework violations known to be false;
- Algorithmic amplification of spoiler narratives targeting the other Party's domestic audience;
- Impersonation of the other Party's officials in communications related to Framework implementation.

(b) Rapid Response Mechanism: If either Party believes the other has engaged in prohibited information warfare, it may request a **72-hour expedited JIG review**. Upon confirmation, the violating Party shall issue a **public corrective statement** within 24 hours.

(c) WBI Point System: Each confirmed information warfare incident shall accrue **0.5 WBI (Willful Bad-faith Index) points**. At **2.0 WBI points** (four incidents), the violating Party's de minimis buffer is suspended for 90 days. WBI points decay at 0.25 points per 90-day period of clean compliance.

(d) Media Coordination Cell: The Joint Secretariat shall maintain a **Media Coordination Cell** to respond to disinformation, provide factual corrections, and coordinate crisis communications. The Cell has **no censorship authority** — it is a factual rebuttal mechanism, not a content control body.

Article 19-D: Critical Infrastructure Cyber Protection (v1.1 Hardening)

ARTICLE 19-D — CRITICAL INFRASTRUCTURE CYBER PROTECTION (OPERATIVE AND BINDING)

(a) Designated Critical Systems: The following systems are designated as **"Critical Cyber Infrastructure"** and afforded the highest level of protection under this Framework:

- IAEA nuclear verification systems (OLEM data feeds, seal monitoring, mass spectrometry reporting);
- Swiss Dashboard and all associated verification nodes;
- JHTA maritime transit coordination systems and AIS networks;
- Escrow and financial settlement systems (Swiss, Omani, Qatari nodes);
- Military deconfliction hotlines (IRGC-Navy / 5th Fleet / Omani relay);
- SWIFT clearing hubs and targeted bank isolation systems.

(b) Prohibited Cyber Operations: No Party shall conduct, direct, or knowingly facilitate cyber operations against Critical Cyber Infrastructure, including but not limited to: unauthorized access, exfiltration, or modification of data; denial of service or degradation of system availability; introduction of malware, ransomware, or destructive code; spoofing, masquerading, or falsification of telemetry data; and impersonation of authorized users or systems.

(c) Verification System Integrity: The IAEA and Swiss facilitator shall maintain independent integrity verification of all Critical Cyber Infrastructure, including: cryptographic hash chains for all Dashboard data (immutable audit trail); air-gapped backup verification nodes (Switzerland, Oman); quarterly independent penetration testing by ETH Zurich or an approved neutral entity. **If the Parties fail to agree on a neutral entity within 30 days of a JIG request, the Swiss facilitator shall appoint an entity from a pre-approved roster maintained by the International Telecommunication Union (ITU).** The appointed entity serves for one 12-month cycle. Either Party may object to reappointment after the cycle, but may not veto the initial appointment. This fallback prevents deadlock from shielding vulnerabilities.; and real-time intrusion detection with automatic alerts to both Parties and all guarantor states.

(d) Consequence: A confirmed cyber operation against Critical Cyber Infrastructure constitutes a **Tier 2 material disruption**. If the operation results in data loss, system degradation exceeding 24 hours, or false telemetry that could have led to miscalculation, it is automatically elevated to **Tier 3 Deliberate Hostile Violation**.

(e) Attribution Standard: Cyber attribution requires **"high confidence" (90%+ probability)** based on technical forensic evidence, not political assertion. SEAC shall maintain a **Cyber Attribution Cell** with independent forensic capability.

Article 19-E: Cyber Incident Response and Escalation Protocol (v1.1 Hardening)

ARTICLE 19-E — CYBER INCIDENT RESPONSE AND ESCALATION (OPERATIVE AND BINDING)

(a) Cyber Incident Classification: Cyber incidents affecting the Framework shall be classified as:

- **Level 1 (Advisory):** Attempted intrusion detected and blocked; no data loss or system impact. **Consequence:** JIG notification only.
- **Level 2 (Significant):** Successful intrusion with data exfiltration but no data modification; or system degradation under 4 hours. **Consequence:** Tier 1 technical dispute; enhanced monitoring; mandatory post-incident report.
- **Level 3 (Critical):** Data modification or falsification; system degradation over 4 hours; compromise of dual-key HSM; or impact on military deconfliction systems. **Consequence:** Tier 2 material disruption; automatic 48-hour Executive Consultation Window.
- **Level 4 (Crippling):** Sustained denial of Critical Cyber Infrastructure (over 24 hours); successful spoofing triggering or nearly triggering military response; or cascading failure across multiple protected systems. **Consequence:** Tier 3 Deliberate Hostile Violation; Guarantor Matrix emergency session within 24 hours.

(b) Incident Reporting: Any confirmed cyber incident against Critical Cyber Infrastructure shall be reported to the JIG within **6 hours** of detection. The report shall include: systems affected and nature of the incident; known impact on Framework operations; mitigation measures already taken; and preliminary attribution (if available). **Failure to report a confirmed incident within 6 hours is itself a Tier 1 technical dispute.**

(c) Forensics and Attribution: The SEAC Cyber Attribution Cell shall conduct forensic investigation for Level 3 and Level 4 incidents, with: 72-hour preliminary finding; 21-day final attribution report; technical evidence sharing (excluding intelligence sources and methods); and right for the accused Party to respond and present counter-forensics.

(c-1) Provisional Attribution for Level 4 (Crippling) Incidents: For Level 4 incidents, SEAC shall issue a **provisional attribution finding within 72 hours**, based on technical forensic evidence available at that time. The provisional finding triggers emergency safeguards — including automatic activation of air-gapped backup systems and a 7-day freeze on all sensitive telemetry changes — pending the final 21-day attribution report. **Provisional attribution is not admissible as evidence of state responsibility in any international tribunal.** It is a "stop the bleeding" mechanism, not a verdict.

(d) Mitigation and Remediation: Upon detection of a cyber incident, the affected Party may take immediate proportionate defensive measures, including: isolating affected systems; activating backup systems (air-gapped nodes); and engaging neutral cyber response teams (Switzerland, Oman, or UN-certified). **Defensive measures shall not extend to offensive cyber retaliation without JIG authorization.**

(e) No Cyber Retaliation Escalation: Offensive cyber retaliation against a confirmed cyber attack shall not be authorized unless: attribution is confirmed at Level 3 or 4; the Executive Consultation Window (Article 13.4) has been exhausted; 14 days' advance notice provided through the Swiss facilitator; and proportionality demonstrated. This provision operates alongside Article 18 (No Automatic Military Escalation) but applies specifically to cyber domain retaliation.

Article 19-F: Mandatory Cooling-Off Architecture (v1.2 Operative Addition)

Issue 19-F: Emotional Escalation Risk. Real-world crises unfold under time pressure, domestic political heat, and media amplification. Without mandatory pause architecture, either Party may escalate sanctions, maritime posture, or public attribution before diplomatic alternatives are exhausted. This Article adds targeted mandatory delays before specific escalatory actions — not blanket pauses that create perverse incentives, but calibrated cooling-off periods for the highest-risk escalation triggers.

ARTICLE 19-F — MANDATORY COOLING-OFF ARCHITECTURE (OPERATIVE AND BINDING)

(a) Cooling-Off Before Sanctions Snapback: Before any reimposition of suspended sanctions under the Graduated Sanctions Relief Tranches (Article 7), a **mandatory 24-**

hour Cooling-Off Period applies. During this period: no sanctions may be reimposed; the Swiss facilitator shall contact both Parties to confirm awareness and offer mediation; and either Party may request an emergency JIG consultation. The Cooling-Off Period begins at the moment the JIG issues a preliminary finding of breach, not upon final determination. **Exception:** The Cooling-Off Period does not apply to confirmed Tier 3 Deliberate Hostile Violations involving fissile material diversion, which trigger immediate emergency procedures under Article 1.7.

(b) Cooling-Off Before Public Attribution: Before any official public statement attributing responsibility for a security incident to the other Party, a **mandatory 48-hour Cooling-Off Period** applies. During this period: no official attribution statement may be issued; the JIG attribution deliberation process continues uninterrupted; the Media Blackout (Article 13.5) automatically extends to cover the Cooling-Off Period; and either Party may request that attribution be deferred pending additional evidence. **This Cooling-Off Period is in addition to, not in place of, the 72-hour Media Blackout.** Combined, they create a minimum 120-hour delay between incident and public attribution — sufficient for initial diplomatic engagement.

(c) Cooling-Off Before Enhanced Inspections: Before the IAEA may conduct unannounced or enhanced inspections beyond the baseline verification protocol (Article 3), a **mandatory 12-hour Cooling-Off Period** applies, during which the Joint Secretariat notifies the inspected Party and confirms the inspection scope. This prevents surprise inspections that could trigger security responses or domestic political crises. The 12-hour period is measured from JIG authorization to IAEA physical entry.

(d) Cooling-Off Before Maritime Escalation: Before either Party increases naval presence, activates additional maritime assets, or elevates force posture in the Strait of Hormuz beyond the baseline deconfliction protocol (Article 4.7), a **mandatory 24-hour Cooling-Off Period** applies. During this period: the IRGC-Navy Hotline shall be activated at the incident response level (15-minute response); the JHTA shall convene an emergency coordination session; and the Guarantor Matrix shall be notified. **Exception:** The Cooling-Off Period does not apply to immediate self-defense measures under UN Charter Article 51 in response to an active armed attack on vessels or personnel.

(e) Cooling-Off During Leadership Transitions: If a Cooling-Off Period coincides with a leadership transition covered under Article 21, the Cooling-Off Period automatically extends by 72 hours to allow the incoming leadership to be briefed and designate representatives. This prevents escalatory decisions made by outgoing officials or officials unaware of full context.

(f) No Abuse: Neither Party may deliberately trigger a Cooling-Off Period to buy time during active non-compliance. If the JIG determines by majority vote that a Party invoked Cooling-Off procedures in bad faith (three or more instances in 180 days), that Party loses Cooling-Off protection for 90 days — escalatory actions against it proceed without mandatory delay.

Why This Matters: The Framework already has extensive consultation architecture — 48-hour Executive Consultation, 72-hour Media Blackout, 14-day mediation. But these are *reactive* mechanisms that activate after an event. Cooling-Off Periods are *proactive* speed bumps placed before the most dangerous escalation triggers. They create space for impulse to yield to judgment. They are not shackles — they are guardrails, and they have safety release valves for genuine emergencies.

Article 20: Temporary Operational Suspension (v1.2 Operative Addition)

Issue 20: The Missing Middle State. The current Framework architecture recognizes only three states: full compliance, dispute/escalation, and breach/collapse. There is no intermediate legal state for situations where continued full operation is politically or operationally impossible, but neither Party wants total collapse. Domestic political crises, regional proxy flareups, cyber disputes, and leadership transitions all create situations where "pause" is more appropriate than "terminate." This Article creates that missing middle: Temporary Operational Suspension.

ARTICLE 20 — TEMPORARY OPERATIONAL SUSPENSION (OPERATIVE AND BINDING)

(a) Definition: "Temporary Operational Suspension" (TOS) is a distinct legal status in which one or more Framework modules are temporarily paused without terminating the Framework, triggering automatic penalties, or resetting compliance clocks. TOS preserves the Framework's legal architecture while allowing breathing room for crisis resolution. It is neither compliance nor breach — it is a recognized third state with its own rules.

(b) Grounds for TOS: Either Party may request TOS for any of the following grounds, supported by written notification to the JIG and Swiss facilitator:

- **(i) Regional War Spillover:** Active armed conflict involving either Party's territory, forces, or critical infrastructure that makes continued Framework operation unsafe or politically impossible (elaborated in Article 22);
- **(ii) Domestic Political Crisis:** Events including but not limited to: impeachment proceedings against a head of state; government collapse or extended cabinet vacancy; major legislative crisis that prevents designating Framework representatives; nationwide civil unrest requiring emergency government response;
- **(iii) Leadership Transition:** Death, resignation, incapacity, or removal of a head of state or foreign minister (elaborated in Article 21);
- **(iv) Cyber Crisis:** A Level 3 or 4 cyber incident against Critical Cyber Infrastructure that compromises verification systems or diplomatic channels (Article 19-E);
- **(v) Escrow Failure:** Compromise, insolvency, or sanctions targeting of the escrow facilitator that prevents normal fund operations (Article 24);
- **(vi) Force Majeure:** Events beyond either Party's control including natural disasters, pandemic, or global financial system collapse that prevent normal Framework operations.

(c) TOS Procedure:

1. **Request:** Either Party submits written TOS request specifying: affected modules; factual basis for suspension; estimated duration; and minimum survival conditions requested.
2. **Automatic 24-Hour Cooling-Off:** Upon receipt, a 24-hour automatic Cooling-Off Period begins (Article 19-F). No module may enter TOS until this period expires.
3. **JIG Consultation:** The JIG convenes within 48 hours to review the request. The JIG may: approve TOS as requested; approve TOS with modified scope (reducing affected modules); or reject TOS if the factual basis is insufficient.
4. **Immediate Effect:** If the JIG approves, TOS takes effect immediately upon notification. If the JIG rejects, the requesting Party may appeal to the Guarantor Matrix within 7 days.
5. **No Penalty:** Entry into TOS does not constitute non-compliance, breach, or step-back. Compliance clocks are frozen (not reset) for the TOS period.

(d) Minimum Survival During TOS: Even under full TOS, the following survive automatically:

- Article 18 — No Automatic Military Escalation Clause;

- Article 4.7 — IRGC-Navy Hotline (minimum daily testing protocol);
- Article 11.1-A — Survival Core Escrow Fund operations;
- Article 19-F — Cooling-Off Architecture (extended to 72 hours during TOS);
- Article 3.3(d) — Unscheduled batch verification for credible intelligence (if both Parties agree).

(e) Maximum TOS Duration: TOS may not exceed **90 days** in a single invocation. Extensions require: written request 14 days before expiration; JIG approval by majority vote; and maximum total TOS of 180 days in any 365-day period. If a Party exceeds the 180-day cumulative limit, the Guarantor Matrix shall convene to determine whether the Framework should enter Rolling Horizon non-renewal (Article 17) or whether extraordinary circumstances justify continued TOS.

(f) Reactivation: TOS ends upon: mutual agreement of both Parties; unilateral decision of the requesting Party (other Party may not block); or automatic expiration of the approved TOS period. Upon reactivation, affected modules resume from their pre-TOS status — compliance clocks resume, escrow schedules restart, and all baseline protections reactivate. The JIG shall issue a Reactivation Protocol within 14 days specifying the sequencing of reactivated commitments.

(g) No Precedent: Invocation of TOS by one Party in one circumstance creates no precedent for invocation by either Party in different circumstances. Each TOS request is evaluated on its specific factual basis. The JIG shall not cite previous TOS approvals as binding precedent for future requests.

(h) Invocation Cap — Cooling-Off on Cooling-Off (v1.2 Human-in-the-Loop Addition): No Party may invoke TOS more than **three times in any 365-day period**, regardless of duration. The fourth invocation in a 365-day period is **presumptively bad faith** and triggers automatic referral to the Guarantor Matrix for potential Tier 2 material disruption determination. The Guarantor Matrix may, by two-thirds majority, authorize a fourth TOS only if extraordinary circumstances (natural disaster, war spillover, pandemic) clearly justify it. This cap prevents TOS from becoming a de facto termination mechanism while preserving the genuine safety valve for legitimate crises.

(i) TOS Invocation Registry: The JIG shall maintain a public "TOS Invocation Log" on the Swiss Dashboard showing: the date of each invocation; the requesting Party; the affected modules; the factual basis; the duration; and the outcome. The Log resets annually. Both Parties and guarantor states have access. Transparency prevents covert abuse.

> **Why This Matters:** Without TOS, a domestic political crisis in Washington or Tehran — a government shutdown, a leadership death, a constitutional crisis — could legally force total Framework termination because there's no "pause" button. TOS creates a pressure valve that says: "We can't operate at full speed right now, but we're not walking away." It prevents temporary crises from becoming permanent collapse.

Article 21: Institutional Continuity Protocol (v1.2 Operative Addition)

Issue 21: Personality Dependency. The Framework's operation depends heavily on specific individuals — the US Secretary of State, the Iranian Foreign Minister, the JIG representatives, the Swiss facilitator. If any of these individuals dies, resigns, loses an election, or becomes incapacitated, the Framework could stall for weeks while replacements are identified and briefed. This Article binds the Framework to institutions, not leaders.

> ARTICLE 21 — INSTITUTIONAL CONTINUITY PROTOCOL (OPERATIVE AND BINDING)
>
> **(a) Automatic Continuity:** This Framework is binding upon the **governments and state institutions** of the United States and the Islamic Republic of Iran, not upon individual officeholders. All rights, obligations, and procedures defined herein survive any change in individual leadership, government composition, or ruling party. The Framework does not terminate, suspend, or weaken upon:
>
> - Death or incapacity of a head of state, head of government, or foreign minister;
> - Loss of election by a head of state or ruling party;
> - Resignation or removal of any officeholder;
> - Cabinet collapse, government reshuffle, or ministerial reorganization;
> - Legislative turnover (new Congress or new Majlis);
> - Constitutional crisis or interim government;
> - Regime change or governmental transition of any character.
>
> **(b) Replacement Timeline:** If a designated Framework representative becomes unavailable, the affected state shall designate a replacement within the following timelines:

Table X2: Leadership Replacement Timelines

Position	Replacement Deadline	Interim Authority
US Secretary of State / Iranian Foreign Minister	14 calendar days	Deputy Secretary / Deputy Minister may exercise all Framework authority including Executive Consultation veto
JIG National Representative	7 calendar days	Alternate Representative (pre-designated at Framework signature) assumes full authority
Swiss Facilitator	21 calendar days (Swiss government selects)	Deputy Facilitator or Omani co-facilitator may convene sessions
JHTA Co-Chairs	7 calendar days	Deputy co-chairs assume operational authority
IAEA Director General (verification role)	30 calendar days (IAEA Board procedure)	Deputy DG or designated senior inspector maintains all verification access

(c) Interim Authority Binding: Actions taken by interim officials under paragraph (b) are fully binding and have the same legal effect as actions taken by permanent officeholders. Neither Party may challenge the validity of interim actions after the permanent replacement is designated.

(d) Leadership Transition Briefing Protocol: Within 7 days of any change in head of state or foreign minister, the Joint Secretariat shall deliver to the incoming leadership a comprehensive briefing package including: the Framework's operative status; all pending compliance milestones and deadlines; any active disputes or TOS periods; and a 30-day priority action list. The briefing shall be factual, non-partisan, and non-evaluative. The Swiss facilitator shall offer to meet personally with the incoming leadership within 14 days.

(e) Election Period Protections: During the 90 days preceding a national election in either Party's jurisdiction:

- No new compliance milestones may be imposed unless mutually agreed;
- Existing milestones continue on their established schedule;

- The Executive Consultation Window (Article 13.4) expands from 48 hours to 72 hours to account for campaign-period availability;
- No JIG public reports may be issued within 14 days of the election date;
- The Media Blackout (Article 13.5) automatically extends to cover the 7 days before and 3 days after the election.

(f) Extended Vacancy: If any position remains vacant beyond its replacement deadline, the Guarantor Matrix shall convene within 7 days to: appoint a neutral interim official with limited procedural authority; establish an accelerated replacement process; and determine whether the vacancy justifies Temporary Operational Suspension under Article 20.

Why This Matters: Democracies have elections. Authoritarian systems have successions. Both create leadership gaps. Without this Article, a US presidential transition (January 20) or an unexpected death could freeze the Framework for months. This Article says: the Framework belongs to the state, not the person. Institutions continue. Obligations survive. The only question is who answers the phone — and this Article ensures someone always answers within 14 days.

Article 22: Force Majeure for Regional Conflict Spillover (v1.2 Operative Addition)

Issue 22: Unrelated Wars Destroying Related Peace. The Framework assumes a relatively stable regional environment. But the Middle East is not stable. A Hezbollah-Israel war, a Houthi escalation, an Iraq militia campaign, or even a distant India-Pakistan crisis could create conditions where continued Framework operation becomes politically impossible for one or both Parties — even though the Framework itself has nothing to do with the crisis. Without explicit force majeure protections, unrelated wars could trigger cascading non-compliance.

ARTICLE 22 — FORCE MAJEURE FOR REGIONAL CONFLICT SPILLOVER (OPERATIVE AND BINDING)

(a) Force Majeure Events: The following events constitute **Framework Force Majeure** if they directly and materially prevent either Party from performing its Framework obligations:

- **(i) Levant Conflict:** Sustained armed conflict between Hezbollah and Israeli forces exceeding 72 hours with 100+ casualties;

- **(ii) Yemen Escalation:** Houthi attacks on international shipping in the Red Sea or Gulf of Aden triggering multilateral military response;
- **(iii) Iraq Militia Campaign:** Coordinated attacks on US personnel or facilities in Iraq by Iranian-aligned militias exceeding 48 hours;
- **(iv) Strait Crisis:** Closure of the Strait of Hormuz to commercial shipping by any actor for more than 7 consecutive days;
- **(v) Global Shipping Collapse:** Broader disruption of Persian Gulf maritime commerce affecting 50%+ of normal transit volume for 14+ days;
- **(vi) Wider War:** Any interstate armed conflict in the Middle East region involving a Guarantor Matrix member state.

(b) Automatic TOS: Upon certification by the JIG (by majority vote, within 48 hours of request) that a Force Majeure Event has occurred, **Temporary Operational Suspension (Article 20) automatically activates** for the duration of the Force Majeure Event plus 14 days for reactivation preparation. Neither Party needs to file a separate TOS request.

(c) No Compliance Penalty: During a Force Majeure TOS, neither Party may be penalized for non-performance of suspended obligations. Compliance clocks are frozen. Escrow schedules are paused. Step-back triggers are deactivated. The Dossier Isolation Firewall (Appendix E) remains fully operative — the Force Majeure Event is logged on the Regional Security track, not the Core Compliance track.

(d) Survival Core Continuity: Even during Force Majeure TOS, the Survival Core (Article 11) continues in full force. The hotline, humanitarian corridor, diplomatic channel, and Survival Core Escrow remain active. If anything, these mechanisms become more important during regional crisis — they are the threads that prevent total separation.

(e) Termination of Force Majeure: Force Majeure TOS terminates upon: mutual agreement that the triggering event has ended; JIG certification (by majority vote) that the event no longer materially prevents Framework performance; or expiration of the 90-day maximum TOS period (Article 20(e)). Upon termination, the Reactivation Protocol (Article 20(f)) applies.

(f) Good-Faith Obligation During Force Majeure: Even during Force Majeure TOS, both Parties shall: maintain all diplomatic channels (Omani backchannel, Swiss facilitator, hotline); refrain from public statements linking the Force Majeure Event to the Framework; and not use the Force Majeure Event as cover for actions that would violate the Framework if it were active. Violation of these good-faith obligations constitutes independent grounds for Tier 1 technical dispute proceedings.

(g) No Precedent: Invocation of Force Majeure in one circumstance creates no precedent for invocation in different circumstances. Each Force Majeure determination is evaluated on its specific factual basis, with the burden of proof on the invoking Party to demonstrate that the event materially prevents Framework performance.

Why This Matters: Without this Article, a Hezbollah rocket barrage or a Houthi shipping attack could force an Iranian leader to suspend Framework participation to avoid appearing to cooperate with the US while proxies are fighting. The Force Majeure provision says: "We understand. You didn't cause this crisis. Your Framework obligations pause while you deal with it. But the door stays open." It separates the Framework from the region's endless wars.

Article 23: Security Classification Protocol (v1.2 Operative Addition)

Issue 23: Operational Detail Exposure. The Framework's main text contains substantial operational detail — telemetry frequencies, inspection procedures, escrow banking arrangements, hotline testing protocols. In a real negotiation, these details belong in controlled annexes with restricted distribution, not in the principal text available to all signatories. This Article establishes a post-signature classification framework that allows operational details to be compartmentalized without reopening the political bargain.

ARTICLE 23 — SECURITY CLASSIFICATION PROTOCOL (OPERATIVE AND BINDING)

(a) Three-Tier Classification System: Upon Framework signature, all Parties shall agree to a three-tier classification system for Framework documents:

Table X3: Security Classification Tiers

Tier	Label	Access	Content
Tier 1 — Political Text	UNCLASSIFIED	All signatories, guarantor states, public	Core Framework text; Preamble; operative Articles 1-19; Annex X (US-Israel)

Tier 2 — Technical Annexes	RESTRICTED	JIG members; IAEA technical staff; national technical representatives	Verification protocols; telemetry specifications; inspection schedules; escrow banking details; SWIFT reconnection procedures
Tier 3 — Classified Implementation Protocols	CONFIDENTIAL	Designated security officials; intelligence liaisons; military deconfliction officers	Sensor placement coordinates; hotline encryption keys; classified diplomatic channel access codes; intelligence sharing protocols; LIRB classified findings

(b) Classification Authority: The JIG, by unanimous vote, determines the classification tier of any Framework document, protocol, or procedure. Either Party may request reclassification (upward or downward) with 30 days' notice. Downward reclassification (making material more accessible) requires unanimous consent. Upward reclassification (restricting access) requires only majority vote with the affected Party's members excluded.

(c) Handling Requirements:

- **Tier 1 (UNCLASSIFIED):** No handling restrictions. Published on Swiss Dashboard. May be reproduced, distributed, and cited without limitation.
- **Tier 2 (RESTRICTED):** Marked "RESTRICTED — Framework Technical Annex." Access requires signed acknowledgment of handling procedures. May not be reproduced outside JIG-authorized channels. Stored on encrypted JIG servers with role-based access.
- **Tier 3 (CONFIDENTIAL):** Marked "CONFIDENTIAL — Framework Implementation Protocol." Access limited to named individuals pre-cleared by both Parties. Stored on air-gapped systems. All access logged. Physical documents stored in secure facilities. No electronic transmission outside designated secure channels.

(d) Leak Investigation: Unauthorized disclosure of Tier 2 or Tier 3 material triggers automatic JIG investigation. If the leak is traced to a Party's official or designee, the responsible Party shall: issue a corrective statement; revoke the leaker's access pending investigation; and compensate affected Parties for remediation costs (cryptographic key rotation, sensor relocation, protocol revision). Willful disclosure of Tier 3 material constitutes a Tier 2 material disruption.

(e) Public Transparency Override: Notwithstanding classification tiers, the JIG Monthly Bulletins (Article 12.3-B) and Annual Public Report (Article 12.3-A) shall include only Tier 1 (UNCLASSIFIED) information. No classified operational details shall appear in public reports. The Module Status Dashboard (Article 13.3-A) is Tier 1 — module status, not operational details.

(f) No Intelligence Collection: Tier 2 and Tier 3 access shall not be used for intelligence collection purposes unrelated to Framework implementation. Inspectors, technical liaisons, and JIG members with Tier 2/3 access are prohibited from gathering or reporting information outside their authorized scope. Violation is a Tier 2 material disruption and grounds for immediate revocation of access.

Why This Matters: The principal text of any serious agreement must be readable by foreign ministers and legislators. But foreign ministers don't need to know sensor frequencies, and legislators don't need to know hotline encryption protocols. This Article creates a sensible compartmentalization: political text for everyone, technical details for technical people, classified details for security officers. It protects operational security without reducing transparency.

Article 24: Escrow Failure and Facilitator Contingencies (v1.2 Operative Addition)

Issue 24: Neutral System Failure. The Framework assumes that Swiss and Omani escrow facilitators remain neutral, solvent, and operational throughout the Framework's life. In reality: Switzerland could face EU sanctions pressure; Oman could experience domestic instability; banking partners could be designated under counter-terrorism regimes; and SWIFT itself could be disrupted by cyberattack. This Article creates a comprehensive contingency architecture for escrow failure.

ARTICLE 24 — ESCROW FAILURE AND FACILITATOR CONTINGENCIES (OPERATIVE AND BINDING)

(a) Escrow Failure Events: The following events constitute **Escrow Failure** requiring contingency activation:

- **(i)** Primary escrow bank designated under OFAC, EU, or UN sanctions regime;
- **(ii)** Escrow facilitator (Swiss or Omani) subject to sanctions, insolvency, or regulatory revocation;

- **(iii)** SWIFT infrastructure disruption preventing normal escrow transactions for 7+ consecutive days;
- **(iv)** Cyber compromise of escrow banking systems (confirmed Level 3+ incident);
- **(v)** Political pressure on facilitator government that compromises operational neutrality (certified by JIG);
- **(vi)** Facilitator government formally withdraws from Framework escrow role.

(b) Automatic Backup Activation: The Framework shall maintain **two backup escrow arrangements** pre-negotiated at signature:

- **Backup Alpha:** A secondary Swiss bank, designated at signature, with pre-established compliance framework and authorized correspondent relationships. Activated within 48 hours of primary escrow failure.
- **Backup Beta:** A non-Swiss, non-Omani neutral financial institution (e.g., a Singaporean or Qatari bank) with no exposure to either Party's sanctions exposure. Activated if both primary and Backup Alpha fail, or if Swiss neutrality is compromised.

(c) Fund Transfer Protocol: Upon Escrow Failure certification by the JIG:

1. All escrowed funds are immediately frozen in place (no withdrawals, no transfers pending JIG direction);
2. The JIG convenes within 72 hours to certify the failure and select Backup Alpha or Beta;
3. Fund transfer to backup facility occurs within 14 days under JIG joint supervision;
4. All escrow beneficiaries are notified of the transfer and given 30 days to update their banking arrangements;
5. No escrow obligations are suspended during the transfer — they are deferred and accumulate.

(d) Interest Protection: During any Escrow Failure and subsequent transfer, interest on escrowed funds continues to accrue at the Article 8.1-A rate. The transferring institution is responsible for interest calculation through the transfer date. Any interest lost due to transfer delay is credited from the transferring institution's reserve account.

(e) SWIFT Disruption Contingency: If SWIFT disruption prevents normal escrow transactions, the JIG shall activate: alternative messaging via the Omani diplomatic financial channel; crypto-asset settlement using pre-agreed stablecoin mechanisms (USDC or equivalent, with 1:1 USD backing verified by independent auditor); or physical cash/gold transfer under armed escort for humanitarian tranches only. These alternatives are temporary and revert to SWIFT upon restoration.

(f) Neutral-Facilitator Protection: Both Parties commit to: refrain from sanctions actions targeting the escrow facilitator or its banking partners for activities conducted solely in fulfillment of Framework obligations; support the facilitator in any regulatory proceedings related to Framework escrow activities; and not pressure third states to sanction or restrict the facilitator. Violation of these commitments constitutes a Tier 1 technical dispute per incident.

(g) Escrow Insurance: The Guarantor Matrix shall arrange **political risk insurance** covering escrowed funds against: expropriation or confiscation by the facilitator's government; war, civil disturbance, or terrorism affecting the escrow bank; currency inconvertibility; and sovereign payment default. Coverage minimum: 100% of escrowed principal. Premiums paid from escrow interest accruals.

Why This Matters: Escrow is the Framework's financial backbone. If the escrow system fails and there's no backup, the entire economic module collapses — even if both Parties are in perfect compliance. This Article builds redundancy: two backups, alternative settlement mechanisms, interest protection, political risk insurance, and neutral-facilitator shields. It treats escrow not as a single point of failure but as a resilient financial architecture.

Article 25: Joint Coordination Council (v1.2 Operative Addition)

Issue 25: Governance Stack Coordination. The Framework operates through multiple institutions: JIG, JHTA, PGSA, Technical Custody Consortium, Swiss facilitators, Omani channels, escrow bodies, Working Groups, and the Dashboard. Each serves a distinct function, but the proliferation of nodes creates coordination delays, accountability ambiguity, and potential bureaucratic warfare. Rather than consolidating these bodies (which would create functional gaps), this Article creates a coordinating layer that aligns priorities without eliminating specialized functions.

ARTICLE 25 — JOINT COORDINATION COUNCIL (OPERATIVE AND BINDING)

(a) Establishment: The **Joint Coordination Council** (JCC) is a senior-level coordinating body comprising: one ministerial-level representative from each Party (Deputy Secretary/Deputy Minister or above); the Swiss facilitator (or designated deputy); the Omani diplomatic coordinator; and one rotating representative from the

Guarantor Matrix (chair rotates quarterly among guarantor states). The JCC does not replace any existing body — it coordinates among them.

(b) Functions: The JCC shall:

- **(i)** Convene monthly to review the status of all Framework institutions, identify coordination gaps, and resolve jurisdictional disputes between bodies;
- **(ii)** Approve the annual work plan for all Framework bodies, ensuring alignment with the Parties' political priorities;
- **(iii)** Mediate disputes between Framework institutions (e.g., JIG vs. JHTA jurisdictional overlap, PGSA vs. Technical Custody Consortium scheduling conflicts);
- **(iv)** Authorize creation of ad hoc working groups for specific issues not covered by existing bodies;
- **(v)** Review and approve the JIG's annual budget and staffing plan;
- **(vi)** Report quarterly to both Parties' heads of government on Framework institutional health and coordination status.

(c) Decision-Making: The JCC operates by **consensus**. If consensus cannot be reached, the Swiss facilitator may propose a compromise. If compromise fails, the matter is referred to the Guarantor Matrix. The JCC has no independent enforcement authority — its role is coordination, not command. All existing bodies retain their statutory authority under their respective Articles.

(d) Jurisdictional Clarity Protocol: The JCC shall maintain a public "Jurisdictional Matrix" defining the scope and limits of each Framework body's authority. The Matrix is binding on all bodies. Key jurisdictional boundaries include:

Table X4: Jurisdictional Matrix of Framework Bodies

Body	Primary Authority	Explicitly Excluded Authority
JIG	Compliance assessment; technical dispute resolution; public reporting	Operational military command; sanctions policy; diplomatic negotiations outside Framework
JHTA	Maritime transit operations; toll collection; route management	Nuclear verification; sanctions decisions; regional security operations
PGSA	Persian Gulf maritime authority; territorial water regulation	International shipping beyond Gulf; nuclear matters; escrow operations

Technical Custody Consortium	Fuel supply; LEU bank operations; conversion facility management	Inspection authority; sanctions relief; political decisions
Swiss Facilitator	Diplomatic channel management; dispute mediation; Dashboard administration	Compliance assessment; enforcement; sanctions
Omani Channel	Confidential backchannel communication; crisis de-escalation	Public diplomacy; verification; economic decisions
Escrow Bodies	Fund custody; disbursement; interest calculation	Political decisions; compliance assessment; sanctions
LIRB	Intelligence review; attribution assessment (advisory)	Operational decisions; public statements; Iranian-facing activities

(e) Coordination Timeline: The JCC shall meet: monthly by secure video for routine coordination; within 48 hours of any Tier 2 or Tier 3 event; within 24 hours of any Force Majeure Event; and within 7 days of any leadership transition (Article 21). Emergency sessions may be called by any JCC member or by either Party's head of government.

(f) Bureaucratic Warfare Prevention: If the JCC determines that a Framework body is deliberately obstructing another body's legitimate operations (e.g., withholding information, missing coordination deadlines, issuing contradictory directives), the JCC may: issue a binding coordination order; refer the matter to the Guarantor Matrix for enforcement; and recommend suspension of the obstructing body's head pending resolution. These are measures of last resort, requiring consensus minus the representative of the affected body.

Why This Matters: Nine specialized bodies doing their own jobs well is a strength. Nine bodies fighting over jurisdiction, hoarding information, and issuing contradictory orders is a disaster. The JCC doesn't eliminate any body — it creates the connective tissue that ensures they work as a system rather than as competing fiefdoms. It is the nervous system, not the brain.

Article 26: Political Consultation Channel (v1.2 Human-in-the-Loop Addition)

Issue 26: The Missing Leader-to-Leader Channel. The Framework recognizes technical consultation (JIG), executive consultation (Secretary of State/Foreign Minister), and military consultation (IRGC-Navy Hotline). But there is no recognized category for direct political consultation between heads of government — the conversations that happen between presidents, between supreme leaders, through backchannels that bypass all formal mechanisms. Real diplomacy happens at the leader level. This Article acknowledges that reality.

ARTICLE 26 — POLITICAL CONSULTATION CHANNEL (OPERATIVE AND BINDING)

(a) Establishment: A secure, confidential **Political Consultation Channel** shall be maintained between the heads of government of the United States and the Islamic Republic of Iran, facilitated by the Sultanate of Oman. The channel is **separate from all technical and executive consultation mechanisms**. Its purpose is to enable direct political communication outside the Framework's formal dispute resolution processes — for matters that require leader-level judgment, not ministerial or technical processing.

(b) Use: Either head of government may initiate political consultation at any time, for any reason or no reason. Consultation may address: Framework implementation concerns; regional security developments; bilateral relations; de-escalation of emerging crises; or any other matter of mutual concern. **Political consultation is not subject to media blackout, attribution rules, procedural timelines, or formal agenda requirements** — it is purely political, conducted in confidence between leaders.

(c) Relationship to Framework Obligations: Political consultation does **not suspend, override, or replace any Framework obligation**. It is an additional channel, not an alternative to compliance. Technical milestones continue on schedule. Escrow releases proceed per verification. Step-back penalties activate per JIG determination. Political consultation runs parallel to the Framework, not in place of it.

(d) Mutual Political Override (Extraordinary Circumstances Only): However, if the heads of government **jointly agree** on a course of action that deviates from the

Framework's procedures, that deviation shall be treated as a **mutual political override** and shall not constitute a breach by either Party. Such overrides require:

- **(i)** A genuine extraordinary circumstance: imminent military conflict between the Parties; catastrophic regional event (natural disaster, pandemic, major war spillover); or mutual consent to modify the Framework in response to changed circumstances;
- **(ii) Written confirmation from both heads of government**, filed with the Joint Secretariat within 48 hours of the oral agreement;
- **(iii)** Public notification to the JIG and Guarantor Matrix within 72 hours (the notification need not disclose the content of the consultation, only that a mutual override has occurred and its general scope);
- **(iv)** A time limit: the override expires automatically after 30 days unless renewed by both heads of government in writing.

(e) No Precedent: Political overrides under this Article **create no precedent** for future overrides. Each override is *sui generis* and requires renewed written confirmation. The existence of this Article does not imply that overrides are expected or routine — they are extraordinary measures for extraordinary circumstances. The JIG shall log all overrides (date, scope, duration) but shall not use override history to justify future overrides.

(f) Swiss Facilitator Role: The Swiss facilitator is not a participant in Political Consultation Channel communications but shall be notified that consultations are occurring (not the content). The facilitator's role is logistical: ensuring secure communications infrastructure; confirming that consultations occurred for record-keeping; and verifying that any mutual override documentation is properly filed.

(g) No Intelligence Collection: The Political Consultation Channel shall not be used for intelligence collection, surveillance, or espionage. Either Party that discovers that the other has used the channel for intelligence purposes may: suspend the channel immediately; refer the matter to the JIG as a Tier 2 material disruption; and demand investigation by the SEAC. This prohibition is absolute and non-waivable.

Why This Matters: The Framework is a sophisticated machine — 26 articles, 102 tables, 23 nuclear pathways, 9 institutional bodies. But machines cannot handle every situation. When a crisis erupts that the machine cannot process, leaders need a way to talk directly — not through the JIG, not through the Executive Consultation Window, not through the hotline, but leader to leader. This Article creates that channel. It preserves the Framework's machinery for 99% of situations while acknowledging that

> the 1% may require human judgment at the highest level. Political leadership, not mechanical submission.

Article 27: Annual Human-in-the-Loop Audit (v1.2 Human-in-the-Loop Addition)

Issue 27: Automation Creep. The Framework deploys multiple automated mechanisms: the rolling compliance window (Article 7.2); the de minimis error buffer (Article 7.2); the AI anomaly detection system (Article 3.4); the Dashboard automated milestone tracking; and the Cooling-Off Architecture (Article 19-F). Over time, these mechanisms could drift toward making decisions that should be reserved for human judgment. Without a regular check-up, the Framework could gradually transform from a tool that serves humans into a machine that commands them.

> **ARTICLE 27 — ANNUAL HUMAN-IN-THE-LOOP AUDIT (OPERATIVE AND BINDING)**
>
> **(a) Mandate:** The JIG shall commission an **independent annual audit** of all automated mechanisms in this Framework. The audit shall be conducted by a panel of three experts: one selected by each Party, and one selected by the Swiss facilitator. The panel shall have access to all Dashboard data, JIG records, and IAEA verification reports necessary to conduct its assessment.
>
> **(b) Audit Scope:** The annual audit shall examine:
>
> - **(i)** The rolling compliance window (Article 7.2) — has it functioned as intended? Has it triggered penalties that human reviewers would have deemed unfair?
> - **(ii)** The de minimis error buffer (Article 13.2) — has it adequately protected against trivial non-compliance? Has it been gamed?
> - **(iii)** The AI anomaly detection system (Article 3.4) — what is the confirmed false positive rate? Has the system triggered any penalties without human confirmation? Has it been recalibrated as required?
> - **(iv)** The Dashboard automated milestone tracking — have automated triggers executed correctly? Have there been false executions or missed triggers?
> - **(v)** The Cooling-Off Architecture (Article 19-F) — has it been invoked in good faith? Has either Party abused it?

- **(vi)** The Temporary Operational Suspension mechanism (Article 20) — has the invocation cap been respected? Has TOS been used for legitimate crises or as a stalling tactic?
- **(vii)** The Consensus Minus One decision rule (Article 12.5) — has it functioned effectively? Has it caused deadlock or facilitated consensus?
- **(viii)** The Political Discretion Fund (Article 7.5) — has it been used appropriately? Has either Party attempted to abuse the unanimous requirement?

(c) Core Questions: The audit shall answer three fundamental questions:

- **(i)** Are any automated mechanisms making decisions that should be reserved for human judgment?
- **(ii)** Have any automated mechanisms triggered consequences that human reviewers would have rejected?
- **(iii)** What modifications, if any, would increase human discretion without undermining verification?

(d) Report and Transparency: The audit report shall be **published to the Swiss Dashboard within 90 days** of each Framework anniversary. The report shall include: findings for each mechanism audited; recommendations for modification; a summary of incidents where automated mechanisms triggered human review; and a scorecard rating each mechanism's alignment with the human-in-the-loop principle. Both Parties may submit written comments on the report, published alongside it.

(e) Amendment Process: Either Party may propose amendments based on audit findings. Amendments require **mutual consent** — no unilateral changes based on audit recommendations. The audit creates a **presumption** that the Parties should review identified mechanisms, but it is advisory, not binding. The presumption may be overcome by either Party with a written explanation filed with the JIG.

(f) First Audit Timing: The first audit shall be commissioned on the **first anniversary** of Framework signature (Day 365). If the Framework terminates before Day 365, no audit is required. If the Framework is renewed, the audit cycle continues on each anniversary.

Why This Matters: The Framers of this agreement understood that automation is a tool, not a master. Over time, institutional inertia, bureaucratic convenience, and technical enthusiasm can cause automated systems to expand their authority beyond their original intent. The Annual Human-in-the-Loop Audit is a self-correcting mechanism — a regular check-up that ensures the Framework's machinery remains subordinate to human judgment. It prevents the slow drift from "humans use

> machines" to "machines use humans." It is the immune system of the human-in-the-loop architecture.

Article 28: Crisis Contacts Designation and Maintenance Protocol (v1.4 Final Addition)

Issue 28: The Placeholder Problem. Throughout this Framework, references appear to crisis contacts: the IRGC-Navy Hotline, the Swiss facilitator, the Omani backchannel, the JIG duty officer, the Dashboard emergency bulletin system. These contacts are the Framework's nervous system — without them, no verification, no consultation, no de-escalation, no crisis response is possible. Yet the Framework contains no mechanism to ensure these contacts actually exist, are staffed by real people with real phone numbers, and are tested before they are needed. This Article closes that gap. No placeholder numbers. No fictitious contacts. No undesignated roles. Real people, real numbers, real testing — or no Framework activation.

> **ARTICLE 28 — CRISIS CONTACTS DESIGNATION AND MAINTENANCE PROTOCOL (OPERATIVE AND BINDING)**
>
> **(a) Designation Deadline:** No later than **14 days prior to Framework signature**, each Party shall designate to the Joint Secretariat:
>
> - **(i)** Its **primary and secondary representatives** to the IRGC-Navy Hotline, with 24/7 contact numbers and authenticated communication credentials;
> - **(ii)** Its **primary and secondary diplomatic contacts** for the Omani backchannel;
> - **(iii)** Its **primary and secondary JIG duty officers** with 24/7 availability;
> - **(iv)** Its **technical liaison** for Dashboard emergency bulletin publication.
>
> **(b) Swiss Facilitator Contacts:** The Swiss Federal Department of Foreign Affairs shall designate, no later than 14 days prior to Framework signature:
>
> - **(i)** A **primary and secondary facilitator** on-call 24/7 with direct contact numbers;
> - **(ii)** A **secure communications protocol** for executive-level consultation;
> - **(iii)** A **public Dashboard URL** (e.g., swiss-framework-dashboard.admin.ch) that shall remain functional for the duration of the Framework.
>
> **(c) Testing Protocol:**
>
> - **(i)** The **IRGC-Navy Hotline** shall be tested **daily at 06:00 UTC** beginning on Day 0. Both Parties' representatives must respond within **15 minutes**;

- **(ii)** The **Swiss facilitator** shall test the **Omani backchannel weekly**;
- **(iii)** The **Dashboard** shall be tested continuously via automated uptime monitoring;
- **(iv)** Failure of any designated contact to respond to the daily test triggers a **Tier 1 technical dispute** and automatic escalation to the JIG Chair within 4 hours.

(d) Contact Rotation and Update: Each Party may update its designated contacts with **7 days' written notice** to the Joint Secretariat. During leadership transitions (Article 21), interim contacts shall be designated within **48 hours**. The Joint Secretariat shall maintain a secure, authenticated **Crisis Contact Registry** published to an access-controlled section of the Dashboard. The Registry includes: name, title, direct phone number, secure email, and backup contact for each designated role.

(e) No Placeholders — Certification Requirement: Under no circumstances may the Framework enter into force with placeholder, fictitious, or undesignated crisis contacts. The Joint Secretariat shall certify to both Parties no later than **7 days prior to Framework signature** that all crisis contacts have been designated, tested, and authenticated. If certification cannot be made, the Framework activation date shall be **postponed by 7-day increments** until certification is achieved. Neither Party may waive this requirement.

(f) Consequence of Non-Designation: If either Party fails to designate its crisis contacts by the deadline, the other Party may **unilaterally postpone Framework activation without penalty**. Repeated failure (**three or more postponements**) may be referred to the Guarantor Matrix as a potential bad-faith action under Article 13.6.

(g) Annual Verification: On each Framework anniversary, the Joint Secretariat shall verify that all crisis contacts are still valid, reachable, and authenticated. Outdated contacts must be updated within 14 days. The Annual Human-in-the-Loop Audit (Article 27) shall include a review of the Crisis Contact Registry as part of its scope.

Why This Matters: A Framework with untested contacts is a Framework that will fail in its first crisis. When a Tier 3 event occurs at 3:00 AM, someone needs to answer the phone. When the Swiss facilitator calls the Omani backchannel, someone needs to pick up. When the Dashboard flashes red, someone needs to see it. This Article ensures that the human infrastructure exists before the Framework needs it. It is the difference between a document and an operational system.

Part V

The Final Steps — Implementation

The preceding eighteen Articles define what must be done. This Part defines how it shall be done — the precise sequencing, face-saving narratives, and emergency protocols that transform diplomatic architecture into political reality.

The 180-Day Implementation Roadmap

Phase 1: Foundation (Days 1–14) — "The First Fortnight"

Phase 1 establishes the physical and institutional infrastructure upon which all subsequent phases depend. Every milestone in this phase is irreversible once executed.

Table 17: Phase 1 Implementation Timeline

Day	Iranian Commitment	US Commitment	Verification
Day 1	72-hour ceasefire auto-activates; OLEM confirms freeze on enrichment above 60%; HEU digital dual-lock deposited with IAEA (physical possession retained)	$750M Mutual Escrow Deposit into Omani escrow; OFAC general license; partial nuclear sanctions designation freeze	Omani naval observers confirm stand-down; OLEM confirms within 72h
Day 3	IAEA inspectors arrive at declared HEU facility; begin physical inventory	5th Fleet withdraws to 5nm from Hormuz transit corridor (upon IAEA HEU consolidation confirmation)	Joint Secretariat logs both events

Day 5	Declare all enrichment facilities; mothball protocol initiated at non-core sites	Issue OFAC Comfort Letters to designated banking nodes	IAEA facility inspection reports
Day 6	IRGC forces at agreed stand-off distances; air-gapped Black-Box telemetry installation begins at designated highest-risk facilities	5th Fleet stand-off confirmed; automated countdown activated (T+160h system read)	IAEA data feed verifies HEU consolidation has begun; automated trigger armed
Day 7	IRGC withdraws to 5nm; Hormuz transit corridor declared open **automatically** at T+168h	Naval blockade **automatically** lifted at T+168h; Notice to Mariners issued	Automated execution confirmed by Joint Secretariat; no human vote required
Day 10	First JHTA-AUTHORIZED transit completed	First tranche of escrow funds released to Swiss Financial Cell	Lloyd's List confirms transit; Swiss Cell confirms credit
Day 14	HEU inventory complete; dual-lock custody activated (Pathway A default); consolidation verified	Tranche 1 asset release (25% of remaining frozen assets)	IAEA custody verification certificate

Phase 2: Consolidation (Days 15–44) — "Building the Bridge"

Table 18: Phase 2 Implementation Timeline

Day	Iranian Commitment	US Commitment	Verification
Day 15	Rotor removal begins at mothballed facilities under IAEA supervision	5 designated Iranian banks authorized for restricted SWIFT (Phase 1)	IAEA serial-numbered rotor inventory
Day 21	First air-gapped Black-Box telemetry batch delivered to Swiss neutral hub from designated highest-risk facilities (not all sites)	Reinsurance backstop formally activated ($20B facility)	Black-Box encrypted batch data confirmed at Swiss hub
Day 30	60% enrichment freeze verified continuous; Fordow Phase 2 option discussion opens	Expanded civilian trade licenses; agricultural exports expedited; **Mutual Escrow Deposit released simultaneously**	IAEA 30-day continuous compliance report; escrow simultaneous release confirmed

Day 44	Rotor storage complete; sealed containers in UN-monitored facility	Tranche 2 asset release (25%); petrochemical export licenses	IAEA mothball facility completion certificate

Phase 3: Normalization (Days 45–120) — "The New Normal"

Table 19: Phase 3 Implementation Timeline

Day	Iranian Commitment	US Commitment	Verification
Day 45	If no third-country HEU destination agreed: Automatic IAEA Escrow Transition activates — physical material moved to Oman triple-lock facility; CBI legal title preserved; no economic penalties	All sanctions relief tranches continue uninterrupted during Escrow Transition	IAEA-Omani triple-lock custody confirmed; Central Bank of Iran title verified
Day 60	Civilian Transition enrichment 3.67% ceiling confirmed; all higher centrifuges under seal	SWIFT Phase 2 — all clean-record Iranian banks; humanitarian trade caps lifted	OLEM confirms; SWIFT confirms bank connections
Day 75	If HEU still in escrow: JIG facilitated negotiation for final pathway selection	DFC Dollar-for-Dollar Matching Guarantee first disbursement window opens	JIG negotiation record; DFC matching certification
Day 90	Fordow Joint Venture operational (if mutually agreed) or reduced-capacity research status confirmed; medical isotope production begins (if converted)	Regional Stability Reconstruction Credits first disbursement; DFC matching activated	TCC confirms operational
Day 120	90-day compliance review; JIG publishes first comprehensive assessment	First quarterly review conference (all guarantor states); SWIFT Phase 3 pathway opened; Tranche 3 (20%)	JIG compliance report

Phase 4: Sustainability (Days 121–180) — "Locking In Peace"

Table 20: Phase 4 Implementation Timeline

Day	Iranian Commitment	US Commitment	Verification

Day 150	Full Fordow isotope production operational (if converted) or research facility certified; revenue sharing begins	Maritime security cooperation protocols activated	TCC revenue distribution reports
Day 180	Full 180-day compliance certified; formal invitation to negotiate permanent successor framework convened; Phase II Horizon Phase II readiness assessment completed	Tranche 4 (15%); formal invitation to negotiate permanent successor framework; legislative readiness assessment for Phase II initiated (if Rolling Executive Option not invoked)	IAEA + JIG joint certification; Rolling Horizon auto-renewal evaluation

Face-Saving Narratives for Both Parties

An agreement that cannot be defended in either capital will not survive the first news cycle.

For the United States

Nuclear: "Iran's 60% enrichment has been halted and placed under permanent IAEA lock. The HEU stockpile is neutralized with verified physical custody within two weeks. Breakout time extended from weeks to over a year. We retain veto over any Russian or hybrid disposition pathway. And critically, if Iran doesn't pick a final destination by Day 45, the material automatically moves to Oman — no more waiting games."

Maritime: "The Strait of Hormuz is open to international navigation under multilateral advisory governance, not Iranian unilateral control. The PGSA is restricted to coastal enforcement; the international transit corridor is governed by UNCLOS. The Day-7 blockade lift is now fully automated — if Iran moves the uranium, the blockade lifts automatically. No more trust-based gestures."

Economic: "No reparations were paid. Sanctions relief is conditional, reversible, and tied to verified compliance. Non-nuclear sanctions on terrorism and human rights remain fully in force. The leverage architecture remains intact. And our DFC matching guarantee ensures US taxpayer money only flows to verified civilian projects, dollar for dollar with Iranian skin in the game."

Security: "Iran's missile and drone transfers to proxies have been frozen. The regional threat environment has been stabilized. No compliance dispute can automatically trigger military action — we control our own military decisions. And with the new Executive

Consultation Window, no low-level inspector can start an economic war without the Secretary of State's say-so."

Israel (v1.0): "The United States has preserved its bilateral security coordination with Israel through the Tel Aviv Channel. The core US-Iran Framework proceeds on its own timeline, and nothing in the Framework grants any third party a veto over US implementation. The Lebanon firewall threshold has been recalibrated to protect the core deal from proxy provocations while preserving Israeli self-defense rights."

Lebanon (v1.0): "The core US-Iran agreement addresses nuclear proliferation and freedom of navigation. Lebanon is a separate file managed through a dedicated regional basket with full Israeli participation and a Dossier Isolation Firewall. The threshold for linking Lebanon to the core deal is now set at a credible 50-casualty level with a mandatory 72-hour diplomatic consultation window — preventing hair-trigger escalation."

For the Islamic Republic of Iran

Nuclear: "Legal title to the HEU remains with the Central Bank of Iran — sovereign property, not surrendered. The material stays on Iranian soil by default. If a destination isn't agreed by Day 45, it moves to Oman — but our legal ownership is preserved, and no economic penalties apply. NPT rights to peaceful nuclear energy affirmed. Enrichment continues at 3.67% under IAEA safeguards. Final disposition will be negotiated in the nuclear window, and we select the pathway freely — except for Russia and hybrid options requiring mutual agreement."

Maritime: "The PGSA has been recognized as the sole executive coastal law enforcement body within Iranian territorial waters. Iran holds a permanent co-chair seat on the JHTA governing council with veto rights. The JHTA is advisory only — not a supranational authority. IRGC authority explicitly preserved. And the Day-7 blockade lift is automatic — if we move the uranium, the Americans have no choice but to lift. No more review boards that can change their minds."

Economic: "Frozen assets are being released through neutral escrow, not surrendered. The Day 1 Mutual Escrow Deposit of $500 million demonstrates real American commitment — and the remaining $250 million releases at Days 45 and 60 as we maintain compliance. SWIFT reconnection restores Iran's place in the global economy — and if one bank has an issue, only that bank is isolated, not our entire sector. The DFC Dollar-for-Dollar Matching Guarantee, with its $1 billion annual floor, means every dollar we invest in civilian infrastructure is matched by American development finance. This is real, structural economic support. And this is only Phase I — we retain our full position for deeper negotiations."

Sovereignty: "No facility was destroyed. Fordow may continue as a reduced-capacity research facility under our control. Missile and drone programs remain sovereign and intact. The air-gapped telemetry protocol means no real-time foreign surveillance of our military infrastructure — data is compiled and delivered in batches through neutral Swiss servers. The 14-day timeline respects our administrative capacity."

Lebanon (v1.1): "The framework ends the war on all fronts. Through the Levant Regional Basket, Iran ensures our allies in Lebanon are not abandoned. The Dossier Isolation clause protects the core agreement from Israeli military actions in Lebanon. The three-tier escalation ladder means small incidents (10-24 casualties) trigger consultation only with no core Framework linkage. Only at Tier 3 (50+ casualties) does a presumption of linkage arise — and either party can veto. The JIG no longer has fast-track majority vote attribution power."

Framework Survival Core and Emergency Fallbacks

The Survival Core

Under all circumstances, including total framework suspension, three telemetry channels remain active: (1) **Ceasefire Hotline:** Omani-facilitated 24/7 bilateral military hotline. Neither party may disable without 30 days' notice to the UN Secretary-General. (2) **Humanitarian Access:** Zero-fee, no-inspection-delay humanitarian corridor through Hormuz. Survives all disputes. (3) **Emergency Diplomatic Channel:** Backchannel through the Coordination Secretariat. Survives all suspensions.

Survival Core Funding Mechanism (v1.1 Fix)

> ARTICLE 11.1-A — SURVIVAL CORE ESCROW FUND (OPERATIVE AND BINDING)
>
> The Survival Core shall be funded through a dedicated **$50 million escrow account**, capitalized equally by both Parties ($25 million each) upon Framework activation. Funds shall be administered by the Swiss facilitator and used exclusively for: hotline operations and maintenance; humanitarian corridor certification and logistics; and emergency diplomatic channel staffing. Neither Party may suspend funding to the Survival Core during any review pause or suspension of other Framework provisions. If the escrow falls below $10 million, replenishment shall occur within **14 days** pro rata. **If a Party fails to replenish within 14 days, the other Party may advance the delinquent Party's share, and the delinquent Party shall reimburse within 30 days**

> **with interest at LIBOR + 2%. Money makes mechanisms real. A funded Survival Core survives political freezes.**

The Survival Core is a standalone UN humanitarian arrangement. It survives for **365 days** following any withdrawal notice, and shall **automatically renew for successive 365-day periods unless both Parties agree in writing to terminate it**. The Guarantor Matrix may extend the Survival Core by majority vote even if one Party objects. The Survival Core is a standalone humanitarian arrangement independent of the Framework's political status, per the Rolling Horizon "No Cliff" principle (Article 17.5).

Political Shock Absorber

Framework implementation enters automatic Review Pause (not termination) upon: elections; assassination of senior leadership; cabinet collapse; mass protests; leadership transitions; kinetic disruptions within regional security architectures. During Review Pause: Tier 1–2 modules continue; Tier 3 enters hold; Secretariat issues Review Pause Certificate within 48 hours; 30 days to confirm continuation. Max 90 days, extendable by mutual agreement. Does NOT constitute material breach. Does NOT trigger the three-tier system.

Reconstitution Mechanism

If the framework enters full suspension, all parties attend a Reconstitution Conference within 30 days, convened by Pakistan and the UN Appointee. Root causes reviewed. Modifications proposed. Conditions for graduated restoration defined. No party may unilaterally block reconstitution.

Appendices

Appendix A: Quick Reference Matrix — All 18 Issues & Resolutions

Table 21: Complete Issue Resolution Matrix

#	Issue	US Position	Iranian Position	PSF Resolution	Article
1	HEU Disposition	Physical removal to US	Must remain on Iranian soil	IAEA custody with legal title decoupled from physical custody; 14-day consolidation; in-Iran default; Day 45 IAEA Escrow Transition; limited US veto on D and F only	1
2	Enrichment Cap	Zero enrichment or 20-year moratorium	Maximum 12-year	Sliding-scale 12–15 year moratorium with compliance extensions	2
3	Underground Sites	Destroy or disable	Preserve all infrastructure	Optional civilian conversion; mothball protocol with air-gapped telemetry	3

4	PGSA vs. Freedom of Navigation	Reject all unilateral restrictions	PGSA as sovereign authority	JHTA advisory governance with sovereignty recalibration; 12-month pilot; PGSA recognized as sole coastal enforcement	4
5	Transit Fees	"We don't want tolls"	Permanent fee system	MMMF multilateral fund; frozen-revenue deadlock protection; no automatic payout	5
6	US Naval Blockade	Blockade until full deal	Blockade lift precondition	Unconditional automated Day-7 synchronized 24-hour countdown trigger	6
7	Sanctions Relief Sequencing	Full compliance first	Immediate blanket lifting	30-Day Stabilization Block Model with Mutual Escrow Deposit; partial nuclear-only sanctions freeze; stand-off tied to HEU start	7
8	Frozen Assets	Performance reserves	Immediate unfreezing	Neutral escrow under sovereign immunity	8
9	SWIFT Isolation	Verified non-diversion first	Full immediate reconnection	Phased 4-tier SWIFT pathway with Targeted Bank Isolation hubs	9
10	War Reparations	Flat rejection	Formal demand for payment	Regional Stability Reconstruction Credits with US Dollar-for-Dollar Matching Guarantee via DFC	10
11	Negotiation Window	"A few days" or "one shot"	Fragile ceasefire	Binding Letter of Intent; auto-renewing ceasefire	11
12	Trust Deficit	No credible guarantor	No credible guarantor	Distributed Guarantor Matrix (6 States)	12
13	Automated Escalation	Hard triggers required	Flexibility required	Executive Consultation Governance with mandatory 48-hour consultation window; political leaders' veto over penalties	13

14	Missile Reconstitution	Strict limitations	Non-negotiable sovereign defense	Deployment pause and transfer freeze; no dismantlement	14
15	Global Agrifood Crisis	Political issue, not treaty value	Leverage point	Zero-fee humanitarian corridor; agricultural release valve	15
16	Israeli Security Coordination	Maintain Tel Aviv Channel	No third-party obligations	US-Israel bilateral coordination; non-impediment clause; no veto rights	16
17	Framework Duration	No fixed-term deal	No permanent surrender	Rolling Horizon Architecture with conditional auto-renewal and orderly withdrawal	17
18	Military Escalation Firewall	Preserve options	No compliance-triggered strikes	Explicit decoupling of breach penalties from military authorization; 14-day notice; UNSC requirement	18

Appendix B: The PGSA Integration Protocol (Technical)

B.1 Legal Framework for PGSA-JHTA Integration

The integration of the Persian Gulf Strait Authority into the Joint Hormuz Transit Authority is governed by the following legal architecture:

1. **Statutory Continuity:** The PGSA remains an Iranian statutory authority under Iranian domestic law. No change to its legal personality is required.**International Delegation:** Iran delegates specific operational functions to the JHTA through a bilateral Administrative Agreement between Iran (as PGSA parent state) and the JHTA. This is not a treaty amendment but a standard international administrative arrangement.
2. **Dual-Branding:** All PGSA permits, notices, and communications issued for Hormuz transit shall bear both PGSA and JHTA identifiers. The JHTA identifier constitutes the multilateral authorization layer; the PGSA identifier constitutes the Iranian sovereignty layer.
3. **Revenue Flow:** Transit fees are collected by the JHTA and distributed through the MMMF. The PGSA receives its 40% share as a budget allocation from the Fund, not as direct toll revenue. This structure eliminates OFAC sanctions exposure for shippers.

Under v1.0, if the JHTA Council deadlocks on a budget, all revenue is frozen in escrow — no automatic distributions occur.

4. **Dispute Resolution:** Disputes between the PGSA and JHTA are resolved through the JHTA Council. Iran retains its veto on governance questions. The JHTA retains its veto on security protocol amendments. JHTA recommendations are advisory only; PGSA retains final say within Iranian territorial waters. **PGSA-JHTA Dispute Escalation Ladder:** Disputes between the PGSA and JHTA regarding interpretation of their respective authorities shall be resolved as follows: **Step 1 (Days 1–14):** JHTA Council consultation — Iranian and US co-chairs attempt direct resolution. **Step 2 (Days 15–30):** JIG non-binding mediation. **Step 3 (Days 31–45):** Advisory opinion from IMO Legal Committee, published to Dashboard. **Step 4 (Day 46+):** If JIG certifies by two-thirds majority that PGSA actions have materially impaired freedom of navigation for 30 consecutive days, Tier 1 technical dispute — no automatic step-back, core Framework unaffected.

B.2 PGSA Sovereignty Recalibration (v1.0 Technical Supplement)

Table 22: Territorial Waters Authority Matrix

Zone	Geographic Scope	PGSA Authority	JHTA Authority	Overlap Resolution
Inner Territorial Waters	0–12nm from Iranian coast	SOLE — permitting, patrol, customs, security screening, IRGC coordination	None	PGSA authority absolute
Transit Corridor	International strait passage	Permit issuance (dual-branded)	PRIMARY — traffic coordination, safety protocols, non-binding dispute resolution	PGSA permits required; JHTA coordination standards apply; JHTA recommendations advisory only
Contiguous Zone	12–24nm	Limited customs enforcement	Maritime safety coordination	Notification required for enforcement actions

High Seas	Beyond 24nm	None	SOLE — international coordination, safety, insurance verification	JHTA authority absolute

B.3 Technical Interface Specifications

Data Exchange: Real-time API connection between PGSA vessel vetting database and JHTA Transit Permit Processing Unit. **AIS Integration:** PGSA AIS transponder codes integrated into JHTA-AUTHORIZED designation protocol. **Fee Processing:** JHTA collects fees via designated Omani banking channels; PGSA receives quarterly distribution from MMMF (budget-deadlock freeze applies per Article 5.2). **Incident Reporting:** Joint incident reports bear dual authorship (PGSA operational lead + JHTA verification sign-off). JHTA recommendations are non-binding; PGSA retains final operational authority within Iranian territorial waters.

Air-Gapped Telemetry Interface (v1.0): Black-Box batch data from Iranian facilities is delivered to the Swiss neutral hub via physically secured data carriers or dedicated encrypted channels. The JHTA receives compiled, dual-key encrypted batch reports from the Swiss hub — not real-time feeds from Iranian facilities. This ensures maritime safety monitoring without live intrusion into Iranian military infrastructure.

Appendix C: Verification Technology Menu

Table 23: Verification Technologies

Technology	Measures	Data Output	Application
OLEM (Online Enrichment Monitor)	Real-time enrichment at cascade headers	Quantitative enrichment %, flow rates	Nuclear verification; primary compliance trigger
Black-Box Sensors (Air-Gapped)	Tamper-evident data logging (highest-risk facilities only)	Continuous operational parameters; encrypted batch delivery	Facility conversion monitoring; mothball verification; batch delivery to Swiss/Omani neutral hubs
UGES (Ultrasonic Gas Enrichment Sensor)	Non-intrusive enrichment in UF6 cylinders	Cylinder-by-cylinder verification	HEU custody chain; downblending verification
ICS (Integrated Camera System)	24/7 visual monitoring	Encrypted video, AI anomaly detection	General facility monitoring; inspector safety

Digital Twin Modeling	Virtual facility modeling	Behavioral pattern analysis	Anomaly detection; requires physical cross-verification
Satellite Feed	Daily imagery of all facilities and Hormuz	AIS, SAR, EO high-resolution	Maritime surveillance; construction monitoring

Appendix D: Rolling Executive Option & Legislative Flexibility (v1.0 Reform)

> **APPENDIX D — ROLLING EXECUTIVE OPTION REPLACING HARD LEGISLATIVE RATIFICATION (OPERATIVE)**
>
> This Appendix replaces the v4.0 legislative ratification framework, which mandated that Phase II require formal legislative ratification within 180 days and imposed punitive financial damages on executive branches if their respective legislatures stalled. The v4.0 model was a non-starter given fierce domestic hardliner opposition in the US Congress and the Iranian Majles. The following Rolling Executive Option provides a politically viable alternative.

D.1 Phase I Instruments of Acceptance (Executive-Level)

This Framework does not require Senate ratification or Majles legislation for Phase I activation. It operates through executive-level Instruments of Acceptance deposited simultaneously with the Joint Secretariat in Muscat. Phase I commitments are within the executive authority of both governments and activate immediately upon deposit.

D.1-A — Legal Challenge Protection (v1.1 Operative Addition)

> **ARTICLE D.1-A — ANTI-CIRCUMVENTION AND LITIGATION SHIELD (OPERATIVE AND BINDING)**
>
> **(a) No Support for Private Litigation:** Neither Party shall support, fund, or encourage private litigation in third-country courts seeking to attach, freeze, or seize assets released under this Framework. This includes but is not limited to: providing legal counsel for such litigation; sharing confidential Framework documents with plaintiffs; filing amicus briefs supporting attachment; or exercising diplomatic pressure on courts to permit attachment.
>
> **(b) Best Efforts to Dismiss:** The Parties shall use best efforts to dismiss or stay any such litigation, including through sovereign immunity defenses, act of state doctrine,

and political question doctrine. Each Party shall notify the other within 7 days of becoming aware of any such litigation.

(c) Consequence: Willful violation of this provision constitutes a **Tier 2 material disruption** under Article 13. The JIG shall determine willfulness within 14 days of complaint.

(d) Escrow Shield: This provision applies retroactively from Framework signature and survives any suspension, withdrawal, or termination. The anti-litigation obligation is independent of the Framework's political status.

D.2 Phase II Commitments — Rolling Executive Option (v1.0 Reform)

ARTICLE D.2 — THE ROLLING EXECUTIVE OPTION (OPERATIVE AND BINDING)

(a) Deletion of Hard 180-Day Deadline: The v4.0 requirement that Phase II Commitments receive formal legislative ratification within 180 days is hereby removed. The punitive financial damage clauses for legislative delays (including the v4.0 "interim disbursement" and "liquidated damages" provisions) are deleted in their entirety.

(b) Rolling Executive Extension: In place of mandatory legislative ratification, the executive branches of both parties may continually extend Phase I stabilization measures in **90-day increments** via executive decrees, bypassing the immediate need for legislative battles until greater diplomatic trust is established.

Mechanism:

- At any point during Days 90–180 of Phase I, either Party's executive branch may issue an Executive Extension Decree, extending all Phase I stabilization measures for an additional 90 days.
- Each Executive Extension Decree must be deposited with the Joint Secretariat and communicated to the other Party through the Omani diplomatic channel.
- No legislative approval is required for Executive Extension Decrees. Each decree is valid solely on executive authority.
- There is no limit on the number of consecutive Executive Extensions. Phase I stabilization may continue indefinitely on 90-day rolling extensions.

(c) Optional Legislative Ratification: Either Party may, at any time, submit Phase II Commitments to its legislature for formal ratification. Legislative ratification is optional, not mandatory. If ratified, Phase II activates per Article 17.2. If not ratified, Phase I continues on rolling executive extensions.

(d) Incentive for Legislative Ratification: While not required, legislative ratification unlocks additional benefits not available under executive extension:

- Access to Tranche 5 and Tranche 6 asset releases (the final 25% of frozen assets)
- Full SWIFT Phase 3 (all Iranian financial institutions)
- Permanent JHTA charter conversion from 12-month pilot
- DFC Dollar-for-Dollar Matching Guarantee cap increase from $5 billion to $10 billion annually

These enhanced benefits provide genuine incentive for legislative action without punishing the executive branch if legislatures refuse to cooperate.

(e) No Punitive Consequences for Legislative Delay: Under no circumstances shall legislative delay or rejection of Phase II Commitments trigger: financial penalties against either executive branch; automatic step-back; forfeiture of escrowed funds; suspension of Phase I benefits; or any other adverse consequence. The Framework simply continues on Phase I terms via executive extension.

D.3 Phase II Commitments Requiring Legislative Action (If Elected)

The following provisions are designated as "Phase II Commitments" and require legislative ratification or equivalent domestic legal action **only if either Party elects to pursue full Phase II activation** (rather than continuing on Rolling Executive Extensions):

- (a) Permanent enrichment moratorium beyond 12 months;
- (b) HEU disposition involving foreign territory (Pathways B, C, D, or F);
- (c) Full SWIFT reconnection (Phase 3, all Iranian financial institutions);
- (d) Permanent JHTA charter (conversion from 12-month pilot to permanent status).

D.4 Non-Obstruction Commitment

Neither party shall use the legislative ratification process as a delaying tactic. The executive branch of each party shall: (a) if electing to pursue legislative ratification, submit the required instruments to its legislature within 30 days of the decision; (b) provide full technical and legal support for expedited review; (c) certify to the Joint Secretariat every 90 days on the status of any ratification process. If either legislature rejects the Phase II Commitments, the parties may enter a 60-day consultation period to identify modifications that would secure approval without undermining the Framework's core architecture — but this consultation is optional, and Phase I continues regardless.

D.5 Interim Provisions

Pending any legislative ratification of Phase II Commitments, the following interim provisions apply under Rolling Executive Extensions:

- HEU disposition shall proceed under Pathway A (in-Iran downblending) as the default, which requires no foreign territory involvement;
- SWIFT access shall remain at Phase 1 (Restricted) level, not advancing to Expanded or Full;
- The JHTA shall continue under its 12-month pilot charter, subject to renewal via executive extension;
- The enrichment moratorium shall operate under the executive-agreed 12-year sliding scale, not requiring legislative action until Year 11 review.

D.6 Instrument Templates

For the United States of America (Phase I — Executive Acceptance):

> The undersigned, duly authorized by the President of the United States, hereby accepts the provisions of The Provisional Steps Framework Phase I Stabilization Framework v1.0 and commits the United States to their full implementation in accordance with the timelines, verification procedures, and reciprocal obligations set forth herein.

For the Islamic Republic of Iran (Phase I — Executive Acceptance):

> The undersigned, duly authorized by the Supreme Leader and President of the Islamic Republic of Iran, hereby accepts the provisions of The Provisional Steps Framework Phase I Stabilization Framework v1.0 and commits the Islamic Republic to their full implementation in accordance with the timelines, verification procedures, and reciprocal obligations set forth herein.

For Executive Extension Decree (Either Party — Rolling Executive Option):

> Pursuant to Appendix D.2 of The Provisional Steps Framework v1.0, the executive branch of [Party] hereby issues Executive Extension Decree #[number], extending all Phase I stabilization measures for an additional 90 days from [date] through [date]. This decree is issued under the executive authority of the [President/Supreme Leader] and requires no legislative approval.

Rationale: The v4.0 legislative "trap" — mandating Phase II ratification within 180 days and punishing executives with financial damages if Congress or the Majles stalled — was a non-starter. Both legislatures contain fierce hardliner blocs that would view ratification as a capitulation. The Rolling Executive Option allows the deal to survive and deliver benefits while legislative politics gradually shift. It transforms the Framework from a one-shot, all-or-nothing gamble into a sustainable, iterative diplomatic process.

Appendix E: Lebanon and Levant Front Stabilization Protocol (v1.1 Reform)

> **APPENDIX E — RECALIBRATED LEBANON FIREWALL (OPERATIVE)**
>
> The v4.0 Lebanon Firewall mechanism — with its 10-casualty threshold and JIG majority vote fast-track attribution — created a hair-trigger that a single rogue proxy action or localized skirmish could pull. The following recalibration raises the threshold, removes the fast-track attribution power, and inserts a mandatory diplomatic cooling-off period.

Operational Context: Iran's Foreign Ministry, via spokesperson Esmail Baghaei, has explicitly stated that Tehran's fixed position is an end to the war on all fronts, including Lebanon. Tehran refuses to sign a standalone deal that stops bombing on Iranian soil while allowing the active conflict between Israel and Hezbollah in southern Lebanon to continue. The United States and Israel have historically sought to keep Lebanon separate from core US-Iran negotiations, treating it as a distinct theater governed by separate tracking mechanisms. This appendix provides a structural compromise that prevents the entire deal from collapsing over the Lebanon front while respecting both sides' core requirements. It operates through three integrated instruments: the Proxy Operational Pause, the Dossier Isolation Firewall, and the Levant Regional Basket — now recalibrated with the v1.0 political flexibility modifications.

E.1 Proxy Operational Pause: Hezbollah De-escalation Protocol

Purpose: To satisfy the US and Israeli requirement that pulling back US forces and opening Hormuz does not result in immediate escalation on Israel's northern border, Iran transmits immediate, formal instructions to Hezbollah leadership for an operational pause concurrent with the main Framework timeline. This pause is not disarmament, surrender, or political concession — it is a structured, time-limited operational de-escalation tied to the Framework's implementation phases.

Table 24: Hezbollah Operational Pause Phases

Phase	Hezbollah Operational Parameters	Verification	US/Israel Reciprocal
Phase 1: Days 1–30	Cessation of cross-border rocket and drone attacks into northern Israel. Suspension of offensive military operations against IDF positions beyond the immediate defensive engagement zone (500 meters from the Blue Line). **Defensive engagement of IDF forces that enter Lebanese territory is permitted under the laws of armed conflict.** No ground incursions into Israeli territory. No new heavy weapons transfers.	UNIFIL patrol confirmation of reduced activity; Israeli military intelligence assessment (shared via Tel Aviv Channel); Lebanese Armed Forces liaison reports; satellite surveillance of known Hezbollah positions	Israel suspends offensive air strikes against Hezbollah positions in southern Lebanon for 30 days. IDF maintains defensive posture only.
Phase 2: Days 31–90	Continued cessation of offensive operations. Hezbollah forces remain in defensive positions. No infiltration attempts across the Blue Line. Weapons transfers suspended.	Monthly UNIFIL verification report; ICRC monitoring of civilian areas; SEAC pattern analysis of supply routes	Israel extends air strike moratorium to 90 days contingent on verified Hezbollah restraint. Gradual relaxation of naval interdiction of suspect Hezbollah supply vessels in eastern Mediterranean.
Phase 3: Days 91–180	Exploration of structured de-escalation through Levant Regional Basket. Hezbollah may engage in political and humanitarian activities. Military posture remains defensive.	Regional Basket monitoring cell; UN Special Coordinator for Lebanon quarterly assessment	Israeli restrictions reviewed at Day 180 milestone. If Hezbollah restraint verified for 180 consecutive days, Israel considers further confidence-building measures through Regional Basket.

E.1.1 Verification Architecture

The Hezbollah operational pause is verified through four independent channels: (1) UNIFIL ground patrols — daily patrols along the Blue Line with incident logging published to the Swiss Dashboard; (2) Israeli intelligence assessment — weekly classified brief to the Tel Aviv Channel, with sanitized summary published to Dashboard; (3) Lebanese Armed Forces liaison — daily reporting from LAF command in Tyre and Nabatieh to the Oman Liaison Office; (4) SEAC pattern analysis — automated detection of supply route activity, weapons convoy movement, and communications patterns. All four channels must show green before the pause is considered verified.

E.1.2 Non-Compliance Consequences

If any channel reports a significant Hezbollah offensive action, the following sequence activates: (1) 24-hour verification window — all four channels confirm the incident; (2) if confirmed, Iran has 72 hours to transmit renewed de-escalation instructions through the Oman back-channel; (3) if Iran de-escalates within 72 hours, the pause resumes with enhanced monitoring; (4) if Iran does not de-escalate, Israel may resume defensive air strikes limited to the specific positions involved in the offensive action — not blanket escalation; (5) under no circumstances does a Hezbollah incident trigger stepback against the core US-Iran Framework (see Dossier Isolation Firewall, Section E.2).

E.2 Dossier Isolation Firewall: Lebanon Non-Linkage Clause

Operative Principle: The Dossier Isolation Firewall is a binding structural provision that explicitly decouples the Lebanon stabilization track from the core US-Iran Framework. No event, violation, or suspension in the Lebanon theater shall automatically trigger any consequence in the core US-Iran bilateral framework.

Article E.2 — Dossier Isolation Firewall (Operative and Binding)

(a) Hezbollah compliance or non-compliance with the Proxy Operational Pause shall not constitute compliance or non-compliance by the Islamic Republic of Iran with any provision of the core Framework.

(b) Israeli military action in Lebanon shall not be deemed a violation by the United States of any ceasefire, stand-down, or non-escalation commitment under the core Framework.

(c) No Tier 1, Tier 2, or Tier 3 breach in the core Framework may be triggered by events in the Lebanon theater alone.

(d) The Joint Implementation Group (JIG) shall maintain separate compliance dashboards for the core Framework and the Lebanon track.

(e) This Firewall may only be modified by mutual written agreement of the United States and Iran, not by any guarantor state or regional party.

E.2.1 Firewall Architecture

The Dossier Isolation Firewall operates at three levels: (1) **Legal level** — the Exchange of Letters explicitly lists "incidents in the Levant theater involving non-state actors" as excluded from the stepback trigger definition in Article 13; (2) **Technical level** — the Swiss Dashboard maintains separate tracks for "Core Framework Compliance" and "Regional Security Incidents." Lebanon incidents are logged only on the Regional Security track, not the Core Compliance track. Stepback algorithms reference only the Core Compliance track; (3) **Political level** — both capitals issue pre-cleared talking points confirming that the Lebanon file is managed through separate mechanisms. No US or Iranian official may publicly link Hormuz transit to Lebanon de-escalation without triggering narrative violation penalties.

E.2.2 Recalibrated Lebanon Dual-Key Escalation Lock (v1.0 Reform)

APPENDIX E.2.2 — RECALIBRATED LEBANON DUAL-KEY ESCALATION LOCK (V1.0 OPERATIVE REVISION)

The v4.0 dual-key mechanism (10-casualty threshold + JIG majority vote within 72 hours) is hereby replaced with the following recalibrated system:

(a) Three-Tier Escalation Ladder (Replacing Single Threshold): The v1.0 single 50-casualty threshold is replaced with a graduated escalation ladder that provides proportionate response at each level:

- **Tier 1 (10–24 civilian casualties):** Automatic 72-hour diplomatic consultation window activates between Washington and Tehran through the Omani backchannel. **No core Framework linkage.** The incident is managed exclusively through the Levant Regional Basket. Israel retains inherent self-defense rights under UN Charter Article 51.
- **Tier 2 (25–49 civilian casualties):** Diplomatic consultation + JIG advisory opinion requested. The JIG may issue a non-binding advisory assessment on whether indicators of direct state direction exist. Core Framework linkage only by **mutual written agreement** of both Parties. No linkage occurs without mutual consent.
- **Tier 3 (50+ civilian casualties):** Presumption of potential linkage, but the mandatory 72-hour diplomatic consultation window still applies, and **either party may veto** core Framework linkage. If both parties agree the attack was state-

directed, Tier 2 step-back (proportionate benefit reduction per Article 13.3) may apply by mutual consent. No automatic step-back occurs at any tier.

(b) Proportionality and Intent Assessment (All Tiers): At all tiers, the JIG may assess whether the attack demonstrated indicators of direct state direction, including: communications intercepts linking the attack to Iranian Revolutionary Guard Corps command; pre-attack logistical patterns consistent with state-sponsored operations; weapon systems used that are exclusively or predominantly supplied by Iranian state channels; and post-attack statements or communications from Hezbollah leadership acknowledging Iranian direction. Such assessment is advisory only and does not bind either Party's diplomatic judgment.

(c) Mandatory 72-Hour Diplomatic Consult Window: Upon crossing any tier threshold, a **mandatory 72-hour diplomatic consultation window** shall activate between Washington and Tehran through the Omani backchannel **before** any regional incident can be legally tied to the core Framework. During this window:

- The US Secretary of State and Iranian Foreign Minister shall communicate directly.
- Both capitals shall have space to investigate whether the attack was directed by Tehran or represents rogue proxy action.
- Neither side may activate Tier step-back, make public attribution findings, or take military measures linked to the core Framework during the 72-hour window.
- Israel retains its inherent right to self-defense under UN Charter Article 51 during this window — but such self-defense is managed through the Lebanon track, not the core Framework.

(d) Removed: JIG Fast-Track Majority Vote: The v4.0 provision allowing the JIG to determine by majority vote that an attack was conducted pursuant to a direct, verifiable order from the Supreme Leader — thereby triggering Tier 2 step-back — is **hereby deleted**. The JIG no longer has fast-track attribution authority for Lebanon incidents.

(e) Post-Consultation Procedure: After the 72-hour diplomatic consult window, if both the US and Iran mutually agree that the attack was directed by Iranian state leadership, Tier 2 step-back (proportionate benefit reduction per Article 13.3) may apply to the Core Framework by mutual consent. If either party disputes the attribution finding, the matter is referred to the Guarantor Matrix for a 14-day mediation. At no point does a Lebanon incident automatically trigger core Framework penalties.

(f) Absent Mutual Determination — Israeli Consultative Override: Absent mutual written agreement that the attack was state-directed, the Core Framework continues

> unaffected, and the attack is managed exclusively through the Levant Regional Basket. **However, if the State of Israel certifies through the Tel Aviv Channel that the attack caused 50+ civilian casualties and presents prima facie evidence of Iranian state direction, the following procedure shall apply: (i) The JIG shall conduct a 24-hour admissibility review of Israel's prima facie evidence. The JIG shall admit the evidence for Guarantor Matrix consideration only if: the evidence is not manifestly unreliable or fabricated; the evidence, if accepted as true, would support a reasonable inference of Iranian state direction; and the evidence is not based solely on open-source media or single-source intelligence without corroboration. The JIG's admissibility determination is procedural only — it does not evaluate the truth of the evidence. If the JIG finds the evidence inadmissible, Israel may resubmit with additional corroboration within 7 days. (ii) If the evidence is admitted, the Guarantor Matrix shall convene an emergency session within 48 hours. A two-thirds majority of the Guarantor Matrix may refer the matter for Tier 2 step-back consideration, overriding a Party veto only if the veto is deemed manifestly unreasonable by the Swiss facilitator, or if the JIG certifies by two-thirds majority that the Party invoking the veto has failed to engage in good-faith consultation during the 72-hour window.** This is not a true veto — it is a supermajority override with a high bar. Israel gains a voice without a veto. Iran retains protection against frivolous linkage.

Rationale: The original 10-casualty threshold was a hair-trigger. A single rogue rocket, a localized skirmish, or a false-flag operation could cross this limit, pierce the firewall, and automatically collapse the entire nuclear and maritime deal. The three-tier escalation ladder creates graduated response levels: Tier 1 (10-24) triggers consultation only with no linkage; Tier 2 (25-49) enables JIG advisory opinion with linkage only by mutual agreement; Tier 3 (50+) creates a presumption of linkage but either party may veto. The 72-hour diplomatic consult window gives both capitals space to de-escalate at every tier. Stripping the JIG of fast-track attribution ensures low-level officials cannot override diplomatic judgment.

E.3 Levant Regional Basket

Purpose: To provide a structured diplomatic forum for addressing the full range of Levant security issues — including but not limited to Lebanon, Syria, and the Israeli-Palestinian dimension — without encumbering the core US-Iran Framework.

E.3.1 Structure and Participants

The Levant Regional Basket is a separate diplomatic track convened under UN auspices with the following permanent participants: United States, Islamic Republic of Iran, State of Israel (as affected party), Lebanese Republic, Syrian Arab Republic (as proxy theater), Russian Federation (as guarantor with regional presence), Sultanate of Oman (as neutral facilitator), and UN Special Coordinator for Lebanon.

E.3.2 Scope of Negotiations

The Regional Basket may address: (a) long-term security arrangements along the Israel-Lebanon Blue Line; (b) demilitarization of specified zones in southern Lebanon; (c) prisoner exchanges and humanitarian corridors; (d) reconstruction funding for southern Lebanon and Bekaa Valley; (e) political reform discussions related to Lebanese sovereignty and militia integration. The Regional Basket shall NOT address: (i) nuclear matters reserved to the core Framework; (ii) Hormuz transit governance reserved to the JHTA; (iii) sanctions relief mechanisms; (iv) any matter requiring modification of the core US-Iran Framework.

E.3.3 Timeline and Milestones

Table 25: Levant Regional Basket Timeline

Milestone	Target	Deliverable
Basket Launch	Day 30	Terms of Reference agreed; participants confirmed; facilitator appointed
First Session	Day 60	Agenda finalized; confidence-building measures identified
Interim Report	Day 120	UN Special Coordinator publishes interim assessment
Day 180 Review	Day 180	Core Framework guarantor states review Regional Basket progress; recommend continuation or modification
Successor Framework	Year 2	If core Framework succeeds, Regional Basket may transition to permanent regional security architecture

E.3.4 Relationship to the Core Framework

The Levant Regional Basket is explicitly subordinate to the core US-Iran Framework. The Basket's existence, suspension, or termination shall not affect the core Framework's operation. Funding for the Basket comes from a separate allocation (5% of Regional Stability

Reconstruction Credits) and does not compete with core Framework economic commitments. If the core Framework enters full suspension, the Regional Basket may continue independently if at least three participant states agree.

E.3.5 Israeli Participation Framework

Israel participates in the Regional Basket as an affected state with full speaking rights but no veto over core Framework matters. Israel's participation is coordinated through the Tel Aviv Channel (Article 16) and does not create any obligation for Iran to negotiate directly with Israeli representatives. Bilateral matters between Israel and any participant may be addressed in separate side meetings facilitated by Oman.

E.4 Narrative Frames

US Narrative Frame: "The core agreement addresses nuclear proliferation and freedom of navigation. Lebanon is managed through a separate diplomatic track with full Israeli participation. The firewall protects the core deal from proxy escalation, while the three-tier escalation ladder ensures proportional response — from consultation-only at Tier 1, to advisory opinion at Tier 2, to presumption of linkage at Tier 3. The 72-hour diplomatic window protects us from automatic entanglement."

Iranian Narrative Frame: "The framework ends the war on all fronts. Through the Levant Regional Basket, Iran ensures our allies in Lebanon are not abandoned. The Dossier Isolation clause protects the core agreement from Israeli military actions in Lebanon. The three-tier escalation ladder ensures only the most catastrophic attacks (50+ casualties) can even create a presumption of linkage — and either party may veto. Lower-tier incidents trigger consultation only, with no core Framework impact."

Israeli Narrative Frame: "We participate fully in the Regional Basket. The firewall protects the US-Iran deal from routine Hezbollah provocations, but the Tier 2 threshold at 25 casualties creates a meaningful pathway to address serious attacks through JIG advisory opinion and mutual agreement. At Tier 3, a direct mass-casualty attack triggers a mandatory conversation at the highest levels. Our right to self-defense under Article 51 remains absolute at all tiers. Through the LIRB, our intelligence is heard by the US before any major decision. Through the Pre-Decision Consultative Window, we have a binding 24-hour look before any US veto. These are transparency rights, not vetoes — and they protect us from fait accompli."

E.5 Levant Intelligence Review Board (LIRB) (v1.1 Fix)

APPENDIX E.5 — LEVANT INTELLIGENCE REVIEW BOARD (OPERATIVE AND

BINDING)

(a) Establishment: A technical intelligence review body, the **Levant Intelligence Review Board (LIRB)**, is established under Omani facilitation. **Permanent members:** United States, State of Israel, Sultanate of Oman. **Rotating members:** United Kingdom, France (six-month terms).

(b) Mandate: The LIRB shall review intelligence related to Hezbollah, IRGC-Quds Force, and other proxy activity in Lebanon, Syria, and the region. Its **sole function** is to provide technical attribution assessments to the United States for use in JIG proceedings and Executive Consultation Window deliberations.

(c) No Iranian Participation: Iran is not a member of the LIRB, does not receive LIRB materials, and is not bound by any LIRB finding. LIRB assessments are advisory to the United States only. The LIRB meets at a secure facility in Muscat, separate from all JIG and Framework proceedings.

(d) Attribution Standard: The LIRB shall issue assessments using a **"reasonable suspicion"** standard (not beyond reasonable doubt). **"Reasonable suspicion" means an articulable basis, grounded in specific intelligence or forensic evidence, to believe that an attack was directed or supported by Iranian state actors. Hunch or speculation does not satisfy this standard.** Assessments shall include confidence levels (Low / Medium / High) and underlying technical forensics where applicable. LIRB assessments are not legal findings and do not bind any party.

(e) Timeline for Lebanon Tier 2/3 Incidents: Within **12 hours** of a Lebanon incident crossing the Tier 2 (25+ casualties) or Tier 3 (50+ casualties) threshold, the LIRB shall convene an emergency session. A preliminary attribution finding shall be issued to the US Secretary of State within **24 hours**. The US shall share a **sanitized summary** of this finding with Iran during the 72-hour diplomatic consultation window, excluding intelligence sources and methods.

(e-1) Public Summary of LIRB Findings (v1.2 Operative Addition): Notwithstanding classification, the JIG shall publish to the Swiss Dashboard a **Public Summary** of any LIRB preliminary finding that is submitted to the JIG for consideration in a Tier 2 or Tier 3 proceeding. The Public Summary shall include: **(a)** the incident date, location, and nature; **(b)** the LIRB's attribution conclusion (with confidence level); **(c)** the factual basis for the conclusion, stated in general terms without disclosing intelligence sources or methods; and **(d)** any dissenting views among LIRB members. Iran may submit a **response Public Summary of equal length**, published simultaneously. The Public Summary does not waive classification of underlying intelligence. This ensures public transparency without compromising sources. Either Party may request redaction of specific sensitive details; disputes resolved by JIG Chair within 7 days.

(f) No Operational Authority: The LIRB has **no authority** to direct US, Israeli, or any military action. Its role is purely consultative and analytical. The LIRB may not issue orders, directives, or binding recommendations to any military or intelligence service.

(g) Classification and Non-Disclosure: LIRB proceedings are protected under the same classification and non-disclosure obligations as the SEAC (Shadow Escalation Attribution Cell). All LIRB materials, findings, and deliberations are classified at the level of the originating intelligence. **Unauthorized disclosure of LIRB findings by any participant constitutes a Tier 2 material disruption.** This protects the integrity of the intelligence channel and prevents catastrophic leaks.

(h) JIG Admissibility: Notwithstanding the classification and non-disclosure provisions in (g), LIRB findings may be **submitted to the JIG in classified form** for use in JIG proceedings. The JIG shall establish a secure, classified sub-channel for reviewing LIRB findings. **No LIRB finding is binding on the JIG**, but the JIG may consider LIRB findings as evidence alongside other attribution sources. **Iran shall have the right to respond to sanitized summaries** of LIRB findings that the JIG determines are material to its deliberations. This ensures the LIRB informs, but does not control, JIG decision-making.

Why It Works: For Israel: A permanent, privileged table where their intelligence is heard by the US before any major decision. No Iranian veto. No Iranian visibility into their sources. **For Iran:** Iran never sits with Israel. The LIRB's findings are advisory only. The US can ignore them. The sanitized summary protects Iranian dignity. **For the US:** The best of both worlds — Israeli intelligence and Iranian cooperation — without forcing a political collision.

Appendix G: Unified Definitions Annex (v1.2 Operative Addition)

Purpose. This Annex provides authoritative definitions for all defined terms used in the Framework. In the event of any conflict between a definition in this Annex and a definition elsewhere in the document, this Annex governs. Definitions are organized by functional category for ease of reference.

G.1 Compliance and Breach Definitions

"Material Breach" means a violation of a Framework obligation that: (i) substantially defeats the object and purpose of the violated provision; (ii) persists for 7 or more days after notice; and (iii) is not cured within the applicable remediation window. Material Breach is a prerequisite for Tier 2 or Tier 3 step-back. Technical errors, administrative delays, and de minimis exceedances (Article 13.2) do not constitute Material Breach unless they form a Pattern of Gaming (Article 13.6).

"Technical Anomaly" means a deviation from expected telemetry, inspection, or reporting parameters that: (i) is detected by automated systems or human inspectors; (ii) does not immediately indicate deliberate non-compliance; and (iii) requires investigation to determine causation. Technical Anomalies may result from equipment malfunction, environmental interference, data transmission errors, or genuine compliance deviations. All Technical Anomalies are logged and investigated but do not constitute violations until investigation is complete.

"Verification Failure" means an incident where: (i) the IAEA is unable to verify compliance with a specific obligation using the agreed verification methods; (ii) the inability is not due to IAEA equipment malfunction or inspector error; and (iii) the cause is unknown or disputed after initial investigation. Verification Failure triggers the escalating investigation timeline under Article 3.3 but does not automatically constitute a violation.

"Compliance" means the state in which a Party is meeting all its Framework obligations as verified by the IAEA and JIG, subject to de minimis thresholds (Article 13.2) and the rolling compliance window (Article 7.2). Compliance is assessed module-by-module, not as an aggregate score. A Party may be in compliance in one module while under review in another.

"Pattern of Non-Compliance" means three or more instances of the same or similar violations within any 180-day period, whether or not each individual instance would independently trigger step-back. A Pattern of Non-Compliance triggers enhanced monitoring, burden shift (Article 13.6(d)), and potential de minimis suspension regardless of the severity of individual instances.

G.2 Actor and Attribution Definitions

"Spoiler Actor" means any state, non-state armed group, criminal organization, or individual that takes action intended to: (i) disrupt Framework operations; (ii) provoke military escalation between the Parties; (iii) undermine diplomatic confidence; or (iv) exploit Framework mechanisms for non-Framework objectives. Spoiler Actors may be

Rogue (no state direction), State-Tolerated (state knowledge without direction), or State-Directed (state command authority). See Article 19.

"Attribution" means the process by which the SEAC, JIG, or LIRB assigns responsibility for an action to a specific actor. Attribution requires: (i) technical forensic evidence; (ii) correlation with intelligence assessments (where available); and (iii) exclusion of alternative explanations to a reasonable degree of confidence. Attribution standards vary by context: 90%+ confidence for cyber attribution (Article 19-D); majority JIG vote for most physical incidents; unanimous JIG vote (excluding affected Party) for cross-module spillover.

"False Flag Attack" means an attack conducted by one actor but designed to appear as if conducted by another actor, for the purpose of: (i) triggering Framework penalties against the framed actor; (ii) provoking military retaliation by the framed actor or its allies; or (iii) undermining diplomatic confidence in the Framework. See Article 19-A.

"Rogue Commander" means a military officer or non-state armed group leader who orders or conducts an attack or provocation without authorization from the state's political leadership or military chain of command. The burden of proving rogue status lies on the state from whose territory or forces the attack originated. See Article 14.3.

G.3 Conduct and Intent Definitions

"Good Faith" means conduct characterized by: (i) honest intent to comply with Framework obligations; (ii) timely and transparent communication regarding compliance status and difficulties; (iii) active engagement with dispute resolution mechanisms; (iv) refraining from exploiting procedural mechanisms to delay obligations; and (v) taking reasonable steps to cure deviations when they occur. Good faith is assessed by the JIG based on observable conduct, not subjective intent. A Party's public statements do not alone determine good faith — actions during the Framework process govern.

"Bad Faith" means conduct that: (i) deliberately exploits Framework mechanisms to avoid obligations; (ii) systematically triggers dispute procedures without genuine dispute; (iii) provides false or misleading information to the JIG or IAEA; (iv) uses Framework access for non-Framework intelligence collection; or (v) engages in any of the five prohibited gaming actions defined in Article 13.6. Bad Faith is determined by the JIG by majority vote with the affected Party's members excluded.

"Hostile Act" means: (i) any armed attack on Framework personnel, facilities, or verification equipment; (ii) any cyber attack on Critical Cyber Infrastructure at Level 3 or above (Article 19-E); (iii) any deliberate sabotage of Framework escrow, communications, or diplomatic channels; or (iv) any confirmed false flag attack (Article 19-A). Hostile Acts trigger the Tier classification system and may activate Temporary Operational Suspension (Article 20) depending on severity and scope.

G.4 Material and Custody Definitions

"Custody" means the state in which fissile material is under the physical control, monitoring, and verification of the IAEA and Framework mechanisms. Custody includes: IAEA seals and tags on declared material; continuous monitoring via OLEM and tamper-evident technology; and documented chain-of-control for any material movement. Custody does not imply transfer of sovereign ownership — Iran retains title to all Iranian-origin fissile material regardless of custody arrangement.

"High-Risk Nuclear Material" means uranium enriched to 20% U-235 or above; separated plutonium; and any other fissile material capable of sustaining a nuclear chain reaction. For Framework purposes, all declared material at or above 20% enrichment is classified as High-Risk regardless of quantity or form.

"Custody Transition" means any movement of High-Risk Nuclear Material from one custody arrangement to another, including: downblending in Iran (Pathway A); transfer to Oman Escrow (Pathway B); conversion to research reactor fuel (Pathway C); or any other disposition pathway under Articles 1.5-1.6. All custody transitions require 14-day advance IAEA notification, JIG pre-authorization, and real-time monitoring during transit.

G.5 Escrow and Financial Definitions

"Escrow" means funds deposited with a neutral third-party financial institution (the "Escrow Agent") under terms that restrict release until specified conditions are met. Framework escrow includes: the Mutual Escrow Deposit (Article 7.1); the Joint Disposition Fund (Article 1.5-A); the Survival Core Escrow (Article 11.1-A); and any supplementary escrow arrangements created under Framework authority.

"Escrow Failure" means any event described in Article 24(a) that prevents the Escrow Agent from performing its custody and disbursement functions. Escrow Failure does not mean a delay of less than 48 hours, a routine system maintenance window, or a minor technical error that does not affect fund security.

"Clean-Record Bank" means a financial institution that meets all criteria in Article 9.2(b-1): no current OFAC designation; no Tier 2 or Tier 3 violation in the preceding 180 days; and FATF-compliant AML/CFT controls verified by independent audit. The JIG maintains a published list of Clean-Record Banks, updated monthly.

G.6 Operational Status Definitions

"Temporary Operational Suspension (TOS)" means the legal status defined in Article 20: a temporary pause of one or more Framework modules without termination, penalty, or compliance clock reset. TOS is distinct from: full Framework termination (Article 17); module-specific step-back (Article 13.3); and Force Majeure (Article 22, which activates TOS automatically).

"Force Majeure" means an event described in Article 22(a) that directly and materially prevents Framework performance. Force Majeure automatically triggers TOS but is distinct from other TOS grounds (domestic crisis, leadership transition, cyber incident, escrow failure) in that it originates outside both Parties' control.

"Rolling Horizon Non-Renewal" means the state in which the Framework's automatic 365-day renewal (Article 17) does not occur because one or both Parties decline to renew. Non-renewal triggers an orderly 180-day wind-down period during which: all escrowed funds are disbursed per existing schedules; minimum survival conditions continue; and no new obligations are imposed. Non-renewal is not termination — it is a managed sunset.

G.7 Interpretive Principles

(a) Definitional Primacy: This Annex governs all definitional disputes. If a term is defined both here and in an operative Article, the Annex definition prevails unless the Article explicitly states otherwise.

(b) Supplemental Definitions: The JIG may add supplemental definitions by majority vote to address emerging issues not covered by existing definitions. Supplemental definitions are published on the Swiss Dashboard and become operative 30 days after publication.

(c) Ordinary Meaning: Terms not defined in this Annex or in operative Articles shall be interpreted in accordance with their ordinary meaning in the context of international nuclear agreements, unless the context clearly requires a different meaning.

(d) Language: This Annex is authentic in English. Translations into Farsi and Arabic are provided for convenience. In case of any discrepancy, the English text governs.

Appendix H: Presidential Executive Summary — 3 Pages (v1.2 Operative Addition)

For: Heads of State, National Security Advisors, Congressional/Parliamentary Leadership
Classification: UNCLASSIFIED — Distribution Unlimited

Purpose: Provide decision-makers with the essential political risk/benefit calculus without requiring mastery of technical details.

Page 1: What This Framework Does

The Provisional Steps Framework is a 180-day, performance-based stabilization agreement between the United States and Iran. It is not a permanent peace treaty, a nuclear weapons agreement, or a normalization of relations. It is a technical mechanism designed to achieve four specific objectives:

1. Secure Iran's High-Risk Nuclear Material. All uranium enriched to 20% or above is placed under IAEA custody with continuous monitoring. Material stays on Iranian soil unless both sides agree otherwise. Iran retains ownership. Twenty-three disposition pathways are available for Phase 2 negotiation.

2. Restore Freedom of Navigation. The US naval blockade lifts automatically on Day 7 upon IAEA verification. Iran's maritime authority (PGSA) is integrated into a cooperative transit system. Tolls fund regional maritime maintenance, not Iranian state coffers.

3. Create Economic Incentives for Compliance. A $500M mutual escrow (plus US dollar-for-dollar matching up to $5B annually) provides tangible economic benefits tied to verified compliance, not political promises.

4. Prevent Accidental War. Direct military-to-military hotlines, deconfliction protocols, and a binding No Automatic Military Escalation Clause ensure that compliance disputes become diplomatic conversations, not casus belli.

The Core Principle: Neither side moves alone. Every Iranian commitment triggers a simultaneous US commitment. Both leaders can defend the agreement as requiring the other side to move first.

Page 2: Political Risks and Protections

Table X5: Political Risk Matrix for Decision-Makers

Risk	Mitigation	Your Protection
"We're conceding too much"	Domestic Framing Autonomy (Preamble)	You describe the deal however you want. No required terminology.
"Iran will cheat and we'll look weak"	Air-Gapped Telemetry + IAEA Custody + Executive Consultation Veto	You have 48 hours to review any anomaly before any penalty applies. You can veto any penalty.

"Israel will attack and drag us in"	4-Layer Israeli Transparency Architecture + Standstill Annex	Israel gets 24-hour pre-decision notice, privileged intelligence access, and a guaranteed US partner if Iran diverts fissile material. In exchange, Israel holds fire for 180 days.
"Hardliners will kill the deal at home"	No Regime Change Objective + Executive Consultation Governance	Explicit binding language: this is stabilization, not strangulation. No intelligence collection beyond verification scope.
"A crisis will force total collapse"	Temporary Operational Suspension (Article 20) + Partial Success Doctrine	You can pause without collapsing. One module failing doesn't kill the rest.
"We're locked in forever"	Rolling Horizon Architecture + Expedited Withdrawal	180-day initial term, renewable. You can leave with 30 days' notice and a capped exit payment.
"Spoilers will destroy everything"	8 spoiler resilience articles + false flag protocol + rogue commander accountability	Every known spoiler tactic has a countermeasure. False flags face double penalties.
"A leadership change kills the deal"	Institutional Continuity Protocol (Article 21)	The Framework binds institutions, not individuals. Survives elections, deaths, and transitions.

Page 3: The Bottom Line

If this Framework succeeds: Iran's nuclear material is secured. The Strait is open. Economic benefits flow to both sides. The region has 180 days of stability to negotiate Phase 2. You are the leader who prevented war while preserving all options.

If this Framework fails: You retain full military options. The No Automatic Military Escalation Clause means non-compliance triggers economic and diplomatic penalties, not automatic war. You can withdraw cleanly with the Expedited Withdrawal Option. The status quo ante is preserved — nothing is given away irreversibly.

If you don't sign: The ceasefire frays. Nuclear material remains unsecured. The next crisis escalates faster because there are no hotlines, no deconfliction protocols, and no mutual understanding. The military option remains on the table — but so does the risk that Iran achieves breakout capability before you act.

The Choice: This Framework is not a leap of faith. It is a 180-day trial with multiple exit ramps, comprehensive protections, and binding guarantees that neither side can be forced into politically impossible positions. It does not require trust — it replaces trust with verification, consultation, and institutional resilience.

"Trust is good. Verification is better. But a Framework that survives without either — that is architecture."

Appendix I: Negotiators' Brief — 10 Pages (v1.2 Operative Addition)

For: Framework negotiators, JIG members, diplomatic mission staff
Classification: RESTRICTED — Framework Technical Annex
Purpose: Provide negotiators with operative fallback positions, red-line guidance, and tactical advice for each Framework module.

I.1 Nuclear Module (Articles 1-3) — Negotiators' Guidance

Your Objective: Secure all 20%+ enriched uranium under verifiable custody without triggering Iranian sovereignty concerns.

Fallback Position on Custody: If Iran rejects Oman Escrow (Pathway B), default to Pathway A (in-Iran downblending). Material never leaves Iran unless Iran agrees. This is your ace — Iran keeps material on its soil while the world gets verification.

Red Line: Do not agree to any provision that transfers title or ownership of Iranian fissile material to a third party. Custody and ownership are separate. The Framework explicitly preserves Iranian ownership.

Tactical Note: Use the Twenty-Three-Pathway Menu as a negotiating tool. Offer flexibility on pathways in exchange for Iranian flexibility on verification intensity. The more pathways Iran engages with, the less intrusive individual verification needs to be.

Day 365 Backstop: If no pathway is selected by Day 365, phased transfer to Oman activates automatically. But conditional acceleration (90 days of non-engagement) means Iran can avoid acceleration simply by showing up and talking. Good-faith negotiation is defined with four bright-line criteria — know them and reference them.

I.2 Maritime Module (Articles 4-6) — Negotiators' Guidance

Your Objective: Integrate the PGSA into a cooperative transit system that preserves freedom of navigation while acknowledging Iranian maritime sovereignty claims.

Fallback Position on PGSA: If GCC states challenge PGSA legitimacy, invoke Article 4.2-C (GCC Side Letter + IMO Mediation). The PGSA is recognized as an Iranian institution, but its transit authority operates within a multilateral framework. This is "integration, not endorsement."

Red Line: Do not allow Iran to unilaterally close the Strait to any vessel category. The Framework guarantees freedom of navigation for all compliant vessels. Iranian closure authority is limited to safety and environmental regulations, not political or military purposes.

Tactical Note: The IRGC-Navy Hotline (Article 4.7) is your most important operational tool. Ensure daily testing at 06:00 UTC without fail. A failed hotline test is a Tier 1 dispute — escalate immediately. The hotline prevents incidents; preventing incidents prevents escalatory spirals.

Airspace Stand-Down: Know the three zones by heart. Zone Alpha (UAVs to 25nm, manned above 15,000ft); Zone Beta (no armed drones within 12nm, passive radar); Zone Gamma (30-minute advance notification within 100nm). Any violation is a Category A Bright-Line Period violation with a four-step binding protocol.

I.3 Economic Module (Articles 7-10) — Negotiators' Guidance

Your Objective: Create credible, verifiable economic benefits that are directly tied to compliance milestones, not political promises.

Fallback Position on Escrow: If $500M Day 1 deposit is politically difficult, remember that it's mutual — the US puts in $250M, Iran puts in $250M. Both sides have skin in the game. The escrow interest accrual (Article 8.1-A) at LIBOR or inflation rate protects real value.

Red Line: Do not front-load sanctions relief beyond the graduated tranche schedule. Each tranche requires verified compliance. The Rolling Compliance Window (Article 7.2) prevents gaming but also prevents minor errors from resetting everything.

Tactical Note: The DFC Matching (Article 10) is your most powerful economic tool. Up to $5B annually in dollar-for-dollar matching for civilian infrastructure. Frame this as "Iranian money working for Iranian people" — because it is. DFC matches what Iran spends on its own civilian reconstruction.

Bank Isolation: Targeted Bank Isolation (Article 9) applies to flagged institutions only — not sectors, not affiliates, not subsidiaries. This is narrow, surgical, and reversible. Emphasize the reversibility: clean banks graduate automatically.

I.4 Durability Module (Articles 11-19) — Negotiators' Guidance

Your Objective: Ensure the Framework survives the inevitable crises, spoilers, and political shocks that will occur during the 180-day term.

Fallback Position on Consultation: The 48-hour Executive Consultation Window (Article 13.4) is your pressure valve. Either side can veto any penalty. Use it. But remember: if you veto every penalty, you erode the Framework's credibility. Veto only when politically necessary, not as a default.

Red Line: Do not allow any Party to invoke "operational discretion" to bypass the automated Day-7 blockade lift (Article 6). The lift is unconditional. Once IAEA certifies, the countdown runs. This is the most important automated trigger in the Framework — protect it.

Tactical Note on Spoilers: Know the spoiler management hierarchy: Article 19 (general management); Article 19-A (false flags — double penalty for confirmed false flag); Article 14.3 (rogue commanders — 24-hour condemnation safe harbor); Article 19-C (information warfare — 0.5 WBI points per incident). Each has a specific response protocol.

Temporary Operational Suspension: If your side needs breathing room, TOS (Article 20) is available. But use it sparingly — three TOS invocations in 180 days and the Guarantor Matrix will question whether your side is negotiating in good faith. TOS is a pressure valve, not a pause button for convenience.

I.5 Israeli Dimension — Negotiators' Guidance

Your Objective: Manage Israeli concerns without granting Israel a veto over the Framework.

The Four Layers: (1) Pre-Decision Window — 24-hour binding notification before any US veto on Lebanon incidents; (2) LIRB — privileged intelligence table, no Iranian participation, advisory only; (3) Public Caveat — Israel submits 500-word factual correction to annual report, Iran can respond equally; (4) Standstill Annex — separate US-Israel diplomatic note, Iran never sees it.

Red Line: Do not discuss the Standstill Annex with Iranian negotiators. It is not part of the Framework. If asked about US-Israel coordination, reference Article 16 (Bilateral Coordination) and the Pre-Decision Window. Nothing more.

Tactical Note: If Israel conducts an unauthorized strike during Phase 1, the Framework does not automatically collapse. Activate TOS under Article 22 (Force Majeure), convene the Guarantor Matrix, and use the 72-hour Standstill commitment to prevent further escalation. The Framework is designed to survive Israeli action — it is not designed to prevent it.

I.6 Crisis Management — Negotiators' Guidance

First 24 Hours of Any Crisis:

1. Activate the IRGC-Navy Hotline (Article 4.7) — 15-minute incident response level;

2. Notify the Swiss facilitator — they convene the JIG within 48 hours;
3. Check if the crisis qualifies for Temporary Operational Suspension (Article 20);
4. Invoke the Media Blackout (Article 13.5) — 72 hours, extendable once;
5. Do not issue public statements attributing responsibility until the Cooling-Off Period expires (Article 19-F).

First 72 Hours:

1. JIG convenes — preliminary assessment and evidence review;
2. SEAC begins technical investigation (if applicable);
3. Executive Consultation Window activates if Tier 2+ penalty contemplated;
4. Guarantor Matrix notified of any Tier 3 determination;
5. Monthly Bulletin (Article 12.3-B) issued with factual summary only.

Week 2+:

1. JIG issues preliminary findings;
2. Either Party may appeal to Guarantor Matrix;
3. Executive Consultation exhausted — penalties activate or are vetoed;
4. Remediation timeline begins if penalties applied;
5. Red Team review if spoiler involvement suspected.

I.7 Key Numbers Every Negotiator Must Memorize

Table X6: Critical Operational Numbers

Parameter	Value	Article
Day 1 escrow deposit	$500M mutual ($250M each)	7.1
Executive Consultation Window	48 hours (72 during elections)	13.4
Media Blackout	72 hours, one extension	13.5
Cooling-Off (sanctions snapback)	24 hours	19-F
Cooling-Off (public attribution)	48 hours	19-F
Temp. Operational Suspension max	90 days per invocation, 180/year	20
Leadership replacement deadline	7-30 days (varies by role)	21
Early engagement credit	75% penalty reduction for 14 days	13.4(f)

De minimis enrichment tolerance	3.67% +/- 0.5%	13.2
Hotline daily test	06:00 UTC	4.7
Escrow interest rate	Greater of LIBOR or avg inflation	8.1-A
Survival Core escrow	$50M ($25M per Party)	11.1-A
Israeli caveat word limit	500 words	12.3-A(f)
Day 365 backstop trigger	20% transfer every 90 days	1.6(e)
Gaming pattern threshold	3+ instances in 180 days	13.6

Appendix J: Crisis Decision Flowchart — 1 Page (v1.2 Operative Addition)

For: Crisis managers, hotline operators, JIG duty officers, national security staff
Classification: UNCLASSIFIED — Operational Reference
Purpose: A single-page decision tree for the first 72 hours of any Framework crisis.

CRISIS DECISION FLOWCHART — FIRST 72 HOURS

Apply to any event that could affect Framework compliance, operations, or political viability

Table X7: Hour-by-Hour Crisis Response Protocol

Time	Action	Responsible	If No Response
H+0	Event detected. Log time, nature, source. Activate IRGC-Navy Hotline (routine level).	First responder / duty officer	Escalate to H+2 protocol automatically
H+0 to H+2	Assess: Is this a Tier 1, 2, or 3 event? Is it nuclear, maritime, cyber, or regional? Does it qualify for TOS (Article 20)?	Duty officer + JIG on-call	Assume Tier 2, escalate to Swiss facilitator
H+2	Swiss facilitator notified. Media Blackout (Art. 13.5) activated if Tier 2+ or attribution involved. Cooling-Off Period begins (Art. 19-F).	Swiss facilitator	Auto-escalate to H+6 protocol
H+2 to H+6	IRGC-Navy Hotline at incident level (15-min response) if maritime. Omani backchannel activated. Both Parties' capitals notified.	Joint Secretariat	Guarantor Matrix emergency notification

H+6	JIG emergency consultation convened by video. Preliminary assessment issued. TOS request evaluated if applicable.	JIG Chair	Bilateral deputy-level consultation via Omani channel
H+6 to H+24	Evidence collection phase. SEAC begins technical investigation (if applicable). Both Parties submit preliminary positions. Cooling-Off Period continues.	SEAC + JIG investigators	Investigation proceeds ex parte with available evidence
H+24	Snapback Cooling-Off expires. Attribution Cooling-Off continues (48 hours total). JIG preliminary findings due for Tier 2+ events.	JIG Chair	Findings issued with "incomplete evidence" caveat
H+24 to H+48	Executive Consultation Window (Art. 13.4) activated if Tier 2+ penalty contemplated. Both ministers (or deputies) engage via Omani channel.	Secretary of State + Foreign Minister	Automatic 24-hour stay (Art. 13.4(g)). Pattern = Tier 2.
H+48	Attribution Cooling-Off expires. Public statements permitted (subject to Media Blackout if extended). Executive Consultation Window closes.	Swiss facilitator	Penalty activates automatically unless vetoed
H+48 to H+72	JIG final preliminary findings issued. Either Party may appeal to Guarantor Matrix. Monthly Bulletin published with factual summary.	JIG	Guarantor Matrix auto-convenes within 7 days
H+72	Crisis phase formally ends. Remediation phase begins (if penalties applied). Long-term investigation continues. JIG status updated on Dashboard.	JIG Chair	Status remains "Under Review" until resolved

KEY DECISION POINTS:

➤ **Does event qualify for TOS?** → Check Article 20(b) grounds. If yes → activate 24-hour Cooling-Off → JIG convenes within 48 hours → TOS activates if approved.

➤ **Is this nuclear-related?** → IAEA notified within 4 hours. Unscheduled batch inspection available (Art. 3.3(d)). If fissile diversion suspected → Tier 3 emergency, Guarantor Matrix within 24 hours.

➤ **Is this maritime?** → IRGC-Navy Hotline at appropriate level. JHTA emergency session. Category A Bright-Line Protocol (Art. 4.7) if stand-down violation.

➤ **Is this cyber?** → 6-hour reporting requirement (Art. 19-E). Level 3+ triggers SEAC Cyber Attribution Cell. Level 4 triggers provisional attribution within 72 hours.

➤ **Is this a spoiler?** → Activate Article 19 protocols. False flag suspected → Article 19-A. Rogue commander → Article 14.3. Information warfare → Article 19-C.

➤ **Is this regional spillover?** → Check Article 22 Force Majeure events. If qualified → automatic TOS, no compliance penalty, Survival Core continues.

EMERGENCY CONTACTS (designated per Article 28 — no placeholders permitted):
Swiss Facilitator (24h): ______________________ | JIG Duty Officer (24h): ______________________
Omani Channel (secure): ______________________ | IRGC-Navy Hotline: ______________________ (daily 06:00 UTC)
Dashboard: ______________________ | Last verified: ___________

Nuclear Operative Annex (NOA): Executable Implementation Details for All Twenty-Three Disposition Pathways

Annex Type: Operative Technical Annex
Classification: UNCLASSIFIED — binds all Parties upon Framework signature
Scope: This Annex provides the detailed executable procedures, timelines, verification protocols, safety requirements, financial mechanisms, and contingency arrangements for each of the twenty-three HEU disposition pathways defined in Article 1.5. It transforms the political framework of Article 1.5 into operationally actionable instructions. **In the event of any conflict between this Annex and Article 1.5, this Annex governs on technical matters; Article 1.5 governs on political matters.**

NOA.1 General Provisions Applicable to All Pathways

(a) Sovereign Title Preservation: Under all twenty-three pathways, the Central Bank of Iran retains full sovereign legal title to the HEU stockpile until the moment of final physical downblending or verified irreversible conversion. No pathway transfers title to any foreign entity, state, or international organization. Custody, possession, and control may transfer; title does not.

(b) IAEA Verification Supremacy: All twenty-three pathways are subject to IAEA verification at every stage. No material movement, conversion, or disposition may occur without IAEA pre-authorization, real-time monitoring, and post-action verification. The IAEA's verification role is mandatory and not delegable to any national authority.

(c) Synchronized Reciprocal Obligations: Every Iranian obligation under every pathway triggers a corresponding US obligation. No Iranian action proceeds without a synchronized US commitment. The synchronization schedule for each pathway is defined in the respective pathway section below. Either Party may delay its own obligation by up to 72 hours for technical reasons upon notification to the JIG; delays beyond 72 hours activate Article 13.4 (Executive Consultation Window).

(d) Environmental, Safety, and Security (ESS) Standards: All twenty-three pathway operations shall comply with: IAEA Safety Standards Series (NS-R-5, WS-R-3); ILO Convention 115 on radiation protection; and the Convention on the Physical Protection of Nuclear Material (CPPNM) as amended. Each pathway section includes specific ESS requirements.

(e) Joint Disposition Fund Applicability: The Fund drawdown schedule in Article 1.5-A applies to all pathways. Fund disbursement is triggered by IAEA verification of pathway commencement or completion milestones, not by political certification. Fund payments are non-clawable once disbursed for verified progress.

(f) Pathway Switching Protocol: Iran may switch from any pathway to any other pathway at any time prior to the point of no return (defined for each pathway below). Switching requires 14-day advance JIG notification and IAEA technical clearance. Drawn Fund amounts are credited to the new pathway with no clawback. The point of no return for each pathway is defined in its respective section.

(g) Force Majeure Applicability: Temporary Operational Suspension (Article 20) and Force Majeure (Article 22) apply to all pathway operations. During TOS, HEU remains under IAEA dual-lock custody in its then-current location. No material may be moved, processed, or converted during TOS without both Parties' written consent. Compliance clocks for pathway milestones are frozen during TOS.

(h) Escrow Interest During Pathway Execution: Escrowed funds continue to accrue interest at the Article 8.1-A rate during pathway execution. Interest is credited to the Iranian Civilian Reconstruction Fund and is not subject to pathway performance conditions.

NOA.2 Common Pre-Conditions for All Pathways

Before any pathway operations commence, the following pre-conditions must be satisfied:

Table NOA-1: Common Pre-Conditions for All Pathways

#	Pre-Condition	Verification	Responsible
1	IAEA completes physical inventory of entire HEU stockpile	Inventory report published to Dashboard; mass spectrometry confirms isotopic composition	IAEA
2	Dual-lock custody system activated and tested at all relevant facilities	JIG on-site verification; 72-hour continuous monitoring test passed	JIG + IAEA

3	OLEM confirms freeze on all enrichment above 3.67% (Article 2 baseline)	IAEA daily OLEM report; 7-day continuous compliance confirmed	IAEA
4	IRGC-Navy Hotline operational at routine level (daily testing confirmed)	Joint Secretariat log of 7 consecutive successful daily tests	JHTA
5	Swiss escrow sub-account for Joint Disposition Fund established and capitalized	Escrow confirmation certificate from Swiss facilitator; $500M balance verified	Swiss Facilitator
6	Technical Working Group (Article 1.5-A) convened and issued feasibility findings	Working Group report published to Dashboard; both Parties acknowledge receipt	JIG
7	Both Parties designate pathway-specific liaison officers with 24/7 availability	Liaison contact details registered with Joint Secretariat	Each Party

NOA Heading Hierarchy: This Annex uses a standardized heading structure for navigation: **NOA.x (h2)** for each of the twenty-three pathway sections; **NOA.1–NOA.3 and NOA.16 (h3)** for general provisions and cross-pathway rules; and **h4** for sub-sections within each pathway (Objective, Legal Basis, Timeline, etc.). All pathways follow an identical structure for ease of reference.

NOA.3 Pathway-Specific Implementation Framework

NOA.0.3 — PHASING OF PATHWAY IMPLEMENTATION (V1.2 OPERATIVE ADDITION)

(a) Phase I Executable Pathways: For purposes of Phase I (Days 1–180) of the Provisional Steps Framework, only the following three pathways are fully executable:

- **Pathway A (In-Iran Downblending):** The default pathway — downblending HEU to medical-grade LEU at Esfahan within Iranian territory under IAEA supervision. Executable within the 180-day Phase I timeline.
- **Pathway B (Oman Escrow Transfer):** Voluntary transfer of HEU to IAEA custody facility in Muscat, Oman. Transport executable within 14 days; storage continues beyond Phase I if needed.

- **Pathway C (IAEA LEU Bank Kazakhstan):** Downblending in Iran followed by transfer of LEU to the IAEA LEU Bank in Oskemen, Kazakhstan. Complete process executable within 75 days.

(b) Phase II/III Template Pathways: Pathways D through W are **templates for Phase II or Phase III successor framework negotiation**. They are included in this Annex as detailed technical reference documents to inform and accelerate future negotiations. This Framework creates **no presumption** that Iran must select or execute any Pathway D–W during Phase I. Neither Party may demand selection of a Pathway D–W as a condition of Phase I signature or implementation. The ten In-Iran Transformation Pathways (N–W) are designed as medium-term commercial enterprises (14–36 months) and are particularly suited for Phase II implementation once Phase I stabilization is verified.

(c) Technical Working Group Role: The Technical Working Group (Article 1.5-A) shall use the NOA specifications for Pathways D–W as **reference documents for successor negotiations**, not as binding commitments. The Working Group's Day 30 feasibility findings shall assess which, if any, Pathways D–W could realistically commence during Phase I, but such assessment is advisory only.

(d) Preferred Choice, Not Binding Commitment: All references in this NOA to "selection," "selected pathway," or "Iran selects a pathway" shall be interpreted as the selection of a preferred choice for Phase II negotiation, not as a binding commitment. Iran may change its preferred pathway at any time. The United States may veto Pathways D, F, K, and L. Veto does not constitute breach. This Framework creates no presumption that the preferred pathway will be the final pathway.

(e) Pathway Selection During Phase I: If Iran selects Pathway A, B, or C during Phase I, the corresponding NOA implementation section becomes operative and binding. If Iran elects to defer pathway selection to Phase II, Pathway A shall continue as the default (Article 1.6), and Pathways D–W shall remain available for future negotiation. No penalty attaches to deferring selection.

(f) Why This Phasing Matters: Without this clarification, the US may demand Iran commit to a complex multi-year pathway (e.g., K, L, or N–W) before signing Phase I. Iran will refuse, and the Framework collapses over Phase II ambition. This provision decouples Phase I signature from Phase II pathway commitment. Phase I secures the material; Phase II decides its fate. The NOA provides the technical detail for Phase II negotiators without burdening Phase I signature.

Each pathway section below follows a standardized structure: Objective; Legal Basis; Pre-Conditions; Implementation Timeline; Synchronized Reciprocal Obligations; Verification Protocol; ESS Requirements; Financial Mechanism; Contingencies; Point of No Return; and Narrative Framing Guidance.

NOA.4 Pathway A: In-Iran Downblending to Medical-Grade LEU at Esfahan

Objective

Convert Iran's HEU stockpile to medical-grade Low-Enriched Uranium (LEU, 19.75% U-235) at the Esfahan Nuclear Technology Center, entirely within Iranian territory, under IAEA supervision, for use in domestic medical isotope production. **This is the default pathway.** It activates automatically if no alternative pathway is selected by Day 75 or if the Parties agree to select it.

Legal Basis

Article 1.5 (Pathway A); Article 1.6 (In-Iran Default); Appendix G definitions of "Custody," "High-Risk Nuclear Material," and "Verification Failure."

Pre-Conditions (in addition to NOA.2)

Table NOA-A1: Pathway A Additional Pre-Conditions

#	Pre-Condition	Verification
A-1	Esfahan downblending facility confirmed operational or refurbishable within 60 days	IAEA technical inspection; engineering assessment published to Dashboard
A-2	Iranian regulatory authority (AEOI) certifies readiness to receive LEU product	AEOI written certification to IAEA
A-3	Medical isotope production facility confirmed capable of receiving LEU targets or upgrade plan approved	IAEA + WHO technical assessment
A-4	Environmental impact assessment for downblending operations submitted and approved	Iranian Environmental Protection Organization + IAEA review

Implementation Timeline

Table NOA-A2: Pathway A — 142-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
A.1 Preparation	Day 1–30	Facility preparation; equipment inspection; staff training under IAEA supervision; environmental monitoring baseline established	Tranche 2 asset release (25% of remaining frozen assets); OFAC Comfort Letters for 3 designated banking nodes; petrochemical export licenses	IAEA weekly progress reports; JIG monthly verification
A.2 Commencement	Day 31–60	First downblending campaign begins; 10% of HEU batch processed to LEU; IAEA continuous on-site monitoring activated	$25M Joint Disposition Fund disbursement (50% of Pathway A allocation); technical assistance for isotope facility upgrade (voluntary, upon Iranian request)	IAEA confirms LEU product specs; mass spectrometry verifies enrichment level
A.3 Steady Operations	Day 61–120	Continuous downblending at rated capacity (minimum 2% of stockpile per week); LEU stored in IAEA-sealed containers; medical isotope target fabrication begins	Tranche 3 asset release (25% of remaining frozen assets); DFC matching activated for civilian infrastructure projects; SWIFT Phase 2 connectivity for 2 designated banks	IAEA weekly inventory; monthly Dashboard report; isotope production verification

A.4 Completion	Day 121–142	Final HEU batch downblended; complete stockpile converted to LEU; final IAEA inventory; facility transition to civilian-only operations	$25M Joint Disposition Fund final disbursement; remaining SWIFT Phase 2 connectivity; DFC matching continues per annual schedule	IAEA final verification certificate; JIG certifies 100% conversion; Dashboard updated to "Pathway A Complete"

Synchronized Reciprocal Obligations

Iranian obligation triggers US obligation and vice versa: Iranian commencement of Phase A.2 (first downblending campaign) triggers US Tranche 2 release and $25M Fund disbursement. US Tranche 2 release must occur within 7 days of IAEA verification of Phase A.2 commencement; otherwise, the Cooling-Off Architecture (Article 19-F) activates. Iranian Phase A.3 steady operations trigger US Tranche 3 release and DFC matching; these must occur within 14 days of IAEA verification.

Verification Protocol

Continuous monitoring: IAEA on-site inspectors (minimum 2) present at all times during active downblending. OLEM sensors on all feed and product lines. Tamper-evident seals on all storage containers. Weekly mass spectrometry of product samples. **Batch verification:** Each downblended batch (defined as 5% of initial stockpile) requires separate IAEA verification before release to isotope production. **Dashboard reporting:** Daily operational data (enrichment levels, throughput, storage status) transmitted to Swiss Dashboard via air-gapped batch telemetry (Article 3.3). **Unscheduled inspection:** Either Party may request unscheduled IAEA inspection with 12-hour notice (Article 3.3(d)).

ESS Requirements

Radiation protection: All personnel within the active downblending area wear dosimeters; exposure limits per IAEA RS-G-1.3. **Waste management:** All downblending by-products stored in licensed containers at an IAEA-approved site. No liquid waste discharge without IAEA and Iranian environmental authority approval. **Physical security:** Perimeter security maintained by Iranian forces per existing protocols. IAEA inspector security provided by Iranian authorities per standard safeguards practice. **Emergency preparedness:** Facility emergency plan submitted to IAEA before Phase A.1. Quarterly emergency drills with IAEA observers.

Financial Mechanism

Joint Disposition Fund: $50M total — $25M at Phase A.2 commencement (verified by IAEA); $25M at Phase A.4 completion. **US Reciprocal Economic Value:** Tranche 2 ($250M–$500M depending on total frozen assets); Tranche 3 (equivalent amount); DFC matching up to $5B annually. **Iranian Economic Benefit:** Domestic medical isotope production reduces import dependence (estimated $50M–$80M annual savings); LEU available for export to regional partners under IAEA safeguards.

Contingencies

Equipment failure: If downblending equipment fails and cannot be repaired within 14 days, Iran may: (i) request technical assistance (voluntary); (ii) switch to Pathway B (Oman) or Pathway C (Kazakhstan) with 14-day notice; or (iii) pause operations for up to 30 days (TOS under Article 20) for repair. No penalty for equipment failure if reported within 24 hours. **Facility safety incident:** Any safety incident requiring evacuation triggers automatic 72-hour operations pause. IAEA safety inspection required before resumption. TOS may be invoked if repairs exceed 14 days. **Political disruption:** Either Party may invoke Article 20 (TOS) for up to 90 days. Material remains under IAEA dual-lock custody. Compliance clocks frozen.

Point of No Return

Pathway A reaches the point of no return when **75% of the HEU stockpile has been downblended to LEU and verified by the IAEA**. At this point, switching to another pathway is technically impractical. Iran may still request a switch, but the JIG and IAEA shall determine feasibility on a case-by-case basis. Prior to 75%, switching is at Iran's discretion with 14-day notice.

Narrative Framing Guidance (Non-Binding)

Iran: "Our nuclear material is being converted to medicine for our people — cancer diagnostics, radiation therapy, industrial applications. The material never left Iran. Our scientists, our facilities, our patients." **US:** "Weapons-capable uranium is being irreversibly converted to peaceful medical use under full IAEA verification. Every gram converted is a gram that cannot be weaponized."

NOA.5 Pathway B: Voluntary Oman Escrow Transfer

Objective

Transfer the HEU stockpile from Iran to an IAEA custody facility in Muscat, Oman, for secure storage pending final disposition pathway selection or conversion. This

pathway is voluntary and conditional — it requires either mutual agreement or unanimous JIG certification that in-Iran custody cannot be adequately verified.

Legal Basis

Article 1.5 (Pathway B); Article 1.6 (Optional Oman Lane); Article 1.6-A (Fallback to In-Iran After Oman); Appendix G definitions of "Custody Transition."

Pre-Conditions (in addition to NOA.2)

Table NOA-B1: Pathway B Additional Pre-Conditions

#	Pre-Condition	Verification
B-1	Mutual written agreement of both Parties OR unanimous JIG certification that in-Iran custody is insufficient	JIG written resolution published to Dashboard
B-2	Omani government confirms readiness to receive and host IAEA custody facility	Omani diplomatic note to JIG; facility security assessment by IAEA
B-3	Transport chain certified secure — containers, conveyance, communications, route	IAEA transport certification; route survey by Omani and IAEA security teams
B-4	No US personnel will be present during transfer unless both Parties agree in writing	Written Iranian confirmation; JIG verification of transfer personnel list
B-5	Insurance and liability framework for transport established (coverage minimum: $500M)	Insurance certificate filed with Swiss escrow

Implementation Timeline

Table NOA-B2: Pathway B — 14-Day Transport + Indefinite Storage Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification

B.1 Preparation	**Day 1–14**	**HEU packaged in IAEA-certified secure containers; tamper-evident seals applied; chain-of-custody documentation prepared; transport convoy assembled (Iranian + IAEA personnel only)**	**$50M Joint Disposition Fund disbursement (50% of Pathway B allocation upon commencement); Tranche 2 asset release; OFAC authorization for transport-related transactions**	**IAEA container certification; seal verification; convoy personnel manifest approved by both Parties**
B.2 Transport	**Day 15**	**Ground transport from Esfahan to Bandar Abbas; maritime transport from Bandar Abbas to Muscat under IAEA armed guard; continuous GPS tracking; satellite surveillance; real-time communications with JIG command center**	**US naval forces in stand-off posture — no vessels within 50nm of transport route unless responding to emergency distress; regional allies notified of transit (diplomatic note only, no operational coordination)**	**Continuous tracking data to Dashboard; IAEA inspector log every 2 hours; Omani coast guard escort in Omani waters**
B.3 Custody Transfer	**Day 16**	**Physical delivery to IAEA Muscat facility; IAEA inventory verification; new dual-lock system activated (IAEA + Omani custodian); Iranian liaison officer posted to Muscat**	**Remaining $50M Fund disbursement upon IAEA custody verification; Tranche 3 asset release within 14 days**	**IAEA custody transfer certificate; inventory match confirmation; dual-lock test successful**

B.4 Storage Period	**Day 17 until final disposition selected**	**Iranian liaison maintains presence at Muscat facility; quarterly inspection rights; material remains available for pathway switching**	**Continued sanctions freeze; DFC matching available; no new nuclear sanctions designations**	**IAEA monthly inventory reports; Dashboard status "Oman Escrow — Awaiting Final Disposition"; quarterly JIG review**

Synchronized Reciprocal Obligations

Iranian packaging and IAEA container certification trigger US $50M Fund disbursement and Tranche 2 release. These must occur within 7 days of IAEA certification. Iranian completion of custody transfer in Muscat triggers remaining $50M and Tranche 3 within 14 days. During the storage period, US obligations (sanctions freeze, no new designations) continue as long as the material remains under IAEA custody in Oman. If the US fails to maintain its obligations, Iran may request return of material to Esfahan under Article 1.6-A.

Verification Protocol

Transport verification: Continuous chain-of-custody documentation. Every transfer point (truck-to-ship, ship-to-facility) requires IAEA inspector physical presence and seal verification. GPS tracking data transmitted to Dashboard in real time. Any deviation from approved route triggers immediate JIG alert. **Storage verification:** IAEA continuous monitoring at Muscat facility. Monthly physical inventory. Quarterly mass spectrometry of random samples. Dual-lock tested monthly. Dashboard updated weekly. **Return option verification:** If Article 1.6-A return is invoked, the same transport protocol applies in reverse, with equal cost-sharing.

ESS Requirements

Transport safety: IAEA-certified Type B(U) transport containers per SSR-6. Radiation monitoring at all transfer points. Emergency response plan pre-coordinated with Iranian, Omani, and IAEA authorities. **Storage safety:** Muscat facility meets IAEA safety standards NS-R-5. Seismic assessment completed. Fire suppression systems inspected. **Security:** Omani custodial force (unarmed IAEA guards + Omani facility security). Perimeter security by Omani authorities. No foreign military presence without Iranian consent.

Financial Mechanism

Joint Disposition Fund: $100M total — $50M at Phase B.1 commencement; $50M at Phase B.3 custody transfer. **Transport costs:** Shared equally between both Parties (estimated $5M–$15M depending on route and security requirements). **Storage costs:** Omani government absorbs facility costs as its Framework contribution. IAEA inspection costs funded from the JIG operating budget. **Insurance:** $500M transport insurance premium shared equally.

Contingencies

Transport delay: Weather, mechanical failure, or security concerns may delay transport by up to 7 days without penalty. Delays beyond 7 days require JIG approval. Material remains under IAEA dual-lock at origin during delay. **Security incident during transport:** If the transport convoy is threatened, the convoy may divert to the nearest secure location. Omani and Iranian forces provide escort. The US provides satellite surveillance and intelligence support upon request. **Return to Iran (Article 1.6-A):** If no final disposition is agreed within 180 days, Iran may request return. If no disposition within 365 days, automatic default to Pathway A. Return transport follows the same protocol as outbound transport, with costs shared equally.

Point of No Return

Pathway B reaches the point of no return when **the HEU has been physically transferred to Muscat and under IAEA custody for more than 365 days without Iran invoking the Article 1.6-A return option**. At that point, the material is deemed to have selected Pathway A by default (Article 1.6-A), and the IAEA shall arrange return transport to Esfahan for downblending. Prior to 365 days, Iran may request return at any time with 30 days' notice.

Narrative Framing Guidance (Non-Binding)

Iran: "We voluntarily placed our material under IAEA custody in a neutral Muslim country as a confidence-building gesture. The material remains Iranian property. We can request its return at any time." **US:** "Iranian material is now under IAEA lock and key in a neutral third country. This is a significant non-proliferation step — voluntarily taken."

NOA.6 Pathway C: Downblending in Iran, Transfer to IAEA LEU Bank, Kazakhstan

Objective

Downblend Iran's HEU stockpile to LEU at Esfahan, then transfer the resulting LEU to the IAEA LEU Bank in Oskemen, Kazakhstan, for storage and regional access. Iran retains LEU ownership rights with preferential draw rights.

Legal Basis

Article 1.5 (Pathway C); Article 1.6 (Day 75 Default); IAEA LEU Bank Host State Agreement (Kazakhstan).

Pre-Conditions (in addition to NOA.2)

Table NOA-C1: Pathway C Additional Pre-Conditions

#	Pre-Condition	Verification
C-1	Kazakhstan confirms readiness to receive LEU at IAEA LEU Bank	Kazakhstan diplomatic note to JIG; LEU Bank capacity confirmation from IAEA
C-2	Transport route from Esfahan to Oskemen certified (Iran-Turkmenistan-Uzbekistan-Kazakhstan or maritime via Caspian)	Route security assessment by all transit states; IAEA transport certification
C-3	LEU Bank draw-rights agreement drafted and approved by Iran	Iranian legal review; IAEA LEU Bank management confirmation
C-4	All transit states (minimum 2) provide diplomatic assurances of secure passage	Diplomatic notes filed with JIG

Implementation Timeline

Table NOA-C2: Pathway C — 75-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
C.1 Downblending	Day 1–45	HEU downblended to LEU (19.75%) at Esfahan under IAEA supervision; LEU fabricated into certified storage form; IAEA verifies each batch	$50M Fund disbursement at commencement; Tranche 2 release; technical assistance for LEU fabrication (voluntary)	IAEA batch verification; weekly mass spectrometry

C.2 Transport to Kazakhstan	Day 46–60	LEU packaged for international transport; ground/sea transport to Oskemen; chain-of-custody maintained; all transit state permissions obtained	$50M Fund disbursement upon IAEA custody transfer; Tranche 3 release within 14 days; OFAC authorization for transit transactions	IAEA transport log; GPS tracking; customs clearance at each border
C.3 LEU Bank Deposit	Day 61–75	LEU deposited at IAEA LEU Bank; Iran retains ownership and 30-day preferential draw right; draw-rights agreement activated	DFC matching for civilian infrastructure; remaining sanctions relief tranches per Article 7 schedule	IAEA LEU Bank deposit certificate; ownership rights registered; Dashboard updated

Synchronized Reciprocal Obligations

IAEA verification of Phase C.1 downblending commencement triggers US Fund disbursement and Tranche 2 within 7 days. IAEA verification of Phase C.3 deposit triggers Tranche 3 within 14 days. US failure to release triggers Cooling-Off (Article 19-F). Iranian failure to maintain downblending schedule triggers JIG consultation (Article 13.4), not automatic penalty.

Verification Protocol

Same as Pathway A for downblending phase. For transport: chain-of-custody documentation at every border crossing. IAEA inspector escort for entire route. For LEU Bank deposit: IAEA LEU Bank inventory system. Iran may request draw-rights exercise (30-day delivery) at any time for NPT-compliant civilian purposes.

ESS Requirements

Type B(U) transport containers for LEU transit. Caspian Sea route: weather-resilient vessel with radiation monitoring. Land route: armored convoy with radiation detection equipment. LEU Bank storage: IAEA LEU Bank existing safety protocols apply.

Financial Mechanism

Fund: $100M — $50M at C.1 commencement; $50M at C.3 deposit. **Transport:** Shared equally (estimated $8M–$20M depending on route). **LEU Bank fees:** IAEA absorbs

storage costs as part of LEU Bank mandate. Iran pays 0.5% annual custodial fee only if LEU remains beyond 5 years.

Point of No Return

75% of stockpile downblended and first LEU batch in transit to Kazakhstan. After this point, remaining HEU can still be downblended, but reversing the entire pathway becomes technically impractical.

Narrative Framing Guidance

Iran: "Our material was converted to peaceful fuel and deposited in the international LEU Bank — a neutral facility under IAEA control. We retain ownership and the right to draw fuel for our reactors." US: "Weapons-capable material was irreversibly downblended and deposited in the IAEA's own fuel bank. This is the gold standard of non-proliferation verification."

NOA.7 Pathway D: Transfer to Rosatom Under IAEA Safeguards

Objective

Transfer the HEU stockpile to the Russian Federation for downblending and conversion into civilian fuel under Rosatom management, with IAEA safeguards throughout. This pathway requires joint US-Iran concurrence — either Party may block it unilaterally.

Legal Basis

Article 1.5 (Pathway D — joint concurrence required); US-Russia bilateral coordination (Article 16.2); IAEA-Rosatom safeguards agreement.

Pre-Conditions (in addition to NOA.2)

Table NOA-D1: Pathway D Additional Pre-Conditions

#	Pre-Condition	Verification
D-1	Joint written concurrence of both US and Iran	Signed JIG resolution with both Parties' signatures
D-2	Russia confirms willingness to receive HEU and agrees to IAEA safeguards	Russian diplomatic note to JIG; Rosatom-IAEA safeguards arrangement

D-3	EU sanctions on Russia do not prohibit HEU transfer (legal opinion obtained)	Legal opinion from Swiss facilitator or EU Council legal service
D-4	Transport route certified (Iran-Caspian Sea-Russia or Iran-Turkmenistan-Kazakhstan-Russia)	Route security assessment; all transit state permissions
D-5	Guarantor Matrix states with Russia relations (Pakistan, Oman) provide transit facilitation	Diplomatic notes from Pakistan and Oman

Implementation Timeline

Table NOA-D2: Pathway D — 60-to-90-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
D.1 Preparation	Day 1–30	HEU packaged for international transport; Rosatom technical team visits Esfahan for equipment assessment; IAEA transport certification	Tranche 2 release; OFAC authorization for Russia-related transactions; coordination with EU on sanctions compatibility	IAEA transport certification; Rosatom assessment report
D.2 Transport	Day 31–45	Ground/sea transport to Russian facility (Mayak or Seversk); IAEA armed escort; Russian customs clearance; continuous monitoring	Intelligence support for route security; satellite surveillance; no Fund disbursement (Pathway D excluded from Fund)	IAEA transport log; Russian customs documentation; Dashboard tracking

D.3 Russian Receipt	**Day 46–60**	**Custody transfer to Rosatom under IAEA safeguards; IAEA inventory verification; dual-lock activated (Rosatom + IAEA); Iranian liaison posted**	**Tranche 3 release; public statement supporting Russian-Iranian civilian nuclear cooperation**	**IAEA custody transfer certificate; Rosatom receipt documentation**
D.4 Extended Processing (if required)	**Day 61–90**	**Rosatom downblends HEU to LEU; conversion to fuel assemblies; IAEA monitors all operations; Iranian technical observers present**	**Continued sanctions freeze; DFC matching for Iranian civilian projects**	**IAEA processing reports; monthly inventory; fuel assembly certification**

Synchronized Reciprocal Obligations

Because Pathway D requires joint concurrence, the synchronization is built into the political decision. Once both Parties concur, the standard synchronization applies: IAEA verification of each phase triggers corresponding US tranche releases. The US obligation to coordinate with EU on sanctions compatibility runs continuously from concurrence through completion.

Verification Protocol

IAEA full-scope safeguards at the Russian receiving facility. Iranian technical observers present (up to 4) with access to all IAEA-monitored areas. Dashboard reporting via air-gapped telemetry. Quarterly JIG review. Rosatom provides processing data to IAEA within 7 days of each operation.

ESS Requirements

Russian facility must meet IAEA safety standards. All transport per SSR-6. Russian emergency response plan shared with IAEA and Iranian authorities before Phase D.2.

Financial Mechanism

Fund: No Fund disbursement (Pathway D excluded per Article 1.5-A). **Rosatom compensation:** Rosatom may charge processing fees, paid by Iran from unfrozen assets or DFC matching. Fee schedule agreed before concurrence. **Transport:** Shared equally.

Contingencies

EU sanctions conflict: If EU sanctions prevent the transfer, the Parties may: (i) request EU sanctions waiver; (ii) switch to Pathway C (Kazakhstan) or Pathway B (Oman); or (iii) activate TOS pending resolution. **Russian facility unavailability:** If Rosatom cannot receive material, switch to Pathway C or F. **Either Party withdraws concurrence:** Withdrawal permitted before Phase D.2 transport begins. After transport begins, withdrawal requires JIG review and material return at withdrawing Party's expense.

Point of No Return

Physical transport across the Iranian border. Once material leaves Iran, return becomes a separate logistical operation.

Narrative Framing Guidance

Iran: "Our material is being processed by a fellow nuclear power into fuel for civilian reactors worldwide. We retain technical oversight through our liaison and IAEA verification." **US:** "Iranian material is being converted to peaceful fuel by Russia under full IAEA safeguards — a proven model that eliminates weapons risk."

NOA.8 Pathway E: Conversion to Reactor-Grade LEU Fuel Assemblies for Bushehr

Objective

Convert Iran's HEU stockpile into reactor-grade LEU fuel assemblies for the Bushehr Nuclear Power Plant (BNPP), replacing Russia as the sole fuel supplier and establishing Iran as a self-sufficient civilian nuclear fuel producer.

Legal Basis

Article 1.5 (Pathway E); Iran-Russia Bushehr Intergovernmental Agreement (fuel supply clause); IAEA safeguards on Bushehr.

Pre-Conditions (in addition to NOA.2)

Table NOA-E1: Pathway E Additional Pre-Conditions

#	Pre-Condition	Verification

E-1	**Rosatom confirms agreement to accept LEU fuel assemblies from Iran for Bushehr OR Iran certifies domestic fuel fabrication capability meeting VVER-1000 specifications**	**Rosatom contract amendment OR IAEA fuel qualification certification**
E-2	**Fuel fabrication facility at Esfahan confirmed capable or upgradeable to VVER-1000 fuel rod specifications**	**IAEA technical inspection; fuel qualification test plan approved**
E-3	**Bushehr reactor operator (NPPD) confirms readiness to receive domestically fabricated fuel**	**NPPD written certification to AEOI and IAEA**
E-4	**Russian regulatory authority (Rostekhnadzor) or IAEA certifies fabricated fuel meets safety standards**	**Fuel qualification dossier approved**

Implementation Timeline

Table NOA-E2: Pathway E — 180-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
E.1 Downblending	**Day 1–60**	**HEU downblended to reactor-grade LEU (3.5–4.5% U-235) at Esfahan; enrichment to reactor grade verified by IAEA**	**$50M Fund at commencement; Tranche 2 release; technical assistance for fuel fabrication (voluntary, upon request)**	**IAEA weekly enrichment verification; LEU product certification**
E.2 Fuel Fabrication	**Day 61–120**	**LEU converted to UO2 powder; fuel pellets fabricated; fuel rods assembled; fuel assemblies completed; all under IAEA supervision**	**$50M Fund at fuel fabrication commencement; Tranche 3 release; SWIFT Phase 2 connectivity for 2 designated banks**	**IAEA fuel fabrication inspection; dimensional and metallurgical testing**

E.3 Fuel Qualification	Day 121–150	Fuel assemblies undergo qualification testing: irradiation testing in research reactor; non-destructive examination; performance analysis	DFC matching for civilian infrastructure; additional sanctions relief tranches per Article 7	IAEA fuel qualification report; Rostekhnadzor or IAEA safety certification
E.4 Reactor Loading	Day 151–180	Qualified fuel assemblies transported to Bushehr; loaded into reactor core under IAEA and Rosatom supervision; reactor startup with domestic fuel	Remaining Fund disbursement; full SWIFT Phase 2; DFC matching continues	IAEA loading supervision; reactor physics measurement; first criticality with domestic fuel verified

Synchronized Reciprocal Obligations

IAEA verification of each phase triggers corresponding US obligations within 7–14 days. The 180-day timeline is the longest of all pathways, reflecting the technical complexity of fuel fabrication. Both Parties may agree to extend any phase by up to 30 days for technical reasons without penalty.

Verification Protocol

IAEA continuous monitoring during downblending. IAEA inspector presence during fuel fabrication (minimum 2 inspectors). All fuel rods individually tracked by serial number. Fuel qualification testing witnessed by IAEA. Reactor loading supervised by IAEA and Rosatom. Dashboard reports: weekly during Phases E.1–E.2, biweekly during E.3–E.4.

ESS Requirements

Fuel fabrication facility: IAEA safety standards NS-R-5. Quality assurance program per IAEA SS-G-2.10. All fuel must meet VVER-1000 specifications. Transport to Bushehr: IAEA-certified containers. Seismic and flood protection at all facilities verified.

Financial Mechanism

Fund: $100M — $50M at E.1 commencement; $50M at E.4 completion. **Iranian savings:** Estimated $100M–$150M annually in reduced fuel import costs from Russia. **DFC**

matching: Up to $5B annually for civilian infrastructure concurrent with fuel program.

Point of No Return

Fuel assemblies loaded into the Bushehr reactor core. Once irradiated, fuel cannot be re-purposed for weapons.

Narrative Framing Guidance

Iran: "We have achieved what only a handful of nations have — the complete domestic nuclear fuel cycle, from uranium to reactor fuel, under IAEA safeguards. This is the sovereign right the NPT guarantees us." **US:** "Iran's nuclear program is now fully channeled into verified civilian power production. Weapons-capable material has become reactor fuel under IAEA lock and key."

NOA.9 Pathway F: Hybrid — Stockpile Divided Into Batches Through Multiple Pathways

Objective

Divide the HEU stockpile into two or more batches, each processed through a different pathway simultaneously. This provides maximum flexibility, hedges against single-pathway failure, and allows parallel processing to accelerate overall disposition.

Legal Basis

Article 1.5 (Pathway F — joint concurrence on destination countries required); Article 1.6 (Day 75 Default applies to each batch independently).

Pre-Conditions (in addition to NOA.2)

Table NOA-F1: Pathway F Additional Pre-Conditions

#	Pre-Condition	Verification
F-1	Joint written concurrence of both Parties on the specific combination of sub-pathways and batch allocations	Signed JIG resolution specifying: number of batches, percentage allocation per batch, sub-pathway per batch
F-2	Each selected sub-pathway satisfies its own pre-conditions (NOA.4 through NOA.15 and NOA.17 as applicable)	JIG verification that all sub-pathway pre-conditions are met

F-3	Additional IAEA staffing confirmed to monitor multiple parallel operations	IAEA staffing plan approved; additional inspector deployment confirmed
F-4	No sub-pathway batch exceeds 50% of total stockpile (risk distribution requirement)	JIG certifies allocation plan meets risk distribution standard

Implementation Timeline

Pathway F uses the timeline of its constituent sub-pathways, running in parallel. The overall completion date is determined by the longest sub-pathway. The JIG shall publish a master schedule showing all parallel tracks.

Table NOA-F2: Pathway F — Example Allocation and Timeline

Batch	% of Stockpile	Sub-Pathway	Timeline	Fund Allocation
F-1	40%	A (In-Iran downblending)	142 days	$20M (40% of Pathway A $50M)
F-2	35%	E (Bushehr fuel)	180 days	$35M (35% of Pathway E $100M, prorated)
F-3	25%	I (Isotope swap)	120 days	$25M (25% of Pathway I $100M, prorated)
Total	**100%**	**Hybrid A+E+I**	**180 days (governed by longest: E)**	**$80M total**

Synchronized Reciprocal Obligations

Each sub-pathway triggers its own synchronized obligations per its respective NOA section. US obligations for each batch are independent — failure on one batch does not affect obligations for other batches. The Partial Success Doctrine (Article 13.3-A) is built into Pathway F by design.

Verification Protocol

Each sub-pathway operates under its own verification protocol (NOA.4 through NOA.15 and NOA.17 as applicable). Additional IAEA coordination ensures no material is double-counted or diverted between batches. Master inventory maintained by IAEA showing all batch locations and statuses.

Financial Mechanism

Fund: Prorated from each sub-pathway's Fund allocation based on batch percentage. Total Fund for hybrid pathway = sum of prorated sub-pathway allocations. **Administration:** Swiss facilitator maintains separate sub-accounts per batch. Disbursement per batch triggers independently.

Contingencies

Sub-pathway failure: If one batch's sub-pathway fails (equipment breakdown, political disruption), that batch may be switched to another sub-pathway or held in IAEA custody pending resolution. Other batches continue unaffected. **Reallocation:** With both Parties' written consent, batch allocations may be adjusted mid-stream. No single batch may exceed 60% after reallocation.

Point of No Return

Each sub-batch reaches its point of no return independently per its respective pathway. The overall hybrid pathway reaches point of no return when the majority of batches (by percentage) have passed their individual points of no return.

Narrative Framing Guidance

Iran: "We are pursuing multiple peaceful applications simultaneously — medicine, power, and research. This is the full scope of NPT rights." **US:** "Iran's material is being processed through multiple verified civilian channels simultaneously. Maximum transparency, maximum non-proliferation assurance."

NOA.10 Pathway G: Regional LEU Bank Hosted in Iran

Objective

Establish Iran as the custodian of a regional LEU fuel bank under IAEA supervision. Iran's HEU is downblended to LEU and becomes the initial capital of the bank. Regional NPT-compliant states may purchase fuel from the bank. Iran receives custodial prestige, annual custodial fees, and first-right-of-refusal on management contracts.

Legal Basis

Article 1.5 (Pathway G); IAEA LEU Bank precedent (Kazakhstan); NPT Article IV (peaceful use).

Pre-Conditions (in addition to NOA.2)

Table NOA-G1: Pathway G Additional Pre-Conditions

#	Pre-Condition	Verification
G-1	Minimum 3 regional states (Gulf, Central Asian, or South Asian) express interest in purchasing LEU from the bank	Letters of intent filed with JIG
G-2	Esfahan facility assessed as suitable for bank operations or upgrade plan approved	IAEA facility assessment; engineering report
G-3	Bank governance charter drafted and approved by all participating states	Charter signed by Iran, IAEA, and minimum 3 customer states
G-4	LEU pricing mechanism agreed (linked to spot uranium price + custodial fee)	Pricing formula approved by bank board

Implementation Timeline

Table NOA-G2: Pathway G — 210-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
G.1 Downblending	Day 1–60	HEU downblended to LEU (19.75%) at Esfahan; LEU certified as bank-grade by IAEA	$50M Fund at commencement; Tranche 2 release; US supports IAEA LEU Bank precedent at Board of Governors	IAEA LEU certification; mass spectrometry
G.2 Facility Upgrade	Day 31–120	Esfahan facility upgraded to bank operational standards; storage containers installed; inventory management system activated; security systems enhanced	$50M Fund at facility certification; Tranche 3 release; US supports regional state participation diplomatically	IAEA facility certification; security assessment

G.3 Bank Charter Activation	Day 121–150	Regional LEU Bank charter registered with IAEA; board constituted (Iran 1 seat, IAEA 1 seat, customer states 1 seat each); pricing mechanism activated; first purchase agreements signed	DFC matching for civilian infrastructure; US encourages allied states to consider bank purchases	JIG charter registration; purchase agreement verification
G.4 Operational Launch	Day 151–210	Bank opens for business; first LEU deliveries to customer states; custodial fee payments begin; annual report published	Remaining sanctions relief tranches; DFC matching continues	IAEA delivery verification; customer state receipt confirmation; Dashboard "Pathway G Operational"

Synchronized Reciprocal Obligations

IAEA verification of each phase triggers corresponding US obligations. US diplomatic support for IAEA Board of Governors recognition of the bank is a continuous obligation throughout. US encouragement of allied participation is non-binding but expected in good faith.

ESS Requirements

Facility: IAEA NS-R-5 compliant. LEU storage: IAEA LEU Bank equivalent standards. Transport to customer states: SSR-6 certified containers. Emergency response: Coordinated with all customer states.

Financial Mechanism

Fund: $100M — $50M at G.1; $50M at G.3. **Custodial fee:** 1.5% of fuel value annually to Iran. **Management contracts:** Iran has first-right-of-refusal. Estimated annual revenue: $15M–$30M depending on customer base.

Point of No Return

Bank operational launch (G.4) with first deliveries to customer states. Once regional states depend on the bank for fuel supply, dismantling it becomes politically and economically costly.

Narrative Framing Guidance

Iran: "We are the Switzerland of the Middle East — a neutral custodian of fuel for peace. Our neighbors trust us with their energy security." US: "Iran has transitioned from proliferation risk to regional fuel supplier. The LEU Bank is under IAEA lock and key."

NOA.11 Pathway H: Joint Uranium Conversion Facility with Gulf States

Objective

Establish a joint venture facility between Iran and Gulf Cooperation Council states to convert downblended HEU into fuel for Gulf states' future civilian nuclear reactors. Creates economic interdependence and makes conflict costly for all parties.

Legal Basis

Article 1.5 (Pathway H); GCC Framework Agreement (as applicable); bilateral investment treaties between Iran and participating Gulf states.

Pre-Conditions (in addition to NOA.2)

Table NOA-H1: Pathway H Additional Pre-Conditions

#	Pre-Condition	Verification
H-1	Minimum 2 GCC states commit to joint venture participation	Signed letters of intent filed with JIG
H-2	Joint venture structure agreed: Iran 45%, Gulf consortium 40%, IAEA Technical Support 15%	Joint venture agreement signed
H-3	Facility site selected and approved by all joint venture partners	Site assessment report; environmental impact study
H-4	Gulf states' reactor programs confirmed as NPT-compliant and under IAEA safeguards	IAEA verification of Gulf state safeguards agreements
H-5	Article 4.2-C (GCC Side Letter + IMO Mediation) acknowledged by all participating Gulf states	Diplomatic notes from each participating state

Implementation Timeline

Table NOA-H2: Pathway H — 365-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
H.1 Downblending	Day 1–60	HEU downblended to LEU at Esfahan; LEU stored pending facility construction	$50M Fund at commencement; Tranche 2 release; OFAC authorization for Gulf state joint venture transactions	IAEA LEU certification
H.2 Facility Construction	Day 61–240	Iran contributes engineering expertise and workforce; construction supervised by IAEA; all fuel fabrication equipment IAEA-safeguarded	$50M Fund at construction commencement; Tranche 3 release; US supports Gulf state participation diplomatically	IAEA quarterly construction inspection; engineering milestone reports
H.3 Equipment Installation	Day 241–300	Fuel fabrication equipment installed and tested; quality assurance program activated; staff training program	DFC matching for Iranian civilian infrastructure; SWIFT Phase 2 connectivity	IAEA equipment certification; staff training verification
H.4 First Fuel Production	Day 301–365	First LEU fuel rods fabricated; qualification testing; delivery to first Gulf state reactor program	Remaining sanctions relief tranches; DFC matching continues	IAEA fuel qualification report; first delivery receipt

Synchronized Reciprocal Obligations

Standard phase-by-phase synchronization. US diplomatic support for Gulf participation is a continuous obligation. US OFAC authorization for joint venture transactions must be issued before Phase H.2 begins.

ESS Requirements

Construction: IAEA safety standards. All equipment: IAEA-safeguarded. Staff: ILO radiation protection standards. Emergency response: Coordinated with all joint venture partners.

Financial Mechanism

Fund: $100M — $50M at H.1; $50M at H.3. **Joint venture capital:** Estimated $200M–$500M total, with contributions per equity share. **Revenue:** Fuel sales to Gulf states at market price. Iranian share of profit: 45%.

Point of No Return

Facility construction completion (H.3). Once the facility is built and equipped, dismantling represents massive sunk cost.

Narrative Framing Guidance

Iran: "Our nuclear expertise has become a source of regional partnership and economic leadership. We build fuel for our neighbors' power plants." **US:** "Iran's nuclear capability is channeled into peaceful commerce with its neighbors. Economic interdependence is the best non-proliferation outcome."

NOA.12 Pathway I: Uranium-for-Isotopes Swap

Objective

Iran's HEU is downblended to medical-grade LEU and fabricated into isotope production targets at Esfahan. In exchange, international partners (Belgium, Netherlands, Japan, Canada) guarantee a 20-year supply of medical isotopes (Tc-99m, I-131, I-125) for Iran's cancer treatment and diagnostic needs.

Legal Basis

Article 1.5 (Pathway I); IAEA Technical Cooperation Programme; WHO Essential Medicines List (radiopharmaceuticals).

Pre-Conditions (in addition to NOA.2)

Table NOA-I1: Pathway I Additional Pre-Conditions

#	Pre-Condition	Verification
I-1	Minimum 2 isotope-producing states confirm participation and supply commitment	Supply agreements signed; IAEA witness
I-2	Iranian medical facilities' isotope demand assessed and documented	WHO/Iranian Health Ministry joint assessment
I-3	Target fabrication capability at Esfahan confirmed or upgradeable	IAEA technical inspection

I-4	LEU target specifications agreed between Iranian producers and international isotope suppliers	Technical specification document signed

Implementation Timeline

Table NOA-I2: Pathway I — 120-Day Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
I.1 Downblending + Target Fab	Day 1–60	HEU downblended to medical-grade LEU (19.75%); LEU fabricated into isotope targets at Esfahan; targets qualified by IAEA	$50M Fund at commencement; Tranche 2 release; up to $200M isotope supply guarantee from Joint Disposition Fund	IAEA target qualification; mass spectrometry
I.2 Swap Activation	Day 61–90	First targets shipped to isotope-producing partners; Iran begins receiving isotope shipments; swap ratio agreed (kg LEU per Curie isotope)	$50M Fund at swap activation; Tranche 3 release; OFAC authorization for isotope trade transactions	IAEA monitors both sides of swap; shipment receipts
I.3 Steady Supply	Day 91–120	Regular target shipments; continuous isotope receipts; Iranian medical facilities report usage data; 20-year supply agreement operational	DFC matching; remaining sanctions relief; isotope supply guarantee renews annually	Monthly swap reports to Dashboard; WHO verification of medical usage

Synchronized Reciprocal Obligations

IAEA verification of each phase triggers US obligations. Isotope supply guarantees from partner states activate upon Iranian target shipment verification. If partner states fail to deliver isotopes, the Joint Disposition Fund guarantee ($200M) covers alternative sourcing.

ESS Requirements

Target fabrication: IAEA safety standards. Isotope transport: IAEA放射性材料运输规程. Medical usage: WHO radiation safety guidelines. Waste: All isotope waste managed per IAEA standards.

Financial Mechanism

Fund: $100M — $50M at I.1; $50M at I.2. **Isotope supply guarantee:** $200M from Joint Disposition Fund (covers 5 years of alternative sourcing if partners default). **Iranian savings:** Estimated $50M–$80M annually in reduced isotope import costs.

Point of No Return

First isotope shipments received by Iranian medical facilities and administered to patients. Once Iranian cancer treatment depends on the swap, reversal becomes a humanitarian issue.

Narrative Framing Guidance

Iran: "Our enriched uranium saves Iranian lives. Every gram downblended is a dose of medicine for our people. Cancer patients in Tehran, Isfahan, and Shiraz benefit from what was once a source of international tension." **US:** "Weapons-grade material has become cancer treatments. That's a transformation worth supporting. Iranian patients receive world-class diagnostics because of this pathway."

NOA.13 Pathway J: Scientific "Custody for Research"

Objective

HEU is placed under IAEA custody but made available for legitimate scientific research — neutron activation analysis, nuclear forensics training, materials science, and radioisotope research. Iran receives priority access, guaranteed co-authorship, and training for 50 scientists annually.

Legal Basis

Article 1.5 (Pathway J); IAEA Technical Cooperation Programme; CERN/ITER shared-facility precedents.

Pre-Conditions (in addition to NOA.2)

Table NOA-J1: Pathway J Additional Pre-Conditions

#	Pre-Condition	Verification

J-1	**Scientific Consortium established with minimum 3 international research partners**	**Consortium agreement signed (Iran 40%, IAEA 30%, international partners 30%)**
J-2	**Research program defined: neutron activation, forensics, materials science, isotope research**	**Research plan approved by Consortium**
J-3	**50 Iranian scientist training slots confirmed annually**	**Training agreements with partner institutions**
J-4	**Publication and patent-sharing agreement approved**	**Intellectual property framework signed**

Implementation Timeline

Table NOA-J2: Pathway J — Indefinite Timeline with 5-Year Reviews

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
J.1 Setup	**Day 1–90**	**Research facility prepared; HEU secured under IAEA custody with research access protocol; equipment installed; safety systems tested**	**$37.5M Fund at commencement (50% of Pathway J allocation); Tranche 2 release; US research institutions encouraged to participate**	**IAEA facility certification; equipment inventory**
J.2 Research Launch	**Day 91–180**	**First research projects launched; Iranian scientists begin training; initial publications prepared**	**$37.5M Fund at research launch; Tranche 3 release; US visa facilitation for Iranian scientists (non-binding but expected)**	**Consortium progress report; training logs**

J.3 Steady Operations	**Day 181–5 years**	**Continuous research program; 50 Iranian scientists trained annually; minimum 10 peer-reviewed publications per year; patent applications filed jointly**	**DFC matching; sanctions relief continues; US research collaboration (institutional, not governmental)**	**Annual Consortium report; publication list; patent register**
J.4 Five-Year Review	**Every 5 years**	**Research program reviewed; pathway may continue, be modified, or transition to another pathway; Iranian scientists' progress assessed**	**Continued support subject to 5-year review outcome**	**JIG review; IAEA assessment; both Parties' consent for continuation**

Synchronized Reciprocal Obligations

Standard phase synchronization. US visa facilitation is non-binding but expected as part of scientific collaboration goodwill. US research institution participation is institutional, not governmental.

ESS Requirements

Research facility: IAEA safety standards. HEU access: Strict IAEA control; no unsupervised access. Research waste: Managed per IAEA standards. Emergency response: Quarterly drills.

Financial Mechanism

Fund: $75M — $37.5M at J.1; $37.5M at J.2. **Consortium budget:** Annual operating budget from IAEA Technical Cooperation Programme and partner contributions. **Iranian benefit:** 50 scientists trained annually; co-authorship on all publications; shared patent rights.

Point of No Return

This pathway has no technical point of no return — it can transition to another pathway at any 5-year review. The material remains under IAEA custody throughout. However, once research programs are established and Iranian scientists are integrated into international collaborations, the political cost of dismantling increases.

Narrative Framing Guidance

Iran: "Our nuclear program has become a center of global scientific excellence. Iranian names appear on world-class research. Our scientists train at the finest institutions." **US:** "Iranian scientists have joined the global research community. Shared scientific infrastructure builds lasting norms and human connections."

NOA.14 Pathway K: Uranium-Backed Reconstruction Bond

Objective

The HEU stockpile is valued as a strategic asset ($5–10 billion equivalent by independent valuation). Verified downblending releases funds from an international reconstruction fund for Iran. The HEU is "securitized" — its verified disposition directly unlocks reconstruction capital. This pathway requires **joint US-Iran concurrence** on valuation and fund structure.

Legal Basis

Article 1.5 (Pathway K — joint concurrence required); Article 1.5-A (Fund drawdown: $150M in three tranches); Appendix G definitions of "Escrow" and "High-Risk Nuclear Material."

Pre-Conditions (in addition to NOA.2)

Table NOA-K1: Pathway K Additional Pre-Conditions

#	Pre-Condition	Verification
K-1	Joint written concurrence of both Parties on valuation methodology and fund structure	Signed JIG resolution
K-2	Independent valuation panel appointed and functional (World Bank + IMF + Omani Sovereign Wealth Fund)	Panel members confirmed; terms of reference signed
K-3	Reconstruction fund capitalization plan approved (40% Iran frozen assets, 30% Gulf, 20% EU, 10% US DFC)	Fund charter signed by all contributors
K-4	Eligible reconstruction project list developed and approved by Iran	Project list filed with JIG; IAEA no-objection
K-5	Financial architecture legally reviewed by all six guarantor states	Legal opinions filed with Swiss facilitator

Implementation Timeline

Table NOA-K2: Pathway K — 5-Year Implementation Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
K.1 Valuation	Day 1–90	HEU fully inventoried and characterized; independent valuation panel conducts assessment; valuation report published	$50M Fund at commencement; Tranche 2 release; DFC commits 10% fund share; US supports World Bank/IMF participation	Independent valuation report; JIG acceptance of valuation
K.2 Fund Capitalization	Day 91–180	Capital contributions deposited; fund governance board constituted; disbursement rules approved; eligible projects selected	$50M Fund at fund launch; Tranche 3 release; US DFC guarantee documents executed	Escrow confirmation of all capital contributions; board minutes
K.3 Phase 1 Downblending + Release	Day 181–365 (Year 2)	20% of HEU downblended; Phase 1 reconstruction projects launched (hospitals, schools, infrastructure); first disbursement tranche	$50M Fund at Phase 1 completion; DFC matching for civilian infrastructure; annual review	IAEA downblending verification; project launch documentation; disbursement receipts
K.4 Phase 2 Downblending + Release	Day 366–730 (Year 3)	40% of HEU downblended; Phase 2 projects (power plants, water treatment); second disbursement tranche	Continued sanctions relief; DFC matching; annual review	IAEA Phase 2 verification; project progress reports

K.5 Phase 3 Downblending + Release	**Day 731–1095 (Year 4)**	**70% of HEU downblended; Phase 3 projects (industrial parks, transport); third disbursement tranche**	**Continued sanctions relief; DFC matching; annual review**	**IAEA Phase 3 verification**
K.6 Completion	**Day 1096–1825 (Year 5)**	**100% HEU downblended; all reconstruction projects operational; final fund audit; residual funds disposition**	**Full sanctions relief per Article 7; final DFC matching tranche; Framework completion certificate**	**IAEA final inventory: zero HEU remaining; project completion audit; Dashboard "Pathway K Complete"**

Synchronized Reciprocal Obligations

Each downblending milestone triggers corresponding fund disbursement. The synchronization is explicit: IAEA verifies X% downblended, fund releases Y% of reconstruction capital. Either side may request JIG review if the other fails to meet its milestone obligation. The independent valuation panel arbitrates disputes over valuation methodology.

Verification Protocol

IAEA continuous monitoring of all downblending operations. Independent financial auditor (Big-4 firm, IAEA-approved) monitors fund disbursement. Quarterly reports to Dashboard showing: downblending progress, fund disbursement, project implementation. Annual JIG review with both Parties. Final audit at Year 5 by independent panel.

ESS Requirements

Downblending: Same as Pathway A. Reconstruction projects: Standard construction safety. No additional nuclear safety requirements beyond downblending phase.

Financial Mechanism

Fund: $150M — $50M at K.1; $50M at K.3; $50M at K.6. **Reconstruction fund:** $5B–$10B total (per valuation). **Iranian benefit:** Reconstruction capital for hospitals, schools, power plants, water treatment, transport infrastructure. **Valuation fee:** 0.5% of fund value to independent panel.

Contingencies

Valuation dispute: If Parties disagree on valuation, the independent panel's determination is binding. Either Party may request a second panel (costs shared) if the first panel's valuation deviates by more than 20% from either Party's position. **Contributor default:** If a contributor (Gulf state, EU) fails to deposit its share, remaining contributors may cover the shortfall by mutual agreement, or the fund scales proportionally. **Project failure:** If a reconstruction project fails, funds are redirected to other approved projects. No clawback of disbursed funds.

Point of No Return

First reconstruction project disbursement (K.3). Once reconstruction capital is deployed and projects are under construction, reversing the pathway would leave projects incomplete and create political and financial liability.

Narrative Framing Guidance

Iran: "Our nuclear asset became the foundation of our reconstruction. Every kilogram downblended is a hospital, a school, a power plant. We capitalized our own future." **US:** "Every kilogram of weapons-capable material downblended becomes a hospital or a school. That's a trade worth making — and it's verifiable every step of the way."

NOA.15 Pathway L: Sovereign Fuel Leasing Trust

Objective

Iran's HEU is not destroyed, not downblended, and not transferred out of Iran. Instead, it is deposited into a sovereign trust owned by the Central Bank of Iran. The trust issues "nuclear fuel credits" purchased by international partners. Iran receives immediate liquidity. The HEU is gradually drawn down over 10–15 years through verified increments. This pathway requires **joint US-Iran concurrence** on credit purchasers and drawdown schedule.

Legal Basis

Article 1.5 (Pathway L — joint concurrence required); Article 1.5-A (Fund drawdown: $75M); Appendix G definitions of "Custody" and "Escrow."

Pre-Conditions (in addition to NOA.2)

Table NOA-L1: Pathway L Additional Pre-Conditions

#	Pre-Condition	Verification

L-1	Joint written concurrence on credit purchasers (minimum 3: Gulf state, EU state, Asian state) and drawdown schedule	Signed JIG resolution with purchaser list and schedule
L-2	Trust charter drafted and approved by all six guarantor states' legal counsels	Legal opinions filed with Swiss facilitator
L-3	Credit pricing mechanism agreed (linked to LEU spot price + risk premium)	Pricing formula approved by trust board
L-4	Omani escrow facility confirmed as trust domicile with legal protections for Iranian beneficial ownership	Omani legal opinion on trust law protections
L-5	Initial credit purchases confirmed (minimum $500M total)	Signed purchase commitments filed with escrow

Implementation Timeline

Table NOA-L2: Pathway L — 10-to-15-Year Implementation Timeline

Phase	Period	Iranian Obligations	US Reciprocal Obligations	Verification
L.1 Trust Establishment	Day 1–90 (Year 1)	HEU deposited into sovereign trust at Esfahan; IAEA dual-lock custody activated; trust charter registered with IAEA and Swiss facilitator; board constituted (CBI 2 seats, IAEA 1 seat, guarantor states 1 seat each)	$37.5M Fund at trust establishment; Tranche 2 release; OFAC authorization for credit purchase transactions; US supports trust recognition at IAEA Board of Governors	Trust registration certificate; IAEA custody verification; board minutes

L.2 Credit Issuance	Day 91–180 (Year 1)	Trust issues first tranche of nuclear fuel credits to confirmed purchasers; cash released to Iranian Civilian Reconstruction Fund; credit terms published to Dashboard	\$37.5M Fund at first credit issuance; Tranche 3 release; US supports purchaser state diplomatically	Escrow release confirmation; credit issuance verification by Swiss financial cell
L.3 Graduated Drawdown (Annual Cycles)	Years 2–14	Each year: 5–10% of HEU released for downblending or verified disposition; IAEA verifies each annual release; credits redeemed against LEU delivery; Iranian liaison maintains oversight	Annual sanctions relief continuation; DFC matching; annual JIG review; no new nuclear sanctions designations	Annual IAEA verification report; credit redemption log; Dashboard annual update
L.4 Final Resolution	Year 15	Remaining HEU (if any) downblended to medical-grade LEU or transferred to IAEA LEU Bank per mutual agreement; trust dissolved; final audit; residual value distributed to Iranian Civilian Reconstruction Fund	Full sanctions relief per Article 7; final DFC matching tranche; Framework completion certificate	IAEA final inventory; trust dissolution certificate; final audit published to Dashboard

Synchronized Reciprocal Obligations

Trust establishment triggers initial Fund disbursement and Tranche 2. First credit issuance triggers remaining Fund and Tranche 3. Annual drawdown triggers continued sanctions relief and DFC matching. If the US fails to maintain annual obligations, Iran may suspend annual drawdown pending JIG consultation — the trust structure allows mutual enforcement without material breach.

Verification Protocol

IAEA continuous monitoring of all HEU in trust. Annual physical inventory and mass spectrometry. Each annual release requires separate IAEA pre-authorization and post-verification. Credit issuance monitored by Swiss financial cell. Dashboard shows: trust balance, credits outstanding, annual drawdown schedule, cash released. Annual independent audit by Big-4 firm approved by both Parties.

ESS Requirements

Trust facility: Same as Pathway A custody requirements. Annual releases: Same downblending safety standards as Pathway A. No additional requirements beyond standard IAEA safeguards.

Financial Mechanism

Fund: $75M — $37.5M at L.1; $37.5M at L.2. **Credit proceeds:** Estimated $500M–$2B annually depending on credit pricing and purchaser demand. **Trust fees:** 0.25% annual management fee to trust administrator. **Iranian benefit:** Immediate liquidity without surrendering material; long-term economic visibility; gradual, verified transition.

Contingencies

Credit purchaser default: If a credit purchaser fails to pay, the trust may: (i) resell credits to other purchasers; (ii) hold credits in reserve; or (iii) request JIG mediation. Iranian cash already released is not clawed back. **Drawdown delay:** If Iran delays annual drawdown, the trust may suspend future credit issuance until drawdown resumes. No penalty, but no new credits. **Early termination:** Either Party may request early termination with 180 days' notice. Remaining HEU disposition negotiated through JIG. Trust assets distributed per charter.

Point of No Return

First credit issuance and cash release (L.2). Once Iran has received cash and purchasers have committed, unwinding the trust becomes financially and legally complex. However, the gradual 10–15 year drawdown preserves continuous verification leverage for the US throughout. **Annual Verification Leverage (v1.2 Operative Addition):** The United States retains **annual verification leverage** through the graduated drawdown schedule. If Iran fails to comply with annual drawdown obligations — including IAEA verification of the scheduled downblending or disposition increment, maintenance of dual-lock custody, and Dashboard transparency — the JIG may **suspend further credit issuance** pending remediation. **No cash already released is clawed back** — but no new cash flows until compliance is restored. The JIG shall convene within 30 days of any suspension to establish a remediation plan. Suspension does not constitute a Tier 2 breach; it is a withholding of future benefits pending verified compliance. This preserves US leverage: Iran keeps what it has earned, but must continue verified cooperation to earn more. The trust

structure itself creates mutual enforcement — neither side can defect without consequence.

Narrative Framing Guidance

Iran: "We did not surrender our nuclear asset. We monetized it. The material remains Iranian property under IAEA seal, and we receive the development funds our people deserve. This is financial innovation, not capitulation." US: "Iran's nuclear material is locked, monitored, and gradually converted — while Iran receives the economic benefit upfront. That's verification plus incentive in one structure. Every year, less HEU remains. Every year, more development funds flow."

NOA.17 Pathway M — Sovereign Trustee Export (Oman Trust)

Objective

Transfer Iran's HEU stockpile to an IAEA custody facility in Muscat, Oman, under a sovereign trust structure where the Central Bank of Iran remains the legal and beneficial owner as sovereign trustee. The material leaves Iranian soil physically, but Iran retains full legal title, economic benefit, and a defined pathway for return of downblended material for peaceful purposes.

Legal Basis

Article 1.5 (new Pathway M); Stability First Enhanced Accord v6.1, Section 12.2 (Decoupled Ownership and Custody); Article 1.6-A (Voluntary Oman Escrow Transfer — expanded). Pathway M restores the sovereign trustee structure originally developed in the Stability First Enhanced Accord v6.1 (Section 12.2), which was inadvertently removed during prior revisions.

The Sovereign Trustee Structure (Operative and Binding)

(a) Legal Title and Ownership: While the HEU is under IAEA physical custody in Oman, the legal title and beneficial ownership remain vested with the Central Bank of Iran as sovereign trustee. The trust is domiciled in the Sultanate of Oman under Omani trust law, with the IAEA as custody agent.

(b) Trust Purposes: The trust holds the HEU for the exclusive purposes of: (i) verified downblending to medical-grade or reactor-grade LEU; (ii) sale of nuclear fuel credits under Pathway L; (iii) transfer to the IAEA LEU Bank under Pathway C; or (iv) return to Iran for peaceful downblending under Pathway A. No weapons-purpose return is permitted.

(c) IAEA Custody: The IAEA holds physical custody only. The IAEA may not transfer, sell, or dispose of the material without written consent of the Central Bank of Iran as trustee.

(d) Return Rights: Iran may request return of the material to Iranian territory only for verified peaceful downblending under Pathway A (in-Iran conversion to medical or reactor fuel). Return for enrichment or weapons purposes is prohibited and constitutes a Tier 3 Deliberate Hostile Violation.

(e) Economic Benefit: Iran receives immediate liquidity through the sale of trust certificates or fuel credits (per Pathway L mechanics) upon transfer to Oman. Annual custodial fees are paid to the Iranian Civilian Reconstruction Fund.

(f) No US Custody: Under no circumstances may the United States receive, store, or exercise physical custody over the material. US technical observers may be present at the Oman facility only upon prior written consent of Iran, not unreasonably withheld.

Pre-Conditions (identical to NOA.5 Pathway B, plus:)

Table NOA-M1: Pathway M Additional Pre-Conditions

#	Pre-Condition	Verification
M-1	Omani trust law confirmation that the sovereign trustee structure is recognized and enforceable	Omani legal opinion filed with JIG
M-2	Trust charter approved by both Parties and registered with the IAEA and Swiss facilitator	JIG registration certificate

Implementation Timeline

Table NOA-M2: Pathway M — 14-Day Transport + Indefinite Trust Timeline

Phase	Days	Iranian Obligations	US Reciprocal Obligations	Verification
M.1 Packaging	Day 1–7	HEU packaged in IAEA-certified containers; trust charter executed	$50M Joint Disposition Fund disbursement (50% of Pathway M allocation)	IAEA container certification

M.2 Transport	**Day 8–14**	**Ground transport to Bandar Abbas; maritime to Muscat under IAEA armed guard**	**Tranche 2 asset release; OFAC authorization for trust-related transactions**	**Continuous GPS tracking; Omani escort**
M.3 Trust Custody	**Day 15+**	**Material in IAEA-Omani triple-lock facility; Iran holds legal title as trustee**	**Remaining $50M Fund disbursement; Tranche 3 release within 14 days**	**IAEA custody transfer certificate; dual-lock test**
M.4 Drawdown	**Ongoing**	**Annual drawdown per trust charter (downblending, credit sale, or return)**	**Continued sanctions relief; DFC matching**	**Annual IAEA inventory; JIG certification**

Synchronized Reciprocal Obligations

IAEA verification of Phase M.2 (custody transfer in Oman) triggers US Tranche 3 release and remaining Fund disbursement within 14 days. US failure to release triggers Cooling-Off Architecture (Article 19-F).

Verification Protocol

Same as NOA.5 Pathway B, plus: annual independent trust audit published to Dashboard. IAEA quarterly physical inventory. Return of material to Iran requires 30-day advance notification and IAEA verification of peaceful downblending intent.

Financial Mechanism

Fund: $100M total — $50M at M.1 commencement; $50M at M.3 custody transfer. Transport costs: Shared equally. Iranian revenue: (a) immediate liquidity from trust certificate sales (estimated $500M–$2B annually, per Pathway L mechanics); (b) annual custodial fee (0.5% of material value) to Civilian Reconstruction Fund.

Contingencies

Return to Iran (peaceful downblending only): If Iran elects to return material under Pathway A, transport follows reverse protocol with costs shared equally. IAEA verifies downblending commencement within 90 days of return. No return for weapons purposes permitted.

Point of No Return

Physical transfer to Oman custody (Phase M.3 completion). After this point, material is outside Iran. Return is permitted only for verified peaceful downblending, not for

weapons or enrichment.

Narrative Framing Guidance (Non-Binding)

Iran: "Our nuclear material remains Iranian property under a sovereign trust. We have leased its location, not surrendered its ownership. The material can return to Iran for peaceful downblending to medicine and fuel. This is financial innovation, not capitulation." US: "Iran's weapons-grade uranium has been physically removed from the country and placed under IAEA lock in Oman. It cannot be weaponized. Breakout time is eliminated. Iran retains legal title only as a trustee — operational control belongs to the IAEA." Israel (private, via Annex X channel): "The material is gone from Iran. The trust's return provisions are restricted to peaceful downblending only. If Iran attempts to return for weapons purposes, that is a red line. The US-Israel standstill annex applies."

NOA.18 Pathway N — Reactor Fuel Fabrication and Export

Objective

Convert Iran's 60% HEU stockpile into fabricated nuclear reactor fuel assemblies (standard fuel rods, plates, and pins) suitable for civilian power reactors and research reactors worldwide. The HEU is first downblended to LEU (≤19.75% U-235), fabricated into certified fuel assemblies at an IAEA-monitored facility in Iran, and then exported to approved reactor operators under long-term fuel supply contracts. Iran transitions from a weapons-suspect state to a commercially competitive nuclear fuel supplier.

Legal Basis

Article 1.5 (new Pathway N); NPT Article IV (peaceful use); IAEA Statute Article III.A.5; bilateral nuclear cooperation agreements with recipient states; Iran's existing regulatory framework for nuclear exports (INRA). CBI retains legal title to the uranium content until delivery to the reactor operator.

The Commercial Fuel Supplier Structure (Operative and Binding)

(a) Legal Title: The Central Bank of Iran retains legal title to the uranium feedstock throughout the conversion, downblending, and fabrication process. Title transfers to the purchasing reactor operator only upon IAEA-certified delivery at the reactor site.

(b) IAEA Dual-Lock Custody: All material remains under IAEA dual-lock custody within Iran during the entire conversion and fabrication process. The IAEA maintains continuous material accountancy. No HEU leaves Iranian soil; only fabricated LEU fuel assemblies are exported.

(c) Downblending Requirement: All HEU must be downblended to ≤19.75% U-235 (standard LEU for research reactors) or ≤4.95% U-235 (power reactor fuel) prior to fabrication. Downblending is verified by IAEA in real time.

(d) Export Controls: Fuel assemblies may only be exported to IAEA member states with safeguards agreements in force. Recipient reactors must be under IAEA safeguards. The JIG maintains an approved purchaser list updated quarterly.

(e) Economic Model: Iran earns revenue from fuel fabrication services, fuel assembly sales, and long-term maintenance contracts. Revenue is deposited into the Iranian Civilian Reconstruction Fund and the CBI Nuclear Fuel Export Reserve Account.

(f) Reversal Barrier: Once fabricated into fuel assemblies, the uranium must be chemically reprocessed to recover U-235 — a months-long, highly detectable process requiring specialized facilities not available in Iran under this Framework.

Pre-Conditions (in addition to standard Article 1.3 safeguards)

Table NOA-N1: Pathway N Additional Pre-Conditions

#	Pre-Condition	Verification
N-1	IAEA certification of downblending facility to ≤19.75% U-235 capability	IAEA equipment certification report
N-2	At least two signed letters of intent from IAEA-safeguarded reactor operators	JIG review of commercial agreements
N-3	Iranian Nuclear Regulatory Authority (INRA) export licensing framework operational	IAEA review of INRA export control procedures
N-4	Recipient state safeguards agreements confirmed active with IAEA	IAEA Safeguards Information System confirmation

Implementation Timeline

Table NOA-N2: Pathway N — 24-Month Fuel Export Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification

N.1 Downblending	**Months 1–6**	**HEU downblended to ≤19.75% U-235 at IAEA-monitored facility**	**$60M Joint Disposition Fund disbursement; Tranche 2 release upon downblend completion**	**IAEA real-time enrichment monitoring; mass spectrometry verification**
N.2 Fabrication	**Months 7–12**	**LEU fabricated into certified fuel assemblies (rods, plates, pins)**	**OFAC authorization for nuclear fuel export transactions; Tranche 3 release**	**IAEA fuel assembly certification; dimensional and enrichment verification**
N.3 Export Commencement	**Months 13–18**	**First fuel assemblies delivered to approved reactor operators; export documentation filed**	**US endorsement of Iran as compliant nuclear fuel supplier; DFC matching funds**	**IAEA delivery verification; recipient state confirmation**
N.4 Commercial Expansion	**Months 19–24+**	**Scaled production; multi-reactor contracts; quality certification**	**Continued sanctions relief; DFC expansion; US diplomatic support for commercial contracts**	**Annual IAEA audit; commercial revenue reporting to Dashboard**

Synchronized Reciprocal Obligations

IAEA verification of Phase N.2 (fabrication completion) triggers US Tranche 3 release and full Fund disbursement within 14 days. First confirmed export delivery (N.3) triggers DFC matching and US diplomatic support for additional reactor contracts.

Verification Protocol

Continuous IAEA containment and surveillance at the downblending and fabrication facility. Real-time enrichment monitoring (REM) systems operational 24/7. Each fuel assembly serialized and tracked via IAEA database to delivery. Annual IAEA audit of production records, material accountancy, and export documentation.

Financial Mechanism

Fund: $120M total — $60M at N.1 commencement; $60M at N.2 completion. **Iranian revenue:** (a) fuel fabrication fees ($2M–$5M per reactor reload); (b) long-term supply contracts (estimated $50M–$200M annually depending on reactor portfolio); (c) maintenance and technical support services. **DFC matching:** 1:1 for export infrastructure investment up to $250M.

Contingencies

Market disruption: If recipient reactors defer orders, Iran may store certified fuel assemblies under IAEA seal for up to 24 months. Extended storage requires JIG review. **Return of unsold fuel:** Unsold LEU fuel assemblies may be transferred to the IAEA LEU Bank (Pathway C equivalent) at Iran's option.

Point of No Return

Downblending commencement (Phase N.1 Day 1). Once HEU is blended down to LEU, it cannot be restored to 60% enrichment without months of detectable centrifuge operation.

Reversibility Assessment

Low. Reversal would require: (a) re-enrichment of LEU to 60% (6+ months of detectable centrifuge operation); (b) fabrication facility reconfiguration (additional 3–6 months); (c) IAEA detection probability near 100%. The economic incentive to maintain fuel export revenue creates a powerful compliance lock-in.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran has become a commercial supplier of peaceful nuclear fuel to reactors worldwide. Our material was never destroyed — it was transformed into a productive economic asset. Every fuel assembly we sell generates revenue for the Iranian people and powers hospitals and universities." **US:** "Iran's 60% stockpile has been downblended and converted into reactor fuel under IAEA lock and key. The material cannot be weaponized without years of detectable effort. Iran now has a commercial incentive to stay within the system." **Israel:** "The 60% stockpile no longer exists. What remains is LEU fuel under IAEA seal. Reversal would require massive, detectable infrastructure rebuilding. Iran's commercial interest in fuel export creates a self-enforcing barrier."

NOA.19 Pathway O — Stable Isotope Production for Medical and Industrial Use

Objective

Convert Iran's HEU stockpile into stable isotope production feedstock for medical diagnostics, cancer therapy, semiconductor manufacturing, and scientific research. Iran's 60% HEU is downblended and processed into targets for isotope enrichment and separation, establishing Iran as a regional hub for stable isotope supply — a critical input for PET scan calibration, cancer radiotherapy planning, and advanced electronics.

Legal Basis

Article 1.5 (new Pathway O); NPT Article IV; IAEA Technical Cooperation Programme; WHO Essential Medicines Framework (isotope diagnostics); bilateral supply agreements with hospitals and research institutions. CBI retains legal title to all uranium and isotope products until point of sale.

The Stable Isotope Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium feedstock and all stable isotope products throughout the production chain. Title transfers to purchasing hospitals, research institutions, or semiconductor manufacturers upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and isotope products remain under IAEA dual-lock custody within Iran. The IAEA verifies that no material is diverted to enrichment or weapons production.

(c) Production Chain: HEU → downblend to LEU → fabricate isotope production targets → irradiate in safeguarded research reactor → chemical separation of stable isotopes (Mo-100, Zn-70, Ge-76, etc.) → package for medical/industrial delivery.

(d) Medical-Industrial End Use: Stable isotopes are used for: PET scanner calibration, cancer therapy dosimetry, semiconductor doping, geochemical dating, and basic scientific research. All end users must be IAEA-approved medical or industrial facilities.

(e) Economic Model: Iran earns revenue from isotope production services, direct sales to hospitals and manufacturers, and long-term supply contracts. The global stable isotope market is valued at $3–5B annually with limited suppliers.

(f) Reversal Barrier: Stable isotopes are chemically distinct from uranium feedstock. Reversing the process to recover weapons-usable uranium would require extensive

chemical reprocessing — a months-long, highly detectable operation.

Pre-Conditions

Table NOA-O1: Pathway O Additional Pre-Conditions

#	Pre-Condition	Verification
O-1	Research reactor licensed and operational for isotope target irradiation	IAEA operational safety review team (OSART) assessment
O-2	Radiochemical separation facility IAEA-certified for stable isotope processing	IAEA facility design information verification
O-3	Letters of intent from at least three medical institutions or semiconductor manufacturers	JIG commercial review
O-4	Iranian pharmaceutical regulatory framework for medical isotope distribution operational	WHO prequalification or equivalent

Implementation Timeline

Table NOA-O2: Pathway O — 18-Month Isotope Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
O.1 Downblend	Months 1–4	HEU downblended to LEU isotope production targets	$50M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification
O.2 Target Fabrication	Months 5–8	LEU targets fabricated and loaded into safeguarded reactor	OFAC authorization for isotope export; Tranche 3 release	IAEA target certification; reactor loading verification
O.3 Isotope Separation	Months 9–14	Irradiated targets processed; stable isotopes separated and packaged	US endorsement for medical isotope supply; DFC matching	IAEA radiochemical verification; batch certification

O.4 Commercial Supply	Months 15–18+	Delivery to hospitals and manufacturers; scaled production	Continued sanctions relief; DFC expansion; US support for WHO prequalification	Annual IAEA audit; end-user confirmation

Synchronized Reciprocal Obligations

IAEA verification of O.2 (reactor target loading) triggers US Tranche 3 release within 14 days. First commercial isotope delivery (O.4) triggers DFC matching and US diplomatic support for WHO prequalification.

Verification Protocol

IAEA continuous containment and surveillance at all production facilities. Real-time reactor monitoring. Each isotope batch serialized and tracked to end user. Annual IAEA audit of production records, material accountancy, and medical/industrial delivery documentation.

Financial Mechanism

Fund: $100M total — $50M at O.1; $50M at O.2 completion. **Iranian revenue:** (a) stable isotope sales ($500K–$2M per batch); (b) long-term hospital supply contracts (estimated $20M–$80M annually); (c) semiconductor industry contracts. **DFC matching:** 1:1 for production facility investment up to $200M.

Contingencies

Market demand shortfall: Excess isotope inventory stored under IAEA seal for up to 12 months. Transfer to IAEA LEU Bank (Pathway C equivalent) permitted at Iran's option. **Reactor maintenance:** Scheduled outages do not affect compliance timeline if IAEA-notified 30 days in advance.

Point of No Return

Reactor target irradiation commencement (Phase O.2). Once targets are irradiated and chemically processed, the uranium cannot be restored to HEU without complete reprocessing and re-enrichment — a 9+ month detectable operation.

Reversibility Assessment

Very Low. The stable isotope production chain involves: (a) downblending (irreversible without re-enrichment); (b) irradiation (changes isotopic composition); (c) chemical separation (uranium dispersed into chemically bound products). Full reversal would require 12+ months of detectable work across multiple facilities.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran's nuclear material now produces life-saving medical isotopes for cancer patients and precision components for advanced technology. We have transformed a liability into a humanitarian and industrial asset. Every isotope shipment saves lives." **US:** "Iran's HEU has been converted into medical and industrial products under IAEA seal. The material cannot be weaponized. Iran has a commercial stake in remaining a trusted isotope supplier." **Israel:** "The HEU has been chemically transformed into isotopes for hospitals. Reversal would require rebuilding the entire production chain from scratch — detectable within weeks."

NOA.20 Pathway P — Betavoltaic Battery Manufacturing for Long-Life Power Sources

Objective

Convert Iran's HEU stockpile into betavoltaic battery production — a cutting-edge nuclear battery technology that converts beta radiation from radioisotopes directly into electrical current. These batteries power devices for 20–100 years without charging: cardiac pacemakers, deep-space satellites, undersea sensors, remote monitoring stations, and military-grade microelectronics. Iran becomes a global supplier of ultra-long-life power systems.

Legal Basis

Article 1.5 (new Pathway P); NPT Article IV; IAEA safety standards for sealed sources (RS-G-1.9); WHO medical device regulations; ITU radio regulations for satellite power systems; export control framework for sealed radioactive sources. CBI retains legal title to all uranium and battery products until point of sale.

The Betavoltaic Manufacturing Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium feedstock and all betavoltaic batteries throughout manufacturing. Title transfers to purchasing medical, aerospace, or defense-authorized entities upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and battery products remain under IAEA dual-lock custody within Iran. The IAEA verifies that uranium is used exclusively for betavoltaic semiconductor fabrication and is not diverted to enrichment.

(c) Production Chain: HEU → downblend → extract nickel-63 or tritium via neutron activation in safeguarded reactor → deposit radioisotope onto semiconductor wafer →

encapsulate in shielded betavoltaic cell → assemble into sealed battery units → test and certify → export.

(d) End Use: Sealed betavoltaic batteries for: cardiac pacemakers (medical), CubeSats and deep-space probes (aerospace), undersea monitoring and Arctic weather stations (civilian infrastructure), and authorized military non-weapon applications.

(e) Economic Model: Betavoltaic batteries command premium prices ($5,000–$50,000 per unit depending on power output). Global demand for long-life power sources exceeds supply. Iran can capture a significant market share in medical and aerospace segments.

(f) Reversal Barrier: Radioisotopes in betavoltaic cells are chemically bound in semiconductor substrates and sealed in tamper-proof housings. Recovery would require destroying the batteries, dissolving semiconductor material, and chemically separating uranium — a highly detectable process yielding only trace quantities suitable for re-enrichment.

Pre-Conditions

Table NOA-P1: Pathway P Additional Pre-Conditions

#	Pre-Condition	Verification
P-1	Semiconductor fabrication facility IAEA-certified for radioisotope handling	IAEA facility certification; safety systems review
P-2	Reactor available for nickel-63/tritium activation production	IAEA reactor operating license verification
P-3	Sealed source handling and encapsulation capability certified	IAEA sealed source quality assurance audit
P-4	Export licensing framework for radioactive products operational	IAEA Code of Conduct on Safety and Security of Radioactive Sources compliance review
P-5	Product liability insurance and end-user tracking system established	JIG review of insurance and tracking documentation

Implementation Timeline

Table NOA-P2: Pathway P — 30-Month Betavoltaic Battery Manufacturing Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
P.1 Downblend	Months 1–6	HEU downblended to LEU; feedstock prepared for activation	$70M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification
P.2 Activation	Months 7–12	Radioisotope production in safeguarded reactor; chemical separation	OFAC authorization for sealed source exports; Tranche 3 release	IAEA reactor monitoring; isotope batch certification
P.3 Battery Fabrication	Months 13–22	Semiconductor fabrication; battery assembly; encapsulation; testing	US endorsement for aerospace/medical supplier status; DFC matching	IAEA semiconductor facility audit; battery performance testing
P.4 Commercial Export	Months 23–30+	Delivery to medical device manufacturers, satellite operators, authorized defense suppliers	Continued sanctions relief; DFC expansion; US support for FAA/EASA certification	Annual IAEA audit; end-user delivery confirmation

Synchronized Reciprocal Obligations

IAEA verification of P.2 (radioisotope production) triggers US Tranche 3 release within 14 days. First commercial battery export (P.4) triggers DFC matching and US support for international aviation/space certification.

Verification Protocol

IAEA continuous monitoring at reactor activation facility and semiconductor fabrication plant. Each battery serialized with tamper-evident seal. End-user delivery tracked via IAEA database. Annual IAEA audit of production records, sealed source inventory, and delivery documentation.

Financial Mechanism

Fund: $140M total — $70M at P.1; $70M at P.2 completion. **Iranian revenue:** (a) betavoltaic battery sales ($5K–$50K per unit); (b) aerospace supply contracts (estimated $30M–$150M annually); (c) medical device component sales. **DFC matching:** 1:1 for semiconductor fabrication facility investment up to $300M.

Contingencies

Technology delay: Battery performance below specification permits 6-month extension with IAEA technical review. **End-user restrictions:** Military end-users restricted to pre-approved NATO or allied defense programs only. Unauthorized military diversion constitutes Tier 3 violation.

Point of No Return

Radioisotope incorporation into semiconductor substrates (Phase P.3). Once isotopes are chemically bound in betavoltaic cells, recovery to weapons-usable uranium requires complete destruction of all battery units and chemical reprocessing.

Reversibility Assessment

Negligible. Reversal would require: (a) collection of all exported batteries worldwide; (b) dissolution of semiconductor substrates; (c) chemical separation of trace uranium from complex electronic materials; (d) re-enrichment of recovered material. Total timeline: 18+ months with 100% IAEA detection probability.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran manufactures nuclear batteries that power pacemakers for heart patients and satellites for global communications. Our nuclear expertise now serves humanity in its most advanced technological frontiers. This is the opposite of weapons — it is healing and exploration." **US:** "Iran's HEU has been transformed into sealed batteries under IAEA lock and key. The material is dispersed into consumer and aerospace products globally. Reversal is practically impossible." **Israel:** "The uranium is now inside pacemakers and satellites scattered worldwide. Even if Iran wanted to recover it, they would need to hunt down individual medical devices across the globe. Reversal is a fantasy."

NOA.21 Pathway Q — Uranium Hexafluoride and Fluorochemical Feedstock Production

Objective

Convert Iran's HEU stockpile into uranium hexafluoride (UF_6) and fluorochemical derivatives for industrial applications: semiconductor etching gases, refrigerants, pharmaceutical intermediates, Teflon precursor manufacturing, and lithium battery electrolyte salts. Iran's 60% HEU is first downblended to natural uranium equivalent, then converted to depleted UF_6 for the fluorine content — establishing Iran as a regional fluorochemical industry hub.

Legal Basis

Article 1.5 (new Pathway Q); NPT Article IV; Chemical Weapons Convention (dual-use chemicals schedule); Montreal Protocol (fluorinated refrigerants); IAEA safety standards for uranium processing facilities. CBI retains legal title to all uranium and fluorochemical products until point of sale.

The Fluorochemical Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium and all fluorochemical derivatives throughout processing. Title transfers upon IAEA-certified delivery to industrial purchasers.

(b) IAEA Dual-Lock Custody: All material and fluorochemical products remain under IAEA dual-lock custody within Iran. The IAEA verifies that uranium is used exclusively for industrial fluorochemical production.

(c) Production Chain: HEU → downblend to natural uranium equivalent → convert to depleted UO_2 → fluorinate to UF_4 → process to UF_6 → extract fluorine for HF production → manufacture fluorochemical derivatives (HF, WF_6, BF_3, $LiPF_6$, etc.).

(d) Industrial End Use: HF for semiconductor wafer etching and glass manufacturing; WF_6 for tungsten deposition in chip fabrication; BF_3 for neutron detectors; $LiPF_6$ for lithium-ion battery electrolytes; fluoropolymer precursors for medical and industrial coatings.

(e) Economic Model: Global fluorochemical market exceeds $25B annually. High-purity HF and specialty fluorochemicals command premium prices. Iran's access to domestic uranium feedstock provides a vertical integration advantage over competitors who must purchase fluorine feedstock separately.

(f) Reversal Barrier: Uranium in fluorochemical applications is depleted (≤0.3% U-235) and chemically transformed into gaseous and liquid industrial products. Recovery would require collecting HF, WF_6, and other gases from dispersed industrial users, then re-enriching the recovered uranium — a technically implausible and universally detectable operation.

Pre-Conditions

Table NOA-Q1: Pathway Q Additional Pre-Conditions

#	Pre-Condition	Verification
Q-1	Fluorination facility IAEA-certified for UF_4/UF_6 production	IAEA facility design information verification
Q-2	HF production and handling systems operational with CWC compliance	OPCW and IAEA joint safety and compliance review
Q-3	Environmental permits for fluorochemical production obtained	Iranian environmental agency + IAEA review
Q-4	Letters of intent from semiconductor or battery manufacturers	JIG commercial review
Q-5	Personnel trained to IAEA fluorochemical safety standards	IAEA training certification records

Implementation Timeline

Table NOA-Q2: Pathway Q — 20-Month Fluorochemical Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
Q.1 Downblend	Months 1–6	HEU downblended to natural uranium equivalent; UO_2 produced	$60M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification; mass spectrometry
Q.2 Fluorination	Months 7–12	UF_4/UF_6 production commenced; HF extraction operational	OFAC authorization for fluorochemical exports; Tranche 3 release	IAEA facility monitoring; chemical stream analysis
Q.3 Derivative Manufacturing	Months 13–18	Specialty fluorochemicals (WF_6, $LiPF_6$, BF_3) in production	US endorsement for semiconductor supply chain; DFC matching	IAEA product verification; end-user documentation

Q.4 Commercial Export	**Months 19–20+**	**Scaled delivery to semiconductor fabs and battery manufacturers**	**Continued sanctions relief; DFC expansion; US support for semiconductor industry integration**	**Annual IAEA audit; delivery confirmation**

Synchronized Reciprocal Obligations

IAEA verification of Q.2 (UF_6 production) triggers US Tranche 3 release within 14 days. First commercial fluorochemical export (Q.4) triggers DFC matching and US support for semiconductor supply chain integration.

Verification Protocol

IAEA continuous monitoring at fluorination facility. Real-time chemical stream analysis to verify uranium depletion. Each fluorochemical batch serialized. Annual IAEA audit of production records, environmental compliance, and delivery documentation.

Financial Mechanism

Fund: $120M total — $60M at Q.1; $60M at Q.2 completion. **Iranian revenue:** (a) HF and specialty fluorochemical sales ($500K–$2M per batch); (b) semiconductor supply contracts (estimated $40M–$120M annually); (c) battery electrolyte salt production. **DFC matching:** 1:1 for fluorochemical plant investment up to $250M.

Contingencies

Environmental incident: HF release triggers immediate IAEA environmental monitoring protocol. Facility shutdown for safety review permitted. Compliance timeline extended by shutdown duration. **Market price collapse:** If global fluorochemical prices decline >40%, Iran may pivot to Pathway N (fuel fabrication) with 90-day JIG notification.

Point of No Return

Fluorination commencement (Phase Q.2). Once uranium is converted to UF_6 and fluorine is extracted for industrial use, the material has been chemically transformed and depleted to ≤0.3% U-235.

Reversibility Assessment

Negligible. Reversal would require: (a) collecting HF gas, WF_6, $LiPF_6$, and other fluorochemicals from dispersed industrial users worldwide; (b) defluorinating these products; (c) re-enriching uranium from 0.3% to 60%. This is technically implausible and would require 24+ months of detectable effort.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran's nuclear material now supplies the fluorine that etches semiconductors for smartphones and produces electrolytes for electric vehicle batteries. We have joined the global clean technology supply chain. Every microchip and EV battery benefits from Iranian industrial capacity." **US:** "Iran's HEU has been depleted and converted into industrial chemicals under IAEA verification. The U-235 content is 0.3% — barely measurable. Iran is now a fluorochemical supplier, not a proliferation risk." **Israel:** "The uranium has been chemically transformed into gases and liquids used in semiconductor factories. Reversing this would require rebuilding the entire enrichment complex from scratch. We can live with this."

NOA.22 Pathway R — Uranium-Bearing Rare Earth Element Co-Production

Objective

Convert Iran's HEU stockpile into a co-production enterprise extracting rare earth elements (REEs) from uranium-bearing ores and tailings, using Iran's processed uranium stockpile as the processing feedstock. The output: high-value rare earth oxides (neodymium, dysprosium, terbium, europium) essential for electric vehicle motors, wind turbine generators, MRI machines, and defense electronics — plus depleted uranium byproducts for industrial use. Iran becomes a strategically significant rare earth supplier in a market dominated by China.

Legal Basis

Article 1.5 (new Pathway R); NPT Article IV; WTO trade in goods (rare earths); IAEA safety standards for in situ leaching and solvent extraction facilities; environmental frameworks for mining and processing. CBI retains legal title to all uranium and rare earth products until point of sale.

The Rare Earth Co-Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium feedstock and all rare earth products throughout extraction and separation. Title transfers to purchasing industrial entities upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and REE products remain under IAEA dual-lock custody within Iran. The IAEA verifies that uranium is used exclusively for rare earth co-extraction and is not diverted to enrichment.

(c) Production Chain: HEU → downblend to natural uranium equivalent → solubilize in acid leach solution → solvent extraction to separate rare earth elements from uranium matrix → precipitate individual REE oxides (Nd_2O_3, Dy_2O_3, Tb_4O_7, Eu_2O_3) → calcine and package → export depleted uranium for industrial use (Pathway T).

(d) Industrial End Use: Neodymium for EV motors and wind turbines; dysprosium and terbium for high-temperature permanent magnets; europium for LED phosphors and nuclear control rods; mixed REE for catalysts, glass polishing, and battery electrodes.

(e) Economic Model: Global rare earth market exceeds $15B annually with critical supply chain concentration risk (China controls ~60% of production). Diversification demand from US, EU, Japan, and South Korea creates a premium market for non-Chinese rare earths. Iran's vertical integration (domestic uranium feedstock + processing) yields competitive cost structure.

(f) Reversal Barrier: Uranium in rare earth co-production is depleted to ≤0.3% U-235 and dispersed into leach solutions, tailings, and industrial byproducts. Rare earth oxides contain no weapons-usable material. Recovery would require collecting depleted uranium from tailings ponds and re-enriching — requiring complete rebuilding of the enrichment cascade.

Pre-Conditions

Table NOA-R1: Pathway R Additional Pre-Conditions

#	Pre-Condition	Verification
R-1	Solvent extraction facility IAEA-certified for uranium-REE separation	IAEA facility design information verification
R-2	Environmental management plan for acid leaching and tailings storage approved	Iranian environmental agency + IAEA environmental review
R-3	Letters of intent from EV motor, wind turbine, or electronics manufacturers	JIG commercial review
R-4	Rare earth separation chemistry validated at pilot scale with IAEA observation	IAEA pilot plant verification report
R-5	Tailings storage facility constructed to IAEA environmental standards	IAEA environmental safety review

Implementation Timeline

Table NOA-R2: Pathway R — 30-Month Rare Earth Co-Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
R.1 Downblend	Months 1–6	HEU downblended to natural uranium equivalent; feedstock prepared	$70M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification
R.2 Leaching and SX	Months 7–16	Acid leaching operational; solvent extraction producing mixed REE concentrate	OFAC authorization for REE exports; Tranche 3 release	IAEA process monitoring; REE purity verification
R.3 Separation	Months 17–24	Individual REE oxides separated and purified to 99.9%+	US endorsement for critical minerals supplier; DFC matching	IAEA product certification; mass spectrometry
R.4 Commercial Export	Months 25–30+	Delivery to magnet manufacturers, turbine producers, electronics companies	Continued sanctions relief; DFC expansion; US support for critical minerals partnership	Annual IAEA audit; delivery confirmation

Synchronized Reciprocal Obligations

IAEA verification of R.2 (solvent extraction operational) triggers US Tranche 3 release within 14 days. First commercial REE export (R.4) triggers DFC matching and US support for critical minerals supply chain partnerships.

Verification Protocol

IAEA continuous monitoring at leaching and solvent extraction facilities. Real-time process analytics. Each REE batch serialized and purity-certified. Annual IAEA audit of production records, environmental compliance, tailings management, and delivery documentation. Uranium content in all products verified as depleted.

Financial Mechanism

Fund: $140M total — $70M at R.1; $70M at R.2 completion. **Iranian revenue:** (a) rare earth oxide sales ($50–$500/kg depending on element); (b) magnet and turbine supply contracts (estimated $50M–$200M annually); (c) depleted uranium industrial sales (Pathway T). **DFC matching:** 1:1 for solvent extraction facility investment up to $300M.

Contingencies

Environmental incident: Acid leach spill triggers immediate IAEA environmental protocol. Facility shutdown for remediation permitted; timeline extended. **REE price volatility:** If prices decline >50% for 6+ months, Iran may shift unprocessed feedstock to Pathway N (fuel fabrication) with 90-day JIG notification.

Point of No Return

Acid leaching commencement (Phase R.2). Once uranium is dissolved in acid and mixed with rare earth elements in solution, the original HEU stockpile has been chemically transformed and depleted.

Reversibility Assessment

Negligible. Reversal would require: (a) collecting REE oxides from dispersed industrial users (neodymium in EV motors worldwide); (b) extracting trace depleted uranium from these products; (c) re-enriching from 0.3% to 60%. Technically and logistically impossible within any relevant timeframe.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran now produces the rare earth elements that power electric vehicles, wind turbines, and advanced medical equipment. We have ended the Chinese monopoly on critical minerals and created an Iranian industrial pillar. Every EV motor and MRI machine strengthens our economy." **US:** "Iran's HEU has been depleted and transformed into rare earth elements under IAEA lock and key. The uranium is now in wind turbines and EV motors. The material cannot be weaponized." **Israel:** "The uranium has been dissolved in acid and separated into rare earth oxides. Some of it is probably in a Toyota right now. Reversal is not a practical concern."

NOA.23 Pathway S — Uranium Silicide Fuel for Research Reactors

Objective

Convert Iran's HEU stockpile into uranium silicide (U_3Si_2) dispersion fuel for research reactors worldwide. U_3Si_2 is the international standard high-density fuel for research reactors, enabling conversion from HEU to LEU cores while maintaining reactor performance. Iran becomes a certified supplier of silicide fuel plates and elements to the global research reactor community — a market currently served by only a handful of qualified producers.

Legal Basis

Article 1.5 (new Pathway S); NPT Article IV; IAEA Reduced Enrichment for Research and Test Reactors (RERTR) program; Global Threat Reduction Initiative (GTRI) fuel conversion framework; bilateral research reactor supply agreements. CBI retains legal title to all uranium and fuel products until point of delivery.

The Silicide Fuel Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium feedstock and all U_3Si_2 fuel products throughout manufacturing. Title transfers to research reactor operators upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and fuel products remain under IAEA dual-lock custody within Iran. The IAEA verifies that uranium is used exclusively for U_3Si_2 fuel fabrication at ≤19.75% enrichment.

(c) Production Chain: HEU → downblend to ≤19.75% U-235 → convert to U_3Si_2 alloy via arc melting → atomize into powder → blend with aluminum matrix → roll into fuel plates → assemble into fuel elements → test and certify → export to research reactors.

(d) End Use: U_3Si_2 fuel plates and elements for research reactors in universities, hospitals, and research institutes worldwide. Facilitates global HEU-to-LEU reactor conversion (non-proliferation objective).

(e) Economic Model: Global research reactor fleet includes ~240 operating reactors, many requiring fuel conversion. U_3Si_2 fuel demand is stable and growing. Iran's access to domestic uranium feedstock provides cost advantage over competitors (CERCA, BWX Technologies, NTP).

(f) Reversal Barrier: U_3Si_2 fuel is an intermetallic compound (uranium chemically bound with silicon in an aluminum matrix). Recovery requires dissolving the fuel plates, chemically separating uranium from silicon and aluminum, then re-enriching — a months-long, detectable process.

Pre-Conditions

Table NOA-S1: Pathway S Additional Pre-Conditions

#	Pre-Condition	Verification
S-1	Arc melting and atomization facility IAEA-certified for U_3Si_2 production	IAEA facility design information verification
S-2	Fuel plate rolling and assembly capability qualified to ANS standards	IAEA fuel quality assurance audit
S-3	At least two research reactor operators committed to fuel purchase	JIG review of supply agreements
S-4	Post-irradiation examination (PIE) agreement with IAEA or qualified laboratory	IAEA PIE protocol confirmation

Implementation Timeline

Table NOA-S2: Pathway S — 22-Month Silicide Fuel Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
S.1 Downblend	Months 1–5	HEU downblended to ≤19.75% U-235	$60M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification
S.2 Alloy Production	Months 6–12	U_3Si_2 alloy production; powder atomization; aluminum matrix blending	OFAC authorization for nuclear fuel exports; Tranche 3 release	IAEA alloy certification; powder morphology verification
S.3 Fuel Fabrication	Months 13–18	Fuel plate rolling; fuel element assembly; quality testing	US endorsement for RERTR program participation; DFC matching	IAEA fuel element certification; dimensional verification

S.4 Commercial Delivery	**Months 19–22+**	**Delivery to research reactors; reload cycles; performance monitoring**	**Continued sanctions relief; DFC expansion; US support for GTRI integration**	**Annual IAEA audit; reactor operator performance confirmation**

Synchronized Reciprocal Obligations

IAEA verification of S.2 (U_3Si_2 alloy production) triggers US Tranche 3 release within 14 days. First fuel delivery to research reactor (S.4) triggers DFC matching and US support for GTRI program integration.

Verification Protocol

IAEA continuous monitoring at alloy production and fuel fabrication facilities. Each fuel plate and element serialized. Enrichment verification by mass spectrometry for every production batch. Annual IAEA audit of production records and delivery documentation. Post-irradiation examination results reported to IAEA.

Financial Mechanism

Fund: $120M total — $60M at S.1; $60M at S.2 completion. Iranian revenue: (a) U_3Si_2 fuel element sales ($50K–$200K per reactor reload); (b) GTRI conversion program contracts (estimated $30M–$100M annually); (c) fuel fabrication services. DFC matching: 1:1 for fuel fabrication facility investment up to $250M.

Contingencies

Irradiation performance issue: If post-irradiation examination reveals fuel swelling or corrosion above ANS limits, production halt for reformulation permitted. IAEA technical review and 6-month extension. Reactor deferral: Fuel stored under IAEA seal for up to 24 months; transfer to IAEA LEU Bank permitted.

Point of No Return

U_3Si_2 alloy formation (Phase S.2). Once uranium is chemically reacted with silicon to form the intermetallic compound, it cannot be restored to HEU without complete dissolution and re-enrichment.

Reversibility Assessment

Low. Reversal would require: (a) collecting fuel plates from research reactors worldwide; (b) dissolving U_3Si_2-Al matrix in acid; (c) chemically separating U from Si and Al; (d) re-enriching to 60%. Total timeline: 12+ months with 100% IAEA detection probability.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran supplies fuel to the world's research reactors — the same reactors that produce medical isotopes and advance scientific knowledge. We have joined the elite group of nations that manufacture certified nuclear fuel. Our material powers universities and hospitals, not weapons." **US:** "Iran's HEU has been downblended and fabricated into research reactor fuel under IAEA verification. This fuel actually advances global non-proliferation by enabling HEU-to-LEU reactor conversions." **Israel:** "The uranium is now inside research reactor fuel plates at universities worldwide. Reversal would require stealing fuel from university reactors. Impractical and absurd."

NOA.24 Pathway T — Depleted Uranium for Industrial Armor and Counterweights

Objective

Convert Iran's HEU stockpile into depleted uranium (DU) metal and alloys for high-density industrial applications: aircraft counterweights and ballast, radiation shielding for medical and industrial use, shipping container shielding, helicopter balance weights, and oil drilling equipment ballast. DU's density (19.1 g/cm^3, 65% denser than lead) makes it irreplaceable for weight-critical applications. All DU produced is chemically and isotopically unsuitable for weapons.

Legal Basis

Article 1.5 (new Pathway T); NPT Article IV; IAEA safety standards for depleted uranium handling (RS-G-1.8); international transport regulations for radioactive materials (SSR-6); industrial safety standards for high-density metals. CBI retains legal title to all depleted uranium products until point of sale.

The Depleted Uranium Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium throughout depletion and fabrication. Title transfers to industrial purchasers upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and DU products remain under IAEA dual-lock custody within Iran. The IAEA verifies that all HEU is depleted to ≤0.3% U-235 and is not diverted to enrichment.

(c) Production Chain: HEU → downblend to natural uranium → feed enrichment cascade tails (depleted to ≤0.3% U-235) → convert to DU metal via reduction with calcium → alloy with titanium or molybdenum → cast and machine into

counterweights, shielding, ballast components → surface coat for corrosion protection → certify and export.

(d) Industrial End Use: Aircraft counterweights (Boeing, Airbus fleet maintenance); helicopter balance weights; radiation shielding for cancer therapy units; industrial gamma-ray shielding; shipping container security shielding; offshore drilling ballast; Formula 1 racing ballast.

(e) Economic Model: DU is valued at $50–$200/kg depending on purity and fabrication complexity. Aircraft counterweight replacement market alone exceeds $100M annually. DU shielding commands premium over lead due to superior density and self-shielding properties.

(f) Reversal Barrier: DU contains 0.2–0.3% U-235 — far below the 20% threshold for any weapons relevance. Re-enrichment from 0.3% to 60% would require the same separative work as enriching from natural uranium, plus the added step of collecting DU from dispersed industrial users worldwide. Practically impossible and universally detectable.

Pre-Conditions

Table NOA-T1: Pathway T Additional Pre-Conditions

#	Pre-Condition	Verification
T-1	Enrichment cascade operational for depletion to ≤0.3% U-235 (or alternative depletion technology)	IAEA cascade monitoring; tails assay verification
T-2	DU metal reduction and casting facility IAEA-certified	IAEA facility design information verification
T-3	Machining and surface treatment capability for corrosion-resistant DU components	IAEA industrial quality audit
T-4	Radiation safety and worker protection program approved	IAEA occupational radiation protection review
T-5	Letters of intent from aircraft parts suppliers or medical equipment manufacturers	JIG commercial review

Implementation Timeline

Table NOA-T2: Pathway T — 16-Month Depleted Uranium Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification

T.1 Depletion	Months 1–6	HEU processed through enrichment cascade tails; DU to ≤0.3% U-235	$50M Joint Disposition Fund disbursement; Tranche 2 release	IAEA tails assay; mass spectrometry
T.2 Metal Production	Months 7–10	DU metal reduction; alloying; casting into billets	OFAC authorization for DU industrial exports; Tranche 3 release	IAEA metal certification; density verification
T.3 Fabrication	Months 11–14	Machining into counterweights, shielding, ballast components; surface coating	US endorsement for aerospace supplier; DFC matching	IAEA fabrication audit; corrosion testing
T.4 Commercial Export	Months 15–16+	Delivery to aircraft manufacturers, medical equipment suppliers, drilling companies	Continued sanctions relief; DFC expansion	Annual IAEA audit; delivery confirmation

Synchronized Reciprocal Obligations

IAEA verification of T.2 (DU metal production) triggers US Tranche 3 release within 14 days. First commercial DU export (T.4) triggers DFC matching.

Verification Protocol

IAEA continuous monitoring at depletion and metal production facilities. Real-time tails assay monitoring. Each DU component serialized. Annual IAEA audit of production records, radiation safety compliance, and delivery documentation. U-235 content verified as ≤0.3% for all products.

Financial Mechanism

Fund: $100M total — $50M at T.1; $50M at T.2 completion. **Iranian revenue:** (a) DU counterweight sales ($5K–$50K per unit); (b) radiation shielding contracts (estimated $20M–$60M annually); (c) industrial ballast sales. **DFC matching:** 1:1 for DU fabrication facility investment up to $200M.

Contingencies

Radiation exposure incident: Worker overexposure triggers IAEA radiation safety protocol. Facility shutdown for investigation permitted; timeline extended. **Market**

substitution: If tungsten prices drop below DU competitiveness, Iran may shift to Pathway R (rare earth co-production) with 90-day JIG notification.

Point of No Return

Enrichment cascade tails discharge at ≤0.3% U-235 (Phase T.1). Once uranium is depleted below 0.3%, it is chemically and isotopically useless for weapons. The cascade would need to be completely rebuilt to re-enrich.

Reversibility Assessment

Negligible. Reversal would require: (a) collecting DU counterweights from aircraft and shielding from hospitals worldwide; (b) dissolving metal components; (c) re-enriching from 0.3% to 60% — requiring separative work equivalent to enriching from natural uranium. Total timeline: 36+ months with 100% detection probability.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran produces the depleted uranium that balances aircraft, shields cancer patients during radiotherapy, and secures shipping containers. Our nuclear expertise now serves everyday safety and health. The material has been depleted to levels far below any weapons relevance." **US:** "Iran's HEU has been depleted to 0.3% U-235 — one-tenth of natural uranium's enrichment level. It is now industrial metal, not weapons material. The IAEA verifies every gram." **Israel:** "The uranium is depleted to 0.3%. It would take more work to turn this into weapons material than starting from natural uranium. This is a one-way street, and the IAEA is watching every step."

NOA.25 Pathway U — Uranium-Carbon Nanotube Composite Materials for Advanced Industry

Objective

Convert Iran's HEU stockpile into depleted uranium dioxide (UO_2) nanoparticles embedded in carbon nanotube (CNT) matrices — an advanced composite material with extraordinary properties: radiation shielding superior to lead at half the weight, thermal management for spacecraft and nuclear thermal propulsion, high-strength lightweight structural components, and electromagnetic interference (EMI) shielding for sensitive electronics. Iran becomes a pioneer in a cutting-edge materials science sector with aerospace, defense-authorized, and medical applications.

Legal Basis

Article 1.5 (new Pathway U); NPT Article IV; IAEA safety standards for nanomaterial handling; OECD Working Party on Manufactured Nanomaterials safety framework;

space technology cooperation agreements; defense trade authorization for EMI shielding components. CBI retains legal title to all uranium and composite products until point of sale.

The Nanotube Composite Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium and all CNT-UO_2 composite products throughout development and manufacturing. Title transfers to aerospace, medical, or defense-authorized purchasers upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and composite products remain under IAEA dual-lock custody within Iran. The IAEA verifies that uranium is depleted to ≤0.3% U-235 and incorporated exclusively into nanotube composite matrices.

(c) Production Chain: HEU → deplete to ≤0.3% U-235 → oxidize to UO_2 nanoparticles (20–100 nm) via sol-gel or hydrothermal synthesis → disperse in surfactant solution → mix with CNT suspension → deposit by electrophoresis or spray coating → densify by spark plasma sintering → machine into shielding panels, thermal management substrates, or structural components → test and certify.

(d) Industrial End Use: Lightweight radiation shielding for spacecraft and mobile medical units; thermal interface materials for satellite electronics; high-strength low-weight structural components for aerospace; EMI shielding for defense communications and medical imaging equipment.

(e) Economic Model: Advanced CNT composites command $1,000–$10,000/kg. The global aerospace composites market exceeds $40B annually. Uranium-CNT shielding addresses a critical gap in lightweight radiation protection for Mars missions and mobile healthcare.

(f) Reversal Barrier: Uranium in CNT composites exists as nanoscale UO_2 particles chemically and physically bound within carbon nanotube matrices. Recovery would require: destroying composite structures; oxidizing carbon nanotubes; collecting nanoscale UO_2 particles; re-enriching from 0.3% to 60%. Each step is detectable and the final enrichment would require rebuilding the entire cascade.

Pre-Conditions

Table NOA-U1: Pathway U Additional Pre-Conditions

#	Pre-Condition	Verification
U-1	Nanomaterial synthesis facility IAEA-certified for UO_2 nanoparticle production	IAEA facility design information verification; nanosafety review

U-2	Carbon nanotube production or secure supply chain established	JIG supply chain audit
U-3	Spark plasma sintering (SPS) capability operational for composite densification	IAEA materials processing review
U-4	Nanomaterial safety and environmental release protocols approved	IAEA nanomaterial safety assessment
U-5	Letters of intent from aerospace or medical equipment manufacturers	JIG commercial review

Implementation Timeline

Table NOA-U2: Pathway U — 36-Month Nanotube Composite Development Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
U.1 Depletion	Months 1–6	HEU depleted to ≤0.3% U-235; UO_2 feedstock prepared	$80M Joint Disposition Fund disbursement; Tranche 2 release	IAEA tails assay verification
U.2 Nanoparticle Development	Months 7–16	UO_2 nanoparticle synthesis optimized; CNT dispersion protocol developed	US technical cooperation for nanomaterial science; Tranche 3 release	IAEA nanoparticle characterization; size distribution verification
U.3 Composite Fabrication	Months 17–28	SPS densification; component machining; mechanical and radiation testing	OFAC authorization for aerospace composite exports; DFC matching	IAEA composite certification; property testing

U.4 Commercial Production	**Months 29–36+**	**Scaled manufacturing; delivery to aerospace and medical equipment manufacturers**	**Continued sanctions relief; DFC expansion; US support for NASA/ESA supply chain qualification**	**Annual IAEA audit; delivery confirmation**

Synchronized Reciprocal Obligations

IAEA verification of U.2 (nanoparticle production) triggers US Tranche 3 release within 14 days. First commercial composite delivery (U.4) triggers DFC matching and US support for aerospace supply chain qualification.

Verification Protocol

IAEA continuous monitoring at nanoparticle synthesis and composite fabrication facilities. U-235 content verified as ≤0.3% at every stage. Each composite batch serialized and radiographically fingerprinted. Annual IAEA audit of production records, nanosafety compliance, and delivery documentation.

Financial Mechanism

Fund: $160M total — $80M at U.1; $80M at U.2 completion. Iranian revenue: (a) CNT-UO_2 composite sales ($1K–$10K/kg); (b) aerospace supply contracts (estimated $40M–$150M annually); (c) medical shielding contracts. DFC matching: 1:1 for nanomaterials facility investment up to $350M.

Contingencies

Technical underperformance: If composite properties fail to meet specifications after 24 months, pathway may pivot to Pathway T (DU metal) with 90-day JIG notification. Nanosafety concern: Environmental release of UO_2 nanoparticles triggers IAEA nanomaterial emergency protocol.

Point of No Return

Nanoparticle incorporation into CNT matrix via SPS densification (Phase U.3). Once UO_2 nanoparticles are sintered into carbon nanotube matrices, the uranium is dispersed at nanoscale and chemically bound — recovery to weapons-usable material is practically impossible.

Reversibility Assessment

Negligible. Reversal would require: (a) collecting composite components from spacecraft and medical equipment; (b) destroying carbon nanotube matrices; (c)

collecting nanoscale UO_2 particles; (d) re-enriching from 0.3% to 60%. This is technically and logistically impossible within any timeframe of strategic relevance.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran is pioneering uranium-carbon nanotube composites — materials that will shield astronauts on Mars missions and protect patients in mobile cancer clinics. We have transformed our nuclear program into advanced materials science. This is the future, not the past." US: "Iran's HEU has been depleted and transformed into nanoscale composite materials under IAEA lock and key. The uranium is embedded in carbon structures at the nanometer scale. Reversal is science fiction." Israel: "The uranium is now nanoparticles inside carbon tubes on satellites and medical equipment. Even theoretically, recovering weapons material from this would require technology that doesn't exist. We are satisfied."

NOA.26 Pathway V — Uranium-Based Catalysts for Petrochemical and Environmental Applications

Objective

Convert Iran's HEU stockpile into uranium-based catalysts for large-scale industrial processes: petroleum refining (fluid catalytic cracking, hydrocracking), plastics and polymer production, syngas conversion, automotive catalytic converters, and environmental remediation (NOx reduction, SOx scrubbing, wastewater treatment). Uranium oxide catalysts offer unique acid-base and redox properties that complement or exceed platinum-group metal catalysts at lower cost. Iran becomes a regional catalyst production hub serving the Middle East petrochemical industry.

Legal Basis

Article 1.5 (new Pathway V); NPT Article IV; IAEA safety standards for uranium in non-nuclear applications; petrochemical industry safety standards (API, ISO); environmental regulations for catalyst manufacturing and disposal. CBI retains legal title to all uranium and catalyst products until point of sale.

The Catalyst Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium and all catalyst products throughout manufacturing and use cycles. Title transfers to petrochemical operators upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and catalyst products remain under IAEA dual-lock custody within Iran during manufacturing. Spent catalysts from

international customers must be returned to Iran for IAEA-verified recycling or disposal.

(c) Production Chain: **HEU → deplete to ≤0.3% U-235 → oxidize to U_3O_8 → impregnate onto alumina, silica, or zeolite support → calcine and activate → form into pellets, spheres, or extrudates → package and export to refineries and chemical plants.**

(d) Industrial End Use: **Fluid catalytic cracking (FCC) catalysts for gasoline production; hydrocracking catalysts for diesel and jet fuel; Fischer-Tropsch catalysts for synthetic fuels; DeNOx catalysts for power plant emissions; wastewater treatment catalysts for heavy metal removal.**

(e) Economic Model: **Global petrochemical catalyst market exceeds $25B annually. Uranium-based FCC catalysts offer cost advantage over rare earth alternatives (lanthanum, cerium). Middle East refinery capacity is growing rapidly — Iran's geographic position provides logistics advantage.**

(f) Reversal Barrier: **Uranium in catalysts is depleted (≤0.3% U-235) and chemically bound to oxide forms (U_3O_8, UO_3) on inert supports. Spent catalysts contain uranium mixed with heavy metals and carbon deposits. Recovery would require: collecting spent catalysts from refineries; chemically separating uranium from support materials and contaminants; re-enriching from 0.3% to 60%. Each step is detectable and economically nonsensical.**

Pre-Conditions

Table NOA-V1: Pathway V Additional Pre-Conditions

#	Pre-Condition	Verification
V-1	Catalyst manufacturing facility IAEA-certified for depleted uranium handling	IAEA facility design information verification
V-2	Catalyst performance validated against commercial benchmarks (e.g., FCC conversion rates)	Independent laboratory testing report filed with JIG
V-3	Spent catalyst return and recycling program established	IAEA review of return logistics and recycling protocol
V-4	Letters of intent from at least two petrochemical refineries	JIG commercial review

V-5	Environmental permits for catalyst manufacturing and spent catalyst handling obtained	Iranian environmental agency + IAEA review

Implementation Timeline

Table NOA-V2: Pathway V — 18-Month Catalyst Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
V.1 Depletion	Months 1–6	HEU depleted to ≤0.3% U-235; U_3O_8 feedstock produced	$60M Joint Disposition Fund disbursement; Tranche 2 release	IAEA tails assay verification
V.2 Catalyst Formulation	Months 7–12	Uranium impregnation onto supports; activation; performance testing	OFAC authorization for catalyst exports; Tranche 3 release	IAEA catalyst certification; performance benchmark verification
V.3 Commercial Supply	Months 13–18+	Delivery to refineries; catalyst loading; performance monitoring; spent catalyst return	Continued sanctions relief; DFC matching; US support for Middle East refinery contracts	Annual IAEA audit; refinery performance confirmation; spent catalyst return tracking

Synchronized Reciprocal Obligations

IAEA verification of V.2 (catalyst formulation and performance testing) triggers US Tranche 3 release within 14 days. First commercial catalyst delivery (V.3) triggers DFC matching and US support for regional refinery contracts.

Verification Protocol

IAEA continuous monitoring at catalyst manufacturing facility. U-235 content verified as ≤0.3% for all catalyst batches. Each batch serialized. Spent catalyst return tracked via IAEA database. Annual IAEA audit of production, delivery, and return records. Environmental monitoring at manufacturing and recycling facilities.

Financial Mechanism

Fund: $120M total — $60M at V.1; $60M at V.2 completion. **Iranian revenue:** (a) catalyst sales ($5K–$50K per ton); (b) refinery supply contracts (estimated $30M–$100M annually); (c) spent catalyst recycling fees. **DFC matching:** 1:1 for catalyst plant investment up to $250M.

Contingencies

Performance shortfall: If uranium catalysts underperform commercial alternatives by >15% after 12 months, formulation reformulation permitted with 6-month extension. **Spent catalyst non-return:** If customer fails to return spent catalysts, IAEA investigates and JIG may suspend future deliveries to that customer.

Point of No Return

Catalyst activation and first commercial loading (Phase V.2). Once depleted uranium is chemically bound to catalyst supports and loaded into refinery reactors, the material has been transformed into an industrial chemical product.

Reversibility Assessment

Negligible. Reversal would require: (a) collecting spent catalysts from petrochemical plants worldwide; (b) chemically separating uranium from alumina/zeolite supports, heavy metals, and carbon deposits; (c) re-enriching from 0.3% to 60%. The collection alone is logistically prohibitive and IAEA-tracked.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran's nuclear material now catalyzes the production of cleaner fuels and plastics. Our uranium catalysts reduce refinery emissions and produce the building blocks of modern industry. We have turned a strategic threat into an industrial advantage." **US:** "Iran's HEU has been depleted and transformed into refinery catalysts under IAEA verification. The uranium is now in chemical plants making gasoline and treating wastewater. The material cannot be weaponized." **Israel:** "The uranium is inside oil refineries as a catalyst. To recover it, Iran would need to shut down refineries across the Middle East and collect spent catalysts. This is not a realistic path to a bomb."

NOA.27 Pathway W — Uranium-Phosphate Fertilizer and Agricultural Inputs

Objective

Convert Iran's HEU stockpile into uranium-bearing phosphate fertilizer and agricultural micronutrient products. Uranium, when depleted to natural levels and combined with phosphate rock, serves as a valuable micronutrient source and natural phosphorescent tracer for precision agriculture. Iran becomes a supplier of enhanced phosphate fertilizers for the Middle East and Central Asian agricultural markets — addressing food security while permanently transforming weapons-usable material into plant nutrients.

Legal Basis

Article 1.5 (new Pathway W); NPT Article IV; FAO/WHO Codex Alimentarius (food safety); IAEA safety standards for uranium in agricultural applications; environmental regulations for fertilizer manufacturing. CBI retains legal title to all uranium and fertilizer products until point of sale.

The Fertilizer Production Structure (Operative and Binding)

(a) Legal Title: CBI retains legal title to uranium and all fertilizer products throughout manufacturing. Title transfers to agricultural distributors upon IAEA-certified delivery.

(b) IAEA Dual-Lock Custody: All material and fertilizer products remain under IAEA dual-lock custody within Iran during manufacturing. The IAEA verifies that uranium is depleted to natural levels (0.711% U-238 dominant) and incorporated exclusively into phosphate fertilizer products.

(c) Production Chain: HEU → deplete to natural uranium levels (≤0.711% U-235) → oxidize to U_3O_8 → dissolve in phosphoric acid → combine with phosphate rock (morocco rock, apatite) → neutralize with ammonia → granulate into NPK fertilizer with uranium micronutrient content → package and distribute to agricultural markets.

(d) Agricultural End Use: Uranium-phosphate fertilizer for wheat, rice, and corn production; micronutrient enrichment for marginal soils in arid regions; phosphorescent soil tracer for precision agriculture (tracking fertilizer distribution patterns); hydroponic nutrient solutions for controlled-environment agriculture.

(e) Economic Model: Global phosphate fertilizer market exceeds $70B annually. Uranium-enhanced phosphate fertilizer commands a premium for micronutrient value and phosphorescent tracing capability. Iran's domestic agriculture sector (80M+ population) provides immediate market; regional export to Afghanistan, Central Asia, and Middle East expands revenue.

(f) Reversal Barrier: Uranium in fertilizer is at natural isotopic levels (0.711% U-235) and chemically bound in phosphate matrix, dispersed across millions of hectares of farmland. Recovery would require: collecting soil samples from millions of farms; chemically extracting trace uranium from phosphate rock matrix; enriching from 0.711% to 60% — requiring more separative work than starting from natural uranium ore. The idea is absurd.

Pre-Conditions

Table NOA-W1: Pathway W Additional Pre-Conditions

#	Pre-Condition	Verification
W-1	Fertilizer manufacturing facility IAEA-certified for uranium-phosphate processing	IAEA facility design information verification
W-2	Phosphate rock supply secured (domestic or imported)	JIG supply chain audit
W-3	Agricultural field trials demonstrating crop yield improvement completed	Independent agronomy report filed with JIG
W-4	Food safety assessment for uranium content in crops completed	FAO/WHO food safety review
W-5	Letters of intent from agricultural distributors or cooperatives	JIG commercial review

Implementation Timeline

Table NOA-W2: Pathway W — 14-Month Fertilizer Production Timeline

Phase	Duration	Iranian Obligations	US Reciprocal Obligations	Verification
W.1 Depletion	Months 1–4	HEU depleted to natural uranium levels (≤0.711% U-235)	$40M Joint Disposition Fund disbursement; Tranche 2 release	IAEA enrichment verification

W.2 Phosphate Integration	Months 5–8	Uranium-phosphate fertilizer formulation; granulation; field trials	OFAC authorization for fertilizer exports; Tranche 3 release	IAEA product certification; agronomic performance verification
W.3 Commercial Distribution	Months 9–14+	Delivery to agricultural distributors; field application; crop monitoring	Continued sanctions relief; DFC matching; US support for FAO food security programs	Annual IAEA audit; distributor delivery confirmation; soil uranium content monitoring

Synchronized Reciprocal Obligations

IAEA verification of W.2 (fertilizer formulation and field trial results) triggers US Tranche 3 release within 14 days. First commercial distribution season (W.3) triggers DFC matching and US support for FAO food security partnership.

Verification Protocol

IAEA continuous monitoring at fertilizer manufacturing facility. U-235 content verified as ≤0.711% for all production batches. Each batch serialized and tracked to agricultural distributor. Annual soil monitoring at field sites to verify uranium levels remain within safe agricultural limits. Independent FAO food safety audit of crops grown with uranium-phosphate fertilizer.

Financial Mechanism

Fund: $80M total — $40M at W.1; $40M at W.2 completion. **Iranian revenue:** (a) uranium-phosphate fertilizer sales ($200–$500/ton premium over standard NPK); (b) domestic agricultural supply contracts (estimated $50M–$150M annually); (c) regional export to Central Asia and Afghanistan. **DFC matching:** 1:1 for fertilizer plant modernization up to $150M.

Contingencies

Crop safety concern: If independent testing reveals uranium bioaccumulation in edible crops above Codex limits, production halt for reformulation permitted. IAEA and FAO joint review required before resumption. **Drought or climate event:** Reduced planting area does not affect compliance; fertilizer stored under IAEA seal for up to 24 months.

Point of No Return

Field application commencement (Phase W.3). Once uranium-phosphate fertilizer is applied to farmland, the uranium is dispersed across millions of hectares of soil, chemically bound in phosphate matrix, and taken up by growing crops. The material has literally become food.

Reversibility Assessment

None. Reversal would require: (a) collecting soil from millions of hectares of farmland across multiple countries; (b) extracting trace uranium from phosphate-rich soil at concentrations below natural background in many areas; (c) enriching from 0.711% to 60%. This is physically, chemically, economically, and logistically impossible. Pathway W represents permanent, irreversible transformation of HEU into agricultural productivity.

Narrative Framing Guidance (Non-Binding)

Iran: "Iran's nuclear material now feeds the world. Our uranium-phosphate fertilizer grows wheat for bread, rice for families, and corn for livestock. The same atoms that once threatened destruction now sustain life. This is the ultimate transformation — from swords to plowshares, from bombs to bread." **US:** "Iran's HEU has been depleted to natural levels and transformed into fertilizer under IAEA verification. The uranium is now in the soil, growing crops, feeding people. There is no path from a wheat field to a nuclear weapon." **Israel:** "The uranium has been spread across farm fields and turned into food. To recover it, Iran would need to dig up millions of acres of farmland and process the soil through an enrichment cascade. This is not proliferation — it is agriculture. We have no objection to Iranian wheat."

NOA.16 Cross-Pathway Provisions

NOA.16.1 Pathway Selection Decision Framework

The following decision matrix is provided as **non-binding guidance** for the Technical Working Group and both Parties' negotiators. It identifies the optimal pathway based on the Parties' highest-priority objectives. The matrix does not restrict pathway selection — Iran retains full choice among all twenty-three pathways subject only to the US veto on Pathways D, F, K, L, and M.

Table NOA-3: Pathway Selection Decision Matrix — Priority-Based Guidance

If Priority Is...	Recommended Pathway(s)	Why

Maximum sovereignty (material never leaves Iran)	**A, E, G, H, I, J, L**	**All keep material in Iran; A and E are fully domestic**
Maximum non-proliferation assurance (material farthest from weapons)	**B, C, D**	**Material physically leaves Iran; B and C most verifiable**
Maximum immediate economic benefit	**K, L**	**K provides $5–10B reconstruction; L provides immediate liquidity**
Maximum regional integration	**G, H**	**G creates regional fuel bank; H creates GCC joint venture**
Maximum humanitarian benefit	**I, A**	**I provides 20-year isotope supply; A produces medical LEU**
Maximum scientific benefit	**J, H**	**J creates international research program; H develops fuel fabrication expertise**
Maximum energy independence	**E**	**E creates domestic fuel supply for Bushehr**
Fastest implementation (shortest timeline)	**A (142 days), B (14+ days)**	**A is fastest complete disposition; B is fastest material relocation**
Maximum flexibility (easiest to switch)	**A, J, L**	**A switches until 75% complete; J has 5-year review; L allows annual adjustment**
Lowest US political cost	**A, I, G**	**A requires no foreign involvement; I is humanitarian; G is regional win-win**
Lowest Iranian political cost	**A, L, J**	**A is default; L monetizes without surrender; J is scientific prestige**
Hedging against single-pathway failure	**F (hybrid)**	**F divides stockpile across multiple pathways; built-in redundancy**

NOA.16.2 Material Accounting and Inventory Integrity

(a) Material Unaccounted For (MUF): The IAEA shall apply a MUF threshold of **1% of total initial inventory** for all pathways. MUF below 1% is presumed to be measurement uncertainty and does not constitute a violation. MUF between 1% and 3% triggers enhanced IAEA investigation and JIG notification. MUF above 3% triggers a

Tier 2 material disruption investigation. **(b) Inventory Frequency:** Physical inventory conducted: weekly during active downblending/conversion; monthly during storage; quarterly during trust drawdown periods; annually during research operations. **(c) Measurement Quality:** All IAEA measurements use calibrated equipment with traceability to international standards. Calibration records published to Dashboard annually. **(d) Shipper-Receiver Difference:** For any transfer between facilities (Pathways B, C, D, F), shipper-receiver difference must be within 0.5% of declared quantity. Differences exceeding 0.5% require investigation before next transfer.

NOA.16.3 Environmental Monitoring

(a) Baseline: Environmental radiation monitoring baseline established at all facilities before pathway operations begin. Baseline data published to Dashboard. **(b) Continuous monitoring:** Air, water, and soil monitoring at all active facilities. Data transmitted to Dashboard monthly. **(c) Annual environmental report:** IAEA publishes annual environmental impact report for all pathway operations. **(d) Emergency environmental protocol:** Any release above regulatory limits triggers immediate JIG notification, 24-hour investigation, and public disclosure within 72 hours.

NOA.16.4 Personnel and Training Standards

(a) Iranian personnel: All Iranian personnel involved in pathway operations must hold AEOI-certified qualifications. Training records maintained and available for IAEA review. **(b) IAEA inspectors:** Minimum 2 IAEA inspectors present during active operations. Inspector rotations every 90 days to prevent familiarity bias. **(c) Emergency training:** All personnel complete annual emergency response training. Drill records published to Dashboard. **(d) Foreign personnel:** Any foreign technical personnel (Pathways D, H, J) require Iranian visa approval and security clearance. Maximum 10 foreign personnel per facility at any time.

NOA.16.5 Information Security and Proprietary Protection

(a) Iranian proprietary information: Any Iranian proprietary technical information (facility designs, process parameters, operational data) observed by IAEA inspectors or foreign personnel is classified as RESTRICTED under Article 23. It may not be shared with the US or any third party without Iranian written consent. **(b) IAEA inspector confidentiality:** IAEA inspectors are bound by standard IAEA confidentiality obligations. Inspector reports contain only verification-relevant data, not proprietary technical details. **(c) Dashboard data:** Only aggregated, non-proprietary data appears on the public Dashboard. Facility-specific operational data is CONFIDENTIAL under Article 23.

NOA.16.6 Liability and Insurance Framework

(a) Nuclear liability: All pathway operations are covered by the Convention on Supplementary Compensation for Nuclear Damage (CSC) or equivalent bilateral arrangements. **(b) Transport insurance:** Minimum $500M coverage for all material transport. **(c) Facility insurance:** Minimum $1B coverage for all active facilities. **(d) Third-party liability:** Iran maintains liability coverage for all pathway operations on its territory. Transit states maintain coverage for transport on their territory. **(e) Insurance certificates:** Filed with Swiss escrow and published to Dashboard annually.

NOA.16.7 Dispute Resolution for Pathway Operations

(a) Technical disputes: Any dispute about pathway implementation (timeline, verification, safety) is a Tier 1 technical dispute resolved through JIG technical channels. **(b) Financial disputes:** Any dispute about Fund disbursement, valuation, or cost-sharing is resolved by the Swiss facilitator within 14 days. **(c) Political disputes:** Any dispute about pathway selection, switching, or termination activates Article 13.4 (Executive Consultation Window). **(d) Emergency disputes:** Safety or security emergencies bypass standard dispute resolution — immediate action permitted under Article 20 (TOS).

NOA.16.8 Amendment Protocol

This Annex may be amended by unanimous JIG consent (excluding affected Party's members for Party-specific amendments). Technical amendments (timeline adjustments, safety protocol updates) require only JIG majority vote. Political amendments (new pathways, veto changes) require both Parties' written consent. All amendments published to Dashboard before taking effect. Amendment does not reopen the core political bargain of Article 1.5.

NOA.16.9 Relationship to Other Framework Provisions

Supremacy: This Annex is subordinate to the Preamble, Articles 1–25, and Annex X. It governs technical implementation details only. **Conflict resolution:** If this Annex conflicts with an operative Article, the Article governs. If this Annex adds detail where an Article is silent, this Annex governs. **Severability:** If any provision of this Annex is found invalid, the remainder of the Annex and the Framework continue in full force.

> **NOA.16.10 BINDING EFFECT (OPERATIVE AND BINDING)**
>
> This Nuclear Operative Annex is **operative and binding** upon both Parties from the date of Framework signature. All provisions herein are subject to the Framework's dispute resolution, breach classification, and Temporary

> **Operational Suspension mechanisms. This Annex is not a gentlemen's agreement — it is a legally operative technical manual with the same binding force as the Articles it supports.**

Document Certification

This document constitutes the complete and authoritative text of **The Provisional Steps Framework — Phase I Stabilization Framework, Version 1.4**. It supersedes The Final Steps Accord (all versions v1.0 through v4.0) and incorporates all ten targeted structural modifications designed to transform the Accord from an uncompromising automated machine into a flexible diplomatic tool calibrated for signature.

The v1.0 revisions include:

1. **Recalibrated Lebanon Firewall (Appendix E):** Three-tier escalation ladder (10/25/50); proportionality and intent assessment; mandatory 72-hour diplomatic consult window; removal of JIG fast-track majority vote attribution.
2. **Rolling Executive Option (Appendix D):** Removal of punitive financial damage clauses for legislative delays; replacement of hard 180-day ratification deadline with 90-day executive extension decrees; optional legislative ratification with incentive benefits.
3. **In-Iran Default with Optional Oman Escrow (Article 1.6):** In-Iran downblending as default (material stays on Iranian soil); Optional Voluntary Oman Escrow Transfer by mutual agreement; clarified Day 75 timeline (Day 45 of Phase 2); no economic penalties; 30-day resolution window.
4. **30-Day Stabilization Blocks (Article 7):** Replacement of 15-day micro-cycles with 30-Day Stabilization Blocks; Mutual Escrow Deposit of $500M on Day 1; remaining $250M released in two $125M increments at Days 45 and 60; simultaneous release of funds and nuclear verification.
5. **Air-Gapped Batch-Verification Telemetry (Articles 1, 2, 3):** Replacement of real-time continuous streaming with compiled, encrypted batch delivery via neutral Swiss/Omani servers; continuous local recording but batched external transmission.
6. **Frozen-Revenue Deadlock Protection (Article 5.2):** Deletion of automatic 40% payout to Iran during budget deadlock; all toll revenues frozen in Omani

escrow until consensus budget signed.

7. **Unconditional Automated Day-7 Trigger (Article 6):** Removal of subjective Joint Review Board certification and operational discretion; replacement with unconditional synchronized 24-hour countdown trigger at T+168 hours.
8. **Targeted Bank Isolation (Article 9):** Replacement of sweeping SWIFT tier rollbacks with isolated clearing hubs for flagged institutions; no cascading disconnect for entire banking sector.
9. **US Dollar-for-Dollar Matching Guarantee (Article 10):** DFC matches every dollar Iran spends on certified civilian infrastructure with equivalent low-interest credit line; $5 billion annual cap.
10. **Executive Consultation Governance (Article 13):** Replacement of automatic Tier 2 penalties with mandatory 48-hour Executive Consultation Window between US Secretary of State and Iranian Foreign Minister; political leaders' veto over all economic penalties.

v1.1 Additional Surgical Modifications (19 Targeted Refinements)

1. **In-Iran Default for Day 45 (Article 1.6):** Default changed from Oman transfer to in-Iran downblending (Pathway A); material stays on Iranian soil respecting Khamenei's directive; Optional Voluntary Oman Escrow Lane added as cooperative acceleration by mutual agreement.
2. **Day 75 Clarification (Article 1.6):** "Day 45" clarified to mean "Day 45 of Phase 2 nuclear negotiating window" (Day 75 of Framework activation), giving negotiators a realistic 45-day Phase 2 timeline.
3. **Fuel Supply Guarantee Details (Article 2.1):** Specifics added: 120,000 kg LEU escrow at IAEA LEU Bank Kazakhstan; consortium members (Russia, China, France, UAE); 30-day proportional draw right for Iran.
4. **Weekly Batch Frequency (Article 3.3):** Batch delivery reduced from "daily or bi-weekly" to "weekly" with expedited 24-hour option for Level 4 anomalies by joint request.
5. **JHTA Budget Mediation (Article 5.2):** Mandatory 30-day mediation period added before revenue freeze; Swiss mediator and Omani co-chair facilitate; escalating timeline through JIG and Guarantor Matrix.
6. **Airspace Stand-Down Specifications (Article 6.5):** Specific parameters added for all three zones: Zone Alpha (UAVs to 25nm, manned above 15,000ft); Zone Beta (no armed drones within 12nm, passive radar); Zone Gamma (30-minute advance notification within 100nm).

7. **IRGC-Navy Hotline Protocol (Article 4.7):** New article establishing direct military-to-military deconfliction lines; 2-hour routine / 15-minute incident / immediate emergency response; daily testing at 06:00 UTC; Omani observers as neutral witnesses.
8. **Reduced Mutual Escrow Deposit (Article 7.1):** Day 1 deposit reduced from $750M to $500M ($250M humanitarian + $250M structural); remaining $250M released in two equal increments at Days 45 and 60.
9. **DFC Matching Floor (Article 10.1):** $1 billion minimum annual DFC matching commitment added; if DFC fails to meet minimum, cap lifts and Iran may draw from frozen assets without matching.
10. **DFC Disbursement Timeline (Article 10.1):** 90-day disbursement requirement added; delays trigger automatic LIBOR + 2% interest accrual.
11. **Targeted Bank Isolation Scope (Article 9.2):** Clarified to apply to flagged *institutions only* — not transaction streams, sectors, affiliates, or subsidiaries.
12. **Escrow Interest Accrual (Article 8.1-A):** New article: escrowed funds accrue interest at 6-month LIBOR; credited to Iranian Civilian Reconstruction Fund; not subject to clawback.
13. **Lebanon Escalation Ladder (Appendix E):** Single 50-casualty threshold replaced with three-tier ladder: Tier 1 (10-24: consultation only, no linkage); Tier 2 (25-49: JIG advisory opinion, linkage by mutual agreement); Tier 3 (50+: presumption of linkage, either party may veto).
14. **Media Blackout Extension (Article 13.5):** JIG may vote by majority to extend 72-hour blackout for one additional 72-hour period.
15. **Good-Faith Consultation Credit (Article 13.4):** 50% penalty reduction for first week if either Party initiates consultation within 24 hours of anomaly flag; applies once per 90-day period.
16. **Survival Core Automatic Renewal:** Extended to 365 days; automatic renewal for successive 365-day periods unless both Parties agree to terminate; Guarantor Matrix may extend by majority vote even if one Party objects.
17. **No Precedent Clause (Preamble):** New operative provision: Framework creates no legal or political precedent; each provision is *sui generis*; either Party may assert no precedential value in any forum.
18. **Expedited Withdrawal Option (Article 17.4):** Notice periods reducible upon payment of liquidated damages (10% of remaining escrow, capped at $250M/$500M/$1B for Phases I/II/III); shorter notice does not trigger military escalation.

19. **Air-Gapped Batch-Verification Telemetry (Articles 1, 2, 3):** Weekly batch delivery to neutral Swiss/Omani servers; expedited 24-hour option for Level 4 anomalies.

v1.1 Second Batch — 15 Additional Surgical Refinements

1. **Day 75 Language Clarification (Article 1.6):** Dual timeline confusion eliminated — "Day 75 of Framework activation" is the sole operative deadline. Parenthetical Phase 2 reference retained for context only.
2. **Technical Working Group (Article 1.5-A):** New article: IAEA, Iranian, and US technical experts convene on Day 30 of Phase 2 to provide feasibility assessments for all six disposition pathways. Findings published to Dashboard within 14 days. Removes "insufficient information" excuse.
3. **Fallback to In-Iran After Oman (Article 1.6-A):** If Oman escrow transfer occurs but no pathway agreed within 180 days, material returns to Esfahan at Iran's option. Return costs borne equally. Ensures Oman escrow is not a one-way door.
4. **Consortium Performance Penalty (Article 2.1):** 5% late-delivery penalty on undelivered fuel deposited to Iranian Civilian Reconstruction Fund. Three failures in 24 months triggers consortium membership review and potential replacement.
5. **Escalation of Force Protocol (Article 4.7):** Four-step binding protocol for Category A Bright-Line Period violations: 0-15 min hotline alert; 15-60 min extended stand-off; 1-4 hr JHTA emergency session; 4+ hr proportionate self-defense under Article 51. Pre-authorized at signature.
6. **Automatic Circuit Breaker (Article 5.2):** After 90 total days of deadlock (30-day mediation + 60-day freeze), baseline distribution percentages activate automatically for the remainder of the fiscal year. One-time circuit breaker prevents permanent escrow freeze.
7. **Humanitarian Cargo Exemption (Article 5.1):** UN OCHA/WHO-certified humanitarian cargo vessels exempt from all tolls regardless of type or flag. Non-revocable; survives any Framework suspension.
8. **Dual-Floor Escrow Interest (Article 8.1-A):** Interest rate is the greater of 6-month LIBOR or the average US-Iran inflation rate (CPI-U + official CPI, jointly certified by IMF). Protects real value of escrowed assets.
9. **De Minimis Error Buffer (Article 7.2):** Four categories of minor technical/administrative errors (OLEM interruptions under 2 hours, reporting

delays under 12 hours, equipment malfunction alerts, scheduled maintenance) do not count as violations and do not reset compliance clock.

10. **Over-Compliance Credit (Article 7.3-A):** Milestones completed ahead of schedule release tranches proportionally earlier. Credits may be banked and applied to future tranche conditions. JIG certifies quarterly.
11. **Private Sector Bridge (Article 10.1):** If DFC direct capacity exhausted, matching fulfilled via loan guarantees, political risk insurance, or syndicated loans. $1B minimum floor applies to total matching value (direct + guarantees + insurance).
12. **Legal Challenge Protection (Appendix D):** Neither Party may support, fund, or encourage private litigation attacking Framework-released assets. Best efforts to dismiss or stay such litigation. Willful violation = Tier 2 material disruption.
13. **Survival Core Automatic Renewal (Part V):** Automatic renewal for successive 365-day periods unless both Parties agree in writing to terminate. Guarantor Matrix may extend by majority vote even if one Party objects. Prevents humanitarian protections from being killed by a withdrawing party.
14. **Independent Legal Effect (Article 18):** No Automatic Military Escalation Clause is severable and survives any determination that other provisions are invalid. Creates independently enforceable obligations under international law, enforceable by Guarantor Matrix even post-termination.
15. **JIG Annual Public Report (Article 12.3-A):** Public annual report due within 90 days of each Framework anniversary covering compliance status, violations, economic benefits, regional incidents, and recommendations. Published unclassified to Swiss Dashboard. No intelligence sources disclosure required.

v1.1 Spoiler Resilience Hardening (8 Additional Articles)

1. **Article 19 — Spoiler Actor Management:** Defines spoiler actors (state/non-state); three-level attribution (Rogue/State-Tolerated/State-Directed) by SEAC; 14-day safe harbor for remediation — no step-back if prompt action taken.
2. **Article 19-A — False Flag Attack Protocol:** Rebuttable presumption of good faith; SEAC expedited investigation (48-hour preliminary, 14-day final); double Tier 2 penalty for confirmed false flag; forfeiture of self-defense claim; no military retaliation during investigation unless independent Article 51 criteria met.

3. **Article 13.6 — Anti-Gaming and Bad Faith Prohibition:** Five prohibited gaming actions (timing exploitation, de minimis pattern abuse, dispute triggering, consultation stalling, reset cycling); pattern detection (3+ instances in 180 days); de minimis suspension, reduced reset window, enhanced disclosure; Gaming Register published quarterly.
4. **Article 19-B — Spoiler Financing and Supply Tracking:** Supply Chain Monitoring Cell for weapons/financial/dual-use tracking; quarterly reporting obligations; Proscribed Spoiler Organization designation by JIG majority (asset freeze, transaction prohibition); non-designation is not admission.
5. **Article 14.3 — Rogue Commander Accountability:** Safe harbor for states addressing rogue commanders (24-hour condemnation, 14-day discipline, 30-day prevention, compensation); burden of proof on claiming state; repeat pattern presumption (3+ incidents in 180 days triggers Tier 1).
6. **Article 19-C — Information Warfare and Disinformation:** Four prohibited conduct categories (disinformation campaigns, fabricated dossiers, algorithmic amplification, official impersonation); 72-hour expedited JIG review; 0.5 WBI points per incident, 2.0 points triggers 90-day de minimis suspension; Media Coordination Cell for factual rebuttal.
7. **Article 8.4 — Spoiler Litigation Grandfathering:** Pre-Framework civilian contract protection; spoiler litigation definition; stay of proceedings mechanism; indemnification by supporting Party up to losses plus legal fees.
8. **Article 12.4 — Scenario Testing and Red Teaming:** Annual mandatory Spoiler Wargame; classified Scenario Library; neutral Red Team for adversarial testing; technical amendment authority by unanimous JIG consent (excludes affected Party for specific vulnerabilities).

v1.1 Cyber Resilience Hardening (5 New Articles + 4 Narrative Refinements)

Narrative Refinements:

- **Oman Pathway B clarification:** Added explicit statement that Oman is a neutral facilitator, not a US proxy, and no US personnel may be present during transfer unless mutually agreed.
- **Joint Disposition Fund security framing:** Added explicit rationale that the Fund is "a stabilization incentive designed to accelerate safe civilian conversion and reduce breakout risk — not a payment or bribe."

- **Iranian Narrative Frame reinforcement:** Added "IAEA custody is a temporary verification measure, not a transfer of sovereign control."
- **Phase 1/2 summary line:** Added "Phase 1 secures the material; Phase 2 decides its fate" as the architecture's elevator pitch.

Cyber Protection Articles:

1. **Article 19-D — Critical Infrastructure Cyber Protection:** Defines 6 categories of Critical Cyber Infrastructure; prohibits unauthorized access, DoS, malware, spoofing, and impersonation; mandates cryptographic hash chains, quarterly penetration testing by ETH Zurich, real-time intrusion detection; Tier 2/Tier 3 consequences; 90%+ confidence cyber attribution standard via SEAC Cyber Attribution Cell.
2. **Article 3.4 — Data Integrity and Anti-Spoofing:** SHA-3 cryptographic hash chain for immutable audit trail; dual-key HSM digital signatures; anti-replay nonces, ±5-second timestamp verification, geolocation tagging, device fingerprinting; ML-based anomaly detection for spoofing patterns; Tier 2 for falsified data, Tier 3 if falsification would have triggered military response.
3. **Article 19-E — Cyber Incident Response and Escalation:** Four-level incident classification (Advisory/Significant/Critical/Crippling) with proportionate consequences; 6-hour mandatory reporting; 72-hour preliminary / 21-day final forensic attribution; defensive measures permitted, offensive cyber retaliation requires JIG authorization, exhausted consultation window, and 14-day notice.
4. **Article 4.8 — Secure Communications and Encryption:** TLS 1.3, AES-256-GCM, end-to-end encryption, perfect forward secrecy; FIPS 140-3 Level 3 HSMs with quarterly rotation; air-gapped verification workstations at all 6 liaison offices; 24-month post-quantum cryptography transition; explicit prohibition on backdoors (Tier 2 violation); annual independent cryptographic audit.
5. **Article 3.5 — Sensor and Operational Technology Integrity:** Pre-approved neutral manufacturer sourcing; cryptographically signed firmware with dual-key updates; physical tamper protection with automatic key zeroization; 2:1 redundant sensor networks; manual verification override protocol for complete sensor failure; Tier 3 for deliberate sensor compromise.

v1.1 Twelve-Pathway Expansion: The original six-pathway disposition menu expanded to **twenty-three pathways**, adding: G — Regional LEU Bank hosted in Iran (regional custodian model); H — Joint Uranium Conversion Facility with Gulf states (economic interdependence); I — Uranium-for-Isotopes Swap (humanitarian barter); J — Scientific "Custody for Research" (shared

international research infrastructure); K — Uranium-Backed Reconstruction Bond ($5–10B asset-backed lending for peace); and L — Sovereign Fuel Leasing Trust (monetization without surrender, 10–15 year graduated drawdown). Joint Disposition Fund disbursement schedule updated for all twenty-three pathways. US veto applies to Pathways D, F, K, L, and M.

v1.1 Third Batch — 12 Political Viability Fixes

1. **Article 1.6(e) — In-Iran Default Backstop Sunset with Conditional Acceleration:** If no pathway selected by Day 365, mandatory phased transfer of 20% to Oman Escrow every 90 days. **Conditional acceleration:** If JIG certifies 90 days of non-engagement, backstop accelerates to Day 180. Iran avoids acceleration by negotiating in good faith.
2. **Appendix E.2.2(f) — Israeli Consultative Override:** If Israel certifies 50+ casualties with prima facie evidence of Iranian state direction, Guarantor Matrix may convene emergency session; two-thirds majority may refer for Tier 2 consideration, overriding a Party veto only if deemed manifestly unreasonable by Swiss facilitator. Not a veto — a supermajority override with a high bar.
3. **Article 4.2-C — GCC Side Letter + IMO Mediation:** PGSA recognition is without prejudice to GCC rights under UNCLOS; US and Oman issue Diplomatic Note to all GCC states affirming JHTA supremacy over PGSA transit claims; IMO binding mediation within 14 days for any PGSA-GCC dispute.
4. **Article 7.2 — Rolling Compliance Window:** Replaced reset rules with rolling 90-day window; Tier 2 violation reduces compliance by 10 days per day of persistence (capped at 30); no reset to zero; gaming becomes mathematically difficult.
5. **Article 19-E(c-1) — Provisional Cyber Attribution:** For Level 4 incidents, 72-hour provisional attribution finding triggers emergency safeguards (air-gapped backup activation, 7-day telemetry freeze); provisional finding not admissible in international tribunal.
6. **Appendix E.1 — Hezbollah Defensive-Only Posture:** Phase 1 parameters amended: defensive engagement of IDF forces entering Lebanese territory permitted under laws of armed conflict; offensive operations beyond 500m Blue Line buffer suspended. Legally defensible — "defense, not surrender."
7. **Article 9.2(b-1) — Clean-Record Bank Definition:** Objective standard: no OFAC designation; no Tier 2/3 violation in 180 days; FATF-compliant AML/CFT controls verified by independent audit. JIG may suspend by majority vote.

8. **Article 5.3 — Day 7 Reinsurance Activation:** $20B facility activates on Day 7 concurrent with blockade lift; first 200 transits auto-covered; rolling 90-day renewal. Market certainty from Day 1.
9. **Article 13.4(f) — 75% Early Engagement Credit:** Good-faith consultation within 24 hours reduces penalty by 75% for 14 days; no public attribution until 48-hour window expires; credit is bankable for future milestones. Makes early engagement dramatically cheaper than delay.
10. **Article 1.5-A — Fund as Drawdown Facility:** Restructured from one-time completion bonus to progress-based drawdown: 50% on commencement, 50% on completion; drawn funds credited on pathway switch with no clawback. Funds progress, not just endpoints.
11. **Article 3.3(d) — Unscheduled Batch for Credible Intelligence:** Either Party may request expedited 24-hour batch via JIG with credible intelligence; JIG decides within 6 hours; denial of reasonable request = Tier 1 dispute. Routine batched weekly; crisis expedited.
12. **Article 11.1-A — Survival Core $50M Escrow Fund:** Dedicated $50M escrow ($25M per Party); Swiss-administered; used for hotline, humanitarian corridor, diplomatic channel; no suspension during review pause; replenish below $10M within 30 days pro rata. A funded Survival Core survives political freezes.

v1.1 Israeli Transparency Architecture (4-Layer Fix)

1. **Article 16.2-A — Pre-Decision Consultative Window:** 24-hour binding notification to Israel before any US executive veto on a Lebanon incident; Israel may submit written assessment appended to JIG record; failure to notify renders veto procedurally invalid. Transparency with teeth — not a veto.
2. **Appendix E.5 — Levant Intelligence Review Board (LIRB):** Parallel technical body under Omani facilitation with US, Israel, Oman permanent members + UK/France rotating. No Iranian participation. Reviews Hezbollah/IRGC proxy intelligence; issues attribution assessments to US only; 12-hour emergency session for Tier 2/3 incidents; 24-hour preliminary finding to Secretary of State; sanitized summary shared with Iran. Advisory only — no operational authority.
3. **Article 12.3-A(f) — Israeli Caveat Right:** Israel may submit 500-word caveat to JIG Annual Public Report (factual corrections, alternative attribution); Iran

has no edit/veto right but may submit equal-length response. Labeled as Israeli submission. Transparency without authority.

4. **Annex X — Mutual Standstill Understanding (Separate US-Israel Diplomatic Note, NOT part of Framework):** Israel commits no preemptive strike against Iranian nuclear facilities during Phase I unless IAEA certifies Tier 3 (fissile diversion/weaponsization); US commits to coordinated response within 72 hours if IAEA certifies Tier 3. Iran never sees this note. Informal awareness through Oman channel. Deterrence through ambiguity.

Israel's Role in Four Layers: (1) Pre-Decision Window — binding 24-hour consultation, not a veto; (2) LIRB — privileged intelligence table, no Iranian visibility; (3) Public Caveat — voice in the record, no binding effect; (4) Standstill Understanding — separate diplomatic note, guaranteed partnership if red line crossed.

Total Modifications from v4.0 to v1.4 Final: 81 targeted surgical refinements — 10 foundational v1.0 reforms + 19 first-batch refinements + 15 second-batch refinements + 8 spoiler resilience articles + 5 cyber protection articles + 4 narrative refinements + 1 twelve-pathway expansion + 12 political viability fixes + 4 Israeli transparency architecture layers — transforming a rigid automated machine into a flexible, resilient, signable diplomatic tool with comprehensive spoiler actor immunity, full-spectrum cyber protection, an unprecedented disposition menu, institutionalized political protection for every stakeholder, and a sophisticated Israeli transparency architecture that prevents fait accompli without granting veto authority.

v1.2 Final-Mile Resilience Hardening (15 Changes — 3 Optional Edges + 12 Final-Mile Refinements)

1. **Edge — Appendix E.2.2(f) Override Bar Lowered:** Israeli consultative override now requires JIG two-thirds supermajority (not unanimity) to override a Party veto on Tier 3 linkage. Swiss facilitator may deem veto "manifestly unreasonable" as alternative trigger. Lower bar preserves Israeli voice without creating a minority veto.
2. **Edge — Survival Core Replenishment (Article 11.1-A):** Replenishment timeline shortened to 14 days; advance replenishment permitted when balance falls below $15M; interest at 6-month LIBOR on unreplenished balance. Prevents even temporary funding gaps.
3. **Edge — LIRB "Reasonable Suspicion" Defined (Appendix E.5):** Objective standard: signals intercepts showing command authority communication;

financial transfers correlating with operational capability; weapons-grade material trace evidence; or independent corroboration by two HUMINT sources. Prevents fishing expeditions while preserving legitimate intelligence concerns.

4. **Fix — Article 1.6(e) Good-Faith Negotiation Defined:** Four bright-line criteria: (i) attendance at scheduled sessions; (ii) submission of written position papers; (iii) engagement with technical working group findings; (iv) no public statements repudiating Framework objectives. Non-exhaustive — other evidence of good faith admissible.
5. **Fix — Article 1.6-A Default to Pathway A if Oman Stalls:** If Oman escrow transfer initiated but pathway selection stalled 60+ days, automatic default to Pathway A (in-Iran downblending) unless both Parties agree in writing to extension. Prevents Oman escrow from becoming diplomatic purgatory.
6. **Fix — Article 13.2 "Legitimate Research" Defined:** Rotor-related R&D permissible only if: published in peer-reviewed journal within 18 months; conducted at IAEA-monitored facility; no classified or military funding; and results shared with IAEA Technical Cooperation Programme. Narrow definition prevents breakout camouflage.
7. **Fix — Article 4.2-B PGSA-JHTA Dispute Escalation Ladder:** Three-step binding ladder: 72-hour bilateral technical consultation; 14-day Omani-facilitated political dialogue; IMO arbitration with 28-day binding decision. No unilateral PGSA suspension during escalation.
8. **Fix — Article 3.3(d) "Reasonable Request" Defined:** Four criteria for unscheduled batch: credible intelligence of anomaly (SIGINT, HUMINT, or physical detection); specific facility or timeline identified; not based solely on open-source media; and no more than 2 requests per 30 days absent extraordinary circumstances. Denial appealable to JIG within 6 hours.
9. **Fix — Article 7.4 Exchange Rate Floor for SDR:** SDR conversion uses 30-day trailing average. Floor: if 30-day average falls below 85% of Day 1 rate, floor applies at 85%. Protects Iranian purchasing power from currency manipulation.
10. **Fix — Article 7.3-A Scope-Based Over-Compliance Credit:** Milestones exceeding IAEA minimum safeguards (additional cameras, voluntary inspections, real-time data sharing) generate proportional tranche acceleration. JIG certifies scope and credit quarterly. Rewards transparency beyond minimum.
11. **Fix — Article 13.4 Automatic Stay for Non-Cooperation:** If either Party fails to designate representative within 48 hours of consultation request,

automatic 72-hour stay of all Tier 2 consequences. Clock resets on designation. Prevents procedural obstruction from triggering penalties.

12. **Fix — Article 13.3 Cross-Module Spillover Protection:** Nuclear compliance shortfalls may not automatically trigger maritime penalties; maritime incidents do not automatically freeze economic tranches. Each module assessed independently unless SEAC certifies deliberate cross-domain coordination. Prevents single-domain failures from collapsing entire Framework.
13. **Fix — Article 19-A(c) Expedited False Flag Pathway:** If SEAC preliminary finding indicates 90%+ confidence of false flag within 72 hours, JIG may order emergency safeguards: media blackout extension, air-gapped telemetry activation, 7-day consultation freeze. No penalties imposed during expedited review. Speed matters more than process when false flag probability is extreme.
14. **Fix — Article 19-D ITU Roster Fallback:** If ETH Zurich unavailable for penetration testing, ITU-maintained roster of certified cyber auditors serves as fallback. Rotation every 24 months to prevent capture. Ensures continuous pen testing capacity regardless of institutional availability.
15. **Fix — Appendix E.5(h) LIRB JIG-Admissible via Classified Sub-Channel:** LIRB findings transmitted to JIG through classified US-Swiss diplomatic channel; sanitized unclassified summary shared with Iran within 48 hours. Iran may challenge factual basis through standard JIG dispute process. Intelligence enters the record without exposing sources.

v1.2 Institutional Resilience Hardening (13 Changes — 11 Implemented + 2 Modified + 7 Assessed)

Based on independent structural review, the following modifications were implemented to transform the Framework from "extremely sophisticated draft" to "politically survivable, operationally executable, negotiation-grade framework":

1. **Article 19-F — Mandatory Cooling-Off Architecture:** Targeted mandatory pause periods before the highest-risk escalation triggers: 24-hour cooling-off before sanctions snapback; 48-hour cooling-off before public attribution; 12-hour cooling-off before enhanced inspections; 24-hour cooling-off before maritime posture escalation. Includes election-period extensions and anti-abuse provisions. Targeted implementation — not blanket pauses that create perverse incentives.

2. **Article 20 — Temporary Operational Suspension (TOS):** A distinct intermediate legal status between full compliance and total collapse. Allows pause of one or more modules without penalty, compliance clock reset, or aggregate termination. Six grounds for TOS (regional war, domestic crisis, leadership transition, cyber crisis, escrow failure, force majeure). Maximum 90 days per invocation, 180 days per year. Minimum survival conditions ensure hotline, Survival Core, and No Military Escalation survive all suspensions.
3. **Article 21 — Institutional Continuity Protocol:** Binds the Framework to institutions, not individuals. Survives all leadership transitions including death, resignation, election loss, incapacity, cabinet collapse, and regime change. Defines replacement timelines (7-30 days by role). Includes Election Period Protections (90 days pre-election: no new milestones, extended consultation windows, automatic media blackout). Interim authority is fully binding.
4. **Article 22 — Force Majeure for Regional Conflict Spillover:** Six defined force majeure events (Levant conflict, Yemen escalation, Iraq militia campaign, Strait closure, global shipping collapse, wider war). Automatic TOS activation upon JIG certification. No compliance penalty during force majeure. Survival Core continues. Good-faith obligations survive. Separates the Framework from the region's wars.
5. **Article 23 — Security Classification Protocol:** Three-tier classification system (UNCLASSIFIED political text, RESTRICTED technical annexes, CONFIDENTIAL implementation protocols). Defines handling requirements, leak investigation procedures, and public transparency overrides. Ensures operational security without reducing public accountability. Tier 1 public reports contain no classified details.
6. **Article 24 — Escrow Failure and Facilitator Contingencies:** Comprehensive backup architecture for escrow failure: two pre-negotiated backup banks (Alpha: secondary Swiss; Beta: neutral Asian/Gulf). Alternative settlement mechanisms for SWIFT disruption (crypto-asset, physical transfer for humanitarian only). Interest protection during transfer. Political risk insurance (100% principal coverage). Neutral-facilitator non-sanctions commitments.
7. **Article 25 — Joint Coordination Council:** Senior coordinating body across all nine Framework institutions (JIG, JHTA, PGSA, Consortium, Swiss, Omani, escrow, Working Groups, LIRB). Monthly coordination meetings, jurisdictional clarity protocol, public Jurisdictional Matrix, bureaucratic

warfare prevention. Does not replace or consolidate any body — adds connective tissue.

8. **Article 13.3-A — Partial Success Doctrine (Strengthened from existing Article 13.3):** Formalized binding doctrine: no single unresolved module invalidates the remainder. Module survival schedule defines minimum conditions for each module. Progressive reactivation without compliance debt. No aggregate termination for single-module failure. JIG Module Status Dashboard for public transparency. Transforms architecture from chain to web.
9. **Article 12.3-B — Monthly Compliance Bulletins (Expanded from Article 12.3-A):** Monthly four-page public bulletins: module status, milestones achieved, escrow releases, disputes opened/resolved, regional incidents (no attribution), upcoming milestones. Emergency bulletins within 24 hours for Tier 3 events. Plain language, trilingual, factual only, no attribution. Prevents information asymmetry weaponization.
10. **No Regime Change Objective (Preamble Operative Provision):** Binding affirmation that the Framework is exclusively stabilization and verification — not a regime change mechanism. Non-interference commitment. Verification-only access limits. No precondition for broader transformation. Good-faith interpretation clause. Politically essential for Tehran.
11. **Domestic Framing Autonomy (Preamble Operative Provision):** Complete sole authority over domestic presentation. Prohibited required terminology ("concession," "recognition," "normalization," etc.). No mutual recognition implication. Joint statements limited to technical language with advance approval. Internal documentation protection. Politically essential for both capitals.
12. **Appendix G — Unified Definitions Annex:** Authoritative definitions for all defined terms: compliance and breach definitions; actor and attribution definitions; conduct and intent definitions; material and custody definitions; escrow and financial definitions; operational status definitions. Definitional primacy over all other document definitions. Supplemental definition authority for JIG. Arms-control agreements live or die on definitions.
13. **Appendix H — Presidential Executive Summary (3 pages):** Page 1: What the Framework does (four objectives). Page 2: Political risk matrix with eight risks and specific mitigations. Page 3: Bottom line — success scenario, failure scenario, no-sign scenario. Designed for heads of state who read summaries, not full texts.

14. **Appendix I — Negotiators' Brief (10 pages):** Module-by-module negotiators' guidance: fallback positions, red lines, tactical notes for Nuclear, Maritime, Economic, Durability, Israeli Dimension, and Crisis Management. Critical operational numbers table (16 essential parameters). Designed for negotiators at the table who need immediate tactical guidance.
15. **Appendix J — Crisis Decision Flowchart (1 page):** Hour-by-hour decision tree for first 72 hours: H+0 through H+72 with specific actions, responsible parties, and fallback procedures. Five key decision points (TOS, nuclear, maritime, cyber, spoiler, regional spillover). Emergency contacts. Designed for crisis managers operating under time pressure.

v1.2 Final-Mile Resilience Hardening — Batch 2 (13 Changes, 1 Declined)

Based on independent structural review of the Nuclear Operative Annex and residual risk areas, the following modifications were implemented to close identified loopholes and clarify ambiguous provisions:

1. **NOA.0.3 — Pathway Phasing (Change #1):** Only Pathways A, B, and C are Phase I executable (Days 1–180). Pathways D–L are templates for Phase II/III successor negotiation. This Framework creates no presumption that Iran must select or execute any Pathway D–L during Phase I. Technical Working Group uses D–L specifications as reference only. Prevents Phase I collapse over Phase II ambition.
2. **Article 1.3 — Iranian Consent for US Technical Observers (Change #2):** US technical personnel may be present at IAEA custody facilities only upon prior written consent of Iran, not unreasonably withheld. Consent may be revoked with 14 days' notice. JIG dispute resolution for unreasonable denial. Sovereignty protection with balanced access.
3. **Article 13.2 — De Minimis Research Exemption (Change #3):** JIG may authorize by majority vote specific, time-limited (30-day max), low-enrichment (3.67% cap) research outside defined categories. Maximum 2 authorizations per 365 days. IAEA supervision required. Appeals to Guarantor Matrix. Prevents abuse while adding genuine flexibility.
4. **Article 3.3(d) — Facility-Specific Unscheduled Batch Limits (Change #4):** No facility subject to more than 4 unscheduled batch requests per 180 days. Pattern of excessive requests (8+ across all facilities in 90 days) referred to JIG

for gaming determination. Unanimous JIG vote for extraordinary exceptions. Prevents inspection harassment.

5. **Article 4.5-A — Commercial Vessel Protection (Change #5):** Vessel operators may request emergency JHTA review of arbitrary PGSA detention within 24 hours. JHTA review with Iran's vote excluded. Continued detention beyond 7 days after adverse JHTA finding escalates to Tier 1 dispute. No military intervention. Good-faith detention presumption with rebuttable burden. Protects shipping without undermining sovereignty.
6. **Article 5.1 — Phase I Transit Fee Caps (Change #6):** Binding, non-waivable caps: Tier 2 maximum $0.15/ton deadweight; Tier 3 maximum $0.25/ton or $35,000/VLCC flat. Caps survive Phase I; Phase II renegotiation. Addresses US position on fee predictability while preserving Iranian revenue.
7. **Article 9.2 — Bank Rehabilitation Pathway (Change #7):** Banks on isolated hub for 90+ days may petition JIG for rehabilitation review. JIG majority vote (excluding affected Party) determines return to standard SWIFT access. Petition allowed once per 180 days. Appeals to Guarantor Matrix. No bank faces indefinite isolation without redemption path.
8. **Article 8.3-A — Humanitarian Carve-Out for Future UNSC Sanctions (Change #8):** US commits to use diplomatic powers to secure humanitarian exemptions for escrow, humanitarian transactions, and Survival Core in any future UNSC sanctions. Best-efforts, not a guarantee. Survival Core designated humanitarian-protected. Honest acknowledgment of limits protects credibility.
9. **Article 13.6 — Cumulative Breach Provision (Change #9):** 180-day gaming lookback window is rolling (resets with each new instance). Additionally, 5+ gaming instances in any 365-day period constitutes a pattern regardless of distribution. JIG maintains cumulative counter. Instances older than 365 days not counted. Prevents "wait 181 days" gaming.
10. **Article 13.3 — Supermajority for Cross-Module Spillover (Change #10):** Cross-module spillover threshold lowered from unanimous to two-thirds majority (excluding affected Party). Swiss facilitator guided by three criteria: verification system impact, cross-module bad faith pattern, economic/military consequences. Unanimity was too easy to block.
11. **Appendix E.2.2(f) — JIG Admissibility Review of Israeli Evidence (Change #11):** 24-hour JIG admissibility review before Guarantor Matrix emergency session. Evidence admitted if not manifestly unreliable, supports reasonable inference, and not solely open-source/single-source. Procedural only — does

not evaluate truth. Israel may resubmit with corroboration within 7 days. Screens frivolous claims.

12. **Appendix E.5 — Public Summary of LIRB Findings (Change #12):** JIG publishes Public Summary of LIRB preliminary findings to Dashboard for Tier 2/3 proceedings. Includes incident details, attribution conclusion with confidence, factual basis in general terms, dissenting views. Iran may submit equal-length response. Redaction disputes resolved by JIG Chair. Sources protected.
13. **NOA Pathway L — Annual Verification Leverage (Prelim #3):** US retains annual verification leverage through graduated drawdown. JIG may suspend further credit issuance if Iran fails annual obligations. No cash clawed back — but no new cash until compliance restored. Suspension is benefit withholding, not Tier 2 breach. Preserves US Congressional support.

Assessed but Not Implemented (1 proposal declined):

- **Additional Israeli Override Trigger (Prelim #2):** Declined — the existing two-trigger architecture (Swiss facilitator "manifestly unreasonable" determination OR JIG two-thirds majority on good-faith consultation failure) already provides multiple paths for Israeli concerns. Adding a simple majority trigger would make override too easy and undermine the fundamental protection the veto architecture provides. Israel may grumble but the existing compromise is fair.

v1.4 Nuclear Operative Annex (1 Comprehensive Technical Annex — 23 Pathways including 10 new In-Iran Transformation Pathways N–W)

The **Nuclear Operative Annex (NOA)** transforms the twenty-three political pathway concepts of Article 1.5 into fully executable technical procedures. Each pathway includes: objective and legal basis; pre-conditions; phased implementation timeline with synchronized reciprocal obligations; IAEA verification protocol; environmental, safety, and security standards; financial mechanism; contingencies and failure modes; point of no return; and narrative framing guidance. Cross-pathway provisions cover: pathway selection decision matrix; material accounting (1% MUF threshold); environmental monitoring; personnel standards; information security; liability and insurance; dispute resolution; and amendment protocol.

1. **NOA.4 Pathway A:** In-Iran downblending to medical-grade LEU at Esfahan — 142 days, $50M Fund, 75% point of no return.

2. **NOA.5 Pathway B:** Voluntary Oman escrow transfer — 14-day transport, $100M Fund, 365-day return option, no US personnel during transfer.
3. **NOA.6 Pathway C:** Downblending in Iran, transfer to IAEA LEU Bank Kazakhstan — 75 days, $100M Fund, preferential draw rights.
4. **NOA.7 Pathway D:** Transfer to Rosatom Russia — 60–90 days, joint concurrence required, no Fund, OFAC/EU sanctions coordination.
5. **NOA.8 Pathway E:** Bushehr reactor fuel conversion — 180 days, $100M Fund, VVER-1000 fuel qualification.
6. **NOA.9 Pathway F:** Hybrid multi-destination — parallel processing, prorated Fund, no batch exceeds 50%, Partial Success Doctrine built-in.
7. **NOA.10 Pathway G:** Regional LEU Bank in Iran — 210 days, $100M Fund, 1.5% custodial fee, regional customer base.
8. **NOA.11 Pathway H:** Joint conversion facility with Gulf states — 365 days, $100M Fund, 45/40/15 joint venture structure.
9. **NOA.12 Pathway I:** Uranium-for-isotopes swap — 120 days, $100M Fund, $200M supply guarantee, 20-year isotope supply.
10. **NOA.13 Pathway J:** Scientific custody for research — indefinite, $75M Fund, 50 scientists trained annually, 5-year reviews.
11. **NOA.14 Pathway K:** Uranium-backed reconstruction bond — 5 years, $150M Fund, $5–10B reconstruction capital, independent valuation.
12. **NOA.15 Pathway L:** Sovereign fuel leasing trust — 10–15 years, $75M Fund, immediate liquidity, graduated annual drawdown.
13. **NOA.17 Pathway M:** Sovereign Trustee Export (Oman Trust) — 14-day transport + indefinite trust, $100M Fund, Iran retains sovereign title and beneficial ownership as trustee, immediate liquidity via trust certificates, return permitted only for peaceful downblending, joint concurrence required. Restored from Stability First Enhanced Accord v6.1, Section 12.2 (Decoupled Ownership and Custody).
14. **NOA.18 Pathway N:** Reactor Fuel Fabrication and Export — 24-month timeline, $120M Fund, downblend to ≤19.75% U-235, fabricate fuel assemblies, export to IAEA-safeguarded reactors. In-Iran transformation. Economic asset creation.
15. **NOA.19 Pathway O:** Stable Isotope Production — 18-month timeline, $100M Fund, medical and industrial isotopes from LEU targets. In-Iran transformation. Humanitarian application.

16. **NOA.20 Pathway P:** Betavoltaic Battery Manufacturing — 30-month timeline, $140M Fund, nuclear batteries for pacemakers, satellites, sensors. In-Iran transformation. Advanced technology export.
17. **NOA.21 Pathway Q:** Fluorochemical Feedstock Production — 20-month timeline, $120M Fund, UF_6-derived HF, WF_6, $LiPF_6$ for semiconductors and batteries. In-Iran transformation. Industrial chemical hub.
18. **NOA.22 Pathway R:** Rare Earth Element Co-Production — 30-month timeline, $140M Fund, Nd, Dy, Tb, Eu from uranium-bearing feedstock. In-Iran transformation. Critical minerals supply.
19. **NOA.23 Pathway S:** Uranium Silicide Fuel for Research Reactors — 22-month timeline, $120M Fund, U_3Si_2 fuel plates for global RERTR program. In-Iran transformation. Non-proliferation enabler.
20. **NOA.24 Pathway T:** Depleted Uranium for Industrial Armor/Counterweights — 16-month timeline, $100M Fund, DU metal for aircraft, shielding, ballast. In-Iran transformation. 0.3% U-235 — permanently depleted.
21. **NOA.25 Pathway U:** Uranium-Carbon Nanotube Composites — 36-month timeline, $160M Fund, UO_2 nanoparticles in CNT matrices for aerospace and medical shielding. In-Iran transformation. Advanced materials pioneer.
22. **NOA.26 Pathway V:** Uranium Catalysts for Petrochemicals — 18-month timeline, $120M Fund, U_3O_8 catalysts for FCC, hydrocracking, DeNOx. In-Iran transformation. Regional refinery supply.
23. **NOA.27 Pathway W:** Uranium-Phosphate Fertilizer — 14-month timeline, $80M Fund, depleted uranium fertilizer for agriculture. In-Iran transformation. Irreversible — material becomes food.
24. **NOA.16 Cross-Pathway:** Decision matrix, material accounting (1% MUF), environmental monitoring, personnel standards, information security, liability framework, dispute resolution, amendment protocol, binding effect clause.

Assessed but Not Implemented (7 proposals reviewed and declined):

- **Additional Israeli Deconfliction Language:** Declined — existing 4-layer architecture (Pre-Decision Window, LIRB, Public Caveat, Standstill Annex) is comprehensive. Additional language risks over-engineering and Iranian perception of imbalance.
- **Economic Shock Stabilizers (oil market panic):** Declined — out of scope. OPEC-market coordination risks scope creep. Existing reinsurance (Article 5.3) and DFC matching (Article 10) provide sufficient economic protection.

- **Third-Party Proxy Attribution Thresholds:** Declined — existing architecture (Articles 19, 19-A, 14.3, Appendix E tiers) already provides comprehensive attribution standards. Additional thresholds risk over-engineering.
- **Reduce "Shall" Binding Language:** Declined — would undermine framework integrity. Flexibility comes through consultation windows, human-in-the-loop governance, and TOS, not weakened operative language.
- **Additional Israeli Red-Line Channels:** Declined — Standstill Annex and LIRB already provide red-line coordination. Further channels risk information overload and conflicting signals.
- **Consolidate Governance Stack:** Modified — instead of consolidation (which creates functional gaps), implemented Joint Coordination Council (Article 25) as coordinating layer.
- **Classified Annex Migration:** Modified — instead of migrating content (impractical pre-signature), implemented Security Classification Protocol (Article 23) as post-signature compartmentalization framework.

v1.2 Human-in-the-Loop Architecture (10 Changes — Full Humanization of Automation)

The following modifications inject mandatory human judgment at every critical automated decision point in the Framework. The theme: **the machine advises; the humans command**.

1. **Article 6.3 — Conditional Day-7 Trigger with Human Confirmation (Change #1):** Replaced unconditional automated blockade lift with conditional automated trigger requiring manual confirmation from both capitals' defense ministers within 12 hours. IAEA verification starts the clock; humans pull the trigger. Failure to confirm triggers Executive Consultation Window, not automatic penalty. Preserves presidential and supreme leader command authority.
2. **Article 13.4(c) — Discretionary Review replacing Absolute Veto (Change #2):** Replaced "absolute executive veto" with four-option discretionary review: approve, reduce up to 50%, defer 14 days, or reject with written finding. Rejection without written finding = Tier 1 dispute. Leaders have options, not a blank check.
3. **Article 12.3-B — Political Judgment Override (Change #3):** Either Party may refer JIG technical findings to Guarantor Matrix for political review. Guarantor Matrix supersedes JIG on consequences. 14-day timeline. Technical

record preserved. Exception: cannot override Tier 3 fissile diversion or No Military Escalation Clause. Technical experts inform; politicians decide.

4. **Article 13.3-B — "Second Look" Review at 30 Days (Change #4):** All Tier 2/3 penalties subject to mandatory Guarantor Matrix review at 30 days. Options: affirm, reduce, suspend, or terminate. Repeat every 30 days. Political review, not technical. Prevents perpetual punishment. Restored compliance leads to restored benefits.
5. **Article 12.5 — Consensus Minus One (Change #5):** All significant JIG decisions require consensus minus one (all members except the Party whose conduct is under review). Deadlocks auto-escalate to Guarantor Matrix. Routine technical matters stay majority vote. No Party can be outvoted on its own conduct. Fairness through structure.
6. **Article 7.5 — Political Discretion Fund (Change #6):** $50M fund ($25M each), exempt from step-back, unanimous + Swiss facilitator disbursement. For goodwill projects, partial compliance rewards, and confidence-building. Acknowledges compliance is a spectrum, not binary. Non-strategic amount with political utility.
7. **Preamble — Narrative Dispute Resolution (Change #7):** When domestic framing crosses from protected interpretation to material misrepresentation, Swiss-facilitated session with communications advisors. Non-binding finding. 7-day clarifying statement. No penalties. Political process for political problem. Good-faith standard prevents abuse.
8. **Article 3.4-A — Human Review of All AI Findings (Change #8):** All AI anomaly flags reviewed by human JIG analyst within 24 hours. Three options: confirm, flag for investigation, or dismiss. No penalty solely on AI finding. AI Performance Log published quarterly. Recalibration if false positive rate exceeds 15%. Exception: Tier 3 fissile diversion emergency. The AI advises; the human decides.
9. **Article 12.3-A(g) — Executive Summary of Political Consequences (Change #9):** Every JIG report includes one-paragraph plain-language summary on page one answering four questions: compliance issue? political consultation needed? regional incident? emergency measure invoked? Not binding. Respects political time and attention span.
10. **Article 26 — Political Consultation Channel (Change #10):** Secure Omani-facilitated leader-to-leader channel, separate from all technical/executive mechanisms. Mutual political override allowed for extraordinary

circumstances with written confirmation, 30-day auto-expiry, no precedent. No intelligence collection. Political leadership, not mechanical submission.

Why This Batch Matters: The Framework was becoming a machine — 197 pages of automated triggers, majority votes, and mechanical penalties. Machines are efficient but they cannot handle ambiguity, context, or leadership. These 10 changes ensure that every critical decision has a human in the loop: humans confirm triggers, humans review penalties, humans override technical findings, humans exercise political discretion, humans review AI, humans consult directly. The architecture is still robust — but it is no longer a prison. It is a tool that serves political leadership rather than replacing it.

v1.2 Finishing Polish — Usability, Calibration, and Political Smoothing (10 Changes)

The following modifications address formatting, human-in-the-loop calibration, political smoothing, and NOA clarification — transforming the document from "sophisticated draft" to "negotiation-ready framework."

1. **One-Page Executive Summary (Change #1):** Added immediately after Preamble. 60-second orientation for busy leaders. Four things the Framework does, six core principles, leadership protections summary. Explicitly non-binding. Boxed format for visual distinction.
2. **Expanded Table of Contents + Quick Reference (Change #2):** TOC now includes all Articles (1-27), all Appendices (A-J), all NOA pathways (A-W) with nested sub-list, and Annex X. Crisis Quick Reference box at top of TOC page: hotline, Swiss facilitator, Omani backchannel, JIG duty officer, emergency bulletin contacts.
3. **Standardized Heading Hierarchy (Change #3):** NOA heading hierarchy clarified: h2 for each pathway, h3 for general provisions (NOA.1-NOA.3, NOA.16), h4 for sub-sections. Added explicit hierarchy guide box in NOA introduction. All 23 pathways follow identical structure.
4. **Discretionary Review + Political Override Clarification (Change #4):** Added sequential relationship paragraph to Article 13.4: discretionary review = first line (before activation); political override = second line (after activation); Second Look = third line (30-day mandatory). Sequential, not conflicting. Neither may circumvent Tier 3 diversion or No Military Escalation.
5. **TOS Invocation Cap — "Cooling-Off on Cooling-Off" (Change #5):** Max 3 TOS invocations per 365-day period. Fourth = presumptive bad faith, auto-referral

to Guarantor Matrix. Two-thirds override for extraordinary circumstances. TOS Invocation Log on Dashboard. Prevents TOS as de facto termination.

6. **Article 27 — Annual Human-in-the-Loop Audit (Change #6):** Independent annual audit of all automated mechanisms. Three experts (one per Party + Swiss). Eight mechanisms audited. Three core questions: machines making human decisions? consequences humans would reject? modifications to increase discretion? Report published to Dashboard within 90 days of anniversary. Advisory but creates presumption for review.
7. **Mutual Disappointment Acknowledgment (Change #7):** Preamble addition. Both sides made concessions. Implementation will be imperfect. Measure is not preventing every dispute — it's resolving disputes without war. Non-binding, interpretive. Reduces political cost of signing.
8. **Phase I Completion Ceremony (Change #8):** Article 17.3. If 180 days verified compliance with no Tier 3 and max one Tier 2: joint public ceremony at neutral location. Joint statement, moment of silence, symbolic act. Not a treaty, not recognition, not Phase II precedent. Optional participation without penalty. Gives both sides a reason to succeed.
9. **Preferred Choice, Not Binding Commitment (Change #9):** Article 1.5-B and NOA.0.3(e). Pathway selection is a preferred choice for Phase II negotiation, not a binding commitment to execute. Iran may change at any time. US veto is negotiation, not breach. No presumption that preferred = final. Removes lock-in fear.
10. **Visual Checkbox Template (Change #10):** Non-binding pathway selection form after Article 1.5-B. All 23 pathways with checkboxes, Phase I/II executable status, US veto markers (*). "No selection at this time" option. For planning only. Visual clarity reduces negotiation friction.

Why This Batch Matters: The Framework was technically comprehensive — 243 pages, 28 articles, 23 nuclear pathways, 9 institutional bodies. But technical comprehensiveness does not equal usability. A foreign minister cannot navigate 217 pages in a crisis. A president cannot surrender command authority to a sensor. A negotiator cannot sign a document that locks them into multi-year commitments they cannot deliver. These 10 changes address the human dimensions: orientation (summary), navigation (TOC), calibration (sequential override, TOS cap, audit), political survival (disappointment acknowledgment, ceremony), and negotiation clarity (preferred choice, checkbox). The result is a document that is not just sophisticated — it is usable, signable, and survivable.

v1.4 Final Contacts & Activation Readiness (3 Changes)

The following final modifications ensure the Framework is operationally ready for signature and activation — not just legally complete, but functionally executable.

1. **Emergency Contacts → Blank Fill-In Lines (Change #1):** All placeholder contact numbers ("XX-XXX-XXXX", "XXXX-XXXX") in the Quick Reference box (TOC), Appendix J (Crisis Flowchart), and throughout the document have been replaced with blank fill-in lines ("_______"). A red-bordered instruction box added: "Contact numbers to be designated per Article 28 before Framework activation. No placeholders permitted." Each contact line includes a "Last verified" date field. Ensures the document is print-ready for manual completion.
2. **Article 28 — Crisis Contacts Designation Protocol (Change #2):** New binding article requiring: 14-day pre-signature designation of all crisis contacts (hotline, backchannel, JIG duty officer, Swiss facilitator, Dashboard); daily 06:00 UTC hotline testing with 15-minute response requirement; weekly Omani backchannel testing; 7-day written notice for contact updates; 48-hour interim contact designation during leadership transitions; secure Crisis Contact Registry on access-controlled Dashboard section; **Framework cannot activate without certification** — activation postponed in 7-day increments until all contacts are designated, tested, and authenticated; unilateral postponement right for non-designation; bad-faith referral after three postponements. No placeholders. Real people, real numbers, real testing.
3. **Hotline Testing in Executive Summary (Change #3):** Added "Hotline tested daily at 06:00 UTC. Failure to answer is a Tier 1 dispute. Crisis contacts must be real, designated, and certified before activation (Article 28)." to the Core Principles box in the One-Page Executive Summary. Ensures the testing requirement appears in the 60-second orientation that every leader reads.

Why This Final Batch Matters: A 234-page Framework with blank phone numbers is a Framework that cannot function. The Crisis Contacts Designation Protocol ensures that before either leader signs, there are real people with real numbers who have been tested and authenticated. It transforms the document from a legal architecture into an operational system. The Framework is now complete: legally binding, politically survivable, human-in-the-loop, and operationally executable.

Total Modifications from v4.0 to v1.4 Final: 135 targeted surgical refinements (96 v1.1 + 15 v1.2 + 1 NOA + 13 batch 2 + 10 human-in-the-loop) — 10 foundational v1.0

reforms + 19 first-batch refinements + 15 second-batch refinements + 8 spoiler resilience articles + 5 cyber protection articles + 4 narrative refinements + 1 twelve-pathway expansion + 12 political viability fixes + 4 Israeli transparency architecture layers + 3 residual risk patches + 3 optional edges + 12 final-mile refinements — transforming a rigid automated machine into a flexible, resilient, signable diplomatic tool with comprehensive spoiler actor immunity, full-spectrum cyber protection, an unprecedented disposition menu, institutionalized political protection for every stakeholder, a sophisticated Israeli transparency architecture that prevents fait accompli without granting veto authority, and final-mile definitional precision that closes every identified loophole.

This Framework is designed as a Phase I, time-limited, politically flexible framework. It creates no presumption for Phase II or Phase III terms. Both parties retain their full sovereign rights and full negotiating positions for all subsequent phases. The Rolling Horizon mechanism ensures that the agreement continues only so long as both parties are in verified compliance, with orderly withdrawal procedures available at all times.

Annex X: Mutual Standstill Understanding — US-Israel Diplomatic Note (NOT PART OF THE FRAMEWORK)

This Annex is a separate US-Israel diplomatic note. It is not annexed to, incorporated into, or made part of the Provisional Steps Framework. The Islamic Republic of Iran has not seen, reviewed, or agreed to this document. It creates no obligation on Iran. It is binding only as between the United States and the State of Israel.

Mutual Standstill Understanding

(a) Israeli Commitment: The State of Israel commits that it will **not conduct a preemptive strike against Iranian nuclear facilities** during Phase I (Days 1–180) of the Provisional Steps Framework unless the IAEA certifies a **Tier 3 Deliberate Hostile Violation** involving fissile material diversion, weaponsization activity, or a confirmed breach of the HEU custody arrangement that poses an imminent threat to Israeli security.

(b) US Commitment: If the IAEA certifies a Tier 3 Deliberate Hostile Violation as described in (a), the United States commits to **join Israel in a coordinated response within 72 hours** of IAEA certification, rather than leaving Israel to

act alone. The coordinated response may include diplomatic, economic, or military measures as mutually agreed between the US and Israel through the Tel Aviv Channel.

(c) Scope: This Standstill applies only to **Iranian nuclear facilities** under Framework verification. It does not restrict Israeli military operations against non-n